# Geometric Representation Theory

**— *EYNTKA* Series —**

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# *Introduction*

## What This Book Is About

This book develops the structure theory and representation theory of algebraic groups over a field, from the definition of a group scheme through the Dynkin diagram classification of reductive groups and the Weyl character formula. It is, at its core, the story the correspondence between geometry and combinatorics: a reductive algebraic group $G$ — an infinite-dimensional geometric object, defined by polynomial equations and carrying a group law — is completely determined by a finite configuration of vectors in a Euclidean space (its root system), and its representation theory is governed by a single explicit formula (the Weyl character formula) that can be evaluated by hand for any given input. The passage from the geometric object to the combinatorial data loses no information, and the combinatorial data has a finite, complete classification: the Dynkin diagrams. This is one of the great classification theorems of twentieth-century mathematics, and the purpose of this book is to give a self-contained, modern account of the ideas and techniques that make it work.

The subject goes by several overlapping names. *Linear algebraic groups* emphasizes the objects (group varieties, or more generally group schemes, that admit faithful linear representations). *Lie theory* emphasizes the infinitesimal technique (the passage from a group to its Lie algebra, which linearizes the group law and makes it amenable to linear-algebraic computation). *Geometric representation theory* emphasizes the application of geometric methods (sheaf cohomology, intersection theory, perverse sheaves) to the study of group actions on vector spaces. This book participates in all three traditions, but its organizing principle is the interplay between *algebra* (Hopf algebras, comodules, the universal enveloping algebra) and *geometry* (the flag variety $G/B$, line bundles, the Borel–Weil theorem). Every major theorem in the book can be understood from both sides, and we try to present both perspectives wherever possible, choosing whichever is cleaner for the argument at hand.

## Motivation: Why Algebraic Groups?

The study of algebraic groups draws its inspiration from many sources, and different readers will find different motivations compelling.

*From linear algebra.* The groups $\mathrm{GL}_n$, $\mathrm{SL}_n$, $\mathrm{SO}_n$, and $\mathrm{Sp}_{2n}$ — the symmetry groups of vector spaces, determinant forms, quadratic forms, and symplectic forms — are the most natural objects in linear algebra. Understanding their structure (subgroups, quotients, conjugacy classes) and their representations (how they act on vector spaces) is a problem that arises organically in almost every branch of mathematics. This book provides the definitive framework for answering these questions, not group by group, but uniformly: the root system encodes the structure of *all* classical groups simultaneously, and the Weyl character formula computes the character of *any* irreducible representation of *any* reductive group in a single formula.

*From number theory.* The theory of automorphic forms — the generalization of classical modular forms to higher-rank groups — is built on the representation theory of reductive groups over local and global fields. The Langlands program, which conjectures a vast web of correspondences between automorphic representations and Galois representations, takes the representation theory of reductive groups as its starting point. The local Langlands correspondence, the Satake isomorphism, and the geometric Satake equivalence all require a thorough understanding of the structure and representation theory developed in this book. While we do not enter the Langlands program itself, the reader who completes this book will have the algebraic foundations needed to begin.

*From algebraic geometry.* Algebraic groups act on algebraic varieties, and the geometry of these actions — orbit structures, quotient constructions, equivariant cohomology — is a central theme in modern algebraic geometry. The flag variety $G/B$ is one of the most studied varieties in all of geometry: its Schubert calculus computes intersection numbers, its line bundles produce representations (the Borel–Weil theorem), and its singularity theory (Schubert varieties, their resolutions, and their intersection cohomology) connects to the important structures in representation theory (Kazhdan–Lusztig polynomials, perverse sheaves). The construction of moduli spaces in algebraic geometry (moduli of bundles, moduli of sheaves, geometric invariant theory) relies heavily on the theory of reductive group actions.

*From topology and differential geometry.* Every compact Lie group is the real form of a complex reductive algebraic group, and the representation theory of compact Lie groups (which governs harmonic analysis on symmetric spaces, the Peter–Weyl theorem, and the Borel–Weil–Bott theorem in differential geometry) is a special case of the algebraic theory developed here. The algebraic approach has the advantage of working uniformly over any field, including fields of positive characteristic where the analytic methods of Lie group theory are unavailable.

*From physics.* The gauge groups of quantum field theory ($\mathrm{SU}(n)$, $\mathrm{SO}(n)$, $\mathrm{Sp}(n)$, and the exceptional groups $G_2$, $F_4$, $E_6$, $E_7$, $E_8$) are real forms of reductive algebraic groups, and the classification of elementary particles in a gauge theory is the classification of irreducible representations of the gauge group. The Dynkin diagram classification — the same classification proved in this book — constrains the possible gauge theories, and the Weyl character formula computes the branching rules (how a representation decomposes when restricted to a subgroup) that govern particle interactions. String theory's reliance on the exceptional group $E_8$ gives the classification a physical significance that was certainly not anticipated by its creators.

*From pure algebra.* The category $\mathrm{Rep}(G)$ of representations of a reductive group is one of the best examples of a rigid symmetric monoidal abelian category, and Tannakian duality asserts that the group $G$ can be *recovered* from $\mathrm{Rep}(G)$ alone (as its tensor-automorphism group). This means that the representation theory is not merely a tool for studying the group, but an *equivalent* description of the group: to know the group is to know its representations, and vice versa. This perspective — category theory as a foundation for algebra — has been enormously influential, leading to quantum groups, tensor categories, and the modern theory of motives.

## Historical Context

The theory of algebraic groups has its roots in three nineteenth-century developments: the theory of *Lie groups* (continuous symmetry groups, studied by Sophus Lie and Wilhelm Killing), the theory of *algebraic equations* (the Galois theory of polynomial equations, which led to the study of permutation groups and eventually matrix groups), and the theory of *invariants* (the study of polynomial functions invariant under group actions, developed by Cayley, Sylvester, Gordan, and Hilbert).

The modern theory begins with Claude Chevalley's work in the 1950s, which established the classification of semisimple algebraic groups over algebraically closed fields by root systems and constructed the "Chevalley groups" over arbitrary fields. Chevalley's approach was algebraic rather than analytic, replacing the Lie-theoretic methods (which require real or complex analysis) with purely algebraic techniques that work in any characteristic. This was extended by Armand Borel, who developed the theory of Borel subgroups and proved the fundamental conjugacy and fixed-point theorems, and by Jacques Tits, who developed the theory of buildings and established the classification over non-algebraically closed fields.

The representation theory was developed in parallel: Hermann Weyl's character formula (1925–1926) was originally proved for compact Lie groups using integration on the group manifold. The algebraic proof, via the Bernstein–Gelfand–Gelfand resolution, came much later (1970s). The geometric approach — the Borel–Weil theorem (1954) and its extension by Raoul Bott (1957) — revealed that representation theory is inseparable from the geometry of the flag variety. The positive-characteristic theory was developed by Robert Steinberg (tensor product theorem, 1963), Jens Carsten Jantzen (the Jantzen filtration, 1970s–1980s), and George Lusztig (Lusztig's conjecture, 1980), with crucial contributions from the Kazhdan–Lusztig theory of Hecke algebras (1979). The recent work of Geordie Williamson (counterexamples to Lusztig's conjecture, 2017) has shown that the positive-characteristic theory is richer and more subtle than previously expected, and the field remains actively evolving.

This book presents the theory in its modern, scheme-theoretic form, following the approach of Milne, Waterhouse, and Jantzen rather than the classical variety-based approach of Borel and Humphreys. The scheme-theoretic language is essential for positive-characteristic phenomena (infinitesimal group schemes, Frobenius kernels) and for the connections to modern algebraic geometry, and we believe it is the correct foundation for a student entering the field today.

## Prerequisites

The reader should have completed a first course in algebraic geometry at the level of Hartshorne's *Algebraic Geometry* the EYNTKA series book *Algebraic Geometry* [5]. Specifically, we assume familiarity with: affine and projective varieties, morphisms and rational maps, dimension theory, smoothness and tangent spaces, the Zariski topology, sheaves and coherent sheaves, and the basic properties of projective varieties (properness, ampleness, the Picard group). For Chapter 10, some acquaintance with sheaf cohomology ($H^i(X, \mathcal{F})$ for coherent sheaves on projective varieties) is needed.

From commutative algebra, we assume: localization, Noetherian rings, the Nullstellensatz, integral extensions, flatness, and the basics of module theory. From general algebra: groups, rings, modules, tensor products, exact sequences, and some homological algebra (Ext, Tor) at a basic level. These prerequisites are covered in the EYNTKA series book *Algebra* [7].

No prior knowledge of Lie groups, Lie algebras, or representation theory is assumed. We develop the Lie algebra of an algebraic group from scratch (Chapter 3), and the representation theory is built from first principles (Chapters 2, 8–11). The reader who has seen Lie algebras before will recognize many constructions (the adjoint representation, root space decompositions, the Weyl character formula), but the proofs here are self-contained and do not rely on the Lie-algebraic theory.

A reader coming from the analytic side (compact Lie groups, differential geometry) will find that the algebraic approach covers the same ground by different methods: every compact Lie group is the real form of a reductive algebraic group, and the representation theory developed here specializes to the classical theory when $k = \mathbb{C}$. The algebraic approach has the advantage of working over any field and of revealing the positive-characteristic phenomena that the analytic theory cannot see.

## The Narrative

The book tells a single story in two acts.

*Part I: Structure Theory.* The question is: *what is a reductive group?* We dismantle it, layer by layer, into combinatorial data. Chapter 1 defines the objects and establishes the dual languages (functorial and Hopf-algebraic). Chapters 2–3 build the basic toolkit (representation theory, Lie algebras). Chapter 4 constructs quotients. Chapter 5 performs the "Jordan decomposition" of the group itself, separating the semisimple atoms (tori) from the unipotent atoms. Chapter 6 reassembles these atoms into solvable groups and Borel subgroups, proving the conjugacy theorems that make the root system well-defined. Chapter 7 extracts the root system and classifies it. The answer: a reductive group is determined by its root datum, which is a finite combinatorial object classified by a Dynkin diagram.

*Part II: Representation Theory.* The question is: *how does a reductive group act?* Chapter 8 classifies the irreducible representations by dominant weights (the "highest weight theorem"). Chapter 9 computes their characters (the Weyl character formula). Chapter 10 constructs them geometrically (the Borel–Weil–Bott theorem on the flag variety $G/B$). Chapter 11 asks what changes in positive characteristic and brings the reader to the frontier of current research (Lusztig's conjecture, Williamson's counterexamples, $p$-Kazhdan–Lusztig theory).

The two parts are connected by a single thread: the root system. In Part I, the root system is extracted from the group. In Part II, the root system governs the representations. The Weyl group, the dominant weights, the character lattice, and the flag variety — all defined in Part I as structural invariants of $G$ — become the computational tools of Part II. The Borel–Weil theorem closes the circle: the flag variety $G/B$, constructed in Part I as a quotient of the group, turns out to be the geometric engine that *produces* every irreducible representation.

A secondary thread runs through the entire book: the role of characteristic. In characteristic 0, the theory is clean: the Lie algebra determines the group (Cartier's theorem), the exponential map provides a diffeomorphism between unipotent groups and nilpotent Lie algebras, every representation is completely reducible, and the Weyl character formula computes everything. In characteristic $p > 0$, each of these statements fails, and the failures are not pathologies but features: they reflect the Frobenius morphism, an arithmetic structure that has no analogue in characteristic 0. The book tracks these failures systematically, from the first appearance of $\mu_p$ and $\alpha_p$ (Chapter 1) through Cartier's theorem and its limits (Chapter 3), the exponential map and its breakdown (Chapter 5), the failure of complete reducibility (Chapter 8), and finally the Steinberg tensor product theorem and Lusztig's conjecture (Chapter 11). The reader finishes the book understanding not just the theorems but the *boundary* of what is known.

## Scope and Limitations

This book covers the representation theory of *split* reductive groups over algebraically closed fields. The "split" hypothesis means our tori are isomorphic to $\mathbb{G}_m^r$ over the ground field (not just over its algebraic closure), and the "algebraically closed" hypothesis avoids the Galois-cohomological subtleties of forms and descent. For most of the structure theory and all of the characteristic-0 representation theory, these hypotheses are harmless: the key theorems (Dynkin classification, highest weight classification, Weyl character formula, Borel–Weil–Bott) hold in this generality. For the arithmetic applications (automorphic forms, Langlands program, rational points), one eventually needs the theory of reductive groups over general fields, as developed by Borel–Tits. We indicate in Chapters 5 and 7 where the non-split theory diverges, but a full treatment is beyond our scope.

We also do not develop:

1. *Infinite-dimensional representations.* The book treats only finite-dimensional (rational) representations. The theory of admissible representations of $p$-adic groups, the category $\mathcal{O}$ of highest-weight modules over Lie algebras, and the representation theory of real reductive groups each require substantial additional machinery (functional analysis, homological algebra, $\mathcal{D}$-modules) and are the subject of separate books.

2. *Quantum groups.* The $q$-deformation $U_q(\mathfrak{g})$ of the universal enveloping algebra is closely related to the characteristic-$p$ theory (via specialization at roots of unity) and has its own rich representation theory (crystal bases, canonical bases, the Lusztig–Kashiwara theory). We mention the connection briefly in Chapter 11 but do not develop it.

3. *Geometric Langlands.* The geometric Satake equivalence and the theory of perverse sheaves on the affine Grassmannian provide a deep geometric interpretation of the representation theory developed here. We indicate this direction in Chapter 11 as motivation, but the full theory requires derived categories, $\ell$-adic sheaves, and the formalism of six-functor categories.

4. *Moduli spaces and geometric invariant theory.* The theory of group actions on varieties (stability conditions, quotient constructions, the Hilbert–Mumford criterion) is a major application of the structure theory of reductive groups, but it is a subject in its own right and we do not pursue it.

## Connections to Other Areas

The theory developed in this book connects to a wide web of mathematics. We briefly indicate the main threads, so that the reader can begin making connections while reading the main material.

*Symmetric functions and combinatorics.* The representation ring of $\mathrm{GL}_n$ is isomorphic to the ring of symmetric polynomials, and the Schur polynomials (the characters of irreducible representations) are fundamental objects in algebraic combinatorics. The Littlewood–Richardson rule for tensor products, the Robinson–Schensted–Knuth correspondence, and the theory of Young tableaux all emerge from the representation theory of $\mathrm{GL}_n$. This connection is developed in Chapter 9.

*Algebraic topology.* The classifying space $BG$ of a compact Lie group $G$ has cohomology ring $H^*(BG;\mathbb{Q}) \cong \mathbb{Q}[x_1, \ldots, x_r]^W$ (a polynomial ring in $r$ variables, invariant under the Weyl group). The characteristic classes of principal $G$-bundles live in this ring, and the Weyl character formula computes the equivariant cohomology of homogeneous spaces. The Borel–Weil–Bott theorem (Chapter 10) can be understood as a computation of equivariant sheaf cohomology on $G/B$.

*Commutative algebra and invariant theory.* The ring of invariants $k[V]^G$ (polynomial functions on a representation $V$ that are invariant under $G$) is a finitely generated $k$-algebra when $G$ is reductive (Hilbert's theorem, generalized by Nagata and Haboush). The structure of this ring — its generators, relations, and singularities — is controlled by the representation theory of $G$. This is the starting point of geometric invariant theory (GIT).

*Number theory and the Langlands program.* The $L$-functions attached to automorphic representations of reductive groups over number fields encode deep arithmetic information (the distribution of primes, the arithmetic of elliptic curves, the structure of Galois representations). The local Langlands correspondence relates representations of $p$-adic reductive groups to Galois representations of local fields, and the Satake isomorphism (a consequence of the structure theory of split reductive groups over $p$-adic fields) is the key tool for relating automorphic representations to $L$-functions.

*Mathematical physics.* Beyond the gauge-theoretic applications mentioned above, the representation theory of reductive groups appears in: conformal field theory (the Wess–Zumino–Witten model, whose space of states is a representation of an affine Lie algebra), integrable systems (the Toda lattice and Calogero–Moser systems, whose symmetries are governed by root systems), and the theory of exactly solvable models in statistical mechanics (where the Yang–Baxter equation is closely related to quantum groups and the $R$-matrix formalism).

## A Note on Conventions

We work over a base field $k$, which is algebraically closed except where stated otherwise. "Algebraic group" means "affine algebraic group scheme of finite type over $k$," unless qualified (e.g. "abelian variety" for the non-affine case). "Representation" means "finite-dimensional rational representation" unless otherwise specified. "Reductive" means "connected reductive" (we always assume connectedness for reductive groups). The characteristic of $k$ is either 0 or a prime $p > 0$, and we are careful to state which hypothesis is in force for each theorem.

For the root system conventions: simple roots are $\alpha_1, \ldots, \alpha_r$, fundamental weights are $\omega_1, \ldots, \omega_r$ (defined by $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$), positive roots are $\Phi^+$ (those in $\mathrm{Lie}(R_u(B))$), and the Weyl vector is $\rho = \frac{1}{2}\sum_{\alpha\in\Phi^+} \alpha = \sum_{i=1}^r \omega_i$. The inner product on $X(T) \otimes \mathbb{R}$ is the one induced by the Killing form, normalized so that long roots have length $\sqrt{2}$ (the "geometric" normalization). These conventions

## How to Read This Book

The book is designed to be read linearly: each chapter depends on the preceding ones, and the narrative builds cumulatively. A reader working through the entire book should expect to spend substantial time on the exercises, which are integral to the exposition (many results used later are first established as exercises, and the solutions are provided).

For a reader with specific goals:

1. *"I want to understand the Dynkin diagram classification."* Read Chapters 1–7 (Part I). Chapters 2–4 can be skimmed on a first pass if the reader is comfortable with representations, Lie algebras, and quotient constructions from other sources.

2. *"I want to learn the representation theory of reductive groups in characteristic* $0$*."* Read Chapters 1–3 (for the basic setup), then Chapters 5 and 7 (for weight spaces and root systems), then Chapters 8–10.

3. *"I want to understand the Borel–Weil–Bott theorem."* Read Chapters 5–7 (for tori, Borel subgroups, and root systems), Chapter 8 (for the highest weight classification), and Chapter 10. Chapter 9 (the Weyl character formula) provides essential computational context but is not logically required for the statement and proof.

4. *"I want to understand what happens in characteristic* $p$*."* Read Chapters 5 and 7 (for the structures that survive in characteristic $p$), Chapter 8 (especially the Weyl modules and the failure of complete reducibility), and Chapter 11.

We have tried to make the exposition self-contained, but the subject is vast and we necessarily cite external references for several results (the existence and isomorphism theorems for root data, the BGG resolution, Lusztig's conjecture, etc.). In each case, we give precise statements and key ideas, so that the reader can understand the *role* of these results in the theory without having read their proofs.

# Part I

# Structure Theory

The goal of Part I is to dismantle a reductive algebraic group into combinatorial data: a root system, encoded by a Dynkin diagram. This is the "anatomy" of the group — we are not yet asking how the group *acts* (that is the subject of Part II), but rather what the group *is*.

The reader is assumed to be familiar with scheme theory at the level of Hartshorne and Vakil [5], and with the definition of a group scheme. We work throughout with affine algebraic groups over a field $k$, and we take the functorial viewpoint seriously: an algebraic group is a representable functor $\mathbf{Alg}_k \to \mathbf{Grp}$, and its coordinate ring is a commutative Hopf algebra. This perspective — developed in chapter 1 alongside the classical matrix-group examples — gives us two complementary languages for every construction: the geometric language of schemes and morphisms, and the algebraic language of Hopf algebras and comodules. We move freely between them, choosing whichever is cleaner for the problem at hand.

The first three chapters build the basic toolkit. Chapter 1 sets up the algebraic group itself: its Hopf algebra, its functor of points, and the linearization theorem that embeds every affine group into $\mathrm{GL}_n$. Chapter 2 turns the group outward, formalizing what it means for an algebraic group to act on a scheme or a vector space, and establishing the category $\mathrm{Rep}(G)$ of representations as a rigid symmetric monoidal category. The chapter culminates in the Closed Orbit Lemma, a geometric result about orbit structures that will be used repeatedly. Chapter 3 extracts the infinitesimal invariant — the Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ — via the dual numbers $k[\varepsilon]$, constructs the adjoint representation, and proves Cartier's theorem: in characteristic 0, the Lie algebra faithfully reflects the group. The chapter closes with a discussion of the failures in characteristic $p$, which is a fundamental reason the scheme-theoretic approach is taken rather than the purely Lie-algebraic one.

Chapter 4 addresses the quotient problem: given a closed subgroup $H \subseteq G$, does the coset space $G/H$ exist as a scheme? Chevalley's theorem provides the answer by realizing $H$ as the stabilizer of a line in a representation, embedding $G/H$ into projective space. This gives us the categorical infrastructure — faithfully flat exact sequences, the isomorphism theorems, and the identification of homogeneous spaces — needed for the rest of the book.

The second half of Part I is the decomposition theory. Chapter 5 begins with the Jordan decomposition, which splits every group element into a semisimple part and a unipotent part, and then studies each piece: tori (classified by their character lattices, with representations decomposing into weight spaces) and unipotent groups (classified in characteristic 0 by their nilpotent Lie algebras via the exponential map). Chapter 6 combines these into the theory of solvable groups. The Lie–Kolchin theorem says that connected solvable groups are "upper-triangular," and Borel's fixed-point theorem converts this linear-algebraic fact into a powerful geometric tool: connected solvable groups have fixed points on complete varieties. Applied to the flag variety $G/B$, this yields the conjugacy of Borel subgroups and maximal tori — the structural theorems that make the root system well-defined.

Chapter 7 brings everything together. The adjoint action of a maximal torus $T$ on $\mathfrak{g}$ decomposes the Lie algebra into root spaces, and the resulting root system $\Phi \subseteq X(T)$ is a finite combinatorial object governed by the Weyl group $W = N_G(T)/Z_G(T)$. The classification of root systems via Dynkin diagrams is a purely combinatorial theorem — there are four infinite families ($A_n$, $B_n$, $C_n$, $D_n$) and five exceptional cases ($G_2$, $F_4$, $E_6$, $E_7$, $E_8$) — and the isomorphism and existence theorems for root data assert that this combinatorial classification is equivalent to the geometric classification of reductive groups. The passage from geometry to combinatorics loses no information: every root datum arises from a group, and groups with isomorphic root data are isomorphic.

A quick word must be given on the characteristic of the fields we are working with. Most theorems hold over arbitrary algebraically closed fields, and we explicitly flag the (few but important) places where characteristic 0 is needed — notably Cartier's theorem and the exponential map for unipotent groups. The infinitesimal group schemes $\mu_p$ and $\alpha_p$ make regular appearances as cautionary examples, reminding us that the Lie algebra is an imperfect tool in characteristic $p$ and that the full scheme-theoretic machinery is not a luxury but a necessity.

# 1

# Algebraic Groups

Algebraic groups represent the synthesis of algebra and algebraic geometry. By endowing a variety or a scheme with a group structure—where the group operations are morphisms of the underlying space—we obtain objects that are rigid enough to be classified completely, yet rich enough to capture the symmetries of almost any geometric or algebraic structure. In this chapter, we establish the fundamental structure of algebraic groups. We will transition from the foundational geometric definitions to their categorical interpretation as functors, explore the algebraic underpinnings via Hopf algebras, and lay the groundwork for the structural decomposition and classification of linear algebraic groups.

## 1.1 Definitions and Properties

Recall [5] that a *group scheme* is an $S$-scheme $G$ for some base $S$ along with morphisms $m : G \times_S G \to G$, $\iota : G \to G$, and $e : \operatorname{Spec} S \to G$ that satisfy the group axioms as commutative diagrams:

$$\begin{tikzcd} G \times_S G \times_S G \arrow[r, "m \times \mathrm{id}_G"] \arrow[d, "\mathrm{id}_G \times m"'] & G \times_S G \arrow[d, "m"] \\ G \times_S G \arrow[r, "m"'] & G \end{tikzcd}$$

Associativity Axiom: $(g \cdot h) \cdot k = g \cdot (h \cdot k)$

$$\begin{tikzcd} G \arrow[r, "\Delta_G"] \arrow[rrd, bend right, "e \circ (G \to S)"'] & G \times_S G \arrow[r, "\mathrm{id}_G \times \iota"] & G \times_S G \arrow[d, "m"] \\ & & G \end{tikzcd}$$

Inverse Axiom: $g \cdot g^{-1} = e$

$$\begin{tikzcd} S \times_S G \arrow[rr, "e\times \mathrm{id}_G"] \arrow[dr, "\cong"'] & & G \times_S G \arrow[dl, "m"] \\ & G & \end{tikzcd} \qquad \begin{tikzcd} G \times_S S \arrow[rr, "\mathrm{id}_G \times e"] \arrow[dr, "\cong"'] & & G \times_S G \arrow[dl, "m"] \\ & G & \end{tikzcd}$$

Left Unit Axiom: $e \cdot g = g$ Right Unit Axiom: $g \cdot e = g$

Equivalently, it is the group object in the category $\mathbf{Sch}_S$. If $G$ is locally of finite type over $S$, then $G$ is called *locally algebraic*, and if it is furthermore quasicompact it is called *algebraic*. Combining these two definitions gets us the common general definition of an algebraic group. In many cases, the definition is restricted to (abstract) varieties, which the following definition accommodates.

**Definition 1.1.1: Algebraic Group**

An *algebraic group* over a field $k$ is a group $k$-scheme $G$ that is of finite type over $k$. Equivalently (by theorem 1.1.5), it is a $k$-variety[a] (or finite-type $k$-scheme) equipped with morphisms of schemes for multiplication, inversion, and the identity:

$$\mu : G \times_k G \to G, \qquad \iota : G \to G, \qquad e : \operatorname{Spec} k \to G$$

satisfying the standard group axioms. The multiplication map $\mu$ is commonly written as $g \cdot h := \mu(g, h)$.

If we say $G$ is an *algebraic group*, we shall implicitly assume it is over a *field*. If we want to be more general and allow an arbitrary base ring $S$, we shall use the term *algebraic group scheme*.

[a] in the abstract sense defined in [5]

When it is clear from context, the fibered product will drop the $k$ and will be written as $G \times G$. We shall mostly be working with groups over a perfect field or groups over an algebraically closed field of characteristic 0. By the Lefschetz principle (see [7, appendix]), if $k$ is algebraically closed of characteristic 0, there is no theoretical issue assuming $k = \mathbb{C}$.

The structure map $e : \operatorname{Spec} k \to G$ is a crucial part of the data. It equips the group with a distinguished $k$-rational point, the identity element. Because algebraic groups over a field are always separated schemes, the image of this identity section is always a closed point. (It is worth noting that for group schemes over a general base $S$, the identity section need not be a closed immersion, as demonstrated in Stacks Project 06E7).

Note that $G$ is *not* a topological group (except in dimension 0) because the scheme-theoretic fiber product $G \times_k G$ carries the *Zariski topology*, which is strictly finer than the product topology on the underlying sets. Indeed, any $T_1$ topological group must be Hausdorff ($T_2$), but algebraic varieties of dimension $> 0$ are never Hausdorff in the Zariski topology.

The group structure provides powerful geometric homogeneity. For any $g \in G(k)$, the left translation map $x \mapsto g \cdot x$ is an isomorphism of varieties, meaning the local geometric properties of $G$ are identical at all $k$-rational points. For example, any variety over a perfect field possesses a dense open subset of smooth points; by translating this smooth locus around the group, we conclude that any *reduced* algebraic group (i.e., an algebraic group variety) is everywhere smooth. (Furthermore, in characteristic 0, Cartier's Theorem guarantees that even non-reduced finite-type group schemes are smooth).

**Proposition 1.1.2: Properties of Algebraic Groups**

Let $G$ be an algebraic group over an algebraically closed field $k = \overline{k}$. Then:

1. $G$ is separated.
2. (Cartier's Theorem) If char $k = 0$, then $G$ is smooth.
3. Only one irreducible component of $G$ contains the identity element $e$.
4. $G$ is connected if and only if $G$ is irreducible. If $G$ is a variety (reduced), then a connected $G$ is an integral scheme.
5. $G$ is equidimensional, meaning $\dim(G) = \dim_g(G)$ for all $g \in G$ (recall $\dim_g(G)$ is the local dimension of $G$ at $g$).
6. A closed subgroup of an algebraic group is an algebraic group.
7. The direct product (i.e., fibered product over $k$) of algebraic groups is an algebraic group.

***Proof***:

1. Consider the morphism $\Psi : G \times_k G \to G$ given by $(g, h) \mapsto g^{-1}h$. Because the identity $\{e\}$ is a closed point (since $G$ is of finite type over a field) and $\Psi$ is continuous, the preimage $\Psi^{-1}(e) = \{(g, g) \mid g \in G\}$ is closed. This set is exactly the diagonal $\Delta_G$, hence $G$ is separated.
2. This will be proven in theorem 3.3.2; it is mentioned here to show one similarity between algebraic groups of characteristic 0 and lie groups.

   Note that for the variety case (reduced group schemes), generic smoothness guarantees the existence of a dense open smooth locus [5]. By translating this locus via the group action, the variety is smooth everywhere. Cartier's theorem extends this to show that in characteristic 0, all group schemes are automatically reduced.
3. Suppose $X_1, \ldots, X_n$ are the irreducible components of $G$ that contain the identity $e$. The finite product $X_1 \times \cdots \times X_n$ is irreducible. The multiplication map $\mu$ is continuous, so the image $Y = X_1 X_2 \cdots X_n$ is irreducible. Since $e \in X_i$ for all $i$, we trivially have $X_i \subseteq Y$ for every $1 \leq i \leq n$. However, because $Y$ is irreducible, it must be contained in *some* irreducible component of $G$, say $X_k$. Thus $X_i \subseteq X_k$ for all $i$, which by the maximality of irreducible components implies $X_1 = X_2 = \cdots = X_n$. Thus, only a single component contains the identity.
4. A topological space is connected if it cannot be partitioned into two disjoint non-empty open sets. By (4), the identity $e$ belongs to a unique irreducible component $G^\circ$, we can translate $G^\circ$ to see that the irreducible components of $G$ are disjoint cosets of $G^\circ$. A space that is a disjoint union of open/closed components is connected if and only if there is exactly one such component. Thus, connected implies irreducible. If $G$ is reduced, being irreducible implies it is integral.

5. For any $g \in G$, left translation $\ell_g : G \to G$ defined by $x \mapsto gx$ is an isomorphism of schemes. In particular, it maps $e$ to $g$ and induces an isomorphism of local rings $\mathcal{O}_{G,g} \cong \mathcal{O}_{G,e}$. Therefore, the Krull dimension of the local rings must be identical, meaning $\dim_g(G) = \dim_e(G)$.

6. Let $H \subseteq G$ be a closed subscheme that is also a subgroup. Because closed immersions are morphisms of finite type, and $G$ is of finite type over $k$, $H$ is finite type over $k$. The restriction of the multiplication and inversion maps of $G$ to $H$ gives morphisms $\mu|_H : H \times_k H \to H$ and $\iota|_H : H \to H$, satisfying the group axioms. Thus $H$ is an algebraic group.

7. Let $G_1, G_2$ be algebraic groups. The fiber product $G_1 \times_k G_2$ is a scheme of finite type over $k$. The group operations are defined component-wise (e.g., multiplication is the composition of the product of the individual multiplication maps with the canonical reordering isomorphism of the factors). Since products and compositions of morphisms are morphisms, $G_1 \times_k G_2$ is an algebraic group.

We give the component with the identity a name:

**Definition 1.1.3: Identity Component**

Let $G$ be an algebraic group. Then the unique irreducible component containing the identity is called the *identity component*. It will be denoted $G^\circ$.

**Proposition 1.1.4: Identity Component and Connectedness**

Let $G$ be an algebraic group. Then:

1. $G^\circ \leqslant G$ is a normal subgroup of finite index in $G$ whose cosets are the connected (and irreducible) components of $G$.

2. Every closed subgroup of finite index in $G$ contains $G^\circ$.

***Proof*:**

1. For any $x \in G^\circ$, the continuous map of inversion $y \mapsto y^{-1}$ maps $G^\circ$ to another irreducible component containing $e^{-1} = e$. Since $G^\circ$ is the unique such component, $(G^\circ)^{-1} = G^\circ$. Similarly, left translation by $x^{-1}$ is a homeomorphism, so $x^{-1}G^\circ$ is an irreducible component containing $x^{-1}x = e$, meaning $x^{-1}G^\circ = G^\circ$. Thus $G^\circ$ is closed under multiplication and inversion, making it a subgroup.

   To see that it is normal, take *any* $x \in G$. Conjugation $y \mapsto xyx^{-1}$ is an automorphism of the variety $G$ that fixes the identity $e$. Therefore, it must map the unique irreducible component containing $e$ to itself, meaning $xG^\circ x^{-1} = G^\circ$.

   Because $G^\circ$ is a subgroup, its cosets partition $G$. Since translation is a homeomorphism, the cosets of $G^\circ$ are exactly the irreducible (and hence connected) components of $G$. Because $G$ is a scheme of finite type, its underlying topological space is Noetherian and therefore has only finitely many irreducible components. Thus, $G^\circ$ is of finite index.

2. Let $H$ be a closed subgroup of finite index. Its complement in $G$ is the union of the other

> left cosets of $H$: $\bigcup_{x \notin H} xH$. Because $H$ is of finite index, this is a finite union. Since translation is a homeomorphism and $H$ is closed, each coset $xH$ is closed. Therefore, the complement of $H$ is a finite union of closed sets, meaning it is closed. This makes $H$ both open and closed (clopen) in $G$.
>
> The left cosets of $H$ partition $G$ into disjoint clopen sets. Because $G^\circ$ is connected, it cannot be divided among multiple disjoint open sets; it must lie entirely within one of them. Since the identity element $e$ is in both $G^\circ$ and $H$, it must be that $G^\circ \subseteq H$.

As the term "irreducible" in representation theory has a different meaning, we shall usually say that $G$ is *connected* if $G = G^\circ$. The above proposition makes this unambiguous: the irreducible components of $G$ are precisely the disjoint cosets of $G^\circ$. Thus, if $G$ is topologically connected, it can only have $G^\circ$ as its single irreducible component. The following deep result will not be proven here, but it is important to know as it guarantees we can always safely work within the realm of quasi-projective geometry:

**Theorem 1.1.5: Algebraic Group implies Quasi-Projective**

Let $G$ be an algebraic group over a field $k$. Then $G$ is *quasi-projective* over $k$.

> ***Proof*:**
> See https://stacks.math.columbia.edu/tag/0BF6.

It is important to take a moment to single out proper algebraic groups. As they are *not* the focus of the book, we must mathematically justify their exclusion:

**Definition 1.1.6: Abelian Variety**

Let $k$ be a field. An *abelian variety* is an algebraic group over $k$ which is also a proper, geometrically integral $k$-variety.

There are many big theorems surrounding abelian varieties. For example, any abelian variety is automatically projective (a theorem of Weil), and the combination of geometric integrality (which gives generic smoothness) and group translation guarantees it is everywhere smooth. Notice that they are not *defined* to be commutative; rather, the geometric rigidity of proper group schemes *forces* their group operation to be abelian.

Most importantly for our purposes, because an abelian variety $A$ is proper and connected, it has no non-constant global functions ($\mathcal{O}(A) = k$). Any finite-dimensional linear representation is a morphism of varieties $\rho : A \to \mathrm{GL}_n(k)$. Because $\mathrm{GL}_n(k)$ is an affine variety, the image of any proper connected scheme inside it must collapse to a single point. Thus, the map $\rho$ must be *trivial*. Consequently, abelian varieties have no non-trivial finite-dimensional linear representations, which is why their representation theory is of little interest to this book.[1]

The category of algebraic groups (or the group-objects in $\mathrm{Var}_k$ or $\mathbf{Sch}_k$) will have groups that are affine, proper, or neither. However, due to a theorem by Chevalley, the study of algebraic groups can be structurally decomposed into "affine" and "abelian" parts. Namely, $G$ has a normal subgroup $H$ such that $H$ is affine and the quotient $G/H$ is projective (i.e., an "abelian variety").

[1] This is "fixed" by studying the line bundles on $A$ and their change under the Fourier–Mukai Transform; this geometrically recovers the notion of characters from the representation theory of finite abelian groups.

**Theorem 1.1.7: Chevalley's Theorem**

Let $k$ be a perfect field and $G$ an algebraic group over $k$. Then there exists a unique normal linear algebraic closed subgroup $H$ in $G$ for which $G/H$ is an abelian variety. That is, there is a unique short exact sequence of algebraic groups

$$1 \to H \to G \to A \to 1$$

with $H$ affine linear algebraic and $A$ an abelian variety. The formation of $H$ commutes with base change to an arbitrary perfect field extension over $k$.

***Proof*:**
See [11].

Thus, this book focuses entirely on the affine case. We can reduce one step further by showing that all such groups are in fact linear:

**Lemma 1.1.8: Affine Algebraic Groups are Linear**

Let $G$ be an affine algebraic group over $k$. Then $G$ is isomorphic to a closed subgroup of the general linear group $\mathrm{GL}_n$ for some $n$.

***Proof*:**
Let $G = \operatorname{Spec} A$ where $A$ is a finitely generated $k$-algebra, say $A \cong k[x_1, \ldots, x_r]/(f_1, \ldots, f_s)$. Define a right-regular action of $G$ on itself: $r_g(x) = xg$. This induces an action on $A = \Gamma(G)$, namely for any $f \in A$:

$$(g \cdot f)(x) = f(xg)$$

giving a homomorphism $\rho : G \to \operatorname{Aut}_k(A)$. If this action can be restricted to a finite-dimensional $G$-stable subspace $V \subset A$, we obtain a morphism $\rho : G \to \mathrm{GL}(V) \cong \mathrm{GL}_n(k)$ where $n = \dim V$. We will show that such a $V$ exists, that $\rho$ is a group homomorphism, and that it is a closed immersion.

First, let us show that the orbit of any function lies in a finite-dimensional space. Given the multiplication $\mu : G \times_k G \to G$, by affineness we have the dual (comultiplication) map $\mu^* : A \to A \otimes_k A$. For any $f \in A$, we can write

$$\mu^*(f) = \sum_{i=1}^{d} f_i \otimes h_i$$

for some $f_i, h_i \in A$. Then for any $x, g \in G$:

$$(g \cdot f)(x) = f(xg) = \mu^*(f)(x, g) = \sum_{i=1}^{d} f_i(x)\, h_i(g)$$

and hence, as functions on $G$,

$$g \cdot f = \sum_{i=1}^{d} h_i(g)\, f_i.$$

So $g \cdot f$ lies in the subspace $W_f$ spanned by $\{f_1, \ldots, f_d\}$, which is finite-dimensional. The key here is that as $g$ ranges over all of $G$, the translated function $g \cdot f$ always lands in the fixed finite-dimensional space $W_f$.

We claim that $W_f$ can be taken to be $G$-stable (i.e. $\mu^*(W_f) \subseteq W_f \otimes_k A$). Without loss of generality, assume the tensor expression $\mu^*(f) = \sum_i f_i \otimes h_i$ is minimal, which guarantees that the set $\{h_i\}$ is linearly independent over $k$. By coassociativity, $(\mu^* \otimes \mathrm{id}) \circ \mu^* = (\mathrm{id} \otimes \mu^*) \circ \mu^*$, so:

$$\sum_{i=1}^{d} \mu^*(f_i) \otimes h_i = \sum_{i=1}^{d} f_i \otimes \mu^*(h_i).$$

The right-hand side clearly lies in $W_f \otimes_k A \otimes_k A$. Because the $h_i$ are linearly independent, we can define dual functionals $\varphi_j \in A^\vee$ such that $\varphi_j(h_i) = \delta_{ij}$. Applying the map $\mathrm{id} \otimes \mathrm{id} \otimes \varphi_j$ to both sides isolates $\mu^*(f_j)$ on the left, proving that $\mu^*(f_j) \in W_f \otimes_k A$ for each $j$. Hence $W_f$ is $G$-stable.

Now, as $A$ is a finitely generated $k$-algebra, choose generators $\alpha_1, \ldots, \alpha_m$. By the above, each $\alpha_j$ is contained in a finite-dimensional $G$-stable subspace $W_{\alpha_j} \subset A$. Let

$$V = \sum_{j=1}^{m} W_{\alpha_j}.$$

Then $V$ is a finite-dimensional $G$-stable subspace of $A$ containing each algebra generator $\alpha_j$, and the action of $G$ restricts to give a representation

$$\rho : G \to \mathrm{GL}(V) \cong \mathrm{GL}_n(k), \quad n = \dim V.$$

Note that $\mathrm{GL}_n(k)$ is an affine algebraic group with coordinate ring

$$k[\mathrm{GL}_n] \cong \frac{k[X_{11}, \ldots, X_{nn}, T]}{\big(T \cdot \det(X) - 1\big)}$$

Let $v_1, \ldots, v_n$ be a basis for $V$. Then for each $g \in G$:

$$\rho(g) \cdot v_j = \sum_{i=1}^{n} M_{ij}(g)\, v_i$$

where $M_{ij} : G \to k$ are regular functions on $G$ (i.e. $M_{ij} \in A$) representing the matrix entries. The pullback $\rho^\# : k[\mathrm{GL}_n] \to A$ is the $k$-algebra homomorphism determined by $\rho^\#(X_{ij}) = M_{ij}$, so $\rho$ is a $k$-morphism of varieties. That $\rho$ is a group homomorphism follows from the definition of the action: for $g_1, g_2 \in G$ and $f \in A$,

$$(g_1 g_2) \cdot f(x) = f(x g_1 g_2) = (g_2 \cdot (g_1 \cdot f))(x),$$

so $\rho(g_1 g_2) = \rho(g_2) \circ \rho(g_1)$ (or with the standard convention reversal, $\rho$ defines a homomorphism $G \to \mathrm{GL}(V)$).

What is left to show is that $\rho$ is a closed immersion, for which it suffices to show that the coordinate ring map $\rho^\# : k[\mathrm{GL}_n] \to A$ is *surjective*. Since $V$ is $G$-stable, the coaction satisfies $\mu^*(v_j) = \sum_{i=1}^{n} v_i \otimes M_{ij}$. Applying the counit axiom $(\epsilon \otimes \mathrm{id}) \circ \mu^* = \mathrm{id}$ gives:

$$v_j = \sum_{i=1}^{n} \epsilon(v_i)\, M_{ij}.$$

> So each basis element $v_j \in V$ is a $k$-linear combination of the matrix entry functions $\{M_{ij}\}$. In particular, each algebra generator $\alpha_l \in V$ lies in the $k$-span of the $M_{ij}$. Since the $\alpha_l$ generate $A$ as a $k$-algebra, the $M_{ij}$ must also generate $A$. As $\rho^{\#}(X_{ij}) = M_{ij}$, the map $\rho^{\#}$ is surjective. Therefore $\rho : G \hookrightarrow \mathrm{GL}_n(k)$ is a closed immersion, and $G$ is a closed subgroup of $\mathrm{GL}_n$.

Hence, we have reduced our study to groups $G$ that can be defined by a finite number of polynomial equations inside a matrix ring!

### 1.1.1 Examples of Algebraic Groups

Starting with the 0-dimensional example: all finite groups are affine algebraic groups. Indeed, by Cayley's Theorem, every finite group $G$ of order $n$ is a subgroup of $S_n$, and $S_n$ is a subgroup of $\mathrm{GL}_n(k)$ by associating a permutation $\sigma \in S_n$ to the corresponding permutation matrix $I_\sigma \in \mathrm{GL}_n(k)$. Since $G$ is a finite set of points in the affine space of matrices, it is a closed algebraic subvariety.

This also means that all finite groups can be interpreted as schemes. Given a finite group $G$ interpreted as a closed algebraic subgroup of $\mathrm{GL}_n(k)$, this scheme is isomorphic to $\mathrm{Spec}\left(\prod_{i=1}^{|G|} k\right)$, which is simply a disjoint union of $|G|$ copies of $\mathrm{Spec}\, k$. This is called a *constant finite group scheme*.

Interestingly, it is *not* the case that finite group schemes are in bijective correspondence with abstract groups; there are *more* group schemes. To see this, consider the group scheme $G = \boldsymbol{\mu}_3$, representing the 3rd roots of unity. Purely on the level of abstract groups, the geometric points form $\boldsymbol{\mu}_3(\overline{\mathbb{Q}}) \cong C_3$ (the cyclic group of order 3). However, $\boldsymbol{\mu}_3$ has further arithmetic structure. Take the underlying $\mathbb{Q}$-scheme $G_{\mathbb{Q}} = \mathrm{Spec}\,\mathbb{Q}[x]/(x^3-1)$. Topologically, $G_{\mathbb{Q}}$ has only 2 points, because the algebra decomposes as $\mathbb{Q}[x]/(x^3-1) \cong \mathbb{Q} \times \mathbb{Q}(\omega)$, where $\omega$ is a primitive 3rd root of unity.

In this scheme, the two non-trivial roots $(\omega, \omega^2)$ are "fused" into a single closed point with residue field $\mathbb{Q}(\omega)$. To recover the full group structure, we must look at the *geometric points* by base-changing to $\overline{\mathbb{Q}}$. This extension of scalars "splits" the degree-2 point into two distinct points. We can geometrically interpret the topological points of the $\mathbb{Q}$-scheme as the $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-orbits of the geometric points.

More generally, if $G$ is a finite étale algebraic group over $k$, then $\Gamma(G)$ is an étale algebra, meaning it is a finite product of separable field extensions:

$$\Gamma(G) \cong L_1 \times \cdots \times L_r$$

(Note that by the group scheme axioms, there must exist an identity section $e : \mathrm{Spec}\, k \to G$, which guarantees that at least one of these $L_i$ factors is exactly $k$). The category of such schemes is equivalent to the category of finite abstract groups equipped with a continuous action of $\mathrm{Gal}(\overline{k}/k)$ by group automorphisms. These are often called *twisted finite algebraic groups*.

The term "twisted" comes directly from Galois cohomology and shares the same geometric intuition as twisted vector bundles. A finite group scheme $G$ over $k$ is called a *twisted form* of a constant group scheme $G_{const}$ if they become isomorphic over the separable algebraic closure $\overline{k}$. That is, there exists an isomorphism of group schemes $\varphi : G_{\overline{k}} \xrightarrow{\sim} (G_{const})_{\overline{k}}$. While $G$ and $G_{const}$ look identical geometrically over $\overline{k}$, the isomorphism $\varphi$ fails to commute with the natural Galois actions on both sides. This failure is measured by the map $a_\sigma = \varphi \circ \sigma \circ \varphi^{-1} \circ \sigma^{-1}$, which defines a 1-cocycle in $H^1(\mathrm{Gal}(\overline{k}/k), \mathrm{Aut}(G_{const}))$. This classification of twisted forms by $H^1$ is identical to the classification of varieties described in [6].

We can interpret this intuitively as a conflict between two Galois actions. The constant group $G_{const}$ possesses a "standard" action $\sigma_{std}$ that acts trivially on the group elements (simply permuting the identical copies of $\overline{k}$). The twisted group $G$ possesses its own intrinsic Galois action derived from its definition over $k$. The cocycle $a_\sigma$ acts as a set of instructions for "gluing" the standard object to itself with a twist: to reconstruct the group $G$ from $G_{const}$, we define a *new* action $\sigma_{new}(x) := a_\sigma(\sigma_{std}(x))$ on the geometric points. The $k$-rational points of the twisted group $G$ are exactly the fixed points of this new, twisted action.

Returning to our example of $G = \boldsymbol{\mu}_3$ and $G_{const} = C_3 = \{1, g, g^2\}$, over $\overline{\mathbb{Q}}$ we have an isomorphism mapping the roots $\{1, \omega, \omega^2\}$ to the abstract elements $\{1, g, g^2\}$. The standard action on $C_3$ fixes every element. However, the intrinsic Galois action on $\boldsymbol{\mu}_3$ (e.g., complex conjugation $\tau$) swaps $\omega$ and $\omega^2$. This mismatch means our cocycle $a_\tau \in \mathrm{Aut}(C_3)$ must be the non-trivial automorphism that swaps $g$ and $g^2$ (namely, the inversion map $x \mapsto x^{-1}$). Thus, while the constant group scheme corresponds to the trivial class in $H^1$, the roots of unity $\boldsymbol{\mu}_3$ correspond to a non-trivial class in $H^1(\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), \mathrm{Aut}(C_3))$, representing a specific arithmetic twist of the constant structure.

For those familiar with group schemes as group functors (covered in section 1.2), we can generalize this idea using the functor of points. If $G = \mathrm{Spec}\,(A)$ is an affine algebraic group, we take the representable functor:

$$R \longmapsto G(R) = \mathrm{Hom}_{k\text{-}\mathbf{Alg}}(A, R)$$

For a constant group scheme like $G_{const} = \mathrm{Spec}\,(k \times k \times k)$, evaluating on $k$-points gives $G_{const}(k) = \mathrm{Hom}_{k\text{-}\mathbf{Alg}}(k^3, k)$. Since a $k$-algebra map to $k$ must pick out exactly one of the three components, there are exactly 3 such maps, so $|G_{const}(k)| = 3$.

By contrast, consider our twisted form $G = \boldsymbol{\mu}_3$ over $\mathbb{Q}$, where $A = \mathbb{Q} \times \mathbb{Q}(\omega)$. If we evaluate the $\mathbb{Q}$-points, $G(\mathbb{Q}) = \mathrm{Hom}_{\mathbb{Q}\text{-}\mathbf{Alg}}(\mathbb{Q} \times \mathbb{Q}(\omega), \mathbb{Q})$. A map must factor through either $\mathbb{Q}$ (yielding 1 map) or $\mathbb{Q}(\omega)$ (yielding 0 maps, as there are no embeddings of $\mathbb{Q}(\omega)$ into $\mathbb{Q}$). Thus, the functor only "sees" the identity element, $G(\mathbb{Q}) = \{1\}$. However, if we evaluate on $L$-points where $L = \mathbb{Q}(\omega)$, we get $G(L) = \mathrm{Hom}_{\mathbb{Q}\text{-}\mathbf{Alg}}(\mathbb{Q} \times L, L)$. The map factoring through $\mathbb{Q}$ gives 1 point, and the map factoring through $L$ gives 2 points (the two Galois embeddings $L \hookrightarrow L$). Thus $|G(L)| = 3$, successfully recovering the full geometric group via the functor of points!

Next, consider the following fundamental 1-dimensional examples:

1. $G = \mathbb{G}_a = \mathbb{A}^1$ (the additive group), where the structure maps are given by $\mu(x, y) = x + y$, $\iota(x) = -x$ and $e = 0$.

2. $G = \mathbb{G}_m = \mathbb{A}^1 \setminus \{0\}$ (the multiplicative group), where $\mu(x, y) = xy$, $\iota(x) = x^{-1}$ and $e = 1$.

Both of these groups are irreducible and 1-dimensional. We shall see in theorem 5.2.4 that (up to isomorphism over $\overline{k}$) these are the *only* irreducible 1-dimensional algebraic groups.

Dimension 2 and higher start yielding non-trivial examples, which will be treated throughout the book.

A natural generalization of $\mathbb{G}_a$ is the affine space $\mathbb{A}^n$ with its natural point-wise additive structure, denoted $\mathbb{G}_a^n$. We cannot simply take $\mathbb{A}^n \setminus \{0\}$ for the multiplicative analog, because elements like $(1, 0, 1)$ have no multiplicative inverse. Instead, we require every coordinate to be non-zero, defining the *standard split torus* of rank $n$: $\mathbb{G}_m^n = (\mathbb{A}^1 \setminus \{0\})^n$.

As all affine algebraic groups are linear, it is vital to understand the general linear group. Let $k$ be a field. The matrix algebra $M_n(k)$ can be interpreted as an affine variety:

$$M_n(k) = \mathrm{Spec}\, k[x_{11}, \ldots, x_{nn}]$$

A (closed) point of $M_n(k)$ is written as a matrix to keep better track of the algebraic structure. Matrix multiplication is associative, has the identity matrix $I$, and the entries of $A \cdot B$ are given by polynomials in the coordinates of $A$ and $B$. Thus, $M_n(k)$ forms an *algebraic monoid*.

The determinant $\det : M_n(k) \to k$ is a regular polynomial function. The principal open subset where $\det \neq 0$ forms an *open submonoid* $\mathrm{GL}_n(k)$, which is a *group* because a matrix is invertible if and only if its determinant is non-zero. To give this open subset the explicit structure of an affine algebraic variety, we use the standard trick of adding a variable $y$ to invert the determinant. We define:

$$\mathrm{GL}_n(k) = \mathrm{Spec}\, \frac{k[x_{11}, \ldots, x_{nn}, y]}{y \cdot \det(X) - 1}$$

where $X = (x_{ij})$. Note that the algebraic group $\mathrm{GL}_n(k)$ is *always* an irreducible scheme over any field $k$. Because $k[x_{ij}]$ is a Unique Factorization Domain, and the polynomial $y \cdot \det(X) - 1$ is primitive and of degree 1 in $y$, Gauss's Lemma guarantees it is irreducible. (Do not confuse the irreducible geometric scheme $\mathrm{GL}_n$ with the discrete set of $k$-rational points $\mathrm{GL}_n(k)$; over a finite field, the set of rational points is a finite discrete group, but the underlying variety remains an irreducible scheme of dimension $n^2$).

By contrast, the orthogonal group $O_n$ is a natural example of a disconnected algebraic group, as the determinant polynomial cuts it into two disjoint components ($\det = 1$ and $\det = -1$).

The following classical algebraic groups will repeatedly be brought up. In the definitions below, let $J$ denote the $n \times n$ anti-diagonal matrix with 1s on the anti-diagonal and 0s elsewhere:

1. The *special linear group* $\mathrm{SL}_n(K)$, defined as the closed subgroup of $\mathrm{GL}_n(K)$ given by the polynomial equation $\det(X) - 1 = 0$.
2. The *symplectic group* $\mathrm{Sp}_{2n}(K)$, consisting of all matrices $x \in \mathrm{GL}_{2n}(K)$ that preserve a non-degenerate alternating bilinear form. Explicitly, it is defined by the equation:
$$^t x \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix} x = \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix}$$
where $^t x$ denotes the transpose of $x$.
3. The *odd special orthogonal group* $\mathrm{SO}_{2n+1}(K)$ (where $\mathrm{char} K \neq 2$), consisting of all $x \in \mathrm{SL}_{2n+1}(K)$ that preserve a non-degenerate symmetric bilinear form, given by:
$$^t x s x = s \qquad \text{where} \qquad s = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & J \\ 0 & J & 0 \end{pmatrix}$$
4. The *even special orthogonal group* $\mathrm{SO}_{2n}(K)$ (where $\mathrm{char} K \neq 2$), consisting of all $x \in \mathrm{SL}_{2n}(K)$ satisfying the same relation $^t x s x = s$, but equipped with the $2n \times 2n$ symmetric matrix:
$$s = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}$$

The classification of *all* abstract subgroups of $\mathrm{GL}_n(K)$ is intractable. For instance, by Cayley's Theorem, every finite group embeds into $\mathrm{GL}_n(K)$ for some $n$. Furthermore, the following famous theorem illustrates just how wildly infinite the abstract subgroups of $\mathrm{GL}_n(K)$ can be:

**Theorem 1.1.9: Tits Alternative**

Let $G$ be a finitely generated linear group (i.e., an abstract subgroup of $\mathrm{GL}_n(k)$) over a field $k$. Then one of the following two mutually exclusive possibilities occurs:

1. $G$ is virtually solvable (i.e., it has a solvable subgroup of finite index), or
2. $G$ contains a non-abelian free group (i.e., it has a subgroup isomorphic to the free group on two generators).

***Proof*:**
See [17].

However, if we restrict our attention from arbitrary abstract subgroups to *closed algebraic* subgroups (and specifically, to connected reductive algebraic groups), the geometric rigidity allows us to fully classify them. See theorem 7.3.6 for details (or the chapter on quiver theory in [7] for a proof using root systems).

Next, let us introduce two more subgroups of $\mathrm{GL}_n$ that are vital for their role in the structural decomposition of any algebraic group $G$. Namely, we have a natural subnormal series:

$$e \trianglelefteq U \trianglelefteq N \trianglelefteq L \trianglelefteq G^\circ \trianglelefteq G$$

where the successive quotients belong to highly restrictive, well-understood geometric categories:

1. $G$ contains a unique connected normal subgroup $G^\circ$ such that $G/G^\circ$ is a finite group (see proposition 1.1.4).
2. $G^\circ$ contains a largest connected affine normal subgroup $L$ (the linear part) such that $G^\circ/L$ is an abelian variety (see theorem 1.1.7).
3. $L$ contains a largest connected affine solvable normal subgroup $N$ (the radical $R(L)$) such that $L/N$ is a semisimple algebraic group (see definition 6.2.10).
4. $N$ contains a largest connected unipotent normal subgroup $U$ (the unipotent radical $R_u(L)$) such that $N/U$ is a torus (see theorem 6.1.7).

Thus, *tori* and *unipotent* groups are singled out for their immediate usefulness as the fundamental building blocks of affine groups:

**Definition 1.1.10: [Algebraic] Torus**

Let $\mathrm{GL}(V)_k$ be the general linear group over a finite-dimensional $k$-vector space $V$, and let $T \subseteq \mathrm{GL}(V)$ be a closed affine subgroup. Let $D \subseteq \mathrm{GL}(V)$ be the subgroup of all diagonal matrices with respect to some basis. Then $T$ is said to be of *multiplicative type* if, when extending to the algebraic closure $\overline{k}$, $T_{\overline{k}}$ is conjugate to a subgroup of $D_{\overline{k}}$. In other words, there exists a basis for $V_{\overline{k}}$ such that $T_{\overline{k}}$ is represented entirely by diagonal matrices. Equivalently, the elements of $T$ are mutually commuting semisimple endomorphisms of $V$.

If $T$ is furthermore connected, it is called a *torus*.

The name comes from the fact that over the complex numbers $\mathbb{C}$, the group of points of an algebraic torus is isomorphic to a subgroup of $(\mathbb{C}^\times)^n$, which contains the standard topological torus $(S^1)^n$ as its maximal compact subgroup.

**Definition 1.1.11: Unipotent Group**

Let $\mathrm{GL}(V)_k$ be the general linear group over a finite-dimensional $k$-vector space $V$, and let $G \subseteq \mathrm{GL}(V)$ be a closed affine subgroup. Then $G$ is called *unipotent* if there exists a basis of $V$ relative to which $G$ is contained in the standard unipotent subgroup $U_n$ of all $n \times n$ upper-triangular matrices with 1s on the diagonal:

$$\begin{pmatrix} 1 & * & \cdots & * & * \\ 0 & 1 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & * \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix}$$

In particular, all elements of a unipotent group are unipotent endomorphisms of $V$.

We will develop the formal definitions and implications of these building blocks—including solvable, reductive, simple, quasi-simple, and semi-simple groups—systematically in the upcoming sections.

### 1.1.2 Hopf Algebras

Let $G = \operatorname{Spec} A$ be an affine algebraic group (over $k$) so that $\mu : G \times_k G \to G$, $\iota : G \to G$ and $e : \operatorname{Spec} k \to G$ are the maps satisfying the group operations. Because the global sections functor $\Gamma(-)$ from affine schemes to $k$-algebras is fully faithful (and contravariant), these maps correspond precisely to "co"-maps on $A$. Namely:

1. multiplication corresponds to comultiplication: $\Delta : A \to A \otimes_k A$ (recall the tensor product is the coproduct in $\mathbf{Alg}_k$, see [7, p. 15.2]).
2. inverse corresponds to the *antipode*: $S : A \to A$.
3. identity corresponds to the *counit*: $\epsilon : A \to k$.

Let us take a moment to internalize the co-versions of the axioms. Recall that the group operation is associative. In terms of maps, the associativity of $\mu : G \times G \to G$ is expressed by the equality:

$$\mu \circ (\mu \times \mathrm{id}_G) = \mu \circ (\mathrm{id}_G \times \mu)$$

as maps $G \times G \times G \to G$. Applying the functor to $A$, we reverse the arrows and replace $\times$ with $\otimes$. Thus, the multiplication $\mu$ becomes comultiplication $\Delta$, and we obtain the *coassociativity* axiom:

$$(\Delta \otimes \mathrm{id}_A) \circ \Delta = (\mathrm{id}_A \otimes \Delta) \circ \Delta$$

This ensures that "splitting" an element once and then splitting the first factor is the same as splitting the second factor.

The identity element $e$ is defined by the property $g \cdot e = g = e \cdot g$. The "unit" of the group is a map $e : \operatorname{Spec} k \to G$, and the corresponding "counit" is the dual map $\epsilon : A \to k$. Just as the unit element allows us to collapse one factor of a product $G \times G \to G$ seamlessly, the counit allows us to collapse one factor of the tensor product $A \otimes A$ back into $k$. The axiom becomes:

$$(\epsilon \otimes \mathrm{id}_A) \circ \Delta = \mathrm{id}_A \quad \text{via the canonical isomorphism } k \otimes A \cong A$$

Finally, we consider the inverse. In a group, the inverse is a map $\iota : G \to G$ satisfying $g \cdot g^{-1} = e$. Let $\Delta_G : G \to G \times G$ be the diagonal map $g \mapsto (g, g)$. The axiom states that the following composition is the trivial map (factoring through $e$):

$$G \xrightarrow{\Delta_G} G \times G \xrightarrow{\mathrm{id} \times \iota} G \times G \xrightarrow{\mu} G \quad \text{equals} \quad G \to \operatorname{Spec} k \xrightarrow{e} G$$

Dualizing this to $A$, the diagonal map on spaces becomes the algebra multiplication map $m : A \otimes A \to A$ (since functions multiply pointwise). The multiplication map $\mu$ on the group becomes the comultiplication $\Delta$. The inverse $\iota$ becomes the *antipode* $S$. Reversing the chain above yields the definition of the antipode:

$$m \circ (\mathrm{id} \otimes S) \circ \Delta = u \circ \epsilon$$

where $u \circ \epsilon$ is the composition $A \xrightarrow{\epsilon} k \xrightarrow{u} A$ (where $u$ is the unit map of the algebra $k \to A$), representing the "constant function" on the group.

Thus, a Hopf Algebra is simply a ring that has enough "co-structure" to simulate being the ring of functions on a group.

**Definition 1.1.12: Hopf Algebra**

Let $k$ be a field. A *Hopf algebra* is a $k$-vector space $A$ equipped with an algebra structure $(A, m, u)$ and a coalgebra structure $(A, \Delta, \epsilon)$ that are compatible (i.e., $\Delta$ and $\epsilon$ are algebra homomorphisms), along with a $k$-linear map $S : A \to A$ called the *antipode* (or *Sweedler map*) such that:

$$m \circ (S \otimes \mathrm{id}_A) \circ \Delta = u \circ \epsilon = m \circ (\mathrm{id}_A \otimes S) \circ \Delta$$

Note that there was no mention of $A$ coming from an algebraic group in the definition. If $G = \operatorname{Spec} A$ is an affine algebraic group, then $A = \Gamma(G)$ is a commutative Hopf algebra, though naturally not necessarily *cocommutative* as the corresponding group multiplication need not be commutative. Given $G$, we can compute $\Delta$ directly, as it must satisfy the commutative diagram:

$$\begin{tikzcd} G \times G \arrow[r, "\mu"] & G \\ A \otimes A & A \arrow[l, "\Delta"'] \arrow[u, "\text{corresponds to}"'] \end{tikzcd}$$

namely:

$$\Delta(f)(a, b) = f(\mu(a, b))$$

**1.1.13 Why $S$ for antipode?** The reader may wonder why we introduce the notation $S$ for the antipode instead of $\iota$, and the strange summation notation for $\Delta$. In the geometric case of an algebraic group $G$, the antipode $S$ is indeed just the pullback of the inverse map $\iota : G \to G$. That is, for a function $f \in \Gamma(G)$, $S(f)(g) = f(g^{-1})$. However, Hopf algebras are more general than algebraic groups. In the study of *Quantum Groups* (non-commutative Hopf algebras), the antipode $S$ is *not* an involution (i.e., $S^2 \neq \mathrm{id}$). Using the symbol $\iota$ (or $f^{-1}$) strongly suggests an inverse that cancels out perfectly or squares to the identity. The symbol $S$ reminds us that this is a linear operator on the algebra, which may have infinite order or behave like a "skew" inverse.

There is also a shorthand called *Sweedler notation* that we will use. The map $\Delta : A \to A \otimes A$ sends an element $a$ to a tensor. A precise notation would be:

$$\Delta(a) = \sum_{i=1}^{n} a_{i,1} \otimes a_{i,2}$$

This becomes unreadable very quickly. For example, the coassociativity axiom $(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta$ would require nested sums and double indices:

$$\sum_i \left( \sum_j (a_{i,1})_{j,1} \otimes (a_{i,1})_{j,2} \right) \otimes a_{i,2}$$

Sweedler's notation drops the dummy summation indices, focusing purely on the "positions" in the tensor product. We write:

$$\Delta(a) = \sum a_{(1)} \otimes a_{(2)}$$

The power of this notation appears when we iterate $\Delta$. By coassociativity, it does not matter how we parenthesize the split into three tensors, so we can seamlessly write:

$$(\Delta \otimes \mathrm{id})\Delta(a) = \sum a_{(1)} \otimes a_{(2)} \otimes a_{(3)}$$

This turns complex diagrammatic proofs into algebraic calculations that visually mimic associativity.

**Example 1.1.14: Hopf Algebras**

1. Let $G = \mathbb{G}_a = \operatorname{Spec} k[x]$ so that $\mu(a, b) = a + b$. Then we need:

$$\Delta(f)(a, b) = f(a + b)$$

As $k[x]$ is generated by $x$, it is enough to define it on $x$ and extend. Namely:

$$\Delta(x)(y_1, y_2) = y_1 + y_2$$

which in $k[x] \otimes_k k[x] \cong k[y_1, y_2]$ is represented by the tensors $x \otimes 1 + 1 \otimes x$. Hence comultiplication is given by:

$$\Delta(x) = x \otimes 1 + 1 \otimes x$$

and so for any polynomial:

$$\Delta(p(x)) = p(x \otimes 1 + 1 \otimes x)$$

For the identity, we have the pullback $\epsilon(f)(a) = f(e(a)) = f(0)$. Therefore on the generator:

$$\epsilon(x) = 0$$

Finally, for $S(f)(a) = f(\iota(a))$, we can evaluate at the generator to get:

$$S(x)(a) = x(-a) = -a \qquad \Rightarrow \qquad S(x) = -x$$

Let us see how these regular functions evaluate on points $a, b \in k$. Treating elements of $k \otimes k$ as functions on $k \times k$, we get:

$$\Delta(x)(a,b) = (x \otimes 1 + 1 \otimes x)(a,b) = a \cdot 1 + 1 \cdot b = a + b$$

Recall that when affine sets were first defined, it was emphasized to forget the vector-space structure of $k^n$ when considering $\mathrm{Hom}_{\mathrm{Reg}}(k^n, k)$. By forcing the coordinate ring to remember this Hopf-algebra structure, we seamlessly recover the vector-space addition!

2. Let $G = \mathbb{G}_m \cong \operatorname{Spec} k[x, x^{-1}]$ where $\mu(a,b) = ab$. Then:

$$\Delta(x)(a,b) = ab \qquad \Rightarrow \qquad \Delta(x) = x \otimes x$$

$$\epsilon(x)(a) = x(1) = 1 \qquad \Rightarrow \qquad \epsilon(x) = 1$$

and similarly for inversion $\iota(a) = a^{-1}$:

$$S(x) = x^{-1}$$

3. Let $G = \mathrm{GL}_n = \operatorname{Spec} k[x_{11}, \ldots, x_{nn}, \det^{-1}] = \operatorname{Spec} A$. Then $A$ is generated by $n^2$ variables $x_{ij}$ and the inverse of the determinant. The function $x_{ij}$ evaluates a matrix $M$ and returns the entry in row $i$, column $j$. i.e., $x_{ij}(M) = M_{ij}$.

Geometrically, the multiplication per-component of two matrices $M, N$ is:

$$(MN)_{ij} = \sum_{k=1}^{n} M_{ik} N_{kj}$$

Thus, evaluating the comultiplication:

$$\Delta(x_{ij})(M,N) = x_{ij}(MN) = (MN)_{ij} = \sum_{k=1}^{n} M_{ik} N_{kj}$$

And so,

$$\Delta(x_{ij}) = \sum_{k=1}^{n} x_{ik} \otimes x_{kj}$$

Similarly, the counit evaluates at the identity matrix $I$:

$$\epsilon(x_{ij}) = x_{ij}(I) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

and the antipode represents the matrix inverse:

$$S(x_{ij}) = \det(x)^{-1} \cdot (\mathrm{Cofactor}_{ji})$$

4. Let $G = \operatorname{Spec} \operatorname{Sym}^\bullet_R(M^\vee)$ be the geometric vector bundle associated to a finitely generated projective $R$-module $M$ over a base ring $R$. Just as $\mathbb{G}_a$ remembered its addition, this relative scheme remembers the fiberwise additive structure of $M$.

   As the ring $\operatorname{Sym}^\bullet_R(M^\vee)$ is generated as an $R$-algebra by the degree-1 elements $m \in M^\vee$, it is enough to define our maps on these linear generators and extend $R$-linearly. Geometrically, addition happens fiberwise, so if $a, b$ are vectors in the same fiber, $\mu(a, b) = a + b$. Then we need:
$$\Delta(m)(a, b) = m(a + b) = m(a) + m(b)$$
   which in $\operatorname{Sym}^\bullet_R(M^\vee) \otimes_R \operatorname{Sym}^\bullet_R(M^\vee)$ is represented by $m \otimes 1 + 1 \otimes m$. Hence:
$$\Delta(m) = m \otimes 1 + 1 \otimes m$$
   For the identity, the zero section $e$ maps to the zero vector in every fiber. We have the pullback $\epsilon(m)(a) = m(e(a)) = m(0) = 0$, giving us:
$$\epsilon(m) = 0$$
   (and $\epsilon(r) = r$ for the degree-0 scalars $r \in R$).

   Finally, for the antipode $S(m) = m \circ \iota$, we can evaluate at a generator to get:
$$S(m)(a) = m(\iota(a)) = m(-a) = -m(a) \qquad \Rightarrow \qquad S(m) = -m$$
   Notice how this is exactly the Hopf-algebra structure of $\mathbb{G}_a$, but promoted to the relative setting over $R$! By taking the symmetric algebra of the dual module $M^\vee$, we didn't just build an arbitrary scheme; we naturally packaged the module's additive structure into a group scheme over $R$ whose fibers perfectly preserve the linear geometry.

## 1.2 Algebraic Groups as Functors

When picking $G = \mathrm{GL}_n(K)$ or $G = \mathrm{SL}_n(K)$, there is a choice of the field (or ring) $K$. We can eliminate this choice by capturing the required properties functorially. Let $S \subseteq k[x_1, \ldots, x_n]$ be a set of polynomials and let $R$ be an arbitrary $k$-algebra. We can look at the solution set of these polynomials in $R$:
$$V(S)(R) = \{(a_1, \ldots, a_n) \in R^n \mid f(a_1, \ldots, a_n) = 0 \text{ for all } f \in S\}$$
Given a $k$-algebra homomorphism $\varphi : R \to R'$, we naturally get a set-theoretic map $V(S)(R) \to V(S)(R')$ simply by applying $\varphi$ to each coordinate. This assignment $R \mapsto V(S)(R)$ therefore defines a functor:
$$\mathbf{Alg}_k \longrightarrow \mathbf{Set}$$
We can thus think of an affine algebraic set not as a static geometric space, but as a functor that eats a $k$-algebra and spits out a set of solutions. Naturally, if our solution sets always form a group, and the transition maps are group homomorphisms, we instead have a functor:
$$\mathbf{Alg}_k \longrightarrow \mathbf{Grp}$$
We can define an *affine algebraic group* abstractly as such a functor. For example, we can define the functor $\mathrm{SL}_n : \mathbf{Alg}_k \to \mathbf{Grp}$ where any $k$-algebra $R$ maps to $\mathrm{SL}_n(R)$, the group of $n \times n$ matrices

with entries in $R$ satisfying the equation $\det(X) - 1 = 0$, where the determinant is the standard polynomial:

$$\det(X) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) X_{1,\sigma(1)} \cdots X_{n,\sigma(n)} \in k[x_{11}, \ldots, x_{nn}]$$

Because our functors must be defined by polynomial equations, this radically restricts the kinds of functors we consider. For example, you cannot have an "indicator" functor such as:

$$F(R) = \begin{cases} (k, +) & \text{if } R = k \\ \{e\} & \text{otherwise} \end{cases}$$

There are no polynomials whose solution set suddenly vanishes when you map $k \hookrightarrow k[x]$.

Let us now formalize this. Recall that if $k$ is Noetherian, then the polynomial ring $k[x_1, \ldots, x_n]$ is also Noetherian (by the Hilbert Basis Theorem, see [7]), meaning every ideal is finitely generated. Thus, our functor $V(S)$ is exactly the same as $V(I)$ where $I$ is the ideal generated by $S$. Let $A = k[x_1, \ldots, x_n]/I$.

A $k$-algebra homomorphism $\psi : A \to R$ is completely determined by where it sends the generators $x_i$. Let $a_i = \psi(x_i) \in R$. Because $\psi(f) = 0$ for all $f \in I$, the tuple $(a_1, \ldots, a_n)$ must be a root of $I$ in $R$. Therefore, the functor sending $R$ to the zero-set of $I$ is canonically isomorphic to the functor:

$$h^A(R) := \operatorname{Hom}_{\mathbf{Alg}_k}(A, R)$$

Functors of the form $h^A$ are called *representable functors*, and $A$ is the representing object. Thus, an affine algebraic group scheme over $k$ is exactly a representable functor $\mathbf{Alg}_k \to \mathbf{Grp}$.

How does the group structure on the functor $h^A$ correspond to the Hopf algebra structure on $A$? Consider a $k$-algebra homomorphism $\Delta : A \to A \otimes_k A$. For any $k$-algebra $R$, we define the multiplication of two points $f_1, f_2 \in h^A(R)$ via the composition:

$$f_1 \cdot f_2 : A \xrightarrow{\Delta} A \otimes_k A \xrightarrow{f_1 \otimes f_2} R \otimes_k R \xrightarrow{m_R} R$$

where $m_R(r_1 \otimes r_2) = r_1 r_2$ is the internal multiplication of the ring $R$. Because $\Delta$ is an algebra homomorphism, the resulting map $f_1 \cdot f_2$ is indeed a valid $k$-algebra homomorphism $A \to R$. This binary operation on $h^A(R)$ is natural in $R$. Therefore, the comultiplication map completely captures the group structure, and we can define an affine algebraic group entirely as a pair $(A, \Delta)$.

**Example 1.2.1: Group Schemes as Functors**

To see how algebraic equations simulate group functors via their representing algebras, consider the following fundamental examples:

1. The Additive Group $\mathbb{G}_a$: The functor assigns $R \mapsto (R, +)$. It is represented by the polynomial ring $A = k[x]$. A map $\varphi \in \operatorname{Hom}(k[x], R)$ is uniquely determined by where it sends $x$, say $\varphi(x) = a \in R$. Given two maps $\varphi_a(x) = a$ and $\varphi_b(x) = b$, we want their product in the group to correspond to $a + b$. Applying the convolution formula with $\Delta(x) = x \otimes 1 + 1 \otimes x$:

$$(\varphi_a \cdot \varphi_b)(x) = m_R \circ (\varphi_a \otimes \varphi_b)(x \otimes 1 + 1 \otimes x) = m_R(a \otimes 1 + 1 \otimes b) = a + b$$

2. The Multiplicative Group $\mathbb{G}_m$: The functor assigns $R \mapsto (R^\times, \times)$, the group of invertible elements of $R$. It is represented by the Laurent polynomial ring $A = k[x, x^{-1}]$. A map $\varphi \in \operatorname{Hom}(k[x, x^{-1}], R)$ must send $x$ to an element $a \in R$ such that $a$ is invertible (because

$x^{-1}$ must map to $a^{-1}$). Thus the points are exactly $R^\times$. With $\Delta(x) = x \otimes x$, the convolution formula yields:

$$(\varphi_a \cdot \varphi_b)(x) = m_R \circ (\varphi_a \otimes \varphi_b)(x \otimes x) = m_R(a \otimes b) = a \cdot b$$

3. The Roots of Unity $\mu_n$: The functor assigns $R \mapsto \{r \in R \big| r^n = 1\}$, which forms a subgroup of $R^\times$ under multiplication. It is represented by the quotient algebra $A = k[x]/(x^n - 1)$. Because it is a quotient of the algebra for $\mathbb{G}_m$, it is a closed subgroup of $\mathbb{G}_m$, and inherits the exact same comultiplication $\Delta(x) = x \otimes x$.
4. The Infinitesimal Additive Group $\alpha_p$: Suppose our base field $k$ has characteristic $p > 0$. Consider the functor $R \mapsto \{r \in R \big| r^p = 0\}$. Because the characteristic is $p$, the binomial expansion $(a + b)^p = a^p + b^p = 0$ holds, meaning this set is actually closed under addition, forming a subgroup of $(R, +)$. It is represented by $A = k[x]/(x^p)$. If we only ever evaluated this functor on fields or reduced rings, the output would trivially be $\{0\}$, and we would think this group is just the trivial group. However, if we evaluate it on the dual numbers $R = k[\epsilon]/(\epsilon^2)$, any multiple $c\epsilon$ satisfies $(c\epsilon)^p = 0$ (since $p \geq 2$). Thus, $\alpha_p(k[\epsilon]/(\epsilon^2)) \cong (k, +)$. The functorial perspective allows us to see the "infinitesimal" structure that classical topology completely misses!

## Yoneda's Lemma and Morphisms

If algebraic groups are functors, what are homomorphisms between them? They are simply natural transformations. The translation between the functorial world and the algebraic world is governed by one of the most pervasive theorems in category theory.

**Theorem 1.2.2: Yoneda Lemma (for $k$-algebras)**

Let $A$ and $B$ be $k$-algebras, and let $h^A, h^B : \mathbf{Alg}_k \to \mathbf{Set}$ be their representable functors. There is a canonical bijection:

$$\mathrm{Nat}(h^B, h^A) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Alg}_k}(A, B)$$

given by evaluating a natural transformation $\eta : h^B \to h^A$ on the identity map $\mathrm{id}_B \in h^B(B)$.

For our purposes, the Yoneda Lemma says that every natural transformation of group functors $G \to H$ comes from a unique Hopf algebra homomorphism $\mathcal{O}(H) \to \mathcal{O}(G)$. This means we can interchangeably prove theorems about maps by drawing commutative diagrams of Hopf algebras, or by doing basic group theory on the $R$-valued points $G(R) \to H(R)$ for an arbitrary test ring $R$. The latter is almost always easier and more intuitive.

### 1.2.1 Arithmetic Tools: Base Change and Restriction

By treating algebraic groups as functors, we gain access to two powerful operations that are indispensable for number theory: moving an algebraic group from one base field to another.

**Definition 1.2.3: Base Change (Extension of Scalars)**

Let $G$ be an affine algebraic group over $k$ represented by the Hopf algebra $A$. Let $K$ be a field extension of $k$ (or more generally, any $k$-algebra). The *base change* of $G$ to $K$, denoted $G_K$, is the algebraic group over $K$ defined by the functor:

$$G_K(R) = G(R)$$

for any $K$-algebra $R$ (where $R$ is viewed as a $k$-algebra via the structure map $k \to K \to R$).

Algebraically, the Hopf algebra representing $G_K$ is simply the tensor product $A_K = A \otimes_k K$. For example, the equations cutting out $G$ might have rational coefficients ($k = \mathbb{Q}$), but we can base change to $\mathbb{C}$ to view the complex solutions $G_{\mathbb{C}}$.

However, in number theory, we often want to go in the *opposite* direction. If we have a group defined over a larger field $K$, can we view it as a group over the smaller field $k$? If the extension is finite, the answer is yes, through a process called Weil restriction.

**Definition 1.2.4: Weil Restriction of Scalars**

Let $K/k$ be a finite field extension (or a finite locally free algebra extension). Let $X$ be an affine algebraic group over $K$. The *Weil restriction of scalars*, denoted $\mathrm{Res}_{K/k}(X)$, is a functor $\mathbf{Alg}_k \to \mathbf{Grp}$ defined by:

$$\mathrm{Res}_{K/k}(X)(R) = X(R \otimes_k K)$$

for any $k$-algebra $R$.

Because $K$ is a finite free $k$-module, one can explicitly write down the coordinates of $R \otimes_k K$ in terms of a $k$-basis. Expanding the polynomial equations of $X$ in this basis guarantees that $\mathrm{Res}_{K/k}(X)$ is representable by a finitely generated $k$-algebra, making it a valid affine algebraic group over $k$. Geometrically, if $X$ has dimension $d$ over $K$, then $\mathrm{Res}_{K/k}(X)$ has dimension $d \cdot [K : k]$ over $k$.

**Example 1.2.5: The Complex Multiplicative Group**

Let $K = \mathbb{C}$ and $k = \mathbb{R}$. Consider the multiplicative group $\mathbb{G}_{m,\mathbb{C}}$ over $\mathbb{C}$. Its $\mathbb{C}$-points are $\mathbb{C}^\times$. We want to construct an algebraic group $G$ over $\mathbb{R}$ such that $G(\mathbb{R}) = \mathbb{C}^\times$.

We apply the Weil restriction: $G = \mathrm{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}})$. By definition, for any $\mathbb{R}$-algebra $R$, $G(R) = (R \otimes_{\mathbb{R}} \mathbb{C})^\times$.

How do we see this as an explicit real variety? Since $\mathbb{C} \cong \mathbb{R}[i]/(i^2+1)$, an element $z \in R \otimes_{\mathbb{R}} \mathbb{C}$ can be uniquely written as $x + iy$ with $x, y \in R$. The element is invertible if and only if its algebraic norm $x^2 + y^2$ is invertible in $R$. Thus, as a real algebraic group, we have:

$$\mathrm{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}}) = \operatorname{Spec} \mathbb{R}\left[x, y, \frac{1}{x^2+y^2}\right]$$

This is a 2-dimensional algebraic group over $\mathbb{R}$. Base changing this group *back* to $\mathbb{C}$ yields $G_{\mathbb{C}} \cong \mathbb{G}_{m,\mathbb{C}} \times \mathbb{G}_{m,\mathbb{C}}$. The interplay between base change and restriction of scalars forms the foundation for the study of algebraic tori over non-algebraically closed fields.

**1.2.6 Remark: Affine Supergroups** For those interested in theoretical physics, the functorial perspective has another application. If we replace the category of commutative $k$-algebras with the category of *supercommutative* (or $\mathbb{Z}/2\mathbb{Z}$-graded) $k$-algebras, the exact same functorial framework produces *Affine Algebraic Supergroups*. These are the geometric objects underlying supersymmetry. The coordinate rings become super-Hopf algebras, containing both bosonic (commuting) and fermionic (anti-commuting) variables.

### 1.2.2 Closed Subgroups and Hopf Ideals

Just as an affine algebraic set is defined by a system of polynomial equations, a closed subspace is defined by simply adding more equations to that system. Algebraically, if $X = \operatorname{Spec}(A)$, a closed subscheme $Y \subseteq X$ corresponds to a quotient ring $A/I$ for some ideal $I \subset A$.

When our scheme is an algebraic group $G = \operatorname{Spec}(A)$, we want to know when a closed subscheme $H$ is actually a *subgroup*. Functorially, the definition is completely natural.

**Definition 1.2.7: Closed Subgroup**

Let $G$ be an affine algebraic group. A *closed subgroup* $H \subseteq G$ is a closed subscheme such that for every $k$-algebra $R$, the subset $H(R) \subseteq G(R)$ is a subgroup.

Because $H$ is a closed subscheme of an affine scheme, $H$ is automatically affine itself. Thus, the terms "closed subgroup" and "affine subgroup" are used interchangeably in this context.

Algebraically, the inclusion map $i : H \hookrightarrow G$ corresponds to a surjective $k$-algebra homomorphism $\pi : A \to A/I$. For $H$ to inherit the group structure, the ideal $I$ cannot be just any ideal; it must be compatible with the comultiplication $\Delta$, the counit $\epsilon$, and the antipode $S$.

**Definition 1.2.8: Hopf Ideal**

Let $(A, \Delta, \epsilon, S)$ be a Hopf algebra. An ideal $I \subseteq A$ is called a *Hopf ideal* if it satisfies:

1. $\Delta(I) \subseteq I \otimes_k A + A \otimes_k I$
2. $\epsilon(I) = 0$
3. $S(I) \subseteq I$

These three conditions perfectly mirror the geometric requirements for $H$ to be a subgroup:

1. The first condition ensures that the multiplication map $H \times H \to G$ factors through $H$. (Note that $I \otimes_k A + A \otimes_k I$ is exactly the kernel of the tensor product projection $A \otimes_k A \to A/I \otimes_k A/I$).
2. The second condition ensures that the identity element $e \in G$ actually lies inside $H$.
3. The third condition ensures that if $h \in H$, its inverse $h^{-1}$ also lies in $H$.

Thus, we have a perfect dictionary: closed subgroups of $G$ correspond exactly to Hopf ideals of $\mathcal{O}(G)$.

**Example 1.2.9: The Special Linear Group as a Subgroup**

Consider $G = \mathrm{GL}_n = \operatorname{Spec} k[x_{11}, \ldots, x_{nn}, \det^{-1}]$. We want to realize $\mathrm{SL}_n$ as a closed subgroup. Geometrically, $\mathrm{SL}_n$ is cut out by the single equation $\det(X) - 1 = 0$.

Algebraically, we consider the ideal $I = (\det(X) - 1)$. Is this a Hopf ideal? Since the determinant of a product is the product of the determinants, we know that $\Delta(\det(X)) = \det(X) \otimes \det(X)$. To check the first condition, we compute:

$$\begin{aligned}\Delta(\det(X) - 1) &= \det(X) \otimes \det(X) - 1 \otimes 1 \\ &= \det(X) \otimes \det(X) - \det(X) \otimes 1 + \det(X) \otimes 1 - 1 \otimes 1 \\ &= \det(X) \otimes (\det(X) - 1) + (\det(X) - 1) \otimes 1\end{aligned}$$

This clearly lies in $A \otimes_k I + I \otimes_k A$, satisfying the first condition.

For the counit, evaluating at the identity matrix gives $\epsilon(\det(X) - 1) = \det(I_n) - 1 = 1 - 1 = 0$.

For the antipode, we have $S(\det(X) - 1) = \det(X)^{-1} - 1$. To see that this belongs to the ideal $I$, we simply factor out the generator:

$$S(\det(X) - 1) = \det(X)^{-1} - 1 = \det(X)^{-1}\big(1 - \det(X)\big) = -\det(X)^{-1}\big(\det(X) - 1\big)$$

Since $-\det(X)^{-1}$ is a regular function in $A$, this expression explicitly belongs to the ideal $I = (\det(X) - 1)$.

Thus, $I$ is a Hopf ideal, and $\mathrm{SL}_n$ is indeed a closed algebraic subgroup of $\mathrm{GL}_n$.

# 2

# *Actions, Representations, and Orbits*

In chapter 1, we built the algebraic group from the inside: its coordinate ring, its Hopf algebra structure, its functor of points. The group $G$ now sits in front of us as a well-defined geometric and algebraic object. But a group in isolation is rarely interesting. The power of group theory — in every branch of mathematics — comes from *what the group acts on.* The symmetric group $S_n$ is interesting because it permutes $\{1, \ldots, n\}$; the Galois group $\mathrm{Gal}(L/k)$ is interesting because it permutes roots; the general linear group $\mathrm{GL}_n$ is interesting because it acts on $n$-dimensional space. In each case, the action reveals the structure of the group.

This chapter formalizes the notion of an algebraic group acting on a scheme, and then specializes to the two most important cases: actions on vector spaces (representations) and actions on varieties (orbits). We will see that the Hopf algebra machinery of chapter 1 translates actions into comodule structures, and that the resulting category of representations $\mathrm{Rep}(G)$ is remarkably well-behaved. We close the chapter by studying the geometry of orbits and proving the Closed Orbit Lemma, a foundational result that will recur throughout the rest of the book.

## 2.1 Group Actions

Let $G$ be an affine algebraic group over a field $k$ and let $X$ be a $k$-scheme. An action of $G$ on $X$ should be a morphism that tells each group element how to move points of $X$, subject to the obvious compatibility: acting by $g$ then by $h$ is the same as acting by $hg$, and acting by the identity does nothing.

**Definition 2.1.1: Left Action**

Let $G$ be an affine algebraic group over $k$ and $X$ a $k$-scheme. A *left action* of $G$ on $X$ is a morphism of $k$-schemes

$$\alpha : G \times_k X \longrightarrow X$$

such that the following diagrams commute:

1. (Associativity) The diagram

$$\begin{tikzcd} G \times_k G \times_k X \arrow[r, "\mu\times\mathrm{id}_X"] \arrow[d, "\mathrm{id}_G\times\alpha"'] & G \times_k X \arrow[d, "\alpha"] \\ G \times_k X \arrow[r, "\alpha"'] & X \end{tikzcd}$$

   commutes, where $\mu : G \times_k G \to G$ is the multiplication of $G$.

2. (Unit) The composition

$$X \xrightarrow{\sim} \operatorname{Spec} k \times_k X \xrightarrow{e\times\mathrm{id}_X} G \times_k X \xrightarrow{\alpha} X$$

   is the identity morphism $\mathrm{id}_X$.

A $k$-scheme $X$ equipped with a left action of $G$ is called a *$G$-scheme*. When $X$ is a variety, we say $X$ is a *$G$-variety*.

There is an obvious dual notion of a *right action* $\alpha : X \times_k G \to X$, where the associativity diagram has $G$ acting on the right. In practice, we will work with left actions unless explicitly stated otherwise, and simply say "action" to mean "left action." Note that every left action $\alpha : G \times_k X \to X$ determines a right action $\beta : X \times_k G \to X$ by precomposing with the inverse: $\beta(x, g) = \alpha(g^{-1}, x)$, and vice versa. We denote $\alpha(g, x)$ by $g \cdot x$ when the action is clear from context.

## The Functorial Perspective

As in chapter 1, the functor of points gives us a second, often more convenient, way to think about actions. If $G$ is an affine algebraic group represented by the Hopf algebra $A = \mathcal{O}(G)$ and $X$ is an arbitrary $k$-scheme, then the action $\alpha : G \times_k X \to X$ is completely determined by what it does on $R$-valued points: for every $k$-algebra $R$, we get a map

$$\alpha_R : G(R) \times X(R) \longrightarrow X(R)$$

that makes $X(R)$ into a $G(R)$-set. The two axioms in definition 2.1.1 translate directly: for all $g, h \in G(R)$ and $x \in X(R)$,

$$(gh) \cdot x = g \cdot (h \cdot x), \qquad e \cdot x = x,$$

and these identities hold *naturally* in $R$, meaning they are compatible with every $k$-algebra homomorphism $R \to R'$. Thus:

**Proposition 2.1.2: Functorial Actions**

Let $G$ be an affine algebraic group over $k$ and $X$ a $k$-scheme. Giving a left action $\alpha : G \times_k X \to X$ is the same as giving, for every $k$-algebra $R$, an abstract group action $G(R) \times X(R) \to X(R)$ that is natural in $R$.

***Proof*:**
This is a direct consequence of the Yoneda Lemma applied to the category of schemes. The action $\alpha$ is a morphism $G \times_k X \to X$, which by Yoneda corresponds to a natural transformation $h^{G \times X} \to h^X$ of set-valued functors. Because the functor of points preserves products, $h^{G \times X}(R) = G(R) \times X(R)$ for any scheme $X$. Thus, an action is precisely a natural family of set maps $G(R) \times X(R) \to X(R)$. The commutativity of the geometric diagrams in definition 2.1.1 is perfectly equivalent to the abstract group action axioms holding for each $R$.

This is extremely useful in practice: to verify that a morphism $\alpha : G \times_k X \to X$ defines a valid action, we can simply check the group axioms on $R$-valued points for an arbitrary test ring $R$. No commutative diagrams of schemes are needed.

## Actions on Affine Schemes: Coactions

Now suppose $X = \operatorname{Spec} B$ is affine as well. The action map $\alpha : G \times_k X \to X$ is a morphism of affine schemes $\operatorname{Spec}(A \otimes_k B) \to \operatorname{Spec} B$. Dualizing this gives us a $k$-algebra homomorphism going in the opposite direction:

$$\rho : B \longrightarrow A \otimes_k B$$

This map $\rho$ is called a *left coaction* of the Hopf algebra $A = \mathcal{O}(G)$ on $B$. The contravariance of the Spec functor preserves the natural "order" of the spaces: $G$ acting on the left of $X$ yields $A$ sitting on the left of the tensor product.

Explicitly, writing $\rho(b) = \sum b_{(-1)} \otimes b_{(0)}$ in Sweedler-type notation (where the subscript $(-1)$ refers to the "$A$-part" and $(0)$ to the "$B$-part"), the geometric left action on points translates as follows. For $g \in G(R) = \operatorname{Hom}_{\mathbf{Alg}_k}(A, R)$ and a "function" $b \in B$:

$$(g \cdot b)(x) = \sum g(b_{(-1)}) \cdot b_{(0)}(x)$$

That is, the coaction decomposes each function $b$ on $X$ into a sum of products: a function on $G$ times a function on $X$.

**Definition 2.1.3: Left Coaction**

Let $(A, \Delta, \epsilon, S)$ be a Hopf algebra over $k$ and $B$ a $k$-algebra. A *left coaction* of $A$ on $B$ is a $k$-algebra homomorphism

$$\rho : B \longrightarrow A \otimes_k B$$

satisfying:

1. (Coassociativity) $(\mathrm{id}_A \otimes \rho) \circ \rho = (\Delta \otimes \mathrm{id}_B) \circ \rho$ as maps $B \to A \otimes_k A \otimes_k B$.
2. (Counit) $(\epsilon \otimes \mathrm{id}_B) \circ \rho = \mathrm{id}_B$ under the canonical isomorphism $k \otimes_k B \cong B$.

A $k$-algebra $B$ equipped with a left coaction of $A$ is called a *left $A$-comodule algebra*.

$$\begin{tikzcd} B \arrow[r, "\rho"] \arrow[d, "\rho"'] & A \otimes B \arrow[d, "\mathrm{id}_A \otimes \rho"] \\ A \otimes B \arrow[r, "\Delta \otimes \mathrm{id}_B"'] & A \otimes A \otimes B \end{tikzcd}$$

Coassociativity

$$\begin{tikzcd} B \arrow[r, "\rho"] \arrow[dr, "\mathrm{id}_B"'] & A \otimes B \arrow[d, "\epsilon \otimes \mathrm{id}_B"] \\ & k \otimes B \cong B \end{tikzcd}$$

Counit

The coassociativity axiom is the exact dual of the associativity of the group action: it says that "decomposing a function on $X$ with respect to $G$, and then further decomposing the $X$-part with respect to $G$" gives the same answer as "decomposing the function with respect to the group multiplication directly." The counit axiom is the dual of $e \cdot x = x$: evaluating the $G$-part at the identity recovers the original function.

The upshot is a clean algebraic dictionary:

$$\left\{ \begin{array}{c} \text{Left Actions of } G \text{ on} \\ \text{an affine scheme } X = \operatorname{Spec} B \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Left coactions of } \mathcal{O}(G) \\ \text{on the } k\text{-algebra } B \end{array} \right\}$$

**Example 2.1.4: The Left Regular Action**

Every group acts on itself by left multiplication: $\alpha(g, x) = gx$. For $G = \operatorname{Spec} A$, the corresponding left coaction is precisely the comultiplication:

$$\rho = \Delta : A \longrightarrow A \otimes_k A$$

To see why, recall from definition 1.1.12 that $\Delta(f)(g, x) = f(gx) = f(\mu(g, x))$, which is exactly the pullback of $f$ along the multiplication map $\mu$. The coassociativity of this coaction is just the coassociativity of $\Delta$ itself, and the counit axiom is $(\epsilon \otimes \mathrm{id}) \circ \Delta = \mathrm{id}$, which is one of the standard Hopf algebra axioms.

**Example 2.1.5: The Conjugation Action**

Let $G$ be an affine algebraic group. Then $G$ acts on itself by *conjugation*:

$$\alpha : G \times G \longrightarrow G, \qquad \alpha(g, x) = gxg^{-1}$$

Functorially, for any $k$-algebra $R$ and elements $g, x \in G(R)$, the action is $g \cdot x = gxg^{-1}$. It is immediate from the group axioms on $G(R)$ that this is a well-defined action. In Hopf algebra terms, the corresponding left coaction $\rho : A \to A \otimes_k A$ is incredibly elegant when written in Sweedler notation:

$$\rho(f) = \sum f_{(1)} S(f_{(3)}) \otimes f_{(2)}$$

where $S$ is the antipode. (Writing this out as a composition of tensor maps requires weaving the antipode and the algebra multiplication map together, which is why Sweedler notation is universally preferred). The conjugation action is the geometric incarnation of "inner automorphisms" and takes center stage in chapter 3, where its derivative yields the *adjoint representation*.

**Example 2.1.6: The Natural Action of $\mathrm{GL}_n$ on $\mathbb{A}^n$**

Let $G = \mathrm{GL}_n$ and $X = \mathbb{A}^n = \operatorname{Spec} k[x_1, \dots, x_n]$. For any $k$-algebra $R$, an element $g = (g_{ij}) \in \mathrm{GL}_n(R)$ acts on a column vector $(a_1, \dots, a_n) \in R^n = \mathbb{A}^n(R)$ by standard matrix multiplication:

$$g \cdot a = \left( \sum_j g_{1j} a_j, \dots, \sum_j g_{nj} a_j \right)$$

The corresponding left coaction $\rho : k[x_1, \dots, x_n] \to \mathcal{O}(\mathrm{GL}_n) \otimes_k k[x_1, \dots, x_n]$ sends each coordinate function to:

$$\rho(x_i) = \sum_{j=1}^{n} t_{ij} \otimes x_j$$

where $t_{ij}$ are the matrix-entry functions on $\mathrm{GL}_n$. Let us verify left coassociativity on the generator $x_i$. On one hand, expanding the $X$-part:

$$(\mathrm{id} \otimes \rho)(\ \rho(x_i)\ ) = \sum_j t_{ij} \otimes \rho(x_j) = \sum_{j,l} t_{ij} \otimes t_{jl} \otimes x_l$$

On the other hand, expanding the $G$-part:

$$(\Delta \otimes \mathrm{id})(\ \rho(x_i)\ ) = \sum_j \Delta(t_{ij}) \otimes x_j = \sum_{j,k} t_{ik} \otimes t_{kj} \otimes x_j$$

where the last step uses the standard comultiplication of $\mathrm{GL}_n$. Changing the dummy summation indices makes the two expressions identical, confirming coassociativity!

**Example 2.1.7: $\mathrm{GL}_n$ Acting on $M_n$ by Conjugation**

Let $G = \mathrm{GL}_n$ act on $X = M_n \cong \mathbb{A}^{n^2}$ by conjugation: $g \cdot A = gAg^{-1}$. This is a perfectly well-defined action on a *non-group* variety. Functorially, for any $k$-algebra $R$, the group $\mathrm{GL}_n(R)$ acts on $M_n(R)$ by conjugation.

Over an algebraically closed field $k = \overline{k}$, the orbits of this action are exactly the *similarity classes* of matrices. By the Jordan normal form theorem, these are parametrized by Jordan types, which are partitions of $n$ decorated with eigenvalues. The stabilizer $\operatorname{stab}(A)$ of a matrix $A$ is its centralizer $C_{\mathrm{GL}_n}(A) = \{g \in \mathrm{GL}_n \mid gA = Ag\}$. We will return to this example in §2.4 to study the orbit structure in detail.

## Equivariant Morphisms

If $G$ acts on two schemes $X$ and $Y$, we need a notion of "maps that respect the action." Functorially, the definition writes itself.

**Definition 2.1.8: $G$-Equivariant Morphism**

Let $G$ be an affine algebraic group acting on $k$-schemes $X$ and $Y$ via $\alpha_X$ and $\alpha_Y$ respectively. A morphism $f : X \to Y$ is called *$G$-equivariant* (or a *morphism of $G$-schemes*) if the diagram

$$\begin{tikzcd} G \times_k X \arrow[r, "\mathrm{id}_G \times f"] \arrow[d, "\alpha_X"'] & G \times_k Y \arrow[d, "\alpha_Y"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

commutes. Equivalently, for every $k$-algebra $R$, every $g \in G(R)$, and every $x \in X(R)$:

$$f(g \cdot x) = g \cdot f(x)$$

When both $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} C$ are affine with left coactions $\rho_B : B \to A \otimes_k B$ and $\rho_C : C \to A \otimes_k C$, a morphism $f : X \to Y$ corresponds to a $k$-algebra homomorphism $f^\# : C \to B$. The equivariance condition translates to the commutativity of:

$$\begin{tikzcd} C \arrow[r, "\rho_C"] \arrow[d, "f^\#"'] & A \otimes_k C \arrow[d, "\mathrm{id}_A \otimes f^\#"] \\ B \arrow[r, "\rho_B"'] & A \otimes_k B \end{tikzcd}$$

That is, $f^\#$ is a *morphism of left $A$-comodule algebras.*

With this notion of morphism, the $G$-schemes over $k$ form a category, and the assignment $X \mapsto \mathcal{O}(X)$ gives a perfect anti-equivalence between *affine* $G$-schemes and left $\mathcal{O}(G)$-comodule algebras.

### Exercise 2.1.1

1. Let $G = \operatorname{Spec} A$ act on itself by twisted right multiplication: $\alpha(g, x) = xg^{-1}$. Show that this is a valid left action, and that the corresponding left coaction $\rho : A \to A \otimes_k A$ is given by $\rho = (S \otimes \mathrm{id}) \circ \tau \circ \Delta$, where $\tau : A \otimes A \to A \otimes A$ is the tensor swap $a \otimes b \mapsto b \otimes a$ and $S$ is the antipode.

2. Show that the *trivial* action of $G$ on $X = \operatorname{Spec} B$ (where $g \cdot x = x$ for all $g, x$) corresponds to the left coaction $\rho(b) = 1_A \otimes b$ for all $b \in B$. Verify the comodule axioms directly.

3. Let $\mathbb{G}_m$ act on $\mathbb{A}^n$ by scalar multiplication: $t \cdot (a_1, \ldots, a_n) = (ta_1, \ldots, ta_n)$ for $t \in \mathbb{G}_m(R) = R^\times$. Write down the left coaction $\rho : k[x_1, \ldots, x_n] \to k[t, t^{-1}] \otimes_k k[x_1, \ldots, x_n]$ explicitly, and verify coassociativity.

4. More generally, let $B$ be a finitely generated $k$-algebra. Show that giving a $\mathbb{G}_m$-action on $\operatorname{Spec} B$ is equivalent to giving a $\mathbb{Z}$-grading $B = \bigoplus_{n \in \mathbb{Z}} B_n$ as a $k$-algebra.

   (*Hint:* Given a left coaction $\rho : B \to k[t, t^{-1}] \otimes_k B$, define $B_n = \{b \in B \mid \rho(b) = t^n \otimes b\}$. Show this gives a direct sum decomposition and that the grading is multiplicative.)

5. Let $G$ act on $k$-schemes $X$ and $Y$. Define the *diagonal action* of $G$ on $X \times_k Y$ via $g \cdot (x, y) = (g \cdot x, g \cdot y)$, and show this is a well-defined action. If $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} C$ are both affine, express the left coaction $\rho : B \otimes_k C \to A \otimes_k (B \otimes_k C)$ in terms of the coactions $\rho_B$ and $\rho_C$ and the multiplication map of $A$.

## 2.2 Representations and Comodules

We now specialize from actions on arbitrary schemes to actions on *vector spaces*. This is the most algebraically tractable setting: the acted-upon object has the rigid linear structure of a $k$-vector space, and the resulting theory is representation theory. In this section, we set up the basic definitions and prove the fundamental finiteness result that makes the theory work: every representation of an affine algebraic group, no matter how large, is built from finite-dimensional pieces.

### Representations as Comodules

**Definition 2.2.1: Representation**

Let $G$ be an affine algebraic group over $k$. A *(rational) representation* of $G$ is a $k$-vector space $V$ together with a homomorphism of algebraic groups

$$\rho : G \longrightarrow \mathrm{GL}(V)$$

When $V$ is finite-dimensional, this means that $\rho$ is a morphism of affine algebraic groups. We call $V$ a *$G$-module* and often suppress $\rho$ from the notation, writing $g \cdot v$ or simply $gv$ for $\rho(g)(v)$.

Functorially, a representation on a finite-dimensional $V$ assigns to each $k$-algebra $R$ a group homomorphism $\rho_R : G(R) \to \mathrm{GL}(V \otimes_k R)$, natural in $R$. As $V \otimes_k R$ is a free $R$-module, choosing a basis $\{v_1, \ldots, v_n\}$ for $V$ identifies $\mathrm{GL}(V \otimes_k R)$ with $\mathrm{GL}_n(R)$, and $\rho$ becomes a morphism of group functors $G \to \mathrm{GL}_n$.

The coaction viewpoint from §2.1 applies cleanly here, but with a notable twist. Recall that a left action on an affine scheme $X = \operatorname{Spec} B$ dualizes to a *left* coaction $B \to \mathcal{O}(G) \otimes_k B$ on its coordinate ring. A representation is a left action on a vector space $V$. However, rather than dualizing to look at the coordinate ring of functions $\mathrm{Sym}^\bullet(V^*)$, we want to keep the comodule structure on $V$ itself. Because $V$ naturally pairs with its dual $V^*$, skipping the dualization to functions effectively "flips" the tensor factors. Thus, a left representation on $V$ corresponds to a *right* comodule structure on $V$!

**Definition 2.2.2: Comodule**

Let $A = \mathcal{O}(G)$ be the coordinate Hopf algebra of an affine algebraic group $G$. A *right $A$-comodule* is a $k$-vector space $V$ equipped with a $k$-linear map

$$\rho : V \longrightarrow V \otimes_k A$$

called the *comodule structure map* (or *coaction*), satisfying:

1. (Coassociativity) $(\rho \otimes \mathrm{id}_A) \circ \rho = (\mathrm{id}_V \otimes \Delta) \circ \rho$,
2. (Counit) $(\mathrm{id}_V \otimes \epsilon) \circ \rho = \mathrm{id}_V$.

The axioms mirror those of algebraic coactions, but now $V$ is a vector space rather than an algebra, and $\rho$ is $k$-linear rather than a $k$-algebra homomorphism. We write $\rho(v) = \sum v_{(0)} \otimes v_{(1)}$ in Sweedler-type notation. The coassociativity axiom then reads:

$$\sum v_{(0)(0)} \otimes v_{(0)(1)} \otimes v_{(1)} = \sum v_{(0)} \otimes v_{(1)(1)} \otimes v_{(1)(2)}$$

which, by the same abuse of notation as for Hopf algebras, we write unambiguously as $\sum v_{(0)} \otimes v_{(1)} \otimes v_{(2)}$.

**Proposition 2.2.3: Representations = Comodules**

Let $G$ be an affine algebraic group over $k$ with coordinate Hopf algebra $A = \mathcal{O}(G)$. There is an equivalence of categories:

$$\left\{ \begin{array}{c} \text{Finite-dimensional} \\ \text{left representations of } G \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Finite-dimensional} \\ \text{right } A\text{-comodules} \end{array} \right\}$$

***Proof*:**
Given a representation $\rho : G \to \mathrm{GL}(V)$, choose a basis $\{v_1, \dots, v_n\}$ for $V$. The representation is determined by the matrix coefficient functions $m_{ij} \in A$ defined by

$$\rho(g)(v_j) = \sum_{i=1}^{n} m_{ij}(g)\, v_i$$

for all $g \in G(R)$ and all $k$-algebras $R$. Define the right coaction $\rho_V : V \to V \otimes_k A$ by

$$\rho_V(v_j) = \sum_{i=1}^{n} v_i \otimes m_{ij}$$

We must verify the comodule axioms. For coassociativity, note that $\rho$ being a group homomorphism means $\rho(gh) = \rho(g)\rho(h)$, which in matrix coefficients reads $m_{ij}(gh) = \sum_l m_{il}(g)\, m_{lj}(h)$. Dualizing: $\Delta(m_{ij}) = \sum_l m_{il} \otimes m_{lj}$. Then:

$$(\mathrm{id}_V \otimes \Delta)(\rho_V(v_j)) = \sum_i v_i \otimes \Delta(m_{ij}) = \sum_{i,l} v_i \otimes m_{il} \otimes m_{lj}$$

while:

$$(\rho_V \otimes \mathrm{id}_A)(\rho_V(v_j)) = \sum_i \rho_V(v_i) \otimes m_{ij} = \sum_{i,l} v_l \otimes m_{li} \otimes m_{ij}$$

Reindexing $l \leftrightarrow i$ in the second expression shows these are equal. For the counit, $\epsilon(m_{ij}) = m_{ij}(e) = \delta_{ij}$ (the identity matrix acts trivially), so $(\mathrm{id}_V \otimes \epsilon)(\rho_V(v_j)) = \sum_i v_i \cdot \delta_{ij} = v_j$.

Conversely, given a comodule $(V, \rho_V)$ with $\rho_V(v_j) = \sum_i v_i \otimes m_{ij}$, the formulas above run in reverse: for each $k$-algebra $R$ and $g \in G(R) = \mathrm{Hom}_{\mathbf{Alg}_k}(A, R)$, define $\rho(g)(v_j) = \sum_i g(m_{ij})\, v_i$. The comodule axioms guarantee this is a group homomorphism. That the two constructions are mutually inverse is immediate from the definitions.

This equivalence means we can work interchangeably in either language: "$G$ acts on $V$" and "$V$ is an $A$-comodule" are the exact same statement. When doing concrete calculations with matrix coefficients, the comodule language is usually cleaner. When checking that a construction is well-defined, the functorial language (does $G(R)$ act on $V \otimes_k R$ for all $R$?) is usually easier.

**Example 2.2.4: Comodule Structures on Low-Dimensional Spaces**

Let $G = \mathrm{GL}_2$ with coordinate ring $A = k[a, b, c, d, (ad - bc)^{-1}]$ and comultiplication $\Delta\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \otimes \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ (matrix multiplication of entries).

1. *The standard representation.* Let $V = k^2$ with basis $\{e_1, e_2\}$. The coaction is $\rho(e_1) = e_1 \otimes a + e_2 \otimes c$ and $\rho(e_2) = e_1 \otimes b + e_2 \otimes d$. This encodes the tautological action of a matrix $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ on column vectors.
2. *The determinant representation.* Let $W = k$ (one-dimensional). The coaction is $\rho(1) = 1 \otimes (ad - bc)$. This is the representation $g \mapsto \det(g) \in \mathrm{GL}_1(R) = R^\times$. As a one-dimensional representation, it is entirely determined by a single *group-like element* $\det = ad - bc \in A$.
3. *The trivial representation.* Let $W = k$ with coaction $\rho(1) = 1 \otimes 1$. Every group element acts as the identity. The element $1 \in A$ is always group-like (since $\Delta(1) = 1 \otimes 1$), so the trivial representation exists for every algebraic group.

It should be noted that we can recover finite group representations in this context:

**Example 2.2.5: The Regular Representation of a Finite Group Scheme**

Let $G = \operatorname{Spec} A$ be a *finite* algebraic group, so that $A$ is a finite-dimensional $k$-vector space. Then $A$ is itself a right $A$-comodule via the comultiplication $\Delta : A \to A \otimes_k A$. This recovers the *regular representation* of $G$ on the space of its own functions (specifically, the left representation induced by right-translation).

When $G$ is a constant finite group scheme (so $A = \prod_{g \in G} k$ with basis $\{e_g\}_{g \in G}$), the coaction sends $e_g$ to $\sum_h e_h \otimes e_{h^{-1}g}$, perfectly recovering the classical regular representation of a finite group on its group algebra.

**2.2.6 Remark: "Rational" Representations** The adjective "rational" in definition 2.2.1 is not a restriction on the field $k$; it distinguishes the representations we study from more exotic possibilities. An abstract group homomorphism $G(k) \to \mathrm{GL}_n(k)$ need not come from a morphism of algebraic groups; for instance, there exist wild field automorphisms of $\mathbb{C}$ that induce abstract group homomorphisms $\mathbb{G}_m(\mathbb{C}) = \mathbb{C}^\times \to \mathbb{C}^\times$ which are not given by any polynomial (or even any measurable function). By requiring $\rho$ to be a morphism of algebraic groups — equivalently, by requiring the matrix coefficients $m_{ij}$ to be regular functions in $\mathcal{O}(G)$ — we mathematically rule out such pathology. In this book, "representation" will always mean "rational representation."

## The Fundamental Finiteness Theorem

In the proof of the linearization theorem (lemma 1.1.8), the essential step was showing that every element of $\mathcal{O}(G)$ lies in a finite-dimensional $G$-stable subspace. That argument used two ingredients: the coaction sends a single element to a *finite* sum of tensors, and coassociativity forces these summands to span a $G$-stable subspace. The same argument works for *any* comodule, and the result is so fundamental that it deserves its own clean statement and proof.

**Theorem 2.2.7: Fundamental Finiteness Theorem**

Let $G$ be an affine algebraic group over $k$ and let $V$ be a (possibly infinite-dimensional) right $\mathcal{O}(G)$-comodule. Then:

1. Every element $v \in V$ is contained in a finite-dimensional sub-comodule of $V$.
2. $V$ is the union (equivalently, the directed colimit) of its finite-dimensional sub-comodules.

***Proof*:**
Let $\rho : V \to V \otimes_k A$ be the coaction, where $A = \mathcal{O}(G)$. Fix $v \in V$ and write

$$\rho(v) = \sum_{i=1}^{d} v_i \otimes a_i$$

with $v_1, \ldots, v_d \in V$ and $a_1, \ldots, a_d \in A$. We may assume the $a_i$ are linearly independent over $k$ (otherwise, consolidate terms to reduce $d$). Let $W = \mathrm{span}_k\{v_1, \ldots, v_d\}$. We will show that $W$ is a sub-comodule of $V$, i.e. $\rho(W) \subseteq W \otimes_k A$.

Apply the coassociativity axiom $(\rho \otimes \mathrm{id}_A) \circ \rho = (\mathrm{id}_V \otimes \Delta) \circ \rho$ to $v$:

$$\sum_{i=1}^{d} \rho(v_i) \otimes a_i = \sum_{i=1}^{d} v_i \otimes \Delta(a_i)$$

The right-hand side lies in $W \otimes_k A \otimes_k A$, since each $v_i \in W$. Now, the $a_i$ are linearly independent in $A$, so viewing the equation in $V \otimes_k A \otimes_k A \cong (V \otimes_k A) \otimes_k A$ and applying an arbitrary functional $\varphi \in A^\vee$ to the rightmost tensor factor, we get:

$$\sum_{i=1}^{d} \rho(v_i) \cdot \varphi(a_i) = \sum_{i=1}^{d} v_i \otimes (\mathrm{id}_A \otimes \varphi)(\Delta(a_i))$$

Since $\varphi$ was arbitrary and the $a_i$ are linearly independent, we may choose $\varphi$ such that $\varphi(a_j) = \delta_{ij}$ for each $j$. This gives $\rho(v_j) \in W \otimes_k A$ for each $j$. Therefore $W$ is a sub-comodule of $V$.

The counit axiom gives $v = (\mathrm{id}_V \otimes \epsilon)(\rho(v)) = \sum_i v_i\, \epsilon(a_i)$, which is a $k$-linear combination of the $v_i$, so $v \in W$. This proves (1). Statement (2) follows immediately: $V = \bigcup_v W_v$ where $W_v$ ranges over the finite-dimensional sub-comodules containing $v$. Since any two elements $v, w \in V$ are contained in the finite-dimensional sub-comodule generated by $W_v + W_w$ (which is itself a sub-comodule by the same argument applied to a spanning set), this union is directed.

The reader who followed the linearization proof in chapter 1 will recognize this as the exact same core idea: the coaction only produces *finite* sums, so coassociativity forces a finiteness constraint. But the consequences here are vastly broader. If you allow an abstract infinite group to act on an infinite-dimensional vector space, it is exceedingly rare that every single vector generates a finite-dimensional stable subspace. The Fundamental Finiteness Theorem shows that the "rationality" of algebraic representations (the fact that the action must arise from a polynomial coaction) imposes a massive, rigid geometric constraint that abstract groups lack. Every representation of an algebraic group, no matter how wild or large, is locally finite.

## Subrepresentations, Quotients, and Irreducibility

With the finiteness theorem in hand, we can set up the basic categorical notions.

**Definition 2.2.8: Morphisms and Sub-comodules**

Let $V$ and $W$ be right $A$-comodules with coactions $\rho_V$ and $\rho_W$.

1. A *morphism of representations* (or *intertwiner*, or *G-equivariant linear map*) is a $k$-linear map $\varphi : V \to W$ satisfying

$$\rho_W \circ \varphi = (\varphi \otimes \mathrm{id}_A) \circ \rho_V$$

   That is, the diagram

$$\begin{tikzcd} V \arrow[r, "\varphi"] \arrow[d, "\rho_V"'] & W \arrow[d, "\rho_W"] \\ V \otimes_k A \arrow[r, "\varphi\otimes\mathrm{id}_A"'] & W \otimes_k A \end{tikzcd}$$

   commutes. We write $\mathrm{Hom}_G(V, W)$ for the $k$-vector space of all such maps.

2. A *subrepresentation* (or *sub-comodule*) is a $k$-linear subspace $U \subseteq V$ such that $\rho_V(U) \subseteq U \otimes_k A$.

Functorially, a morphism of representations $\varphi : V \to W$ is simply a $k$-linear map such that $\varphi(g \cdot v) = g \cdot \varphi(v)$ for all $k$-algebras $R$, all $g \in G(R)$, and all $v \in V \otimes_k R$. A subrepresentation $U \subseteq V$ is a subspace such that $U \otimes_k R$ is stable under $G(R)$ for every $R$.

Given a subrepresentation $U \subseteq V$, the quotient space $V/U$ inherits a comodule structure. This requires a brief check:

**Proposition 2.2.9: Quotient Representations**

Let $V$ be a representation of $G$ and $U \subseteq V$ a subrepresentation. Then the quotient $V/U$ carries a unique comodule structure making the projection $\pi : V \to V/U$ a morphism of representations.

***Proof*:**
Since $\rho_V(U) \subseteq U \otimes_k A$, the coaction $\rho_V : V \to V \otimes_k A$ descends to a well-defined $k$-linear map $\overline{\rho} : V/U \to (V/U) \otimes_k A$ via $\overline{\rho}(v + U) = (\pi \otimes \mathrm{id}_A)(\rho_V(v))$. This is well-defined because if $v' = v + u$ for some $u \in U$, then $\rho_V(v') = \rho_V(v) + \rho_V(u)$, and $(\pi \otimes \mathrm{id}_A)(\rho_V(u)) = 0$ since $\rho_V(u) \in U \otimes_k A$. The comodule axioms for $\overline{\rho}$ follow from those of $\rho_V$ by applying $\pi \otimes \mathrm{id}_A$ (or $\pi \otimes \mathrm{id}_A \otimes \mathrm{id}_A$) throughout.

**Definition 2.2.10: Irreducible Representation**

A nonzero representation $V$ of $G$ is called *irreducible* (or *simple*) if it has no sub-comodules other than 0 and $V$ itself.

By the Fundamental Finiteness Theorem (theorem 2.2.7), every nonzero comodule contains a nonzero *finite-dimensional* sub-comodule. In particular, an irreducible representation must itself be finite-dimensional. This is a highly nontrivial statement: it says that the notion of irreducibility for algebraic group representations automatically implies finite-dimensionality, without our having to impose it as a hypothesis! Contrast this to, for example, $p$-adic representation theory [1], where the vast majority of smooth irreducible representations are infinite-dimensional, and the only finite-dimensional irreducible representations are trivial 1-dimensional characters.

A natural question is whether every representation decomposes as a direct sum of irreducible ones (i.e., whether $\mathrm{Rep}(G)$ is *semisimple*). For finite groups in characteristic zero, this is Maschke's theorem. For algebraic groups, the answer depends heavily on the group:

**Example 2.2.11: Failure of Complete Reducibility**
Let $G = \mathbb{G}_a$ over an arbitrary field $k$, and consider the two-dimensional representation $\rho : \mathbb{G}_a \to \mathrm{GL}_2$ defined on $R$-valued points by

$$\rho(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad t \in \mathbb{G}_a(R) = R$$

The subspace $U = k \cdot e_1$ is a subrepresentation (it is fixed by $G$), but there is no complementary subrepresentation. Indeed, a complement would be a line $k \cdot (ae_1 + be_2)$ with $b \neq 0$ that is $G$-stable. But $\rho(t)(ae_1 + be_2) = (a + tb)e_1 + be_2$, which lies in the line $k \cdot (ae_1 + be_2)$ only if the multiple is 1 (since the $e_2$ coordinate is preserved). This implies $tb = 0$ for all $t \in R$ and all $R$. Since $b \neq 0$, this is clearly false. So $V$ is indecomposable but not irreducible.

The group $\mathbb{G}_a$ is unipotent, and this failure of complete reducibility is a general phenomenon for unipotent groups in any characteristic: their only irreducible representation is the trivial one, but they typically admit non-trivial extensions like the one above. (Note that for *reductive* groups, complete reducibility holds in characteristic 0, but can fail in characteristic $p > 0$).

## Invariants and Fixed Points

Given a representation $V$ of $G$, the most basic question is: which vectors are fixed by the entire group? These form a subrepresentation on which $G$ acts trivially.

**Definition 2.2.12: $G$-Invariants**

Let $V$ be a representation of an affine algebraic group $G$ with right coaction $\rho : V \to V \otimes_k A$. The subspace of *$G$-invariants* is

$$V^G := \{v \in V \mid \rho(v) = v \otimes 1\}$$

Equivalently, $V^G = \{v \in V \mid g \cdot v = v \text{ for all } g \in G(R) \text{ and all } R\}$.

A representation $V$ is trivial if and only if $V^G = V$. The invariant subspace functor $V \mapsto V^G$ is left-exact but not exact in general: if $0 \to U \to V \to W \to 0$ is a short exact sequence of representations, the induced sequence $0 \to U^G \to V^G \to W^G$ is exact, but $V^G \to W^G$ need not be surjective. The failure of surjectivity is measured by the first rational cohomology group $H^1(G, U)$ (see, for example, [12]).

When $G$ acts on an affine scheme $X = \operatorname{Spec} B$ (so $B$ is a left comodule *algebra*), the invariant subalgebra $B^G$ is the *ring of invariants*. It governs the geometry of the "quotient" $X//G$ (the categorical quotient), which we will construct in chapter 4.

**Example 2.2.13: Invariants of $\mathrm{GL}_n$ Acting on Matrices**

Consider the conjugation action of $\mathrm{GL}_n$ on $M_n \cong \mathbb{A}^{n^2}$ from example 2.1.7. The ring of invariants $k[M_n]^{\mathrm{GL}_n}$ consists of all polynomial functions on matrices that are constant on conjugacy classes. By a classical theorem, these are exactly the symmetric functions of the eigenvalues, or equivalently, the coefficients of the characteristic polynomial:

$$k[M_n]^{\mathrm{GL}_n} = k[\sigma_1, \ldots, \sigma_n]$$

where $\sigma_i(A) = \operatorname{tr}(\wedge^i A)$ is the $i$-th coefficient of $\det(tI + A)$. This is the Chevalley restriction theorem for $\mathrm{GL}_n$ (see [7]), and its generalization to arbitrary reductive groups is a central result of Part II.

### Exercise 2.2.1

1. Let $G = \mathrm{SL}_2$ with coordinate ring $A = k[a, b, c, d]/(ad - bc - 1)$. Write down the coaction for the standard two-dimensional representation $V = k^2$. Verify both comodule axioms (coassociativity and counit) explicitly.

2. Show that one-dimensional representations of $G$ correspond bijectively to *group-like elements* of $\mathcal{O}(G)$, that is, elements $c \in \mathcal{O}(G)$ satisfying $\Delta(c) = c \otimes c$ and $\epsilon(c) = 1$.

3. Let $V$ be a representation of $G$ and let $\varphi : V \to W$ be a morphism of representations. Show that $\ker(\varphi)$ and $\operatorname{im}(\varphi)$ are sub-comodules of $V$ and $W$ respectively.

4. Let $V = k^2$ be the standard representation of $\mathrm{GL}_2$. The $d$-th symmetric power $\mathrm{Sym}^d(V)$ has basis $\{x^{d-i}y^i | 0 \le i \le d\}$ (monomials in two formal variables). Write down the coaction

$\rho : \mathrm{Sym}^d(V) \to \mathrm{Sym}^d(V) \otimes_k \mathcal{O}(\mathrm{GL}_2)$ explicitly. Show that when $k$ has characteristic 0, the representations $\{\mathrm{Sym}^d(V)\}_{d\geq 0}$ are all irreducible. (*Hint*: for irreducibility, show that a nonzero sub-comodule must contain a highest-degree or lowest-degree monomial, then use the coaction to generate everything.)

5. (Schur's Lemma for algebraic groups) Let $V$ and $W$ be finite-dimensional irreducible representations of $G$ over an algebraically closed field $k$. Show that $\mathrm{Hom}_G(V, W)$ is either 0 (if $V \ncong W$) or $k$ (if $V \cong W$). (*Hint*: use exercise 2.2.1 (3) and that $k$ is algebraically closed.)

## 2.3 The Category Rep($G$)

In the previous section, we defined representations, morphisms between them, and proved that every representation is a filtration of finite-dimensional pieces. We now assemble these ingredients into a single algebraic structure: the category Rep($G$) of all finite-dimensional representations of $G$. This category carries far more structure than a typical abelian category — it has tensor products, duals, and an identity object — making it a *symmetric monoidal category*. Understanding this structure is essential: in Part II, the decomposition of tensor products into irreducibles will be governed by the combinatorics of root systems and highest weights.

### Rep($G$) is Abelian

**Definition 2.3.1: The Category Rep($G$)**

Let $G$ be an affine algebraic group over $k$. We define Rep($G$) to be the category whose objects are finite-dimensional (rational) representations of $G$ and whose morphisms are $G$-equivariant $k$-linear maps (intertwiners). Equivalently, Rep($G$) is the category of finite-dimensional right $\mathcal{O}(G)$-comodules.

The first structural property we need is that Rep($G$) behaves like the category of modules over a ring: we can form kernels, cokernels, direct sums, and exact sequences.

**Theorem 2.3.2: Rep($G$) is Abelian**

The category Rep($G$) is a $k$-linear abelian category. Explicitly:

1. Each hom-set $\mathrm{Hom}_G(V, W)$ is a $k$-vector space, and composition is $k$-bilinear.
2. Finite direct sums exist: $V \oplus W$ with the diagonal coaction $\rho_{V\oplus W}(v, w) = (\rho_V(v), \rho_W(w))$ is a representation.
3. Every morphism $\varphi : V \to W$ in Rep($G$) has a kernel and cokernel in Rep($G$).
4. Every monomorphism is a kernel and every epimorphism is a cokernel.

***Proof*:**
Statement (1) is clear: $\mathrm{Hom}_G(V, W)$ is a $k$-subspace of $\mathrm{Hom}_k(V, W)$ (the equivariance condition is $k$-linear), and composition of equivariant maps is equivariant.

For (2), the direct sum $V \oplus W$ as a $k$-vector space has coaction $\rho(v, w) = \rho_V(v) + \rho_W(w) \in (V \oplus W) \otimes_k A$, where we identify $(V \oplus W) \otimes_k A \cong (V \otimes_k A) \oplus (W \otimes_k A)$. The comodule axioms follow componentwise.

For (3), let $\varphi : V \to W$ be a morphism in $\mathrm{Rep}(G)$. By exercise 2.2.1 (3), $\ker(\varphi) \subseteq V$ is a sub-comodule, hence an object of $\mathrm{Rep}(G)$, and the inclusion $\ker(\varphi) \hookrightarrow V$ is the categorical kernel. For the cokernel, $W/\mathrm{im}(\varphi)$ inherits a comodule structure by proposition 2.2.9 (applied to the sub-comodule $\mathrm{im}(\varphi) \subseteq W$), and the projection $W \to W/\mathrm{im}(\varphi)$ is the categorical cokernel.

Statement (4) follows from the corresponding fact in $\mathrm{Vec}_k$: a monomorphism $\varphi : V \hookrightarrow W$ in $\mathrm{Rep}(G)$ has a categorical kernel $K \hookrightarrow V$. Because $\varphi \circ i = 0 = \varphi \circ 0$, the monomorphism property implies $i = 0$, so $K = 0$. Since the categorical kernel is exactly the underlying vector space kernel, $\varphi$ is injective as a $k$-linear map, hence is the kernel of $W \to W/\varphi(V)$. Dually, an epimorphism is surjective (as can be verified on underlying vector spaces via its cokernel) and is the cokernel of its kernel.

The fact that $\mathrm{Rep}(G)$ is abelian means we can do homological algebra inside it: we can speak of short exact sequences $0 \to U \to V \to W \to 0$ of representations, of extensions, and (eventually) of derived functors. One also has the forgetful functor

$$\omega : \mathrm{Rep}(G) \longrightarrow \mathrm{Vec}_k, \qquad V \mapsto V$$

which forgets the $G$-action and remembers only the underlying vector space. This functor is exact and faithful (it detects zero maps), so it is a *fiber functor*. We will return to this point momentarily.

## Tensor Products and the Monoidal Structure

By using the comultiplication of the Hopf algebra $\mathcal{O}(G)$, the category $\mathrm{Rep}(G)$ has a multiplication structure by taking tensor product of representations. Let $V$ and $W$ be representations of $G$ with coactions $\rho_V : V \to V \otimes_k A$ and $\rho_W : W \to W \otimes_k A$. The tensor product $V \otimes_k W$ as a $k$-vector space carries a coaction defined by:

$$\rho_{V\otimes W} : V \otimes_k W \xrightarrow{\rho_V \otimes \rho_W} V \otimes_k A \otimes_k W \otimes_k A \xrightarrow{\mathrm{id}_V \otimes \tau \otimes \mathrm{id}_A} V \otimes_k W \otimes_k A \otimes_k A \xrightarrow{\mathrm{id}_{V\otimes W} \otimes m_A} V \otimes_k W \otimes_k A$$

where $\tau : A \otimes_k W \to W \otimes_k A$ is the tensor swap and $m_A : A \otimes_k A \to A$ is the multiplication of $A$. In Sweedler notation, for $v \in V$ and $w \in W$:

$$\rho_{V\otimes W}(v \otimes w) = \sum v_{(0)} \otimes w_{(0)} \otimes v_{(1)} w_{(1)}$$

Functorially, the formula is transparent: for $g \in G(R)$, the action on $V \otimes_k W$ is simply $g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w)$, the *diagonal action*. The coaction formula above is just the Hopf-algebraic encoding of this: to "act by $g$" on both factors, we must "split $g$ into two copies" using the comultiplication $\Delta$.

**Proposition 2.3.3: Tensor Product Comodule**

Let $V$ and $W$ be right $A$-comodules. Then $V \otimes_k W$ with the coaction $\rho_{V\otimes W}$ defined above is a right $A$-comodule.

***Proof*:**
We verify coassociativity. Using Sweedler notation, we need:

$$(\rho_{V\otimes W} \otimes \mathrm{id}_A) \circ \rho_{V\otimes W} = (\mathrm{id}_{V\otimes W} \otimes \Delta) \circ \rho_{V\otimes W}$$

The left side applied to $v \otimes w$ gives:

$$\sum v_{(0)(0)} \otimes w_{(0)(0)} \otimes v_{(0)(1)} w_{(0)(1)} \otimes v_{(1)} w_{(1)}$$

By coassociativity of $\rho_V$ and $\rho_W$, this equals:

$$\sum v_{(0)} \otimes w_{(0)} \otimes v_{(1)} w_{(1)} \otimes v_{(2)} w_{(2)}$$

The right side gives:

$$\sum v_{(0)} \otimes w_{(0)} \otimes \Delta(v_{(1)} w_{(1)})$$

Since $\Delta$ is an algebra homomorphism, $\Delta(v_{(1)}w_{(1)}) = \Delta(v_{(1)})\Delta(w_{(1)}) = \sum v_{(1)}w_{(1)} \otimes v_{(2)}w_{(2)}$. The two sides agree. The counit axiom is similar: $\epsilon(v_{(1)}w_{(1)}) = \epsilon(v_{(1)})\epsilon(w_{(1)}) = 1 \cdot 1 = 1$ since $\epsilon$ is an algebra homomorphism, so $(\mathrm{id} \otimes \epsilon) \circ \rho_{V\otimes W}(v \otimes w) = v \otimes w$.

The *trivial representation* $\mathbf{1} = k$ (with coaction $\rho(1) = 1 \otimes 1$) serves as the unit for this tensor product. The canonical isomorphisms $k \otimes_k V \cong V \cong V \otimes_k k$ are $G$-equivariant, and the associativity and symmetry isomorphisms

$$(U \otimes V) \otimes W \cong U \otimes (V \otimes W), \qquad V \otimes W \cong W \otimes V$$

are all $G$-equivariant (because they are natural transformations of the underlying vector space functors, and the coaction is defined in terms of the vector space structure). Thus Rep($G$) is a *symmetric monoidal category* with unit object $\mathbf{1}$.

## Duals and Rigidity

Every finite-dimensional representation has a *dual* (contragredient) representation, and this dual behaves as an “inverse” with respect to the tensor product in a precise categorical sense.

**Definition 2.3.4: Dual Representation**

Let $V$ be a finite-dimensional representation of $G$ with coaction $\rho_V(v_j) = \sum_i v_i \otimes m_{ij}$. The *dual* (or *contragredient*) representation is $V^\vee = \mathrm{Hom}_k(V, k)$ with coaction defined by:

$$\rho_{V^\vee}(v_j^*) = \sum_i v_i^* \otimes S(m_{ji})$$

where $\{v_j^*\}$ is the dual basis and $S$ is the antipode of $\mathcal{O}(G)$.

Functorially, the action of $g \in G(R)$ on $f \in V^\vee \otimes_k R = \mathrm{Hom}_R(V \otimes_k R, R)$ is $(g \cdot f)(v) = f(g^{-1} \cdot v)$, which is the usual contragredient action. The appearance of the antipode $S$ in the coaction formula is exactly the algebraic encoding of "act by the inverse": the antipode dualizes the group inversion $\iota : G \to G$.

**Proposition 2.3.5: Dual Comodule Axioms**

Let $V$ be a finite-dimensional right $A$-comodule. Then $V^\vee$ with the coaction in definition 2.3.4 is a right $A$-comodule.

***Proof*:**
We need two properties of the antipode. First, from the Hopf algebra axioms, $S$ is an *anti-homomorphism* for the comultiplication: $\Delta \circ S = \tau \circ (S \otimes S) \circ \Delta$, where $\tau$ is the tensor swap. Second, $\epsilon \circ S = \epsilon$. (Both are standard; see for instance [7].)

For coassociativity of $\rho_{V^\vee}$, we compute $(\rho_{V^\vee} \otimes \mathrm{id}) \circ \rho_{V^\vee}$ on $v_j^*$:

$$\sum_i \rho_{V^\vee}(v_i^*) \otimes S(m_{ji}) = \sum_{i,l} v_l^* \otimes S(m_{li}) \otimes S(m_{ji})$$

Meanwhile, $(\mathrm{id} \otimes \Delta) \circ \rho_{V^\vee}(v_j^*)$:

$$\sum_i v_i^* \otimes \Delta(S(m_{ji})) = \sum_i v_i^* \otimes \tau(S \otimes S)(\Delta(m_{ji})) = \sum_{i,l} v_i^* \otimes S(m_{jl}) \otimes S(m_{li})$$

using $\Delta(m_{ji}) = \sum_l m_{jl} \otimes m_{li}$ and the anti-homomorphism property. Reindexing $i \leftrightarrow l$ in the first expression gives agreement. For the counit: $(\mathrm{id} \otimes \epsilon)(v_i^* \otimes S(m_{ji})) = v_i^* \epsilon(S(m_{ji})) = v_i^* \epsilon(m_{ji}) = v_i^* \delta_{ji} = v_j^*$.

The dual representation satisfies the key property that makes $\mathrm{Rep}(G)$ a *rigid* monoidal category: the evaluation and coevaluation maps

$$\mathrm{ev} : V^\vee \otimes_k V \longrightarrow k, \qquad f \otimes v \longmapsto f(v)$$

$$\mathrm{coev} : k \longrightarrow V \otimes_k V^\vee, \qquad 1 \longmapsto \sum_i v_i \otimes v_i^*$$

are both $G$-equivariant. This is a direct consequence of the antipode axiom $m \circ (\mathrm{id} \otimes S) \circ \Delta = u \circ \epsilon$:

it ensures that $\mathrm{ev}(g \cdot f \otimes g \cdot v) = f(g^{-1}g \cdot v) = f(v)$ for all $g \in G(R)$, so ev intertwines the $G$-action on $V^\vee \otimes V$ with the trivial action on $k$. The verification for coev is dual.

Summarizing, we have shown:

**Theorem 2.3.6: Structure of Rep($G$)**

Let $G$ be an affine algebraic group over $k$. The category $\mathrm{Rep}(G)$ of finite-dimensional representations of $G$ is a $k$-linear, abelian, rigid symmetric monoidal category with unit object $\mathbf{1} = k$ (the trivial representation). The tensor product is given by the diagonal $G$-action, and the duality is the contragredient.

## Internal Hom and the Algebra of Matrix Coefficients

The rigidity of $\mathrm{Rep}(G)$ gives us an *internal hom*: for representations $V$ and $W$, the vector space $\mathrm{Hom}_k(V, W) \cong W \otimes_k V^\vee$ carries a natural $G$-action by conjugation:

$$(g \cdot \varphi)(v) = g \cdot \varphi(g^{-1} \cdot v), \qquad \varphi \in \mathrm{Hom}_k(V, W)$$

The $G$-invariants of this representation are precisely the intertwiners:

$$\mathrm{Hom}_k(V, W)^G = \mathrm{Hom}_G(V, W)$$

This shows that the hom-spaces in $\mathrm{Rep}(G)$ arise naturally from the monoidal structure, linking the "external" categorical structure to the "internal" representation-theoretic one.

The representations of $G$ also produce important elements in the coordinate ring $\mathcal{O}(G)$. If $V$ is a representation with coaction $\rho(v_j) = \sum_i v_i \otimes m_{ij}$, the functions $m_{ij} \in \mathcal{O}(G)$ are called *matrix coefficients*. More invariantly, for $v \in V$ and $f \in V^\vee$, the *matrix coefficient* $c_{f,v} \in \mathcal{O}(G)$ is defined by

$$c_{f,v}(g) = f(g \cdot v)$$

The span of all matrix coefficients from all finite-dimensional representations of $G$ is a subalgebra of $\mathcal{O}(G)$. In fact, by the Fundamental Finiteness Theorem (theorem 2.2.7), every element of $\mathcal{O}(G)$ lies in a finite-dimensional sub-comodule of the regular representation, and hence is a matrix coefficient. Thus:

**Corollary 2.3.7: $\mathcal{O}(G)$ is Spanned by Matrix Coefficients**

Let $G$ be an affine algebraic group over $k$. Then $\mathcal{O}(G)$ is spanned as a $k$-vector space by the matrix coefficients of all finite-dimensional representations of $G$. In particular, $\mathcal{O}(G) = \sum_{V \in \mathrm{Rep}(G)} c(V)$, where $c(V) \subseteq \mathcal{O}(G)$ denotes the subspace of matrix coefficients of $V$.

**2.3.8 Remark: The Tannakian Perspective** The structure of Rep$(G)$ described in theorem 2.3.6 is not just a convenient packaging of the representation theory. The *Tannaka–Krein duality* (in the algebraic setting, due to Saavedra-Rivano and Deligne–Milne) asserts the converse: the fiber functor $\omega : \mathrm{Rep}(G) \to \mathrm{Vec}_k$ together with the monoidal structure *completely determines* $G$. That is, $G$ can be reconstructed as the group of monoidal automorphisms of $\omega$. This means the category Rep$(G)$ is just as good a definition of $G$ as the Hopf algebra $\mathcal{O}(G)$ or the functor of points $R \mapsto G(R)$. We shall not develop this theory further, but the reader should keep this principle in mind: *an algebraic group is determined by its representations.*

**Example 2.3.9: Tensor Decomposition for $\mathrm{GL}_2$**

Let $V = k^2$ be the standard representation of $\mathrm{GL}_2$. The tensor square $V \otimes_k V$ is four-dimensional, and it decomposes (when $\mathrm{char}\, k \neq 2$) as

$$V \otimes V \cong \mathrm{Sym}^2(V) \oplus \bigwedge^2(V)$$

where $\mathrm{Sym}^2(V)$ is the three-dimensional symmetric square (spanned by $e_1 \otimes e_1$, $e_1 \otimes e_2 + e_2 \otimes e_1$, $e_2 \otimes e_2$) and $\bigwedge^2(V)$ is the one-dimensional exterior square (spanned by $e_1 \otimes e_2 - e_2 \otimes e_1$). This decomposition is $G$-equivariant: both summands are sub-comodules.

As a representation, $\bigwedge^2(V)$ is one-dimensional: $g$ acts by $\det(g)$. So $\bigwedge^2(V)$ is the determinant representation from example 2.2.4. Restricting to $\mathrm{SL}_2$ (where $\det = 1$), $\bigwedge^2(V)$ becomes the trivial representation, and we get the decomposition $V \otimes V \cong \mathrm{Sym}^2(V) \oplus \mathbf{1}$.

In characteristic 2, the decomposition fails: the map $v \otimes w - w \otimes v$ lands in $\mathrm{Sym}^2(V)$ (since $-1 = 1$), so $\bigwedge^2(V)$ is *not* a direct summand but rather a submodule of $\mathrm{Sym}^2(V)$. This is a first instance of a recurring theme: tensor product decompositions become more subtle in positive characteristic.

**Example 2.3.10: The Regular Representation Revisited**

Let $G = \mathrm{Spec}\, A$ be a finite algebraic group, so $\dim_k A < \infty$. The regular representation $A$ (with coaction $\Delta : A \to A \otimes_k A$) is an object of Rep$(G)$. By corollary 2.3.7, $A$ contains all matrix coefficients, so every irreducible representation of $G$ appears as a sub-comodule of $A$.

This generalizes a classical fact from finite group theory: if $G$ is a finite group and $k$ is algebraically closed with $\mathrm{char}\, k \nmid |G|$, then the regular representation $k[G]$ decomposes as $\bigoplus_V V^{\oplus \dim V}$, where the sum runs over all irreducible representations. For group *schemes* in positive characteristic, the regular representation may be indecomposable (for instance, the coordinate ring of $\alpha_p = \mathrm{Spec}\, k[t]/(t^p)$ is a local ring and hence indecomposable as a module over itself).

## Exercise 2.3.1

1. Let $V$ and $W$ be representations of $G = \mathbb{G}_m$ with coactions $\rho_V(v) = v \otimes t^n$ and $\rho_W(w) = w \otimes t^m$ (so $V$ and $W$ are one-dimensional representations on which $\mathbb{G}_m$ acts by the $n$-th and $m$-th power respectively). Compute the coaction on $V \otimes_k W$ and on $V^\vee$. Conclude that $\mathbb{G}_m$-representations

are "graded pieces": the tensor product of the $n$-th and $m$-th power representations is the $(n+m)$-th, and the dual of the $n$-th is the $(-n)$-th.

2. Let $0 \to U \xrightarrow{\iota} V \xrightarrow{\pi} W \to 0$ be a short exact sequence in $\mathrm{Rep}(G)$. Show that this sequence *splits* (i.e. $V \cong U \oplus W$ as representations) if and only if there exists a $G$-equivariant section $\sigma : W \to V$ with $\pi \circ \sigma = \mathrm{id}_W$. Give an example where the sequence splits as vector spaces but not as representations. (*Hint*: revisit example 2.2.11.)

3. Let $V$ be an $n$-dimensional representation of $G$. Show that the exterior powers $\bigwedge^d(V)$ for $0 \le d \le n$ are naturally representations of $G$. Show that $\bigwedge^n(V)$ is one-dimensional and corresponds to the group-like element $\det(\rho) \in \mathcal{O}(G)$, where $\rho : G \to \mathrm{GL}(V)$ is the representation map. (*Hint*: for the coaction, use the fact that the wedge product $v_1 \wedge \cdots \wedge v_d$ is the antisymmetrization of $v_1 \otimes \cdots \otimes v_d$, and the tensor power is already a representation.)

4. Let $V$ be a finite-dimensional representation of $G$. Verify directly that the evaluation map $\mathrm{ev} : V^\vee \otimes V \to k$ defined by $f \otimes v \mapsto f(v)$ is $G$-equivariant. That is, show

$$\sum \epsilon(S(m_{ji}) \cdot m_{il}) = \delta_{jl}$$

where $\rho(v_i) = \sum_l v_l \otimes m_{li}$, using the antipode axiom of the Hopf algebra.

5. Let $V$ and $W$ be finite-dimensional representations of $G$. Show that the natural isomorphism $\mathrm{Hom}_k(V, W) \cong W \otimes_k V^\vee$ is an isomorphism of representations (where the left side has the conjugation action and the right side has the tensor product coaction). Deduce that $\dim_k \mathrm{Hom}_G(V, W) = \dim_k(W \otimes V^\vee)^G$.

## 2.4 $G$-Varieties and the Closed Orbit Lemma

We now turn from the linear world of representations to the geometric world of group actions on varieties. The central objects of study are *orbits* — the subsets traced out by the action of $G$ on a single point — and *stabilizers* — the subgroups that fix a point. The interplay between these two objects is governed by a dimension formula that parallels the orbit-stabilizer theorem of finite group theory, but now the "sizes" are measured by dimension rather than cardinality.

The main result of this section is the *Closed Orbit Lemma*, which asserts that orbits are always well-behaved geometric objects (smooth, locally closed) and that among all orbits, those of minimal dimension must be closed. In practice, this is one of the most frequently invoked tools in the structure theory of algebraic groups.

Throughout this section, $k$ denotes an algebraically closed field and $G$ a connected affine algebraic group over $k$, unless otherwise stated. All varieties are over $k$.

### Orbit Maps and Stabilizers

Let $G$ act on a variety $X$ via $\alpha : G \times_k X \to X$. For a closed point $x \in X$, the action determines a morphism that records how $G$ moves $x$:

**Definition 2.4.1: Orbit Map**

Let $G$ act on a variety $X$ and let $x \in X(k)$ be a $k$-rational point. The *orbit map* at $x$ is the morphism

$$\varphi_x : G \longrightarrow X, \qquad g \longmapsto g \cdot x$$

obtained by restricting the action $\alpha : G \times_k X \to X$ to $G \times_k \{x\} \cong G$.

The image $\varphi_x(G) = \{g \cdot x \mid g \in G(k)\} \subseteq X$ is the *orbit* of $x$, denoted $G \cdot x$ or $O(x)$. The fiber over $x$ itself — the set of group elements that fix $x$ — is the stabilizer:

**Definition 2.4.2: Stabilizer**

Let $G$ act on a variety $X$ and let $x \in X(k)$. The *stabilizer* (or *isotropy group*) of $x$ is

$$G_x := \varphi_x^{-1}(x) = \{g \in G \mid g \cdot x = x\}$$

Equivalently, $G_x$ is the fiber product $G \times_X \operatorname{Spec} k(x)$, where $G \to X$ is the orbit map and $\operatorname{Spec} k(x) \to X$ is the inclusion of the point $x$.

**Proposition 2.4.3: The Stabilizer is a Closed Subgroup**

Let $G$ act on a variety $X$ and let $x \in X(k)$. Then $G_x$ is a closed subgroup of $G$.

***Proof*:**
Since $X$ is separated (it is a variety), the point $\{x\} \subseteq X$ is closed. The orbit map $\varphi_x : G \to X$ is a morphism of varieties, hence continuous, so $G_x = \varphi_x^{-1}(\{x\})$ is a closed subset of $G$. It remains to check that $G_x$ is a subgroup, but this is immediate from the functorial perspective: for any $k$-algebra $R$, the set $G_x(R) = \{g \in G(R) \mid g \cdot x = x\}$ is a subgroup of $G(R)$ (the identity fixes $x$, if $g$ and $h$ fix $x$ then so does $gh$, and if $g$ fixes $x$ then so does $g^{-1}$).

At the level of points, the orbit of $x$ is a *homogeneous space* for $G$: the group $G$ acts transitively on $G \cdot x$, and the orbit map induces a bijection of sets $G(k)/G_x(k) \xrightarrow{\sim} G \cdot x$. We would like to promote this to an isomorphism of varieties $G/G_x \cong G \cdot x$, but this requires constructing the quotient $G/G_x$ as a variety, which is a nontrivial problem that we defer to chapter 4. For now, we content ourselves with the set-theoretic picture and the following dimension formula.

**Proposition 2.4.4: Orbit-Stabilizer Dimension Formula**

Let $G$ act on a variety $X$ and let $x \in X(k)$. Then

$$\dim(G \cdot x) = \dim(G) - \dim(G_x)$$

where $\dim(G \cdot x)$ means the dimension of the closure $\overline{G \cdot x}$ (equivalently, of $G \cdot x$ itself as a locally closed subset, as we shall see shortly).

***Proof*:**
Consider the orbit map $\varphi_x : G \to \overline{G \cdot x}$, which is dominant by definition. For any point $y = g \cdot x$ in the image $G \cdot x$, the fiber $\varphi_x^{-1}(y)$ is exactly the left coset $gG_x$. Because translation by $g$ is an isomorphism of varieties, every non-empty fiber of this map is isomorphic to $G_x$, and hence has dimension $\dim(G_x)$. By the standard theorem on the dimension of fibers in algebraic geometry (see [5]), for a dominant morphism of varieties, the dimension of the generic fiber is the difference of the dimensions of the source and target. Since all fibers over the dense set $G \cdot x$ share the exact same dimension, we obtain $\dim(G_x) = \dim(G) - \dim(\overline{G \cdot x})$. Rearranging yields the formula.

**Example 2.4.5: Orbits of $\mathrm{GL}_n$ on $\mathbb{A}^n$**
Let $\mathrm{GL}_n$ act on $\mathbb{A}^n$ by the standard action (matrix multiplication on column vectors). There are exactly two orbits:

1. The *origin* $\{0\}$: The stabilizer is all of $\mathrm{GL}_n$, so $\dim(\{0\}) = n^2 - n^2 = 0$.
2. The *complement* $\mathbb{A}^n \setminus \{0\}$: Given any nonzero vector $v$, extend it to a basis $\{v, v_2, \ldots, v_n\}$ of $k^n$. The matrix with these columns lies in $\mathrm{GL}_n$ and sends $e_1$ to $v$. So $\mathrm{GL}_n$ acts transitively on $\mathbb{A}^n \setminus \{0\}$. The stabilizer of $e_1 = (1, 0, \ldots, 0)^T$ consists of matrices whose first column is $e_1$, which is isomorphic to $\mathbb{A}^{n-1} \rtimes \mathrm{GL}_{n-1}$ (of dimension $n^2 - n$). Thus $\dim(\mathbb{A}^n \setminus \{0\}) = n^2 - (n^2 - n) = n$.

The orbit $\mathbb{A}^n \setminus \{0\}$ is open (hence locally closed), and its closure is all of $\mathbb{A}^n$. The boundary is $\{0\}$, which is indeed a single orbit of strictly smaller dimension.

**Example 2.4.6: Conjugacy Classes in $M_n$**
We revisit example 2.1.7: $\mathrm{GL}_n$ acts on $M_n$ by conjugation, $g \cdot A = gAg^{-1}$. The orbits are conjugacy classes, and the stabilizer of $A$ is its centralizer $C_{\mathrm{GL}_n}(A)$. Consider $n = 2$ over $k = \overline{k}$ with $\mathrm{char} k = 0$. There are three types of orbits:

1. *Scalar matrices* $\lambda I$: The orbit is a single point $\{\lambda I\}$ and the stabilizer is all of $\mathrm{GL}_2$, so $\dim = 4 - 4 = 0$.
2. *Diagonalizable with distinct eigenvalues* $\mathrm{diag}(\lambda, \mu)$, $\lambda \neq \mu$: The stabilizer is the diagonal torus $T \cong \mathbb{G}_m \times \mathbb{G}_m$ (of dimension 2), so $\dim = 4 - 2 = 2$.
3. *Non-diagonalizable (Jordan block)* $\left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)$: The stabilizer consists of matrices of the form $\left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right)$ with $a \neq 0$ (of dimension 2), so $\dim = 4 - 2 = 2$.

The 0-dimensional orbits (scalar matrices) are closed. Surprisingly, for $\lambda \neq \mu$, the 2-dimensional orbit $O(\mathrm{diag}(\lambda, \mu))$ is *also* closed. This is because the characteristic polynomial is constant on orbits, and the only matrices with characteristic polynomial $(x - \lambda)(x - \mu)$ are the diagonalizable ones.

However, the orbit of a Jordan block is locally closed but *not* closed. The closure of the Jordan block orbit $\overline{O(J_\lambda)}$ contains the scalar matrix $\lambda I$. To see this, conjugate $J_\lambda$ by a diagonal matrix:

$$\begin{pmatrix} t & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} t^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \lambda & t \\ 0 & \lambda \end{pmatrix}$$

As we let $t \to 0$ in the closure, the off-diagonal entry vanishes, leaving $\lambda I$. The precise orbit structure for matrices with a single eigenvalue $\lambda$ is thus:

$$\overline{O(J_\lambda)} = O(J_\lambda) \cup \{\lambda I\}$$

This "degeneration" pattern — generic orbits specializing to more degenerate, smaller orbits — is typical of algebraic group actions, and understanding it precisely leads to the theory of orbit closures and singularities in geometric invariant theory.

## Orbits are Constructible

What kind of subset of $X$ is an orbit $G \cdot x$? It is the image of the orbit map $\varphi_x : G \to X$, and a fundamental theorem of algebraic geometry controls the images of morphisms.

Recall from [5] that a subset $S \subseteq X$ of a variety is called *constructible* if it is a finite Boolean combination of open and closed subsets, or equivalently, a finite disjoint union of locally closed subsets. Chevalley's theorem states that the image of a constructible set under a morphism of finite type is again constructible.

**Proposition 2.4.7: Orbits are Constructible**

Let $G$ act on a variety $X$ and let $x \in X(k)$. Then the orbit $G \cdot x$ is a constructible subset of $X$.

***Proof*:**
The orbit map $\varphi_x : G \to X$ is a morphism of varieties (hence of finite type). The group $G$ is itself constructible in $G$ (it is the whole space). By Chevalley's theorem on constructible images (see [5]), $G \cdot x = \varphi_x(G)$ is constructible in $X$.

Constructibility alone does not yet give us the clean geometric picture we want. The key step is to combine constructibility with the transitivity of the $G$-action. This will be the engine of the Closed Orbit Lemma.

First, we extract a general topological fact about constructible subsets that are homogeneous spaces.

**Lemma 2.4.8: Orbits are Open in their Closure**

Let $G$ be an algebraic group acting on a variety $X$ and let $Y = G \cdot x$ be an orbit. Then $Y$ is open in its closure $\overline{Y}$.

***Proof*:**
By the general theory of constructible sets in Noetherian spaces (see [5]), any non-empty constructible set contains a non-empty open subset of its closure. Since $Y$ is constructible (proposition 2.4.7), there exists a non-empty subset $U \subseteq Y$ that is open in the subspace topology of $\overline{Y}$.

The group $G$ acts on $X$ by homeomorphisms, and since $Y$ is stable under $G$, so is its closure $\overline{Y}$. Therefore, for every element $g \in G(k)$, the translation map $y \mapsto g \cdot y$ is a homeomorphism of

$\overline{Y}$ to itself. This implies that the translate $g \cdot U$ is also an open subset of $\overline{Y}$, and it is clearly contained in $Y$.

Because $Y$ is a single orbit, $G$ acts transitively on it. Thus, we can cover $Y$ by sweeping $U$ around with group elements:

$$Y = \bigcup_{g \in G(k)} g \cdot U$$

Since $Y$ is a union of open subsets of $\overline{Y}$, $Y$ itself is open in $\overline{Y}$.

We record the clean geometric consequence:

**Corollary 2.4.9: Orbits are Locally Closed**

Let $G$ be an algebraic group acting on a variety $X$. Then every orbit $G \cdot x$ is locally closed in $X$ (which is exactly the definition of being open in its closure).

In fact, we can say more about the geometric quality of orbits:

**Proposition 2.4.10: Orbits are Smooth**

Let $G$ be an algebraic group acting on a variety $X$ over an algebraically closed field $k$. Then every orbit $G \cdot x$ is a smooth variety.

***Proof*:**
The orbit $G \cdot x$ is locally closed in $X$, hence inherits the structure of a (reduced, separated) variety. The translation map $\ell_g : G \cdot x \to G \cdot x$ given by $y \mapsto g \cdot y$ is an isomorphism of varieties for every $g \in G(k)$ (with inverse $\ell_{g^{-1}}$). In particular, all points of $G \cdot x$ are geometrically identical: any local property that holds at one point holds at all points. Since $G \cdot x$ is a variety over an algebraically closed field, it has at least one smooth point (the smooth locus is non-empty and open; see [5]). By the translation argument, every point must therefore be smooth.

This is the exact same translation argument used in proposition 1.1.2 to show that algebraic groups themselves are smooth: homogeneity forces all points to look alike.

## Transitive Actions and Homogeneous Spaces

**Definition 2.4.11: Transitive Actions**

An action of $G$ on a variety $X$ is called *transitive* if there is a single orbit: $G \cdot x = X$ for some (equivalently, every) $x \in X(k)$. A variety equipped with a transitive $G$-action is called a *homogeneous space* (or *$G$-homogeneous variety*).

If $G$ acts transitively on $X$ with stabilizer $G_x$ at a point $x \in X(k)$, then the orbit map $\varphi_x : G \to X$ is surjective and its fibers are the left cosets $gG_x$. Morally, $X$ "is" the quotient $G/G_x$. At the level of closed points this is literally true: $\varphi_x$ induces a bijection $G(k)/G_x(k) \xrightarrow{\sim} X(k)$. However, promoting this to an isomorphism of *varieties* requires constructing $G/G_x$ as a scheme and verifying that $\varphi_x$ is an isomorphism onto it. This is a nontrivial task — the quotient $G/H$ of an algebraic group by a closed subgroup need not be affine, and its construction requires careful attention to faithfully flat descent — which we will carry out in chapter 4.

For now, we record the key consequences that will suffice for our purposes:

**Proposition 2.4.12: Properties of Homogeneous Spaces**

Let $G$ act transitively on a variety $X$, and let $x \in X(k)$ with stabilizer $H = G_x$. Assuming $\mathrm{char} k = 0$, we have:

1. $X$ is smooth and equidimensional of dimension $\dim(G) - \dim(H)$.
2. The orbit map $\varphi_x : G \to X$ is a smooth, surjective morphism.
3. For any other point $y \in X(k)$, the stabilizer $G_y$ is conjugate to $H$: if $y = g \cdot x$, then $G_y = gHg^{-1}$.

***Proof*:**

1. Smoothness follows from proposition 2.4.10, and $\dim(X) = \dim(G) - \dim(H)$ from the orbit-stabilizer formula (proposition 2.4.4).
2. By generic flatness and $G$-equivariant translation, the morphism $\varphi_x$ is flat everywhere. Furthermore, $\varphi_x$ is surjective (by transitivity) and all of its scheme-theoretic fibers are isomorphic to translates of the stabilizer $H$. Because $\mathrm{char} k = 0$, Cartier's Theorem guarantees that the algebraic group $H$ is automatically reduced, and hence smooth. A flat morphism whose fibers are smooth varieties is a smooth morphism (see [5]).
3. If $y = g \cdot x$ and $h \in G_y$, then $h \cdot (g \cdot x) = g \cdot x$, so $(g^{-1}hg) \cdot x = x$, meaning $g^{-1}hg \in H$. Hence $G_y = gHg^{-1}$.

Statement (3) is the geometric analogue of a basic fact in group theory: in a transitive action, all stabilizers are conjugate. This means that the "type" of the stabilizer (its isomorphism class, or more precisely its conjugacy class in $G$) is an invariant of the homogeneous space, not of the chosen basepoint.

**Example 2.4.13: Projective Space as a Homogeneous Space**

The group $\mathrm{GL}_n$ acts on $\mathbb{P}^{n-1}$ via its action on lines: $g \cdot [v] = [gv]$. This action is transitive (any nonzero vector can be mapped to any other by an invertible matrix). The stabilizer of the point $[e_1] = [1 : 0 : \cdots : 0]$ consists of matrices of the form

$$G_{[e_1]} = \left\{ \begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & * & \cdots & * \end{pmatrix} \in \mathrm{GL}_n \right\}$$

This is the *standard parabolic subgroup* $P_1$ (the stabilizer of the line $\langle e_1 \rangle$). We thus have a bijection $\mathrm{GL}_n(k)/P_1(k) \cong \mathbb{P}^{n-1}(k)$, and the quotient construction of chapter 4 will promote this to an isomorphism of varieties $\mathrm{GL}_n/P_1 \cong \mathbb{P}^{n-1}$. Note that $\mathbb{P}^{n-1}$ is *projective*, so $G/P_1$ is not affine — this illustrates why the quotient $G/H$ is generally harder than the affine theory we have developed so far.

**Example 2.4.14: The Grassmannian**
More generally, $\mathrm{GL}_n$ acts transitively on the Grassmannian $\mathrm{Gr}(d, n)$ of $d$-dimensional subspaces of $k^n$. The stabilizer of the standard $d$-plane $\langle e_1, \ldots, e_d \rangle$ is the parabolic subgroup

$$P_d = \left\{ \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} : A \in \mathrm{GL}_d,\ D \in \mathrm{GL}_{n-d},\ B \in M_{d\times(n-d)} \right\}$$

and so $\mathrm{Gr}(d, n) \cong \mathrm{GL}_n/P_d$. The Grassmannian has dimension $\dim(\mathrm{GL}_n) - \dim(P_d) = n^2 - (d^2 + (n-d)^2 + d(n-d)) = d(n-d)$. These quotients are central examples of *flag varieties*, which will play a pivotal role in the structure theory of Chapters 6 and 7.

## The Closed Orbit Lemma

We now prove the main geometric result of the chapter. The idea is that the boundary of an orbit — the "stuff that gets added when you take the closure" — is itself a union of strictly smaller orbits. Since dimension cannot decrease forever, this forces the existence of closed orbits.

**Theorem 2.4.15: Closed Orbit Lemma**

Let $G$ be a connected algebraic group acting on a variety $X$. Then:

1. Every orbit $G \cdot x$ is a smooth, locally closed subvariety of $X$.
2. The boundary $\partial(G \cdot x) := \overline{G \cdot x} \setminus G \cdot x$ is a closed, $G$-stable subset of $X$ that is a union of orbits of strictly lower dimension:
$$\dim(G \cdot y) < \dim(G \cdot x) \quad \text{for every } y \in \partial(G \cdot x)$$
3. The set of orbits admits a partial order by closure: $G \cdot y \preceq G \cdot x$ if $G \cdot y \subseteq \overline{G \cdot x}$.
4. Orbits of minimal dimension are closed. In particular, closed orbits always exist.

***Proof*:**
Statement (1) is corollary 2.4.9 and proposition 2.4.10.

For (2), since $G \cdot x$ is open in $\overline{G \cdot x}$ (by (1)), the boundary $\partial(G \cdot x) = \overline{G \cdot x} \setminus G \cdot x$ is closed in $\overline{G \cdot x}$, hence closed in $X$. The boundary is $G$-stable: if $y \in \partial(G \cdot x)$ and $g \in G(k)$, then $g \cdot y \in \overline{G \cdot x}$ (since $\overline{G \cdot x}$ is $G$-stable, as the closure of a $G$-stable set under a continuous action). And $g \cdot y \notin G \cdot x$, for otherwise $y = g^{-1} \cdot (g \cdot y) \in G \cdot x$, a contradiction. So $g \cdot y \in \partial(G \cdot x)$.

Now let $G \cdot y$ be an orbit contained in $\partial(G \cdot x)$. Then $\overline{G \cdot y} \subseteq \overline{G \cdot x}$ and $G \cdot y \cap G \cdot x = \emptyset$. Since $G \cdot y$ is open in $\overline{G \cdot y}$ and $G \cdot x$ is dense in $\overline{G \cdot x}$, we must have $\overline{G \cdot y} \subsetneq \overline{G \cdot x}$ (if they were equal, then $G \cdot y$ would be a non-empty open subset of $\overline{G \cdot x}$ disjoint from the dense subset $G \cdot x$, which is impossible since $\overline{G \cdot x}$ is irreducible — here we use that $G$ is connected, so $G \cdot x = \varphi_x(G)$ is the image of an irreducible variety and hence irreducible, and its closure is also irreducible).

Therefore:

$$\dim(G \cdot y) \le \dim(\overline{G \cdot y}) < \dim(\overline{G \cdot x}) = \dim(G \cdot x)$$

Statement (3) is a restatement: $G \cdot y \preceq G \cdot x$ if and only if $G \cdot y \subseteq \overline{G \cdot x}$. This is a partial order (reflexive, transitive, and antisymmetric by (2)).

For (4), since $X$ is a Noetherian topological space, the set of orbit dimensions $\{\dim(G \cdot x) : x \in X(k)\}$ is bounded below by 0. Any orbit of minimal dimension has empty boundary (by (2)), hence is closed. If $X$ is non-empty, there exist orbits (take any point), and the orbit of smallest dimension among all orbits is closed.

**2.4.16 Remark: The Role of Connectedness** The assumption that $G$ is connected is used in (2) to guarantee that $\overline{G \cdot x}$ is irreducible. Without it, the orbit $G \cdot x$ might not be irreducible, and the dimension argument would need modification. For a disconnected group, one works instead with the connected component $G^\circ$ and uses the fact that $G$ has finitely many connected components (proposition 1.1.4).

**Example 2.4.17: Orbit Closures for $\mathbb{G}_m$ Acting on $\mathbb{A}^2$**

Let $\mathbb{G}_m$ act on $\mathbb{A}^2$ by $t \cdot (x, y) = (tx, t^{-1}y)$. The orbits are:

1. The origin $\{(0, 0)\}$: a closed orbit of dimension 0.
2. The punctured $x$-axis $\{(x, 0) : x \neq 0\}$: an orbit of dimension 1 (the stabilizer is trivial). Its closure is the full $x$-axis $\{y = 0\}$, with boundary $\{(0, 0)\}$.
3. The punctured $y$-axis $\{(0, y) : y \neq 0\}$: similarly, dimension 1 with boundary $\{(0, 0)\}$.
4. For each $c \neq 0$, the *hyperbola* $\{xy = c\} = \{(t, c/t) : t \neq 0\}$: an orbit of dimension 1 with stabilizer the trivial group. Its closure is $\{xy = c\}$, which is already closed in $\mathbb{A}^2$ (it is a level set of the polynomial $xy$). So these orbits are closed.

The orbit structure is consistent with the Closed Orbit Lemma: the boundary of each non-closed orbit (the punctured axes) is the origin, which is the unique 0-dimensional orbit. The invariant ring is $k[\mathbb{A}^2]^{\mathbb{G}_m} = k[xy]$ (the polynomial $xy$ is the unique independent invariant), and the "quotient map" $\mathbb{A}^2 \to \mathbb{A}^1$ sending $(x, y) \mapsto xy$ collapses the three orbits $\{(0, 0)\}$, $\{x\text{-axis}\}$, and $\{y\text{-axis}\}$ to the single point $0 \in \mathbb{A}^1$. This non-separation of orbits is a preview of the subtleties of geometric invariant theory.

An important special case of the Closed Orbit Lemma arises when the variety $X$ is *complete* (or *projective*). In a complete variety, any closed subset is itself complete. Because the Closed Orbit Lemma guarantees that closed orbits always exist, it provides us with the following crucial structural tool:

**Corollary 2.4.18: Projective Homogeneous Spaces**

Let $G$ be an affine algebraic group acting on a non-empty projective variety $X$. Then $X$ contains at least one closed orbit $G \cdot x$. Furthermore, this closed orbit is a projective homogeneous space $G/G_x$. If this closed orbit happens to be a single point, then $G$ has a *fixed point* in $X$.

We will see in chapter 6 that connected *solvable* groups cannot have non-trivial projective homogeneous spaces. Therefore, by the corollary above, any connected solvable group acting on a projective variety must have a fixed point (Borel's Fixed-Point Theorem). This entire geometric theory, including the Lie–Kolchin theorem and the theory of Borel subgroups, rests firmly on the Closed Orbit Lemma.

**Example 2.4.19: Orbit Closure on $\mathbb{P}^1$**

Let $G = \mathbb{G}_m$ act on $\mathbb{P}^1$ by $t \cdot [x : y] = [tx : y]$. The orbits are:

1. $\{[1 : 0]\}$: a fixed point (dimension 0, closed).
2. $\{[0 : 1]\}$: a fixed point (dimension 0, closed).
3. $\{[x : y] : x \neq 0,\ y \neq 0\} \cong \mathbb{G}_m$: an orbit of dimension 1.

The closure of the 1-dimensional orbit is all of $\mathbb{P}^1$, and its boundary is $\{[1 : 0]\} \cup \{[0 : 1]\}$. Contrast this with the action of $\mathbb{G}_m$ on $\mathbb{A}^2$ from example 2.4.17: there, some 1-dimensional orbits (the hyperbolas) were closed. On $\mathbb{P}^1$, the open orbit $\mathbb{G}_m$ is an affine variety, so it cannot possibly be projective (closed). The only closed orbits are the 0-dimensional fixed points, perfectly supplying the projective homogeneous spaces guaranteed by corollary 2.4.18.

## Exercise 2.4.1

1. Let $\mathrm{GL}_n$ act on the space $\mathrm{Sym}^2(k^n)^\vee$ of symmetric bilinear forms by $g \cdot \beta(v, w) = \beta(g^{-1}v, g^{-1}w)$. Compute the stabilizer of the standard form $\beta_0(v, w) = \sum_i v_i w_i$ (assuming $\mathrm{char} k \neq 2$). What familiar group is this? What is the dimension of the orbit $\mathrm{GL}_n \cdot \beta_0$?
2. Let $G = \mathrm{GL}_3$ act on $M_3$ by conjugation (over an algebraically closed field of characteristic 0). List all possible Jordan types for $3 \times 3$ matrices with a single eigenvalue $\lambda$, compute the dimension of each conjugacy class, and determine the closure partial order among these orbits.
3. Let $G$ act on an affine variety $X$. Show that $X$ contains a closed orbit. (*Hint*: $X$ is Noetherian, so among all orbit closures, choose one of minimal dimension.)
4. Let $\mathbb{G}_m$ act on $\mathbb{A}^3$ by $t \cdot (x, y, z) = (t^2 x, ty, t^{-1} z)$. Determine all orbits, their dimensions, and which orbits are closed. Verify that the boundary of each non-closed orbit is a union of orbits of strictly lower dimension.
5. Let $G$ be a connected algebraic group acting transitively on a variety $X$. Show that $X$ is irreducible. (*Hint*: $X$ is the image of the connected, hence irreducible, group $G$ under the orbit map.)
6. Let $G$ act on a variety $X$ and let $x \in X(k)$. Show that the stabilizer $G_x$ contains $(G_{\overline{G \cdot x}})^\circ$, where $G_{\overline{G \cdot x}} = \left\{ g \in G \mid g \cdot \overline{G \cdot x} = \overline{G \cdot x} \right\}$ is the setwise stabilizer of the closure. Conclude

that $G_x$ has finite index in $G_{\overline{G \cdot x}}$.

# 3

# *Infinitesimal Structure*

Every algebraic group $G$ carries a finite-dimensional Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ that encodes the "infinitesimal" structure of $G$ near the identity. In the classical theory over $\mathbb{R}$ or $\mathbb{C}$, the Lie algebra is constructed by differentiating curves through the identity, and the exponential map provides a local inverse: the group and its Lie algebra determine each other in a neighborhood of $e$. In the algebraic setting, we have no convergent power series and no analytic neighborhoods, but we have something better: the *ring of dual numbers* $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$, which serves as an algebraic substitute for "infinitesimally close to the identity." The functor of points, developed in chapter 1, makes this completely rigorous: we simply evaluate the group functor on $k[\varepsilon]$ and extract the kernel of the reduction map.

This chapter introduces the Lie algebra, computes it for the classical groups, and establishes the adjoint representation — the action of $G$ on its own Lie algebra by conjugation. This is the representation that will eventually decompose $\mathfrak{g}$ into root spaces and drive the classification in chapter 7. We close by proving Cartier's theorem, which guarantees that in characteristic 0 the Lie algebra faithfully reflects the group, and by examining the explicit failures that arise in characteristic $p$.

## 3.1 The Lie Algebra of an Algebraic Group

Let $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$ be the ring of *dual numbers* over $k$. As a $k$-algebra, it is two-dimensional with basis $\{1, \varepsilon\}$, and the only interesting multiplication rule is $\varepsilon^2 = 0$. Geometrically, $\operatorname{Spec} k[\varepsilon]$ is a "fat point": it consists of a single (closed) point with a one-dimensional tangent direction attached. There is a canonical surjection $\pi : k[\varepsilon] \to k$ sending $\varepsilon \mapsto 0$ ("forgetting the tangent direction") and a section $\iota : k \to k[\varepsilon]$ sending $a \mapsto a$ ("the point itself").

If $G$ is an affine algebraic group over $k$, evaluating the functor of points on $k[\varepsilon]$ gives the group $G(k[\varepsilon])$, whose elements are "points of $G$ with an infinitesimal perturbation." The surjection $\pi$

induces a group homomorphism $G(\pi) : G(k[\varepsilon]) \to G(k)$ that "forgets the perturbation." The kernel of this map is exactly the set of infinitesimal elements: group elements that reduce to the identity modulo $\varepsilon$.

## Definition via Dual Numbers

**Definition 3.1.1: Lie Algebra of an Algebraic Group**

Let $G$ be an affine algebraic group over $k$. The *Lie algebra* of $G$ is

$$\mathrm{Lie}(G) := \ker\big(G(k[\varepsilon]) \xrightarrow{G(\pi)} G(k)\big)$$

the kernel of the reduction map. We write $\mathfrak{g} = \mathrm{Lie}(G)$ and call it the Lie algebra of $G$.

Let us unpack this definition concretely. An element of $G(k[\varepsilon]) = \mathrm{Hom}_{\mathbf{Alg}_k}(\mathcal{O}(G), k[\varepsilon])$ is a $k$-algebra homomorphism $\varphi : \mathcal{O}(G) \to k[\varepsilon]$. Since every element of $k[\varepsilon]$ is uniquely of the form $a + b\varepsilon$ with $a, b \in k$, we can compose $\varphi$ with the two $k$-linear projection maps to decompose it into $\varphi = \varphi_0 + \varphi_1\varepsilon$ where $\varphi_0 : \mathcal{O}(G) \to k$ and $\varphi_1 : \mathcal{O}(G) \to k$ are $k$-linear maps (notice linear, not algebraic!). The condition that $\varphi$ is a $k$-algebra homomorphism means:

$$\varphi_0(fg) + \varphi_1(fg)\varepsilon = (\varphi_0(f) + \varphi_1(f)\varepsilon)(\varphi_0(g) + \varphi_1(g)\varepsilon) = \varphi_0(f)\varphi_0(g) + (\varphi_0(f)\varphi_1(g) + \varphi_1(f)\varphi_0(g))\varepsilon$$

where the $\varepsilon^2$ term vanishes. Comparing coefficients:

1. $\varphi_0(fg) = \varphi_0(f)\varphi_0(g)$, so $\varphi_0$ is a $k$-algebra homomorphism $\mathcal{O}(G) \to k$, i.e. a point of $G(k)$.
2. $\varphi_1(fg) = \varphi_0(f)\varphi_1(g) + \varphi_1(f)\varphi_0(g)$, so $\varphi_1$ is a *derivation at* $\varphi_0$: it satisfies the Leibniz rule with respect to the evaluation $\varphi_0$.

The reduction map $G(\pi)$ sends $\varphi$ to $\varphi_0$. So $\varphi \in \ker(G(\pi))$ means $\varphi_0 = \epsilon$ (the counit, corresponding to the identity $e \in G(k)$), and $\varphi_1$ is a derivation at $\epsilon$:

$$\varphi_1(fg) = \epsilon(f)\varphi_1(g) + \varphi_1(f)\epsilon(g) = f(e)\varphi_1(g) + \varphi_1(f)g(e)$$

Writing $A = \mathcal{O}(G)$ and $\mathfrak{m}_e = \ker(\epsilon)$ for the maximal ideal at the identity, a derivation at $\epsilon$ is a $k$-linear map $D : A \to k$ satisfying $D(fg) = f(e)D(g) + D(f)g(e)$. Note that such a $D$ vanishes on $k$ (since $D(1) = D(1 \cdot 1) = 1 \cdot D(1) + D(1) \cdot 1 = 2D(1)$, so $D(1) = 0$) and is determined by its restriction to $\mathfrak{m}_e$ (since any $f \in A$ can be written $f = f(e) + (f - f(e))$ with $f - f(e) \in \mathfrak{m}_e$). Moreover, $D$ vanishes on $\mathfrak{m}_e^2$: if $f, g \in \mathfrak{m}_e$, then $D(fg) = f(e)D(g) + D(f)g(e) = 0$ since $f(e) = g(e) = 0$. Therefore $D$ factors through a linear functional on $\mathfrak{m}_e/\mathfrak{m}_e^2$:

**Proposition 3.1.2: Lie Algebra as Tangent Space**

Let $G = \mathrm{Spec}\, A$ be an affine algebraic group over $k$ with identity $e$ and maximal ideal $\mathfrak{m}_e = \ker(\epsilon : A \to k)$. Then there is a canonical isomorphism of $k$-vector spaces:

$$\mathrm{Lie}(G) \cong (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee = \mathrm{Hom}_k(\mathfrak{m}_e/\mathfrak{m}_e^2,\, k)$$

That is, the Lie algebra is the *Zariski tangent space* of $G$ at the identity.

***Proof*:**
By the discussion above, the map $\mathrm{Lie}(G) \to (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$ sending a derivation $D$ to its restriction $D|_{\mathfrak{m}_e}$ (which factors through $\mathfrak{m}_e/\mathfrak{m}_e^2$) is well-defined and injective (since $D$ is determined by $D|_{\mathfrak{m}_e}$). For surjectivity, given any linear functional $\lambda : \mathfrak{m}_e/\mathfrak{m}_e^2 \to k$, define $D : A \to k$ by $D(f) = \lambda(f - f(e) \bmod \mathfrak{m}_e^2)$. Then $D$ is a derivation at $\epsilon$: for $f, g \in A$,

$$\begin{aligned} D(fg) &= \lambda\big(fg - f(e)g(e) \bmod \mathfrak{m}_e^2\big) \\ &= \lambda\big((f - f(e))g(e) + f(e)(g - g(e)) + (f - f(e))(g - g(e)) \bmod \mathfrak{m}_e^2\big) \\ &= g(e)\lambda(f - f(e)) + f(e)\lambda(g - g(e)) \\ &= g(e)D(f) + f(e)D(g) \end{aligned}$$

where the $(f - f(e))(g - g(e))$ term vanishes since it lies in $\mathfrak{m}_e^2$.

This is exactly the tangent space $T_eG$ from algebraic geometry (see [5]). The dimension of $\mathrm{Lie}(G)$ is the *embedding dimension* of the local ring $\mathcal{O}_{G,e}$, which equals $\dim_k(\mathfrak{m}_e/\mathfrak{m}_e^2)$. We will see in §3.3 that this dimension is always at least $\dim G$, with equality if and only if $G$ is smooth at $e$.

## The Lie Bracket

We have shown that $\mathrm{Lie}(G)$ is a $k$-vector space. It also carries a bilinear operation — the *Lie bracket* — that encodes the non-commutativity of $G$ at the infinitesimal level.

The idea is simple: if $X, Y \in \mathrm{Lie}(G)$ are two infinitesimal elements, then the group commutator $XYX^{-1}Y^{-1}$ measures how much $X$ and $Y$ fail to commute. Since $X$ and $Y$ are "$\varepsilon$-close" to the identity, the commutator is "$\varepsilon^2$-close" to the identity. To capture this second-order information, we need to work over a ring with *two* independent nilpotent directions.

Let $k[\varepsilon_1, \varepsilon_2] = k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1^2, \varepsilon_2^2)$ be the ring of "double dual numbers." Note that $\varepsilon_1\varepsilon_2 \neq 0$ in this ring (it is killed only by further multiplication by either $\varepsilon_i$), so this ring has a richer nilpotent structure than $k[\varepsilon]$.

Given $X, Y \in \mathrm{Lie}(G) = \ker(G(k[\varepsilon]) \to G(k))$, we can "embed" them into $G(k[\varepsilon_1, \varepsilon_2])$ by the two algebra maps $k[\varepsilon] \to k[\varepsilon_1, \varepsilon_2]$ sending $\varepsilon \mapsto \varepsilon_1$ and $\varepsilon \mapsto \varepsilon_2$ respectively. Call the resulting elements $X_1, Y_2 \in G(k[\varepsilon_1, \varepsilon_2])$. Since $\varepsilon_1^2 = \varepsilon_2^2 = 0$, both $X_1$ and $Y_2$ reduce to $e$ modulo $(\varepsilon_1, \varepsilon_2)$.

Now form the group commutator:

$$[X_1, Y_2] := X_1Y_2X_1^{-1}Y_2^{-1} \in G(k[\varepsilon_1, \varepsilon_2])$$

This element reduces to $e$ modulo $(\varepsilon_1, \varepsilon_2)$, and in fact reduces to $e$ modulo $(\varepsilon_1)$ alone (since $Y_2$ and $Y_2^{-1}$ cancel when $\varepsilon_2 = 0$) and modulo $(\varepsilon_2)$ alone (by the same argument with $X_1$). A careful computation shows that $[X_1, Y_2]$ lies in the kernel of the reduction to any ring where $\varepsilon_1\varepsilon_2 = 0$, and hence is determined by its "$\varepsilon_1\varepsilon_2$-component."

The algebra map $k[\varepsilon_1, \varepsilon_2] \to k[\varepsilon]$ sending $\varepsilon_1 \mapsto \varepsilon$, $\varepsilon_2 \mapsto \varepsilon$ would collapse our two directions, but the map $k[\varepsilon_1, \varepsilon_2] \to k[\varepsilon]$ sending $\varepsilon_1\varepsilon_2 \mapsto \varepsilon$ (and $\varepsilon_1, \varepsilon_2 \mapsto 0$ individually) extracts exactly the $\varepsilon_1\varepsilon_2$-coefficient. This yields an element of $\mathrm{Lie}(G)$.

**Definition 3.1.3: Lie Bracket**

Let $G$ be an affine algebraic group and $X, Y \in \mathfrak{g} = \mathrm{Lie}(G)$. The *Lie bracket* $[X, Y] \in \mathfrak{g}$ is the element defined by the following procedure:

1. Embed $X$ and $Y$ into $G(k[\varepsilon_1, \varepsilon_2])$ via $\varepsilon \mapsto \varepsilon_1$ and $\varepsilon \mapsto \varepsilon_2$ respectively.
2. Form the group commutator $X_1 Y_2 X_1^{-1} Y_2^{-1} \in G(k[\varepsilon_1, \varepsilon_2])$.
3. Extract the $\varepsilon_1\varepsilon_2$-component to obtain $[X, Y] \in \mathrm{Lie}(G)$.

That this is well-defined (in particular, $k$-bilinear) follows from the explicit computation, which we will carry out for $\mathrm{GL}_n$ momentarily. The key properties follow from the group axioms:

**Proposition 3.1.4: Lie Algebra Axioms**

Let $\mathfrak{g} = \mathrm{Lie}(G)$. The bracket $[\,\cdot\,,\,\cdot\,] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ satisfies:

1. (Bilinearity) $[aX + bY, Z] = a[X, Z] + b[Y, Z]$ and $[X, aY + bZ] = a[X, Y] + b[X, Z]$ for all $a, b \in k$.
2. (Antisymmetry) $[X, Y] = -[Y, X]$ for all $X, Y \in \mathfrak{g}$.
3. (Jacobi Identity) $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ for all $X, Y, Z \in \mathfrak{g}$.

In particular, $(\mathfrak{g}, [\,\cdot\,,\,\cdot\,])$ is a Lie algebra in the abstract algebraic sense.

Using anti-symmetry, the Jacobi identity can be written as:

$$[X, [Y, Z]] = [[X, Y], Z] + [Y, [X, Z]]$$

Notice how this looks like the Leibniz product. In particular, if $Y \cdot Z := [Y, Z]$ so that multiplication is defined via the lie bracket, and $D_X(-) = [X, -]$ represents the multiplication by $X$-action, then:

$$D_X(Y \cdot Z) = D_X(Y) \cdot Z + Y \cdot D_X(Z)$$

***Proof*:**

Antisymmetry follows from the fact that swapping $X_1$ and $Y_2$ inverts the commutator: $Y_2 X_1 Y_2^{-1} X_1^{-1} = (X_1 Y_2 X_1^{-1} Y_2^{-1})^{-1}$. At the infinitesimal level, inverting an element $e + Z\varepsilon$ gives $e - Z\varepsilon$, so the $\varepsilon_1\varepsilon_2$-component changes sign.

Bilinearity requires a computation. We verify it for $\mathrm{GL}_n$ in example 3.1.6 below (where the bracket is $[X, Y] = XY - YX$, visibly bilinear). For a general affine group $G \hookrightarrow \mathrm{GL}_n$, the Lie bracket on $\mathrm{Lie}(G)$ is the restriction of the bracket on $\mathfrak{gl}_n$, hence bilinear.

The Jacobi identity follows from the same embedding principle: any identity that holds in $\mathfrak{gl}_n$ restricts to $\mathrm{Lie}(G) \subseteq \mathfrak{gl}_n$. For $\mathfrak{gl}_n$, the Jacobi identity is a direct (and classical) computation using $[X, Y] = XY - YX$.

**3.1.5 Remark: The Bracket in Characteristic** 2 In characteristic 2, the antisymmetry condition $[X, Y] = -[Y, X]$ becomes $[X, Y] = [Y, X]$, which no longer implies $[X, X] = 0$. The "correct" structure in characteristic $p$ is that of a *restricted Lie algebra* (or *p-Lie algebra*), which carries an additional operation $X \mapsto X^{[p]}$ satisfying $\mathrm{ad}(X^{[p]}) = \mathrm{ad}(X)^p$. This $p$-operation arises naturally from the Frobenius endomorphism. We will encounter it briefly in §3.3, but a full treatment belongs to the representation theory of Part II.

## Computing Lie Algebras of Classical Groups

The power of the functorial definition is that computing $\mathrm{Lie}(G)$ reduces to elementary linear algebra: write down what $G(k[\varepsilon])$ looks like, and take the kernel of the reduction.

**Example 3.1.6:** $\mathrm{Lie}(\mathrm{GL}_n) = \mathfrak{gl}_n$
An element of $\mathrm{GL}_n(k[\varepsilon])$ is an invertible $n \times n$ matrix with entries in $k[\varepsilon]$. Write such a matrix as $g = A + B\varepsilon$ where $A, B \in M_n(k)$. For $g$ to be invertible, we need $\det(A + B\varepsilon) \in k[\varepsilon]^\times$. Since

$$\det(A + B\varepsilon) = \det(A) + \mathrm{tr}(\mathrm{adj}(A) \cdot B)\varepsilon$$

(where $\mathrm{adj}(A)$ is the adjugate, see [20]), the determinant is a unit in $k[\varepsilon]$ if and only if $\det(A) \neq 0$ in $k$. So $g$ is invertible if and only if $A \in \mathrm{GL}_n(k)$.

The reduction map sends $A + B\varepsilon \mapsto A$. The kernel consists of elements $I + X\varepsilon$ with $X \in M_n(k)$ arbitrary (no condition on $X$, since $\det(I) = 1 \neq 0$). Therefore:

$$\mathrm{Lie}(\mathrm{GL}_n) = \{I + X\varepsilon \mid X \in M_n(k)\} \cong M_n(k) = \mathfrak{gl}_n$$

as a $k$-vector space of dimension $n^2$.

To compute the Lie bracket, take $X_1 = I + X\varepsilon_1$ and $Y_2 = I + Y\varepsilon_2$ in $\mathrm{GL}_n(k[\varepsilon_1, \varepsilon_2])$. Their inverses are $X_1^{-1} = I - X\varepsilon_1$ and $Y_2^{-1} = I - Y\varepsilon_2$ (using $\varepsilon_i^2 = 0$). The commutator is:

$$\begin{aligned} X_1 Y_2 X_1^{-1} Y_2^{-1} &= (I + X\varepsilon_1)(I + Y\varepsilon_2)(I - X\varepsilon_1)(I - Y\varepsilon_2) \\ &= (I + X\varepsilon_1 + Y\varepsilon_2 + XY\varepsilon_1\varepsilon_2)(I - X\varepsilon_1 - Y\varepsilon_2 + XY\varepsilon_1\varepsilon_2) \\ &= I + (XY - YX)\varepsilon_1\varepsilon_2 \end{aligned}$$

where all $\varepsilon_i^2$ terms vanish. Extracting the $\varepsilon_1\varepsilon_2$-component:

$$[X, Y] = XY - YX$$

This is the usual matrix commutator, confirming that $\mathfrak{gl}_n$ with the bracket $[X, Y] = XY - YX$ is the Lie algebra of $\mathrm{GL}_n$.

**Example 3.1.7:** Lie($\mathrm{SL}_n$) = $\mathfrak{sl}_n$
An element of $\mathrm{SL}_n(k[\varepsilon])$ is a matrix $I + X\varepsilon$ (reducing to $I$ in the kernel) with $\det(I + X\varepsilon) = 1$. Using the expansion:

$$\det(I + X\varepsilon) = 1 + \mathrm{tr}(X)\varepsilon$$

(this follows from the general formula $\det(I + tM) = 1 + t\mathrm{tr}(M) + O(t^2)$, and $\varepsilon^2 = 0$ kills all higher terms). The condition $\det = 1$ forces $\mathrm{tr}(X) = 0$. Therefore:

$$\mathrm{Lie}(\mathrm{SL}_n) = \{X \in M_n(k) \mid \mathrm{tr}(X) = 0\} = \mathfrak{sl}_n$$

This is an $(n^2 - 1)$-dimensional Lie subalgebra of $\mathfrak{gl}_n$. The bracket is the restriction of the matrix commutator, which is well-defined since $\mathrm{tr}([X, Y]) = \mathrm{tr}(XY) - \mathrm{tr}(YX) = 0$.

**Example 3.1.8:** Lie($\mathrm{Sp}_{2n}$) = $\mathfrak{sp}_{2n}$
Recall from §1.1.1 that $\mathrm{Sp}_{2n}$ consists of matrices $g \in \mathrm{GL}_{2n}$ satisfying ${}^t g\, \Omega\, g = \Omega$ where $\Omega = \left(\begin{smallmatrix} 0 & J \\ -J & 0 \end{smallmatrix}\right)$. Setting $g = I + X\varepsilon$ and expanding:

$${}^t(I + X\varepsilon)\, \Omega\, (I + X\varepsilon) = \Omega + ({}^t X\, \Omega + \Omega\, X)\varepsilon$$

The condition becomes ${}^t X\, \Omega + \Omega\, X = 0$, i.e. the matrix $\Omega X$ is symmetric (since ${}^t(\Omega X) = {}^t X\, {}^t\Omega = -{}^t X \Omega = \Omega X$). Therefore:

$$\mathrm{Lie}(\mathrm{Sp}_{2n}) = \left\{X \in M_{2n}(k) \;\middle|\; {}^t X\Omega + \Omega X = 0\right\} = \mathfrak{sp}_{2n}$$

The dimension is $n(2n + 1)$.

**Example 3.1.9:** Lie($\mathrm{SO}_n$) = $\mathfrak{so}_n$
The orthogonal group $O_n$ consists of $g \in \mathrm{GL}_n$ with ${}^t g\, s\, g = s$ for a symmetric form $s$. Setting $g = I + X\varepsilon$:

$${}^t(I + X\varepsilon)\, s\, (I + X\varepsilon) = s + ({}^t X\, s + s\, X)\varepsilon$$

The condition is ${}^t X s + sX = 0$, meaning $sX$ is skew-symmetric ($\mathrm{char} k \neq 2$). Thus:

$$\mathrm{Lie}(O_n) = \left\{X \in M_n(k) \;\middle|\; {}^t X s + sX = 0\right\} = \mathfrak{so}_n$$

The dimension is $\binom{n}{2}$. Note that $O_n$ and $\mathrm{SO}_n$ have the *same* Lie algebra: the determinant condition $\det(g) = 1$ imposes $\mathrm{tr}(X) = 0$, but for any $X \in \mathfrak{so}_n$ the skew-symmetry of $sX$ already implies $\mathrm{tr}(sX) = 0$, and when $s = I$ this gives $\mathrm{tr}(X) = 0$ directly. Thus $\mathrm{Lie}(\mathrm{SO}_n) = \mathrm{Lie}(O_n) = \mathfrak{so}_n$. The Lie algebra cannot see the difference between $O_n$ and $\mathrm{SO}_n$ because they share the same identity component (recall from definition 1.1.3 that $\mathrm{SO}_n = O_n^\circ$).

The last observation is an instance of a general principle:

**Proposition 3.1.10: Lie Algebra Depends Only on $G^\circ$**

Let $G$ be an affine algebraic group. Then $\mathrm{Lie}(G) = \mathrm{Lie}(G^\circ)$.

***Proof*:**
The identity $e$ lies in $G^\circ$, and $\mathrm{Lie}(G)$ depends only on the local structure of $G$ at $e$ (it is the tangent space $T_eG$). Since $G^\circ$ is an open subgroup of $G$ containing $e$, the local rings $\mathcal{O}_{G,e}$ and $\mathcal{O}_{G^\circ,e}$ are isomorphic, hence so are the tangent spaces.

## The Functor Lie

The assignment $G \mapsto \mathrm{Lie}(G)$ is not just a construction on individual groups; it is a *functor* from affine algebraic groups to Lie algebras.

**Proposition 3.1.11: Functoriality of Lie**

Let $\varphi : G \to H$ be a homomorphism of affine algebraic groups. Then the map

$$d\varphi := \mathrm{Lie}(\varphi) : \mathrm{Lie}(G) \longrightarrow \mathrm{Lie}(H)$$

defined by $(d\varphi)(X) = \varphi_{k[\varepsilon]}(X)$ (applying $\varphi$ on $k[\varepsilon]$-valued points) is a homomorphism of Lie algebras. That is, $d\varphi$ is $k$-linear and $d\varphi([X, Y]) = [d\varphi(X), d\varphi(Y)]$.

***Proof*:**
Since $\varphi$ is a natural transformation of group functors, $\varphi_{k[\varepsilon]} : G(k[\varepsilon]) \to H(k[\varepsilon])$ is a group homomorphism. It sends $\ker(G(k[\varepsilon]) \to G(k))$ into $\ker(H(k[\varepsilon]) \to H(k))$ by naturality of $\varphi$ with respect to the map $k[\varepsilon] \to k$. So $d\varphi$ is a well-defined $k$-linear map $\mathfrak{g} \to \mathfrak{h}$. Since $\varphi$ preserves the group operation, it preserves commutators: $\varphi(XYX^{-1}Y^{-1}) = \varphi(X)\varphi(Y)\varphi(X)^{-1}\varphi(Y)^{-1}$. Extracting the $\varepsilon_1\varepsilon_2$-component gives $d\varphi([X, Y]) = [d\varphi(X), d\varphi(Y)]$.

The map $d\varphi$ is the algebraic-group analogue of the "differential" or "derivative" of a Lie group homomorphism. We can now ask: to what extent does $d\varphi$ determine $\varphi$?

**Proposition 3.1.12: Lie Preserves Closed Immersions**

Let $H \hookrightarrow G$ be a closed immersion of affine algebraic groups. Then $d\iota : \mathrm{Lie}(H) \hookrightarrow \mathrm{Lie}(G)$ is injective.

***Proof*:**
A closed immersion $H \hookrightarrow G$ corresponds to a surjection $\mathcal{O}(G) \twoheadrightarrow \mathcal{O}(H)$ of Hopf algebras. On tangent spaces at the identity, a surjection of local rings induces an injection of cotangent spaces $\mathfrak{m}_{e,H}/\mathfrak{m}_{e,H}^2 \hookrightarrow \mathfrak{m}_{e,G}/\mathfrak{m}_{e,G}^2$ (or rather, a surjection of cotangent spaces and an injection of tangent spaces). Concretely, if $X \in \mathrm{Lie}(H)$ satisfies $d\iota(X) = 0$, then $X$ acts as the zero derivation on all functions pulled back from $G$. Since $\mathcal{O}(H)$ is a quotient of $\mathcal{O}(G)$, these functions generate $\mathcal{O}(H)$, so $X = 0$.

This means that the functor Lie embeds the lattice of closed subgroups of $G$ into the lattice of Lie subalgebras of $\mathfrak{g}$. However, the embedding is generally *not* surjective: there are Lie subalgebras of $\mathfrak{g}$ that do not arise as $\mathrm{Lie}(H)$ for any closed subgroup $H$. Whether or not every subalgebra integrates to a subgroup depends on the characteristic; in characteristic 0, this is true (by a theorem that will follow from Cartier's theorem in §3.3), but in characteristic $p$ it can fail.

### The Lie Algebra via Derivations and Left-Invariant Vector Fields

There is a third description of $\mathrm{Lie}(G)$ that is sometimes useful, especially when connecting to the differential-geometric theory. A *left-invariant derivation* of $\mathcal{O}(G)$ is a $k$-linear map $D : \mathcal{O}(G) \to \mathcal{O}(G)$ satisfying the Leibniz rule $D(fg) = fD(g) + D(f)g$ and commuting with the left action of $G$ on $\mathcal{O}(G)$. In Hopf algebra terms, left-invariance means:

$$\Delta \circ D = (\mathrm{id} \otimes D) \circ \Delta$$

That is, $D$ acts only on the "right factor" of the comultiplication.

**Proposition 3.1.13: Lie Algebra as Left-Invariant Derivations**

There is a canonical isomorphism between $\mathrm{Lie}(G)$ and the space of left-invariant derivations of $\mathcal{O}(G)$.

***Proof*:**
Given a derivation $D_e : A \to k$ at the identity (i.e. an element of $\mathrm{Lie}(G) = (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$), define $D : A \to A$ by $D = (\mathrm{id} \otimes D_e) \circ \Delta$. Explicitly, for $f \in A$ with $\Delta(f) = \sum f_{(1)} \otimes f_{(2)}$:

$$D(f) = \sum f_{(1)} D_e(f_{(2)})$$

We verify the Leibniz rule. Since $\Delta$ is an algebra homomorphism, $\Delta(fg) = \Delta(f)\Delta(g) = \sum f_{(1)}g_{(1)} \otimes f_{(2)}g_{(2)}$. Then:

$$D(fg) = \sum f_{(1)}g_{(1)} D_e(f_{(2)}g_{(2)}) = \sum f_{(1)}g_{(1)} \big(\epsilon(f_{(2)})D_e(g_{(2)}) + D_e(f_{(2)})\epsilon(g_{(2)})\big)$$

Using the counit axiom $\sum f_{(1)}\epsilon(f_{(2)}) = f$:

$$D(fg) = f \cdot D(g) + D(f) \cdot g$$

Left-invariance $\Delta \circ D = (\mathrm{id} \otimes D) \circ \Delta$ follows from coassociativity. The inverse map sends a left-invariant $D : A \to A$ to $D_e = \epsilon \circ D : A \to k$, which is a derivation at $\epsilon$.

For the reader familiar with differential geometry (see [20]), this is the exact algebraic analogue of the classical fact that the Lie algebra of a Lie group is the space of left-invariant vector fields. The left translation by $g$ on a Lie group is replaced by the coaction $\Delta$, and "invariance under left translation" is replaced by $\Delta \circ D = (\mathrm{id} \otimes D) \circ \Delta$.

### Exercise 3.1.1

1. Compute $\mathrm{Lie}(\mathbb{G}_a)$ and $\mathrm{Lie}(\mathbb{G}_m)$. For $\mathbb{G}_m$, show that the Lie algebra is one-dimensional and abelian (i.e. the bracket is identically zero).

2. Let $B_n \subseteq \mathrm{GL}_n$ be the Borel subgroup of upper-triangular invertible matrices, and let $U_n \subseteq B_n$ be the unipotent subgroup of upper-triangular matrices with 1's on the diagonal. Compute $\mathrm{Lie}(B_n)$ and $\mathrm{Lie}(U_n)$ using the dual numbers definition. What are their dimensions?

3. Let $G$ and $H$ be affine algebraic groups. Show that $\mathrm{Lie}(G \times H) \cong \mathrm{Lie}(G) \oplus \mathrm{Lie}(H)$ as Lie algebras, where the direct sum has bracket $[(X_1, Y_1), (X_2, Y_2)] = ([X_1, X_2], [Y_1, Y_2])$.

4. Let $\varphi : G \to H$ be a homomorphism of affine algebraic groups with (scheme-theoretic) kernel $K = \ker(\varphi)$. Show that $\mathrm{Lie}(K) = \ker(d\varphi)$. Deduce that if $\varphi$ is surjective and $\dim G = \dim H$, then $K$ is finite if and only if $d\varphi$ is an isomorphism (assuming both groups are smooth).

5. Let $G = \mathrm{GL}_2$ and let $D_e \in \mathrm{Lie}(\mathrm{GL}_2)$ correspond to the matrix $X = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Compute the left-invariant derivation $D = (\mathrm{id} \otimes D_e) \circ \Delta : \mathcal{O}(\mathrm{GL}_2) \to \mathcal{O}(\mathrm{GL}_2)$ by evaluating $D$ on the coordinate functions $a, b, c, d$ (the matrix entries). (*Hint*: use $\Delta(a) = a \otimes a + b \otimes c$, etc. from example 1.1.14.)

6. Let $K/k$ be a finite field extension and $G$ an algebraic group over $K$. Show that $\mathrm{Lie}(\mathrm{Res}_{K/k}(G)) \cong \mathrm{Lie}(G)$ as $k$-vector spaces (where the right side is viewed as a $k$-vector space via restriction of scalars). (*Hint*: use the definition $\mathrm{Res}_{K/k}(G)(R) = G(R \otimes_k K)$ from definition 1.2.4 and compute on $k[\varepsilon]$.)

## 3.2 The Adjoint Representation

In chapter 2, we introduced the conjugation action of $G$ on itself (example 2.1.5): a group element $g$ acts on $G$ by $x \mapsto gxg^{-1}$. This action preserves the identity (since $geg^{-1} = e$), and therefore it acts on the infinitesimal structure near $e$ — that is, on the Lie algebra $\mathfrak{g}$. This induced action of $G$ on $\mathfrak{g}$ is the *adjoint representation*, and it is arguably the single most important representation in the structure theory of algebraic groups. When we decompose $\mathfrak{g}$ into eigenspaces for the adjoint action of a maximal torus in chapter 7, we will obtain the *root space decomposition*, which is the foundation of the entire classification.

### The Adjoint Representation of $G$

Let $G$ be an affine algebraic group over $k$. For each $g \in G(R)$ (where $R$ is any $k$-algebra), the *inner automorphism* $\mathrm{Inn}_g : G_R \to G_R$ is the morphism $x \mapsto gxg^{-1}$. This is an automorphism of $G_R$ as a group scheme over $R$, and it fixes the identity. By the functoriality of Lie (proposition 3.1.11), it induces a Lie algebra automorphism $d(\mathrm{Inn}_g) : \mathfrak{g} \otimes_k R \to \mathfrak{g} \otimes_k R$.

**Definition 3.2.1: The Adjoint Representation**

Let $G$ be an affine algebraic group with Lie algebra $\mathfrak{g}$. The *adjoint representation* is the homomorphism of algebraic groups

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}(\mathfrak{g})$$

defined on $R$-valued points by: for $g \in G(R)$ and $X \in \mathfrak{g} \otimes_k R$,

$$\mathrm{Ad}(g)(X) = d(\mathrm{Inn}_g)(X)$$

That is, $\mathrm{Ad}(g)$ is the differential of the conjugation $x \mapsto gxg^{-1}$ at the identity.

Let us make this concrete. An element $X \in \mathfrak{g} = \mathrm{Lie}(G)$ is an element of $G(k[\varepsilon])$ that reduces to $e$ modulo $\varepsilon$: it has the form $e + X\varepsilon$ (in a matrix group) or, more generally, it is a $k$-algebra map $\mathcal{O}(G) \to k[\varepsilon]$ with reduction $\epsilon$. The conjugation $\mathrm{Inn}_g$ acts on $G(k[\varepsilon])$ by:

$$\mathrm{Inn}_g(e + X\varepsilon) = g(e + X\varepsilon)g^{-1} = e + (gXg^{-1})\varepsilon$$

where in the last step we used the fact that $g \in G(k) \subseteq G(k[\varepsilon])$ (viewing $k$ as a subring of $k[\varepsilon]$), so the computation takes place in $G(k[\varepsilon])$. Extracting the $\varepsilon$-coefficient, we get $\mathrm{Ad}(g)(X) = gXg^{-1}$.

More precisely, to see that Ad is a morphism of *algebraic* groups (not just a map on $k$-points), we work over an arbitrary $k$-algebra $R$. An element $g \in G(R)$ and $X \in \mathfrak{g} \otimes_k R \subseteq G(R[\varepsilon])$ (where $R[\varepsilon] = R \otimes_k k[\varepsilon]$) satisfy:

$$\mathrm{Ad}(g)(X) = gXg^{-1} \in \mathfrak{g} \otimes_k R$$

The naturality in $R$ ensures that Ad is a natural transformation of functors $G \to \mathrm{GL}(\mathfrak{g})$, hence a morphism of algebraic groups by the Yoneda Lemma (theorem 1.2.2).

That Ad is a group homomorphism is immediate from the functorial viewpoint: for $g, h \in G(R)$,

$$\mathrm{Ad}(gh)(X) = (gh)X(gh)^{-1} = g(hXh^{-1})g^{-1} = \mathrm{Ad}(g)(\mathrm{Ad}(h)(X))$$

**Example 3.2.2:** Ad for $\mathrm{GL}_n$

For $G = \mathrm{GL}_n$, the Lie algebra is $\mathfrak{g} = \mathfrak{gl}_n = M_n(k)$, and the adjoint representation is simply conjugation of matrices:

$$\mathrm{Ad}(g)(X) = gXg^{-1}, \qquad g \in \mathrm{GL}_n(R),\ X \in M_n(R)$$

This is a representation $\mathrm{GL}_n \to \mathrm{GL}(\mathfrak{gl}_n) = \mathrm{GL}_{n^2}$, but it is *not* the standard $n^2$-dimensional representation of $\mathrm{GL}_n$: the action is by simultaneous left- and right-multiplication (conjugation), not by left-multiplication alone. In the language of chapter 2, $\mathfrak{gl}_n$ as a $\mathrm{GL}_n$-module is isomorphic to $V \otimes V^\vee$, where $V = k^n$ is the standard representation and $V^\vee$ is the contragredient. This is because $\mathrm{Hom}_k(V, V) \cong V \otimes V^\vee$ as representations (exercise 2.3.1 (5)), and conjugation is precisely the action on $\mathrm{Hom}_k(V, V)$.

**Example 3.2.3: Ad for $\mathrm{SL}_n$**

For $G = \mathrm{SL}_n$, the Lie algebra $\mathfrak{sl}_n = \{X \in M_n(k) : \mathrm{tr}(X) = 0\}$ is a codimension-1 subspace of $\mathfrak{gl}_n$. The adjoint representation is again $\mathrm{Ad}(g)(X) = gXg^{-1}$, but now $g \in \mathrm{SL}_n(R)$ and $X \in \mathfrak{sl}_n \otimes R$. Since conjugation preserves the trace ($\mathrm{tr}(gXg^{-1}) = \mathrm{tr}(X)$), this is well-defined. As a representation of $\mathrm{SL}_n$, the Lie algebra $\mathfrak{sl}_n$ is the quotient of $V \otimes V^\vee \cong \mathfrak{gl}_n$ by the trivial subrepresentation $k \cdot I_n$ (the scalar matrices, which have trace $n$ and are fixed by conjugation). When $\mathrm{char} k \nmid n$, the trace map $\mathfrak{gl}_n \to k$ splits and $\mathfrak{gl}_n \cong \mathfrak{sl}_n \oplus k$ as $\mathrm{SL}_n$-representations. When $\mathrm{char} k \big| n$, the scalar matrix $I_n$ has trace 0 and so $k \cdot I_n \subseteq \mathfrak{sl}_n$; this subrepresentation does not split off, and the representation theory becomes more subtle.

## The Kernel of Ad: The Center

The kernel of a representation is always a closed normal subgroup. For the adjoint representation, this kernel has a particularly natural description.

**Proposition 3.2.4: Kernel of Ad is the Center**

Let $G$ be an affine algebraic group. The kernel of $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ contains the *scheme-theoretic center*

$$Z(G) := \{g \in G \mid gx = xg \text{ for all } x \in G\}$$

and if $G$ is smooth, then $\ker(\mathrm{Ad}) = Z(G)$.

***Proof*:**

Functorially, $g \in Z(G)(R)$ means $gxg^{-1} = x$ for all $x \in G(R')$ and all $R$-algebras $R'$. In particular, taking $R' = R[\varepsilon]$ and $x \in \mathfrak{g} \otimes_k R \subseteq G(R[\varepsilon])$, we get $\mathrm{Ad}(g)(X) = gXg^{-1} = X$, so $g \in \ker(\mathrm{Ad})(R)$. Thus $Z(G) \subseteq \ker(\mathrm{Ad})$.

For the reverse inclusion when $G$ is smooth, suppose $g \in \ker(\mathrm{Ad})(R)$, so that $\mathrm{Ad}(g)$ acts trivially on $\mathfrak{g} \otimes R$. The inner automorphism $\mathrm{Inn}_g : G_R \to G_R$ is a group automorphism whose differential $d(\mathrm{Inn}_g) = \mathrm{Ad}(g)$ is the identity on $\mathfrak{g} \otimes R$. When $G$ is smooth and connected, a morphism of smooth algebraic groups whose differential is the identity must be the identity itself (this follows from the fact that in characteristic 0, or more generally for smooth groups, the differential is faithful on connected groups — we will make this precise in §3.3). For disconnected $G$, the argument still works on $G^\circ$, and one checks separately that the center respects connected components.

Without the smoothness hypothesis, the inclusion can be strict. For example, over a field of characteristic $p$, the group scheme $\mu_p = \mathrm{Spec}\, k[t]/(t^p - 1)$ has trivial Lie algebra action but a nontrivial center; we will see such phenomena in §3.3.

**Example 3.2.5: Center of $\mathrm{GL}_n$**

The center of $\mathrm{GL}_n$ consists of scalar matrices: $Z(\mathrm{GL}_n) = \{aI : a \in k^\times\} \cong \mathbb{G}_m$. Indeed, $\mathrm{Ad}(aI)(X) = (aI)X(aI)^{-1} = X$ for all $X \in \mathfrak{gl}_n$. Conversely, if $gXg^{-1} = X$ for all $X \in M_n(k)$, then $g$ commutes with all matrices, and a standard linear algebra argument (or Schur's lemma applied to the standard representation) shows $g$ is scalar.

For $\mathrm{SL}_n$, the center is $Z(\mathrm{SL}_n) = \mu_n = \{aI : a^n = 1\}$, the $n$-th roots of unity (as a group scheme). The quotient $\mathrm{SL}_n/Z(\mathrm{SL}_n) = \mathrm{PGL}_n$ is the *projective general linear group*, and the adjoint representation of $\mathrm{SL}_n$ factors through $\mathrm{PGL}_n$. Since $Z(\mathrm{SL}_n)$ acts trivially on $\mathfrak{sl}_n$, we have $\mathrm{Ad} : \mathrm{SL}_n \to \mathrm{GL}(\mathfrak{sl}_n)$ with $\ker = \mu_n$. In characteristic $p$ dividing $n$, the group scheme $\mu_n$ is not reduced (it contains the infinitesimal subgroup $\mu_p$), a subtlety that the adjoint representation detects.

## The Infinitesimal Adjoint: ad

The adjoint representation $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ is a homomorphism of algebraic groups, so we can apply the functor Lie to it. The result is a homomorphism of Lie algebras

$$\mathrm{ad} := \mathrm{Lie}(\mathrm{Ad}) : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathfrak{g})$$

where $\mathfrak{gl}(\mathfrak{g}) = \mathrm{Lie}(\mathrm{GL}(\mathfrak{g})) = \mathrm{End}_k(\mathfrak{g})$ with the commutator bracket. This map ad sends each element $X \in \mathfrak{g}$ to a linear endomorphism $\mathrm{ad}(X) : \mathfrak{g} \to \mathfrak{g}$.

**Proposition 3.2.6: ad is the Lie Bracket**

For all $X, Y \in \mathfrak{g}$, we have $\mathrm{ad}(X)(Y) = [X, Y]$.

***Proof*:**
We compute using dual numbers. By definition, $\mathrm{ad}(X)$ is the differential of Ad in the direction $X$. Concretely, let $X \in \mathfrak{g}$ and consider the element $e + X\varepsilon_1 \in G(k[\varepsilon_1])$. The adjoint representation sends this to $\mathrm{Ad}(e + X\varepsilon_1) \in \mathrm{GL}(\mathfrak{g})(k[\varepsilon_1]) = \mathrm{GL}(\mathfrak{g} \otimes_k k[\varepsilon_1])$.

To evaluate $\mathrm{Ad}(e + X\varepsilon_1)$ on an element $Y \in \mathfrak{g}$, we embed $Y$ into $G(k[\varepsilon_1, \varepsilon_2])$ as $e + Y\varepsilon_2$ and conjugate:

$$\begin{aligned}\mathrm{Ad}(e + X\varepsilon_1)(e + Y\varepsilon_2) &= (e + X\varepsilon_1)(e + Y\varepsilon_2)(e + X\varepsilon_1)^{-1} \\ &= (e + X\varepsilon_1)(e + Y\varepsilon_2)(e - X\varepsilon_1) \\ &= e + Y\varepsilon_2 + (XY - YX)\varepsilon_1\varepsilon_2\end{aligned}$$

The $\varepsilon_2$-coefficient of the result gives the action of $\mathrm{Ad}(e + X\varepsilon_1)$ on $Y$. This is $Y + [X, Y]\varepsilon_1$, meaning:

$$\mathrm{Ad}(e + X\varepsilon_1) = \mathrm{id}_{\mathfrak{g}} + \mathrm{ad}(X) \cdot \varepsilon_1$$

where $\mathrm{ad}(X)(Y) = XY - YX = [X, Y]$. (Here the computation is in a matrix group; for a general group the same argument works by embedding $G \hookrightarrow \mathrm{GL}_n$ and using the linearization theorem, lemma 1.1.8.)

The map $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ is thus a Lie algebra homomorphism from $\mathfrak{g}$ to the endomorphism algebra of itself. The fact that ad is a Lie algebra homomorphism is equivalent to the Jacobi identity:

$$\mathrm{ad}([X, Y]) = [\mathrm{ad}(X), \mathrm{ad}(Y)] \iff [[X, Y], Z] = [X, [Y, Z]] - [Y, [X, Z]]$$

This is sometimes used as a way to *prove* the Jacobi identity: it follows formally from the fact that Ad is a group homomorphism, which is obvious.

**Definition 3.2.7: Adjoint Representation of the Lie Algebra**

Let $\mathfrak{g}$ be the Lie algebra of an affine algebraic group $G$. The *adjoint representation of the Lie algebra* is the map $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ defined by $\mathrm{ad}(X)(Y) = [X, Y]$. Its kernel is the *center of the Lie algebra*:

$$\mathfrak{z}(\mathfrak{g}) := \ker(\mathrm{ad}) = \{X \in \mathfrak{g} \mid [X, Y] = 0 \text{ for all } Y \in \mathfrak{g}\}$$

The center of the Lie algebra and the Lie algebra of the center are related but not identical:

**Proposition 3.2.8: Center Comparison**

Let $G$ be an affine algebraic group. Then:

$$\mathrm{Lie}(Z(G)) \subseteq \mathfrak{z}(\mathfrak{g}) = \ker(\mathrm{ad})$$

If $G$ is smooth and connected (and $\mathrm{char} k = 0$), equality holds.

***Proof*:**
By proposition 3.1.12, the closed immersion $Z(G) \hookrightarrow G$ gives an injection $\mathrm{Lie}(Z(G)) \hookrightarrow \mathfrak{g}$. An element $X \in \mathrm{Lie}(Z(G)) \subseteq \mathfrak{g}$ corresponds to a derivation at $\epsilon$ that factors through $\mathcal{O}(Z(G))$. Since $Z(G) \subseteq \ker(\mathrm{Ad})$, the image of $X$ under $\mathrm{ad} = \mathrm{Lie}(\mathrm{Ad})$ is zero, so $X \in \mathfrak{z}(\mathfrak{g})$.

For equality in characteristic 0, one uses the fact that Lie is exact on smooth groups (which we establish in §3.3): the sequence $1 \to Z(G) \to G \xrightarrow{\mathrm{Ad}} \mathrm{GL}(\mathfrak{g})$ is exact, so applying Lie gives an exact sequence $0 \to \mathrm{Lie}(Z(G)) \to \mathfrak{g} \xrightarrow{\mathrm{ad}} \mathfrak{gl}(\mathfrak{g})$, and the kernel of ad is exactly $\mathrm{Lie}(Z(G))$.

**Example 3.2.9: The Adjoint Representation of $\mathrm{SL}_2$**
Let $G = \mathrm{SL}_2$ over a field $k$ with $\mathrm{char} k \neq 2$. The Lie algebra $\mathfrak{sl}_2$ has the standard basis:

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The Lie brackets are $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. The adjoint representation $\mathrm{ad} : \mathfrak{sl}_2 \to \mathfrak{gl}(\mathfrak{sl}_2) \cong \mathfrak{gl}_3$ is determined by the matrices of $\mathrm{ad}(e)$, $\mathrm{ad}(f)$, $\mathrm{ad}(h)$ in the basis $\{e, f, h\}$:

$$\mathrm{ad}(h) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathrm{ad}(e) = \begin{pmatrix} 0 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathrm{ad}(f) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ -1 & 0 & 0 \end{pmatrix}$$

(The reader should verify these from the bracket relations.) The matrix $\mathrm{ad}(h)$ is diagonal, and its eigenvalues $\{2, -2, 0\}$ are the *roots* of $\mathrm{SL}_2$ with respect to the Cartan subalgebra $\mathfrak{t} = k \cdot h$. The eigenspaces are:

$$\mathfrak{g}_2 = k \cdot e, \qquad \mathfrak{g}_{-2} = k \cdot f, \qquad \mathfrak{g}_0 = k \cdot h = \mathfrak{t}$$

This is the simplest example of a *root space decomposition* $\mathfrak{g} = \mathfrak{g}_{-2} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_2$. The general theory of chapter 7 will show that every semisimple Lie algebra admits such a decomposition, with the roots forming a combinatorial structure (a *root system*) that completely determines the group.

The center $\mathfrak{z}(\mathfrak{sl}_2) = 0$ (since $\mathrm{ad}(h)$ has no kernel on the full space when $\mathrm{char} k \neq 2$), confirming that $\mathfrak{sl}_2$ is a *semisimple* Lie algebra.

At the group level, $\mathrm{Ad} : \mathrm{SL}_2 \to \mathrm{GL}_3$ sends a matrix $g \in \mathrm{SL}_2$ to the $3 \times 3$ matrix representing conjugation $X \mapsto gXg^{-1}$ on $\mathfrak{sl}_2$. The image lands in $\mathrm{SO}_3$ (the adjoint representation preserves the Killing form, which is a non-degenerate symmetric bilinear form on $\mathfrak{sl}_2$). The kernel is $Z(\mathrm{SL}_2) = \{\pm I\} = \mu_2$. Thus Ad realizes the classical isomorphism $\mathrm{SL}_2/\mu_2 \cong \mathrm{SO}_3$ (when $\mathrm{char} k \neq 2$).

**Example 3.2.10: Abelian Groups Have Trivial Adjoint**

If $G$ is a commutative algebraic group ($\mathbb{G}_a$, $\mathbb{G}_m$, an abelian variety, a torus, ...), then the conjugation action is trivial: $gxg^{-1} = x$ for all $g, x$. Therefore $\mathrm{Ad}(g) = \mathrm{id}_{\mathfrak{g}}$ for all $g$, and $\mathrm{ad} = 0$, meaning the Lie bracket is identically zero. The Lie algebra of a commutative group is always *abelian*.

The converse is false: the Lie algebra can be abelian even when the group is not. For example, in characteristic $p$, the group scheme $\alpha_p \rtimes \mathbb{G}_m$ (where $\mathbb{G}_m$ acts on $\alpha_p$ by scaling) can have abelian Lie algebra while being non-commutative, depending on the specific action. More strikingly, any non-commutative group $G$ with $\dim G = 1$ necessarily has a one-dimensional (hence abelian) Lie algebra, but the group itself need not be commutative at the scheme-theoretic level.

## The Adjoint Action on the Coordinate Ring

The adjoint representation gives $G$ an action on the finite-dimensional vector space $\mathfrak{g}$. By the theory of chapter 2, this is a comodule: Ad equips $\mathfrak{g}$ with a coaction $\rho_{\mathrm{Ad}} : \mathfrak{g} \to \mathfrak{g} \otimes_k \mathcal{O}(G)$. More explicitly, if $\{X_1, \ldots, X_n\}$ is a basis for $\mathfrak{g}$, there exist matrix coefficient functions $a_{ij} \in \mathcal{O}(G)$ such that:

$$\mathrm{Ad}(g)(X_j) = \sum_{i=1}^{n} a_{ij}(g) X_i, \qquad \rho_{\mathrm{Ad}}(X_j) = \sum_{i=1}^{n} X_i \otimes a_{ij}$$

These functions $a_{ij}$ are the *matrix coefficients of the adjoint representation.* They generate a Hopf subalgebra of $\mathcal{O}(G)$ (the image of $\mathrm{Ad}^* : \mathcal{O}(\mathrm{GL}(\mathfrak{g})) \to \mathcal{O}(G)$), whose spectrum is the image $\mathrm{Ad}(G) \subseteq \mathrm{GL}(\mathfrak{g})$.

There is also a natural extension to symmetric and exterior powers. Since $\mathfrak{g}$ is a representation of $G$, so are $\mathrm{Sym}^d(\mathfrak{g})$, $\bigwedge^d(\mathfrak{g})$, and $\mathfrak{g}^{\otimes d}$ (by the constructions of §2.3). Of particular importance is the *coadjoint representation*: the dual $\mathfrak{g}^\vee$ with the contragredient action $\mathrm{Ad}^*(g)(\xi) = \xi \circ \mathrm{Ad}(g^{-1})$.

**Proposition 3.2.11: Invariant Bilinear Forms**

Let $G$ be an affine algebraic group acting on its Lie algebra via Ad. A bilinear form $B : \mathfrak{g} \times \mathfrak{g} \to k$ is called *$G$-invariant* (or Ad-*invariant*) if $B(\mathrm{Ad}(g)(X), \mathrm{Ad}(g)(Y)) = B(X, Y)$ for all $g \in G$ and $X, Y \in \mathfrak{g}$.
If $B$ is $G$-invariant, differentiating the invariance condition yields the corresponding *infinitesimal invariance* condition on $\mathfrak{g}$:

$$B([Z, X], Y) + B(X, [Z, Y]) = 0 \qquad \text{for all } X, Y, Z \in \mathfrak{g}$$

Furthermore, if $G$ is connected and $\mathrm{char}(k) = 0$, these two conditions are equivalent.

***Proof*:**
Fix $X, Y \in \mathfrak{g}$ and consider the function $\varphi : G \to k$ defined by $\varphi(g) = B(\mathrm{Ad}(g)(X), \mathrm{Ad}(g)(Y))$. If $B$ is $G$-invariant, then $\varphi$ is constant, so its differential at $e$ vanishes. Evaluating in the direction $Z \in \mathfrak{g}$ (i.e. replacing $g$ by $e + Z\varepsilon$):

$$\begin{aligned}\varphi(e + Z\varepsilon) &= B(\mathrm{Ad}(e + Z\varepsilon)(X),\ \mathrm{Ad}(e + Z\varepsilon)(Y)) \\ &= B(X + [Z, X]\varepsilon,\ Y + [Z, Y]\varepsilon) \\ &= B(X, Y) + \big(B([Z, X], Y) + B(X, [Z, Y])\big)\varepsilon\end{aligned}$$

The constant term is $B(X, Y) = \varphi(e)$, and the $\varepsilon$-coefficient must vanish, giving the infinitesimal invariance condition.

For the converse, the fact that infinitesimal invariance implies global invariance (under the assumptions that $G$ is connected and $\mathrm{char}(k) = 0$) follows from the faithfulness of Lie in that setting, which we establish in §3.3.

The most important example of such a form is the *Killing form* $\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}(X) \circ \mathrm{ad}(Y))$, which we will study in the context of semisimplicity. The $\mathrm{SL}_2$ example above already demonstrated that the adjoint representation preserves a non-degenerate form, which is a manifestation of the Killing form.

## Exercise 3.2.1

1. Let $G = \mathrm{GL}_2$ with Lie algebra $\mathfrak{gl}_2$, basis $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ (the elementary matrices). Write down the $4 \times 4$ matrix representing $\mathrm{Ad}(g)$ for a general element $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2$. Verify that $\mathrm{Ad}(g)$ has determinant 1 for all $g$ (that is, the adjoint representation of $\mathrm{GL}_2$ lands in $\mathrm{SL}_4$, not just $\mathrm{GL}_4$).

2. Show that the scheme-theoretic center $Z(\mathrm{SL}_n)$ is the group scheme $\mu_n$ of $n$-th roots of unity, and verify directly that $\mathrm{Lie}(\mu_n) = 0$ when $\mathrm{char} k \nmid n$. What happens when $\mathrm{char} k \big| n$?

3. Compute the Killing form $\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}(X) \circ \mathrm{ad}(Y))$ for $\mathfrak{sl}_2$ (with $\mathrm{char} k \neq 2$) in the basis $\{e, f, h\}$, and show it is non-degenerate. Verify that $\kappa$ satisfies the ad-invariance condition $\kappa([Z, X], Y) + \kappa(X, [Z, Y]) = 0$.

4. Let $B_2 \subseteq \mathrm{GL}_2$ be the Borel subgroup of upper-triangular matrices. Compute the adjoint representation $\mathrm{Ad} : B_2 \to \mathrm{GL}(\mathrm{Lie}(B_2))$ explicitly. Show that Ad is *not* faithful (i.e. $\ker(\mathrm{Ad}) \supsetneq$

$Z(B_2)$ — wait, this is wrong; verify instead that $\ker(\mathrm{Ad}) = Z(B_2) = \mathbb{G}_m$ embedded as scalar matrices).

5. Let $G = H \rtimes K$ be a semidirect product of algebraic groups (so $H \trianglelefteq G$ and $G/H \cong K$). Show that $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{k}$ as vector spaces, and that the Lie bracket satisfies $[\mathfrak{k}, \mathfrak{h}] \subseteq \mathfrak{h}$ (i.e. $\mathfrak{h}$ is a Lie ideal). Describe $\mathrm{Ad}|_K : K \to \mathrm{GL}(\mathfrak{h})$ in terms of the conjugation action of $K$ on $H$.

## 3.3 Cartier's Theorem and the Limits of the Lie Algebra

How faithfully does the Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ reflect the algebraic group $G$? In the classical theory over $\mathbb{R}$ or $\mathbb{C}$, the answer is "almost perfectly": the exponential map provides a local diffeomorphism between $\mathfrak{g}$ and a neighborhood of $e$ in $G$, and connected, simply connected Lie groups are completely determined by their Lie algebras. In the algebraic setting, this correspondence is governed by the *smoothness* of $G$. Smooth groups are well-behaved: their Lie algebras have the "right" dimension and the functor Lie is well-behaved. Non-smooth groups — which can only arise in positive characteristic — are the source of all pathology.

The main theorem of this section, due to Cartier, states that in characteristic 0, every algebraic group scheme is automatically smooth. This is a remarkable consequence of the Hopf algebra structure: the coalgebra axioms force the local ring at the identity to be regular, something that is far from automatic for a general scheme. In characteristic $p$, the theorem fails, and we will study the two fundamental counterexamples in detail.

### The Dimension Inequality

Recall from proposition 3.1.2 that $\mathrm{Lie}(G) \cong (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$, the Zariski tangent space at the identity. The dimension of this space is an *upper bound* for the Krull dimension of the local ring $\mathcal{O}_{G,e}$, with equality if and only if the local ring is regular:

**Proposition 3.3.1: Dimension Inequality**

Let $G$ be an affine algebraic group scheme over $k$. Then:

$$\dim \mathrm{Lie}(G) = \dim_k(\mathfrak{m}_e/\mathfrak{m}_e^2) \geq \dim G$$

with equality if and only if $G$ is smooth at the identity $e$. Moreover, if $G$ is smooth at $e$, then $G$ is smooth everywhere.

***Proof*:**
The inequality $\dim_k(\mathfrak{m}_e/\mathfrak{m}_e^2) \geq \dim \mathcal{O}_{G,e} = \dim_e G$ is a general fact of commutative algebra: the embedding dimension of a Noetherian local ring is at least its Krull dimension, with equality characterizing regular local rings (see [7]). And $\dim_e G = \dim G$ since $G$ is equidimensional (all irreducible components have the same dimension, by the translation argument of proposition 1.1.2).

For the "moreover" statement: if $G$ is smooth at $e$, then for any other point $g \in G(k)$, the left translation $\ell_g : G \to G$ is an isomorphism carrying $e$ to $g$. Since smoothness is preserved by isomorphisms, $G$ is smooth at $g$. As $g$ was arbitrary, $G$ is smooth everywhere. (For a group scheme over a non-algebraically closed field, one must also check smoothness at non-rational points, but this follows from the same translation argument after base change to $\overline{k}$.)

In chapter 1 (proposition 1.1.2), we stated that algebraic groups are smooth, but the argument relied on $G$ being a *variety* (a reduced, separated scheme of finite type). The crucial word was "reduced": a reduced scheme of finite type over a perfect field is generically smooth, and homogeneity then forces smoothness everywhere. But group *schemes* need not be reduced, and the question of whether the group structure alone forces smoothness is far more delicate. This is what Cartier's theorem addresses.

## Cartier's Theorem

**Theorem 3.3.2: Cartier's Theorem**

Let $G$ be an affine algebraic group scheme of finite type over a field $k$ of characteristic 0. Then $G$ is smooth over $k$. In particular, $\dim \operatorname{Lie}(G) = \dim G$.

The theorem says that in characteristic 0, the Hopf algebra structure is rigid enough that it forces $\mathcal{O}(G)$ to be geometrically reduced (no nilpotent elements after base change to $\overline{k}$). This is false for general $k$-algebras: the ring $k[x]/(x^2)$ is a perfectly good $k$-algebra with nilpotents. But it *cannot* be the coordinate ring of a group scheme in characteristic 0.

***Proof*:**
Let $A = \mathcal{O}(G)$ with comultiplication $\Delta$, counit $\epsilon$, and antipode $S$. We need to show that $A$ is geometrically reduced, i.e. $A \otimes_k \overline{k}$ has no nonzero nilpotent elements. Since this condition is checked after base change, we may assume $k = \overline{k}$.

We will show that $A$ is reduced (has no nonzero nilpotents). Let $I = \operatorname{Nil}(A)$ be the nilradical. Since $A$ is a finitely generated $k$-algebra, $A_{\text{red}} = A/I$ is a reduced $k$-algebra and $G_{\text{red}} = \operatorname{Spec}(A_{\text{red}})$ is the underlying reduced scheme. We first show that the Hopf algebra structure descends to $A_{\text{red}}$.

*Claim 1: $I = \operatorname{Nil}(A)$ is a Hopf ideal.* We need $\Delta(I) \subseteq \operatorname{Nil}(A \otimes A)$, $\epsilon(I) = 0$, and $S(I) \subseteq I$.

For the counit: $\epsilon : A \to k$ is a $k$-algebra homomorphism, and $k$ is a field (hence has no nilpotents), so $\epsilon(I) = 0$.

For the antipode: $S : A \to A$ is a $k$-algebra anti-homomorphism (i.e. $S(ab) = S(b)S(a)$). If $a \in I$ is nilpotent ($a^n = 0$), then $S(a)^n = S(a^n) = S(0) = 0$ (noting that $S$ is an algebra homomorphism when $A$ is commutative, which it is). So $S(I) \subseteq I$.

For the comultiplication: $\Delta : A \to A \otimes_k A$ is an algebra homomorphism. If $a \in I$ is nilpotent with $a^n = 0$, then $\Delta(a)^n = \Delta(a^n) = \Delta(0) = 0$, so $\Delta(a)$ is nilpotent in $A \otimes A$. Thus $\Delta(I) \subseteq \operatorname{Nil}(A \otimes A)$.

Therefore $A_{\text{red}} = A/I$ inherits a Hopf algebra structure, and $G_{\text{red}} = \operatorname{Spec}(A_{\text{red}})$ is a group scheme — in fact an algebraic group (since $A_{\text{red}}$ is reduced and of finite type, $G_{\text{red}}$ is a variety). By the standard argument for algebraic groups (generic smoothness of a variety plus translation by group elements), $G_{\text{red}}$ is smooth everywhere.

*Claim 2: Nilpotent elements are contained in the augmentation ideal.* Let $\mathfrak{m}_e = \ker(\epsilon)$ be the maximal ideal at the identity. If $a \in I$ with $a^n = 0$, then $\epsilon(a)^n = \epsilon(a^n) = 0$ in $k$. Since $k$ is a field, $\epsilon(a) = 0$. Thus, $I \subseteq \mathfrak{m}_e$.

*Claim 3:* $I \subseteq \mathfrak{m}_e^2$. This is the critical step where characteristic 0 is used. Suppose for contradiction that $I \not\subseteq \mathfrak{m}_e^2$. Then there exists some nilpotent element $a \in I$ that is not in $\mathfrak{m}_e^2$. Because $a$ is nilpotent, there exists a minimal integer $m \geq 2$ such that $a^m \in \mathfrak{m}_e^2$.

Let $b = a^{m-1}$. By our choice of $m$, $b \notin \mathfrak{m}_e^2$. However, $b^2 = a^{2m-2}$. Since $m \geq 2$, we have $2m - 2 \geq m$, which implies $b^2 \in \mathfrak{m}_e^2$.

Apply $\Delta$ to $b$. We can write $\Delta(b) = b \otimes 1 + 1 \otimes b + w$ with $w \in \mathfrak{m}_e \otimes \mathfrak{m}_e$. In the quotient $(A \otimes A)/(\mathfrak{m}_e^2 \otimes A + A \otimes \mathfrak{m}_e^2)$, we have $\Delta(b) \equiv \bar{b} \otimes 1 + 1 \otimes \bar{b}$, where $\bar{b}$ is the nonzero image of $b$ in $\mathfrak{m}_e/\mathfrak{m}_e^2$.

Squaring this yields $\Delta(b)^2 \equiv \bar{b}^2 \otimes 1 + 2\bar{b} \otimes \bar{b} + 1 \otimes \bar{b}^2$. Since $b^2 \in \mathfrak{m}_e^2$, its image $\bar{b}^2$ is 0, and because $\Delta$ is an algebra homomorphism, $\Delta(b^2)$ must also vanish in this quotient. Therefore:

$$0 = 2\bar{b} \otimes \bar{b} \quad \text{in} \quad (\mathfrak{m}_e/\mathfrak{m}_e^2) \otimes_k (\mathfrak{m}_e/\mathfrak{m}_e^2)$$

In characteristic 0, the integer 2 is invertible, so we can divide by 2 to get $\bar{b} \otimes \bar{b} = 0$. Since tensoring a nonzero vector space element with itself over a field cannot be zero, this forces $\bar{b} = 0$. Thus $b \in \mathfrak{m}_e^2$, contradicting our assumption. Therefore, $I \subseteq \mathfrak{m}_e^2$.

*Conclusion.* The surjection $\pi : A \twoheadrightarrow A_{\text{red}}$ sends $\mathfrak{m}_e$ to $\mathfrak{m}_{e,\text{red}}$. Because the kernel of this map is $I$, and we just proved $I \subseteq \mathfrak{m}_e^2$, modding out by $I$ does not kill any linear terms. Consequently, $\pi$ induces an isomorphism on the cotangent spaces at the identity:

$$\mathfrak{m}_e/\mathfrak{m}_e^2 \xrightarrow{\sim} \mathfrak{m}_{e,\text{red}}/\mathfrak{m}_{e,\text{red}}^2$$

Passing to the dual spaces (the tangent spaces), we have an equality of dimensions:

$$\dim \operatorname{Lie}(G) = \dim_k(\mathfrak{m}_e/\mathfrak{m}_e^2) = \dim_k(\mathfrak{m}_{e,\text{red}}/\mathfrak{m}_{e,\text{red}}^2) = \dim \operatorname{Lie}(G_{\text{red}})$$

Because $G_{\text{red}}$ is smooth, $\dim \operatorname{Lie}(G_{\text{red}}) = \dim G_{\text{red}}$. Since $G$ and $G_{\text{red}}$ share the same underlying topological space, $\dim G_{\text{red}} = \dim G$.

Stringing these equalities together yields $\dim \operatorname{Lie}(G) = \dim G$. This means the local ring $\mathcal{O}_{G,e}$ is a regular local ring. By a standard result in commutative algebra, every regular local ring is an integral domain, and therefore reduced. Thus, $\operatorname{Nil}(\mathcal{O}_{G,e}) = 0$. By translation, the local rings at all points are reduced, meaning $A$ itself is reduced.

## Failure in Characteristic $p$: Two Fundamental Counterexamples

Over a field of characteristic $p > 0$, there exist group schemes that are *not* smooth: their coordinate rings contain nilpotents, their Lie algebras are "too large," and the functor Lie loses information. The two most basic examples are $\mu_p$ and $\alpha_p$.

**Example 3.3.3: The Group Scheme $\mu_p$**

Let $k$ be a field of characteristic $p > 0$ and define:

$$\mu_p = \operatorname{Spec} k[t]/(t^p - 1)$$

with the group structure inherited from $\mathbb{G}_m$: the comultiplication is $\Delta(t) = t \otimes t$, the counit is $\epsilon(t) = 1$, and the antipode is $S(t) = t^{-1} = t^{p-1}$ (since $t^p = 1$). Functorially, $\mu_p(R) = \{r \in R : r^p = 1\}$ for any $k$-algebra $R$.

In characteristic $p$, we have $t^p - 1 = (t-1)^p$ in $k[t]$. So $\mu_p = \operatorname{Spec} k[t]/(t-1)^p = \operatorname{Spec} k[s]/(s^p)$ where $s = t - 1$. The coordinate ring $k[s]/(s^p)$ is a *local* ring with maximal ideal $(s)$ and nilpotent element $s$ (with $s^p = 0$ but $s \neq 0$). Thus:

1. $\mu_p$ has a *single* topological point (it is the spectrum of a local ring).
2. $\dim \mu_p = 0$ (the Krull dimension of $k[s]/(s^p)$ is 0).
3. $\operatorname{Lie}(\mu_p) = (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$ where $\mathfrak{m}_e = (s)$ and $\mathfrak{m}_e^2 = (s^2)$. So $\mathfrak{m}_e/\mathfrak{m}_e^2 = ks$ is one-dimensional, giving $\operatorname{Lie}(\mu_p) = k$.

We have $\dim \operatorname{Lie}(\mu_p) = 1 > 0 = \dim \mu_p$. The Lie algebra is "too big": it sees an infinitesimal direction that does not correspond to any actual geometric dimension. The group scheme is non-smooth (the local ring $k[s]/(s^p)$ is not regular).

Compare with the situation in characteristic 0: there $t^p - 1 = (t-1)(t^{p-1} + t^{p-2} + \cdots + 1)$ with the second factor having no common root with $t - 1$, so $\mu_p = \operatorname{Spec} k[t]/(t^p - 1)$ is étale (a disjoint union of $p$ reduced points over $\overline{k}$), and $\operatorname{Lie}(\mu_p) = 0$. Cartier's theorem is satisfied.

**Example 3.3.4: The Group Scheme $\alpha_p$**

Over a field $k$ of characteristic $p > 0$, define:

$$\alpha_p = \operatorname{Spec} k[t]/(t^p)$$

with the group structure inherited from $\mathbb{G}_a$: the comultiplication is $\Delta(t) = t \otimes 1 + 1 \otimes t$, the counit is $\epsilon(t) = 0$, and the antipode is $S(t) = -t$. Functorially, $\alpha_p(R) = \{r \in R : r^p = 0\}$ for any $k$-algebra $R$: the set of $p$-nilpotent elements of $R$, with addition as the group operation. (This is a subgroup of $(R, +) = \mathbb{G}_a(R)$ because $(r_1 + r_2)^p = r_1^p + r_2^p = 0$ in characteristic $p$.)

As a scheme, $\alpha_p$ is identical to $\mu_p$ as a topological space: both are Spec of a local $k$-algebra of dimension $p$. The coordinate ring is $k[t]/(t^p)$, the maximal ideal is $(t)$, and $t^p = 0$. Therefore:

1. $\dim \alpha_p = 0$.
2. $\operatorname{Lie}(\alpha_p) = k$ (one-dimensional, by the same tangent space computation as for $\mu_p$).

Again, $\dim \operatorname{Lie}(\alpha_p) = 1 > 0 = \dim \alpha_p$, and $\alpha_p$ is non-smooth.

The examples $\mu_p$ and $\alpha_p$ are the two simplest *infinitesimal group schemes*: connected group schemes of dimension 0 (i.e. Spec of a local Artinian $k$-algebra). Their Lie algebras are isomorphic — both are the one-dimensional abelian Lie algebra $k$ — but the group schemes are *not* isomorphic. The distinction is visible only in the Hopf algebra structure:

| | $\mu_p = \operatorname{Spec} k[s]/(s^p)$ | $\alpha_p = \operatorname{Spec} k[t]/(t^p)$ |
|---|---|---|
| Comultiplication | $\Delta(s) = s \otimes 1 + 1 \otimes s + s \otimes s$ | $\Delta(t) = t \otimes 1 + 1 \otimes t$ |
| Origin | multiplicative ($\mathbb{G}_m$-like) | additive ($\mathbb{G}_a$-like) |
| Lie algebra | $k$ | $k$ |

(Here we wrote $\mu_p$ in terms of $s = t - 1$ so that $\epsilon(s) = 0$, matching the convention that the augmentation ideal is $\ker(\epsilon)$. The comultiplication $\Delta(s) = s \otimes 1 + 1 \otimes s + s \otimes s$ comes from $\Delta(t) = t \otimes t$ via $s = t - 1$.)

This is a stark illustration of the limitations of the Lie algebra in characteristic $p$: two non-isomorphic group schemes with isomorphic Lie algebras. The functor Lie is *not faithful* on the category of group schemes in positive characteristic.

**3.3.5 Remark: The $p$-Lie Algebra Structure** What *does* distinguish $\mu_p$ and $\alpha_p$ at the infinitesimal level? The answer is the *restricted Lie algebra* (or *$p$-Lie algebra*) structure. In characteristic $p$, the Lie algebra of a group scheme carries an additional operation $X \mapsto X^{[p]}$ satisfying $\operatorname{ad}(X^{[p]}) = \operatorname{ad}(X)^p$, which encodes the $p$-th power map in the group. For $\mu_p$, the $p$-operation on the generator $X$ of $\operatorname{Lie}(\mu_p)$ satisfies $X^{[p]} = X$ (reflecting the multiplicative structure: $t^p = t$ in $\mu_p$ after identifying $t^p = 1$). For $\alpha_p$, the $p$-operation satisfies $X^{[p]} = 0$ (reflecting the additive structure: $t^p = 0$ in $\alpha_p$). These are genuinely different restricted Lie algebras, and the restricted Lie algebra functor *is* faithful on infinitesimal group schemes of height 1. A full development of this theory belongs to Part II.

## Consequences of Cartier's Theorem

In characteristic 0, Cartier's theorem has several powerful corollaries that justify treating algebraic groups through their Lie algebras.

**Corollary 3.3.6: Subgroup Correspondence in Characteristic 0**

Let $G$ be a connected affine algebraic group over a field $k$ of characteristic 0. Then the map

$$H \longmapsto \operatorname{Lie}(H) \subseteq \mathfrak{g}$$

is an injection from the set of closed connected subgroup schemes of $G$ to the set of Lie subalgebras of $\mathfrak{g}$.

***Proof*:**
Let $H_1, H_2$ be closed connected subgroup schemes of $G$ with $\operatorname{Lie}(H_1) = \operatorname{Lie}(H_2)$. By Cartier's theorem, both $H_1$ and $H_2$ are smooth, so $\dim H_i = \dim \operatorname{Lie}(H_i)$. Since $\operatorname{Lie}(H_1) = \operatorname{Lie}(H_2)$, we have $\dim H_1 = \dim H_2$. The intersection $H_1 \cap H_2$ is a closed subgroup scheme of both $H_1$ and $H_2$, and $\operatorname{Lie}(H_1 \cap H_2) \supseteq \operatorname{Lie}(H_1) \cap \operatorname{Lie}(H_2) = \operatorname{Lie}(H_1)$ (the intersection of equal subspaces). By Cartier's theorem, $H_1 \cap H_2$ is smooth of dimension $\dim \operatorname{Lie}(H_1 \cap H_2) \geq \dim \operatorname{Lie}(H_1) = \dim H_1$. Since $H_1 \cap H_2 \subseteq H_1$ and $\dim(H_1 \cap H_2) \geq \dim H_1$, both being connected and irreducible, we get $H_1 \cap H_2 = H_1$, i.e. $H_1 \subseteq H_2$. By symmetry, $H_2 \subseteq H_1$.

**Corollary 3.3.7: Exactness of $\mathrm{Lie}$ in Characteristic $0$**

Let $1 \to H \xrightarrow{\iota} G \xrightarrow{\pi} Q \to 1$ be an exact sequence of affine algebraic groups over a field of characteristic 0 (meaning $\iota$ is a closed immersion, $\pi$ is faithfully flat, and $H = \ker(\pi)$). Then the induced sequence of Lie algebras

$$0 \to \mathrm{Lie}(H) \xrightarrow{d\iota} \mathrm{Lie}(G) \xrightarrow{d\pi} \mathrm{Lie}(Q) \to 0$$

is exact.

***Proof*:**
Exactness on the left ($d\iota$ is injective) follows from proposition 3.1.12. Exactness in the middle ($\ker(d\pi) = \mathrm{im}(d\iota)$) follows from exercise 3.1.1 (4): $\ker(d\pi) = \mathrm{Lie}(\ker(\pi)) = \mathrm{Lie}(H) = \mathrm{im}(d\iota)$. Exactness on the right ($d\pi$ is surjective) is a dimension count: by Cartier's theorem, all three groups are smooth, so $\dim \mathrm{Lie}(G) = \dim G = \dim H + \dim Q = \dim \mathrm{Lie}(H) + \dim \mathrm{Lie}(Q)$, and since $d\iota$ is injective, $d\pi$ must be surjective.

This exactness is one of the most practically useful consequences of Cartier's theorem: it means that in characteristic 0, short exact sequences of groups induce short exact sequences of Lie algebras, allowing us to study extensions by purely Lie-algebraic methods. In characteristic $p$, the sequence $0 \to \mathrm{Lie}(H) \to \mathrm{Lie}(G) \to \mathrm{Lie}(Q)$ is always left-exact (proposition 3.1.12), but it can fail to be right-exact (the surjection $\pi$ may have a "purely infinitesimal" kernel that Lie misses).

**Corollary 3.3.8: Faithful $\mathrm{Ad}$ in Characteristic $0$**

Let $G$ be a connected affine algebraic group over a field of characteristic 0. Then $\ker(\mathrm{Ad}) = Z(G)$.

***Proof*:**
We already showed $Z(G) \subseteq \ker(\mathrm{Ad})$ in proposition 3.2.4. For the reverse, let $K = \ker(\mathrm{Ad})$. Since $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ is a homomorphism, $K$ is a closed normal subgroup of $G$. By Cartier's theorem, $K$ is smooth, and $\mathrm{Lie}(K) = \ker(\mathrm{ad}) = \mathfrak{z}(\mathfrak{g})$. Similarly, $Z(G)$ is smooth with $\mathrm{Lie}(Z(G)) \subseteq \mathfrak{z}(\mathfrak{g})$.

Now $Z(G) \subseteq K$ and both are smooth and connected (replacing by connected components if necessary). We have $\mathrm{Lie}(Z(G)) \subseteq \mathrm{Lie}(K) = \mathfrak{z}(\mathfrak{g})$. But also $\mathrm{Lie}(Z(G)) \supseteq \mathfrak{z}(\mathfrak{g})$: any $X \in \mathfrak{z}(\mathfrak{g})$ satisfies $[X, Y] = 0$ for all $Y \in \mathfrak{g}$, meaning $\mathrm{ad}(X) = 0$, so $X \in \mathrm{Lie}(K)$. By corollary 3.3.6, $Z(G)^\circ = K^\circ$. Since both $Z(G)$ and $K$ are defined as kernels (of the commutator map and of Ad respectively), they agree on connected components as well.

## The Moral

We close this chapter by summarizing what the Lie algebra can and cannot do, as a guide for the rest of the book.

In characteristic 0:

1. The functor Lie is exact and faithful on connected algebraic groups (corollary 3.3.7, corollary 3.3.6).
2. $\dim \mathrm{Lie}(G) = \dim G$ always, and every algebraic group scheme is smooth (theorem 3.3.2).
3. The kernel of Ad is exactly the center $Z(G)$ (corollary 3.3.8).
4. Questions about connected subgroups, exact sequences, and the center can be reduced to Lie algebra computations.

In characteristic $p > 0$:

1. The functor Lie is *not* faithful: non-isomorphic group schemes ($\mu_p$ vs. $\alpha_p$) can have isomorphic Lie algebras.
2. $\dim \mathrm{Lie}(G) \geq \dim G$, with possible strict inequality for non-smooth group schemes.
3. The kernel of Ad can be strictly larger than $Z(G)$.
4. The category of group schemes is richer than the category of Lie algebras. The additional structure is partially captured by the restricted ($p$-)Lie algebra, and fully captured only by the Hopf algebra $\mathcal{O}(G)$ or the functor of points.

This is the fundamental reason that this book takes the scheme-theoretic and functorial approach rather than the Lie-algebraic one: we want our theorems to hold in arbitrary characteristic, and the Lie algebra alone is insufficient for this.

Despite these limitations, the Lie algebra remains indispensable even in characteristic $p$: it still detects dimensions of smooth groups, computes tangent spaces, and (via the adjoint representation) decomposes into root spaces. The structure theory of Chapters 5–7 will use the Lie algebra as a *tool* throughout, but the foundational arguments (quotient constructions, Borel's fixed-point theorem, the classification) will be carried out at the level of the group scheme, falling back to the Lie algebra only when it is provably reliable.

## Exercise 3.3.1

1. Let $k$ be a field of characteristic $p > 0$ and let $n = p^r m$ with $\gcd(p, m) = 1$. Show that $\mu_n \cong \mu_{p^r} \times \mu_m$ as group schemes, that $\mu_m$ is étale (and in particular smooth), and that $\mu_{p^r}$ is the non-smooth part. Compute $\mathrm{Lie}(\mu_n)$ and compare with $\dim \mu_n$.
2. Verify directly that the Hopf algebra structure on $k[t]/(t^p)$ (with $\Delta(t) = t \otimes 1 + 1 \otimes t$, $\epsilon(t) = 0$, $S(t) = -t$) satisfies all the Hopf algebra axioms. In particular, check that $\Delta$ is a well-defined map $k[t]/(t^p) \to k[t]/(t^p) \otimes_k k[t]/(t^p)$ by verifying $\Delta(t^p) = (t \otimes 1 + 1 \otimes t)^p = 0$ in the target (use the fact that $\binom{p}{i} \equiv 0 \pmod{p}$ for $0 < i < p$).
3. Prove that $\mu_p$ and $\alpha_p$ are not isomorphic as group schemes. (*Hint*: show that the Hopf algebras $k[s]/(s^p)$ with $\Delta(s) = s \otimes 1 + 1 \otimes s + s \otimes s$ and $k[t]/(t^p)$ with $\Delta(t) = t \otimes 1 + 1 \otimes t$ are not isomorphic by considering the set of group-like elements.)
4. Let $G$ be a smooth connected algebraic group over an algebraically closed field $k$ of characteristic $p > 0$. The *Frobenius morphism* $F : G \to G^{(p)}$ is a purely inseparable isogeny of degree $p^{\dim G}$. Show that $\ker(F)$ is an infinitesimal group scheme of order $p^{\dim G}$ with $\mathrm{Lie}(\ker(F)) = \mathfrak{g}$ (the full Lie algebra). This shows that even for a smooth group, the Frobenius kernel is non-smooth and exhibits the "too-large Lie algebra" phenomenon.

5. Give an example of a short exact sequence of group schemes $1 \to H \to G \to Q \to 1$ over a field of characteristic $p > 0$ such that the induced sequence $0 \to \mathrm{Lie}(H) \to \mathrm{Lie}(G) \to \mathrm{Lie}(Q) \to 0$ is *not* right-exact. (*Hint*: consider $1 \to \mu_p \to \mathbb{G}_m \xrightarrow{t \mapsto t^p} \mathbb{G}_m \to 1$.)

# 4

# *Quotients and Exact Sequences*

In the first three chapters, we built algebraic groups, studied their actions and representations, and extracted their infinitesimal invariants. Throughout, we repeatedly encountered the *quotient problem*: given an algebraic group $G$ and a closed subgroup $H$, does the coset space $G/H$ exist as a scheme, and does it have good geometric properties? In chapter 2, we showed that a transitive action of $G$ on a variety $X$ with stabilizer $H$ at a point $x$ gives a set-theoretic bijection $G(k)/H(k) \cong X(k)$, and promised that this could be promoted to an isomorphism of varieties. In chapter 3, we used the exact sequence $1 \to \ker(\mathrm{Ad}) \to G \to \mathrm{GL}(\mathfrak{g})$ to study the center of $G$, relying on a notion of exactness that we had not yet made precise.

This chapter settles these debts. We first explain why the naive notion of "surjectivity on points" is inadequate for algebraic groups and introduce the correct categorical framework: faithfully flat morphisms and short exact sequences. We then construct the quotient $G/H$ for any closed subgroup $H$ using Chevalley's theorem, which realizes $H$ as the stabilizer of a line in a representation of $G$. This embeds $G/H$ into projective space, showing that quotients are always quasi-projective. Finally, we establish the fundamental properties of quotients — the isomorphism theorems, the identification of homogeneous spaces, and the group structure on $G/N$ when $N$ is normal — completing the categorical infrastructure needed for the structure theory of Part I.

## 4.1 Quotient Maps and Exact Sequences

### The Problem of Surjectivity

In the theory of abstract groups, a surjective homomorphism $\varphi : G \to Q$ with kernel $N = \ker(\varphi)$ gives a short exact sequence $1 \to N \to G \to Q \to 1$, and $Q \cong G/N$. Everything is straightforward because surjectivity is a simple condition: every element of $Q$ is hit by some element of $G$. For algebraic groups, the situation is more subtle. Consider the homomorphism $\varphi : \mathbb{G}_m \to \mathbb{G}_m$ defined by $\varphi(t) = t^n$

for some integer $n \geq 2$. On $k$-valued points (with $k = \overline{k}$), this map is surjective: every nonzero element of $k$ has an $n$-th root. The kernel consists of the $n$-th roots of unity $\mu_n(k) = \{t \in k^\times : t^n = 1\}$, a finite group of order $n$. So far, so good.

But now consider what happens on $R$-valued points for a general $k$-algebra $R$. The map $\varphi_R : R^\times \to R^\times$ sends $t \mapsto t^n$. Is this surjective? Certainly not: for $R = k[x]$, the element $x$ is a unit only if it is a nonzero constant, and $x$ itself has no $n$-th root in $k[x]^\times$. More strikingly, if $\operatorname{char} k = p$ and $n = p$, then the kernel $\mu_p$ is an infinitesimal group scheme (example 3.3.3): over $k = \overline{k}$, the only $p$-th root of unity is 1, so $\mu_p(k) = \{1\}$ is trivial! The kernel is invisible to $k$-points, yet it is a nontrivial group scheme. The map $\varphi$ is "surjective" in any reasonable geometric sense, but the kernel is detected only by the functor of points, not by classical points.

An even more dramatic example arises from the Frobenius endomorphism. Over a field $k$ of characteristic $p > 0$, the $p$-th power map $F : \mathbb{G}_a \to \mathbb{G}_a$ defined by $F(t) = t^p$ is a homomorphism of algebraic groups (since $(s + t)^p = s^p + t^p$ in characteristic $p$). On $k$-valued points, $F_k : k \to k$ is the Frobenius of the field, which is injective (and surjective if $k$ is perfect). The kernel is $\alpha_p = \operatorname{Spec} k[t]/(t^p)$ (example 3.3.4), an infinitesimal group scheme with $\alpha_p(k) = \{0\}$. Again, the kernel exists only at the scheme-theoretic level.

These examples show that "surjectivity on $k$-points" (or even on $\overline{k}$-points) is not the right notion. The correct replacement comes from commutative algebra:

**Definition 4.1.1: Faithfully Flat Homomorphism**

A homomorphism $\varphi : G \to Q$ of affine algebraic groups is called *faithfully flat* if the corresponding map of coordinate rings $\varphi^* : \mathcal{O}(Q) \to \mathcal{O}(G)$ is a faithfully flat ring homomorphism. That is, $\mathcal{O}(G)$ is a faithfully flat $\mathcal{O}(Q)$-module via $\varphi^*$.

Recall from commutative algebra (see [7]) that a ring homomorphism $A \to B$ is *flat* if the functor $- \otimes_A B$ is exact, and *faithfully flat* if it is flat and additionally $M \otimes_A B = 0$ implies $M = 0$ for every $A$-module $M$. Geometrically, faithful flatness means the morphism is surjective on the level of spectra and has "uniform" fibers (no fiber degenerates to the empty set, and the fiber dimensions are controlled).

Why is faithful flatness the right notion? Because it is equivalent to saying that the functor of points $\varphi_R : G(R) \to Q(R)$ is "as surjective as possible in families": while $\varphi_R$ need not be surjective for any individual $R$, every element of $Q(R)$ can be lifted to $G(R')$ after a faithfully flat base change $R \to R'$. In practice, this is exactly the condition needed to ensure that $G \to Q$ behaves like a "quotient map" in the scheme-theoretic sense.

**4.1.2 Remark: Faithfully Flat = fpqc Surjective** For morphisms of algebraic groups (which are of finite type), faithful flatness of $\varphi : G \to Q$ is equivalent to $\varphi$ being surjective as a morphism of schemes (i.e. surjective on the underlying topological spaces). It is also equivalent to the weaker-sounding condition that $\varphi_{\overline{k}} : G(\overline{k}) \to Q(\overline{k})$ is surjective on geometric points. The reader who finds "faithfully flat" intimidating can think "surjective on geometric points" in all cases that arise in this book, though the flatness condition carries essential algebraic content (it controls the fibers of $\varphi$, ensuring they are all translates of $\ker(\varphi)$).

Before defining exact sequences, we quickly box the scheme-theoretic notion of a normal subgroup.

**Definition 4.1.3: Normal Subgroup**

Let $G$ be an affine algebraic group and $N \subseteq G$ a closed subgroup. We say $N$ is *normal* in $G$ (written $N \trianglelefteq G$) if for every $k$-algebra $R$, the subgroup $N(R) \trianglelefteq G(R)$ is a normal subgroup of $G(R)$. Equivalently, the conjugation action of $G$ on itself restricts to an action on $N$:

$$\mathrm{Inn} : G \times N \longrightarrow N, \qquad (g, n) \longmapsto gng^{-1}$$

is a well-defined morphism.

Algebraically, if $I \subseteq \mathcal{O}(G)$ is the Hopf ideal defining $N$, normality means that the conjugation coaction $\rho : \mathcal{O}(G) \to \mathcal{O}(G) \otimes_k \mathcal{O}(G)$ (from example 2.1.5) satisfies $\rho(I) \subseteq I \otimes_k \mathcal{O}(G)$. In practice, normality is usually verified functorially: one checks $gng^{-1} \in N(R)$ for all $g \in G(R)$ and $n \in N(R)$.

**Example 4.1.4: Standard Normal Subgroups**

1. $\mathrm{SL}_n \trianglelefteq \mathrm{GL}_n$: For any $k$-algebra $R$, if $g \in \mathrm{GL}_n(R)$ and $A \in \mathrm{SL}_n(R)$, then $\det(gAg^{-1}) = \det(g)\det(A)\det(g)^{-1} = \det(A) = 1$, so $gAg^{-1} \in \mathrm{SL}_n(R)$.
2. The center $Z(G) \trianglelefteq G$: By definition, $Z(G)(R) = \{g \in G(R) : gx = xg \text{ for all } x \in G(R'), \text{ all } R\text{-algebras } R'\}$. If $z \in Z(G)(R)$ and $g \in G(R)$, then $gzg^{-1} = z \in Z(G)(R)$.
3. The kernel $\ker(\varphi)$ of any homomorphism $\varphi : G \to H$: If $n \in \ker(\varphi)(R)$ and $g \in G(R)$, then $\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} = \varphi(g) \cdot e \cdot \varphi(g)^{-1} = e$, so $gng^{-1} \in \ker(\varphi)(R)$.
4. $\mu_n \trianglelefteq \mathbb{G}_m$: This is a special case of (3), with $\mu_n = \ker(\mathbb{G}_m \xrightarrow{t \mapsto t^n} \mathbb{G}_m)$.

**Definition 4.1.5: Short Exact Sequence of Algebraic Groups**

A *short exact sequence* of affine algebraic groups is a diagram

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \longrightarrow 1$$

where:

1. $\iota : N \hookrightarrow G$ is a closed immersion identifying $N$ with a normal closed subgroup of $G$,
2. $\pi : G \to Q$ is a faithfully flat homomorphism, and
3. $N = \ker(\pi)$ (as group schemes, i.e. $\iota$ identifies $N$ with the scheme-theoretic fiber $\pi^{-1}(e_Q)$).

The three conditions say, respectively, that $N$ injects into $G$ (and is normal), that $\pi$ is "surjective" (in the faithfully flat sense), and that the image of $N$ is exactly the kernel of $\pi$. Functorially, for each $k$-algebra $R$, we get a sequence of groups

$$1 \longrightarrow N(R) \xrightarrow{\iota_R} G(R) \xrightarrow{\pi_R} Q(R)$$

which is exact on the left and in the middle (i.e. $\iota_R$ is injective and $\mathrm{im}(\iota_R) = \ker(\pi_R)$), but the map $\pi_R$ need not be surjective for any particular $R$.

**Example 4.1.6: The $n$-th Power Sequence**

The sequence
$$1 \longrightarrow \mu_n \xrightarrow{\iota} \mathbb{G}_m \xrightarrow{t \mapsto t^n} \mathbb{G}_m \longrightarrow 1$$
is a short exact sequence. The map $\iota$ is the inclusion of the $n$-th roots of unity. The map $\pi : t \mapsto t^n$ corresponds to the ring homomorphism $\pi^* : k[t, t^{-1}] \to k[t, t^{-1}]$ sending $t \mapsto t^n$. This makes $k[t, t^{-1}]$ into a free (hence faithfully flat) module of rank $n$ over $\pi^*(k[t, t^{-1}]) = k[t^n, t^{-n}]$, with basis $\{1, t, t^2, \dots, t^{n-1}\}$.

On $R$-valued points, the sequence reads $1 \to \mu_n(R) \to R^\times \xrightarrow{t \mapsto t^n} R^\times$. The last map is not surjective in general (not every unit in $R$ is an $n$-th power), but it is "surjective after faithfully flat base change": for any $u \in R^\times$, the ring $R' = R[x]/(x^n - u)$ is a faithfully flat $R$-algebra in which $u = x^n$ becomes an $n$-th power.

In characteristic $p$ with $n = p$, the sequence becomes $1 \to \mu_p \to \mathbb{G}_m \xrightarrow{t \mapsto t^p} \mathbb{G}_m \to 1$. The kernel $\mu_p$ is the infinitesimal group scheme from example 3.3.3, and as we computed in exercise 3.3.1 (5), the induced Lie algebra sequence is *not* right-exact: the differential $d\pi$ is the zero map $k \xrightarrow{0} k$.

**Example 4.1.7: The Determinant Sequence**

The sequence
$$1 \longrightarrow \mathrm{SL}_n \xrightarrow{\iota} \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m \longrightarrow 1$$
is a short exact sequence. The inclusion $\mathrm{SL}_n \hookrightarrow \mathrm{GL}_n$ is a closed immersion (cut out by $\det = 1$, which is a Hopf ideal by example 1.2.9). The determinant map is surjective on $k$-points (given $a \in k^\times$, the diagonal matrix $\mathrm{diag}(a, 1, \dots, 1) \in \mathrm{GL}_n(k)$ maps to $a$), and one checks that $\det^* : k[t, t^{-1}] \to \mathcal{O}(\mathrm{GL}_n)$ is faithfully flat.

On Lie algebras: $d(\det) : \mathfrak{gl}_n \to k$ sends $X \mapsto \mathrm{tr}(X)$. This is the trace map, and it is surjective (as we used in example 3.1.7). In characteristic 0, the Lie algebra sequence $0 \to \mathfrak{sl}_n \to \mathfrak{gl}_n \xrightarrow{\mathrm{tr}} k \to 0$ is exact (corollary 3.3.7), reflecting the group-level exactness perfectly.

**Example 4.1.8: The Frobenius Sequence**

Let $k$ be a perfect field of characteristic $p > 0$. The *Frobenius homomorphism* $F : \mathbb{G}_a \to \mathbb{G}_a$ defined by $F(t) = t^p$ gives a short exact sequence
$$1 \longrightarrow \alpha_p \xrightarrow{\iota} \mathbb{G}_a \xrightarrow{F} \mathbb{G}_a \longrightarrow 1$$
The map $F^* : k[t] \to k[t]$ sends $t \mapsto t^p$, making $k[t]$ a free module of rank $p$ over $F^*(k[t]) = k[t^p]$ with basis $\{1, t, \dots, t^{p-1}\}$. So $F$ is faithfully flat. The kernel is $\alpha_p = \operatorname{Spec} k[t]/(t^p)$, the infinitesimal additive group from example 3.3.4.

This sequence illustrates a phenomenon impossible in characteristic 0: a surjective endomorphism of a connected group with a nontrivial infinitesimal kernel.

**Example 4.1.9: The Adjoint Quotient**

Let $G$ be a connected affine algebraic group over a field of characteristic 0, and let $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ be the adjoint representation. By corollary 3.3.8, $\ker(\mathrm{Ad}) = Z(G)$. Therefore, we obtain a short exact sequence

$$1 \longrightarrow Z(G) \longrightarrow G \xrightarrow{\mathrm{Ad}} \mathrm{Ad}(G) \longrightarrow 1$$

where $\mathrm{Ad}(G)$ is the image, a closed subgroup of $\mathrm{GL}(\mathfrak{g})$. The quotient $G/Z(G)$ is isomorphic to $\mathrm{Ad}(G)$ as an algebraic group. For $G = \mathrm{SL}_n$, this gives $\mathrm{SL}_n/\mu_n \cong \mathrm{PGL}_n$, and for $G = \mathrm{SL}_2$ the isomorphism $\mathrm{SL}_2/\mu_2 \cong \mathrm{SO}_3$ from example 3.2.9.

## The Central Problem

Given a closed subgroup $H \subseteq G$ (not necessarily normal), we want to construct the quotient $G/H$ as a $k$-scheme with the following properties:

1. There is a morphism $\pi : G \to G/H$ that is faithfully flat.
2. The fibers of $\pi$ are exactly the left cosets $gH$.
3. $G/H$ has a universal property: any morphism $f : G \to X$ that is constant on left cosets factors uniquely through $\pi$.
4. If $H = N$ is normal, then $G/N$ inherits the structure of an algebraic group.

When $H$ is not normal, $G/H$ will not be a group, but it will be a variety with a transitive $G$-action — a *homogeneous space*.

The construction is not formal: unlike the case of abstract groups, where $G/H$ is simply the set of cosets, we must build a scheme structure on $G/H$ and verify that it has the right geometric properties. The key tool is Chevalley's theorem, which provides a representation-theoretic handle on $H$ and allows us to embed $G/H$ into projective space.

### Exercise 4.1.1

1. Show that the multiplication-by-$n$ map $[n] : \mathbb{G}_a \to \mathbb{G}_a$ (defined by $t \mapsto nt$) is an isomorphism when $\mathrm{char} k \nmid n$, and is the zero map when $\mathrm{char} k | n$. In the latter case, what is $\ker([n])$?
2. Let $\varphi : \mathrm{GL}_n \to \mathrm{GL}_n$ be defined by $\varphi(g) = g^T$ (transpose). Show that $\varphi$ is an isomorphism of algebraic groups (hence faithfully flat). What is the kernel? (*Hint*: the transpose is an anti-homomorphism for matrix multiplication; how can you fix this to get a homomorphism?)
3. Let $G = \mathrm{GL}_2$ and $H = \{\left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right) : t \in \mathbb{G}_a\} \cong \mathbb{G}_a$ (the upper-triangular unipotent subgroup). Show that $H$ is *not* normal in $G$ by finding $g \in \mathrm{GL}_2(k)$ and $h \in H(k)$ with $ghg^{-1} \notin H(k)$. Then show that $H$ is normal in the Borel subgroup $B_2$ of upper-triangular matrices.
4. Show that there is a short exact sequence $1 \to \mathrm{Sp}_{2n} \to \mathrm{SL}_{2n} \to X \to 1$ if and only if $\mathrm{Sp}_{2n}$ is normal in $\mathrm{SL}_{2n}$. Is $\mathrm{Sp}_{2n}$ normal in $\mathrm{SL}_{2n}$? (*Hint*: for $n \geq 2$, consider $\mathrm{Lie}(\mathrm{Sp}_{2n})$ as a subspace of $\mathfrak{sl}_{2n}$ and check whether it is an ideal.)
5. Let $1 \to N \to G \to Q \to 1$ be a short exact sequence of algebraic groups. Show that for any $k$-algebra $R$, the sequence $1 \to N(R) \to G(R) \to Q(R)$ is exact (i.e. $N(R) = \ker(G(R) \to Q(R))$).

Give an example where $G(R) \to Q(R)$ is not surjective. (*Hint*: for the first part, use the fiber product description of the kernel; for the second, take $R = k$ in the $n$-th power sequence with $n > 1$.)

## 4.2 The Construction of $G/H$

We now turn to the main theorem of this chapter: the existence and basic properties of the quotient $G/H$ for any closed subgroup $H$ of an affine algebraic group $G$. The strategy, due to Chevalley, is very geometric: we find a representation of $G$ containing a line whose stabilizer is exactly $H$, and then the orbit of that line in projective space gives a concrete model for $G/H$. The key results we use are the Fundamental Finiteness Theorem (theorem 2.2.7) from chapter 2, the Closed Orbit Lemma (theorem 2.4.15), and the Hopf ideal description of closed subgroups (definition 1.2.8) from chapter 1.

### Chevalley's Theorem on Stabilizers

The key input is a theorem that converts the algebraic data defining $H$ (a Hopf ideal $I \subseteq \mathcal{O}(G)$) into geometric data (a line in a representation). The idea is simple: the ideal $I$ is a subspace of $\mathcal{O}(G)$, and the Fundamental Finiteness Theorem guarantees it is contained in a finite-dimensional $G$-stable subspace. The Hopf ideal conditions on $I$ then translate into the statement that $H$ is exactly the stabilizer of a certain subspace.

**Theorem 4.2.1: Chevalley's Theorem**

Let $G$ be an affine algebraic group over $k$ and let $H \subseteq G$ be a closed subgroup. Then there exists a finite-dimensional representation $V$ of $G$ and a one-dimensional subspace $L \subseteq V$ (a "line") such that:

$$H = \mathrm{Stab}_G(L) := \{g \in G \mid g \cdot L = L\}$$

That is, $H$ is the stabilizer of the line $L$ in the $G$-representation $V$.

***Proof*:**
Let $A = \mathcal{O}(G)$ and let $I \subseteq A$ be the Hopf ideal defining $H$, so that $\mathcal{O}(H) = A/I$. Since $A$ is a finitely generated $k$-algebra, the ideal $I$ is finitely generated: say $I = (f_1, \ldots, f_r)$.

*Step 1: Find a finite-dimensional $G$-stable subspace containing $I$.* The coordinate ring $A$ is a right $A$-comodule via the comultiplication $\Delta : A \to A \otimes_k A$ (this corresponds to the right translation action of $G$ on itself). By the Fundamental Finiteness Theorem (theorem 2.2.7), each generator $f_i$ is contained in a finite-dimensional sub-comodule $W_i \subseteq A$. Let $W = W_1 + \cdots + W_r$, a finite-dimensional $G$-stable subspace of $A$ containing all the generators of $I$.

*Step 2: Extract a subspace whose stabilizer is $H$.* Define $V_0 = W \cap I$. This is a subspace of the finite-dimensional space $W$, hence finite-dimensional. We claim that $V_0$ is $H$-stable but *not* $G$-stable (unless $H = G$), and that $H$ is exactly the subgroup of $G$ that preserves $V_0$.

To see this, we use the coaction. The $G$-action on $A$ is given by the comultiplication $\Delta : A \to A \otimes_k A$. For $f \in I$, the Hopf ideal condition $\Delta(I) \subseteq I \otimes_k A + A \otimes_k I$ tells us that:

$$\Delta(f) \equiv \sum f_i' \otimes a_i \pmod{A \otimes_k I}$$

where $f_i' \in I$ and $a_i \in A$. The action of an element $g \in G(k) = \mathrm{Hom}_{\mathbf{Alg}_k}(A, k)$ on $f$ is $g \cdot f = (\mathrm{id} \otimes g)(\Delta(f))$. If $g \in H(k)$, then $g$ annihilates $I$ modulo $k$ (i.e. $g$ factors through $A \to A/I$), and the terms in $A \otimes_k I$ evaluate to elements of $I$ under $\mathrm{id} \otimes g$. More precisely, for $g \in H(R)$, the map $g : A \to R$ sends $I$ into $IR$, so $(\mathrm{id} \otimes g)(\Delta(f))$ lies in $I \cdot R$ when $f \in I$. Thus $H$ preserves $I$, hence preserves $V_0 = W \cap I$.

Conversely, suppose $g \in G(k)$ preserves $V_0$. Since $V_0$ contains the generators $f_1, \ldots, f_r$ of $I$ (they lie in $W \cap I$ by construction), and $g$ preserves $V_0 \subseteq I$, the action of $g$ sends each generator to an element of $I$. Because the $G$-action is an algebra automorphism, $g$ preserves the entire ideal $I$, meaning $g \cdot f \in I$ for all $f \in I$.

Geometrically, the action of $g$ on a function $f \in A$ corresponds to right translation: $(g \cdot f)(x) = f(xg)$. Because $g \cdot f \in I$, this new function must vanish on $H$. Evaluating at the identity $e \in H(k)$ gives:

$$(g \cdot f)(e) = f(eg) = f(g) = 0$$

Since $f(g) = 0$ for all $f \in I$, the element $g$ must belong to the vanishing locus of the ideal $I$, which is exactly $H(k)$. Therefore, $g \in H(k)$.

*Step 3: Reduce to a line.* We have found a finite-dimensional $G$-representation $W$ and a subspace $V_0 \subseteq W$ with $\mathrm{Stab}_G(V_0) = H$. To reduce to a *line*, we use the exterior power. Let $d = \dim V_0$ and consider the representation $V = \bigwedge^d W$. The subspace $V_0$ determines a line

$$L = \bigwedge^d V_0 \subseteq \bigwedge^d W = V$$

which is one-dimensional (choosing a basis $\{v_1, \ldots, v_d\}$ for $V_0$, the line $L$ is spanned by $v_1 \wedge \cdots \wedge v_d$).

An element $g \in G$ preserves $V_0$ if and only if $g$ preserves $L$: the forward direction is clear (if $g(V_0) = V_0$, then $g(v_1 \wedge \cdots \wedge v_d) = g(v_1) \wedge \cdots \wedge g(v_d) \in \bigwedge^d V_0 = L$). For the converse, if $g$ preserves $L = \bigwedge^d V_0$, then $g$ preserves the subspace $V_0 \subseteq W$ (because $\bigwedge^d$ detects $d$-dimensional subspaces: a $d$-dimensional subspace $U \subseteq W$ satisfies $\bigwedge^d U = L$ if and only if $U = V_0$, since $L$ is a line in the Plücker embedding). Therefore $\mathrm{Stab}_G(L) = \mathrm{Stab}_G(V_0) = H$.

The proof has a clean geometric interpretation: the Hopf ideal $I$ defining $H$ lives inside the "world of functions" $\mathcal{O}(G)$. The finiteness theorem converts $I$ into finite-dimensional linear algebra, and the exterior power trick converts a subspace stabilizer into a line stabilizer. The line $L$ then provides a point $[L] \in \mathbb{P}(V)$, and the orbit $G \cdot [L]$ will be our model for $G/H$.

## The Quotient as a Quasi-Projective Variety

With Chevalley's theorem in hand, we can now construct $G/H$.

**Theorem 4.2.2: Existence of the Quotient**

Let $G$ be an affine algebraic group over $k$ and let $H \subseteq G$ be a closed subgroup. Then there exists a quasi-projective $k$-variety $G/H$ and a faithfully flat morphism $\pi : G \to G/H$ such that:

1. The fibers of $\pi$ are exactly the left cosets $gH$. That is, $\pi(g_1) = \pi(g_2)$ if and only if $g_1^{-1}g_2 \in H$.
2. (Universal property) For any morphism $f : G \to X$ of $k$-schemes that is constant on left cosets (i.e. $f(gh) = f(g)$ for all $g \in G(R)$ and $h \in H(R)$, all $R$), there exists a unique morphism $\overline{f} : G/H \to X$ with $f = \overline{f} \circ \pi$.
3. $G/H$ is smooth and of dimension $\dim G - \dim H$.
4. $G$ acts on $G/H$ by left translation, and this action is transitive.

***Proof*:**
By Chevalley's theorem (theorem 4.2.1), there exists a representation $V$ of $G$ and a line $L \subseteq V$ with $H = \mathrm{Stab}_G(L)$. Consider the induced action of $G$ on the projective space $\mathbb{P}(V)$. The point $[L] \in \mathbb{P}(V)$ has stabilizer $H$, and the orbit map $\varphi : G \to \mathbb{P}(V)$ defined by $\varphi(g) = g \cdot [L]$ has the property that $\varphi(g_1) = \varphi(g_2)$ if and only if $g_1^{-1}g_2 \in H$.

By the Closed Orbit Lemma (theorem 2.4.15), the orbit $\Omega = G \cdot [L] \subseteq \mathbb{P}(V)$ is a smooth, locally closed subvariety of $\mathbb{P}(V)$. Since $\mathbb{P}(V)$ is projective, a locally closed subvariety of a projective variety is quasi-projective. We set $G/H := \Omega$ with the reduced scheme structure (or, if $G$ is smooth, the natural scheme structure coming from the orbit).

*(1): Fiber description.* The orbit map $\varphi : G \to \Omega$ satisfies $\varphi(g_1) = \varphi(g_2)$ iff $g_1 \cdot [L] = g_2 \cdot [L]$. Multiplying on the left by $g_1^{-1}$, this is equivalent to $[L] = g_1^{-1}g_2 \cdot [L]$, which happens iff $g_1^{-1}g_2 \in \mathrm{Stab}_G([L]) = H$. If $g_1^{-1}g_2 = h \in H$, then $g_2 = g_1 h$, meaning $g_1 H = g_2 H$. So the fibers of $\varphi$ are exactly the left cosets of $H$ in $G$.

*(2): Universal property.* Let $f : G \to X$ be constant on left cosets. Define $\overline{f} : \Omega \to X$ by $\overline{f}(g \cdot [L]) = f(g)$. This is well-defined because $g_1 \cdot [L] = g_2 \cdot [L]$ implies $g_1 H = g_2 H$, which implies $f(g_1) = f(g_2)$. That $\overline{f}$ is a morphism of $k$-schemes (not just a set map) follows from the fact that $\varphi : G \to \Omega$ is a faithfully flat surjection (see below) and faithfully flat descent: a morphism $G \to X$ that is constant on the fibers of a faithfully flat map $G \to \Omega$ descends uniquely to a morphism $\Omega \to X$ (see [5]).

*(3): Smoothness and dimension.* The orbit $\Omega$ is smooth by proposition 2.4.10 (the translation argument: all points of an orbit look alike, and at least one is smooth). The dimension is $\dim \Omega = \dim G - \dim H$ by the orbit-stabilizer formula (proposition 2.4.4).

*(4): Transitive $G$-action.* The group $G$ acts on $\Omega \subseteq \mathbb{P}(V)$ by restriction of its action on $\mathbb{P}(V)$, and $\Omega$ is a single orbit, so the action is transitive by definition.

*Faithful flatness of $\varphi$.* The morphism $\varphi : G \to \Omega$ is surjective (since $\Omega = \varphi(G)$) and has all fibers isomorphic to $H$ (by the coset description). Since $G$ and $\Omega$ are both smooth and the fibers are equidimensional of constant dimension $\dim H$, the morphism $\varphi$ is flat by the miracle flatness theorem (a morphism from a Cohen–Macaulay scheme to a smooth scheme with equidimensional fibers is flat; see [5]). A flat surjective morphism of finite type is faithfully flat.

The construction shows that $G/H$ is always *quasi-projective*: it is a locally closed subvariety of a projective space. It is not always projective (it need not be closed in $\mathbb{P}(V)$), and it is not always affine (for instance, $\mathrm{GL}_n/B_n$ is projective, as we will see in chapter 6). The question of when $G/H$ is projective will turn out to be intimately related to the structure of $H$: the quotient $G/P$ is projective if and only if $P$ is a *parabolic subgroup*, a topic we take up in chapter 6.

**4.2.3 Remark: Uniqueness of the Quotient** The universal property (2) determines $G/H$ up to unique isomorphism, as is standard for objects defined by universal properties. In particular, the construction does not depend on the choice of representation $V$ and line $L$ in Chevalley's theorem. Different choices give isomorphic quotients, and the isomorphism is canonical (it is the unique map making the obvious triangle commute).

**Example 4.2.4: $\mathrm{GL}_n/B_n$ and the Complete Flag Variety**

Let $G = \mathrm{GL}_n$ and $H = B_n$ (the upper-triangular Borel subgroup). We apply the construction. The Borel $B_n$ is the stabilizer of the standard flag $0 \subset \langle e_1\rangle \subset \langle e_1, e_2\rangle \subset \cdots \subset k^n$. To realize $B_n$ as the stabilizer of a *line*, we use the exterior power trick from the proof of Chevalley's theorem, combined with a tensor product.

Consider the representation $V = \bigotimes_{d=1}^{n-1} \bigwedge^d(k^n)$ and the line $L$ spanned by $e_1 \otimes (e_1 \wedge e_2) \otimes \cdots \otimes (e_1 \wedge \cdots \wedge e_{n-1})$. The stabilizer of $L$ in $\mathrm{GL}_n$ consists of matrices that preserve each subspace $\langle e_1, \ldots, e_d\rangle$ for $1 \le d \le n-1$ (scaling the tensor factors by the determinant of each block), which is exactly $B_n$.

The orbit $\mathrm{GL}_n \cdot [L] \subseteq \mathbb{P}(V)$ is the *complete flag variety* $\mathrm{Fl}(1, 2, \ldots, n-1; n)$, parametrizing complete flags $0 \subset V_1 \subset V_2 \subset \cdots \subset V_{n-1} \subset k^n$ with $\dim V_d = d$. This is a smooth projective variety of dimension $\dim \mathrm{GL}_n - \dim B_n = n^2 - \binom{n+1}{2} = \binom{n}{2}$. The Plücker embedding followed by the Segre embedding realizes this flag variety as a closed subvariety of the product of Grassmannians $\prod_{d=1}^{n-1} \mathrm{Gr}(d, n)$, and hence inside the single projective space $\mathbb{P}(V)$.

**Example 4.2.5: $\mathrm{SL}_2/B$ and $\mathbb{P}^1$**

The simplest nontrivial case: let $G = \mathrm{SL}_2$ and $B = \left\{\begin{pmatrix} a & b \\ 0 & a^{-1}\end{pmatrix}\right\}$ (the upper-triangular Borel in $\mathrm{SL}_2$). The Borel $B$ is the stabilizer of the line $\langle e_1\rangle$ in the standard representation $V = k^2$. The orbit of $[e_1] \in \mathbb{P}^1$ under $\mathrm{SL}_2$ is all of $\mathbb{P}^1$ (since $\mathrm{SL}_2$ acts transitively on lines in $k^2$). Therefore:

$$\mathrm{SL}_2/B \cong \mathbb{P}^1$$

This is a smooth projective variety of dimension $\dim \mathrm{SL}_2 - \dim B = 3 - 2 = 1$. The quotient map $\pi : \mathrm{SL}_2 \to \mathbb{P}^1$ sends $g \mapsto g \cdot [e_1]$, and the fiber over $[e_1]$ is $B$ itself.

This example is the prototype for all of chapter 6: the quotient $G/B$ for a Borel subgroup $B$ is always a smooth projective variety (the *flag variety*), and its geometry encodes the structure of $G$.

**Example 4.2.6: Quasi-Affine Quotients:** $\mathrm{GL}_n/\operatorname{Stab}(e_1)$

Not all quotients are projective. Let $G = \mathrm{GL}_n$ and let $H \subseteq \mathrm{GL}_n$ be the stabilizer of the standard basis vector $e_1 \in k^n$. Matrices in $H$ have $e_1$ as their first column, so $H$ is the semi-direct product $k^{n-1} \rtimes \mathrm{GL}_{n-1}$ of the form $\{\left(\begin{smallmatrix} 1 & v \\ 0 & A \end{smallmatrix}\right)\}$.

The orbit of $e_1$ under the standard action of $\mathrm{GL}_n$ on $k^n$ is all non-zero vectors, which is $k^n \setminus \{0\} = \mathbb{A}^n \setminus \{0\}$ (as computed in example 2.4.5). Therefore:

$$\mathrm{GL}_n/H \cong \mathbb{A}^n \setminus \{0\}$$

This is a smooth, quasi-affine variety of dimension $\dim \mathrm{GL}_n - \dim H = n^2 - ((n-1)^2 + n - 1) = n$. The orbit is not closed in $\mathbb{A}^n$ (its closure includes the origin), so for $n \geq 2$, $\mathrm{GL}_n/H$ is strictly quasi-affine and not affine.

*Remark:* If we had instead quotiented by the reductive subgroup $\mathrm{GL}_{n-1}$ directly, Matsushima's Theorem guarantees the resulting quotient $\mathrm{GL}_n/\mathrm{GL}_{n-1}$ would be an affine variety (specifically, the affine variety of vector-covector pairs $(v, \lambda)$ such that $\lambda(v) = 1$). The presence of the unipotent radical $k^{n-1}$ in $H$ is what breaks the affineness.

## The Quotient Map is a Principal $H$-Bundle

The quotient map $\pi : G \to G/H$ has an important structural property that goes beyond faithful flatness: it is a *principal $H$-bundle* (or *$H$-torsor*) over $G/H$. This means that $H$ acts freely on $G$ (by right multiplication), the orbits of this action are exactly the fibers of $\pi$, and the map $\pi$ is locally trivial in a suitable topology.

**Proposition 4.2.7: $G \to G/H$ is an $H$-Torsor**

Let $G$ be an affine algebraic group and $H \subseteq G$ a closed subgroup. The morphism

$$\psi : G \times H \longrightarrow G \times_{G/H} G, \qquad (g, h) \longmapsto (g, gh)$$

is an isomorphism. In other words, the action map $G \times H \to G$ given by right multiplication exhibits $G$ as a (right) $H$-torsor over $G/H$.

***Proof*:**

This follows immediately from constructing an explicit inverse morphism of schemes. Consider the morphism:

$$\varphi : G \times_{G/H} G \longrightarrow G \times H, \qquad (g_1, g_2) \longmapsto (g_1, g_1^{-1} g_2)$$

This map is well-defined because the fiber product $G \times_{G/H} G$ is exactly the subscheme of $G \times G$ defined by the relation $\pi(g_1) = \pi(g_2)$, which is equivalent to $g_1^{-1} g_2 \in H$.

The compositions are trivial to compute: $\varphi(\psi(g, h)) = \varphi(g, gh) = (g, g^{-1}gh) = (g, h)$, and $\psi(\varphi(g_1, g_2)) = \psi(g_1, g_1^{-1} g_2) = (g_1, g_1(g_1^{-1} g_2)) = (g_1, g_2)$. Since $\psi$ and $\varphi$ are mutually inverse morphisms of schemes, $\psi$ is an isomorphism.

The torsor property has a concrete consequence: *locally* on $G/H$ (in the flat, or étale, or fppf topology), the quotient map $G \to G/H$ looks like the projection $U \times H \to U$ for an open subset

$U \subseteq G/H$. In particular, $\pi$ is smooth of relative dimension $\dim H$.

**Corollary 4.2.8: Properties of the Quotient Map**

The quotient map $\pi : G \to G/H$ is faithfully flat and smooth. In particular:

1. The map $\pi$ is open (images of open sets are open).
2. For any field extension $K/k$, the base change $\pi_K : G_K \to (G/H)_K$ is the quotient map $G_K \to G_K/H_K$.
3. If $f : G/H \to X$ is a morphism, then $f$ is flat (resp. smooth, étale, an isomorphism) if and only if $f \circ \pi : G \to X$ has the same property.

***Proof*:**
Statement (1) follows from the general fact that flat morphisms of finite presentation are open (see [5]). Statement (2) follows from the universal property: the base-changed map $G_K \to (G/H)_K$ satisfies the same universal property as $G_K \to G_K/H_K$, so they are canonically isomorphic. Statement (3) is a consequence of faithfully flat descent: a property that can be checked after faithfully flat base change holds if and only if it holds after pulling back along $\pi$.

**Example 4.2.9: Local Triviality for $\mathrm{SL}_2 \to \mathbb{P}^1$**
Consider $\pi : \mathrm{SL}_2 \to \mathbb{P}^1$ from example 4.2.5. Over the standard open $U_0 = \{[x : y] : x \neq 0\} \cong \mathbb{A}^1$ (with coordinate $t = y/x$), the fiber $\pi^{-1}(U_0)$ consists of matrices $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2$ with $a \neq 0$. We can construct an explicit trivialization $\pi^{-1}(U_0) \xrightarrow{\sim} U_0 \times B$ using a section.

Define the section $\sigma : U_0 \to \mathrm{SL}_2$ by $t \mapsto \left(\begin{smallmatrix} 1 & 0 \\ t & 1 \end{smallmatrix}\right)$. It satisfies $\pi(\sigma(t)) = \sigma(t) \cdot [e_1] = [1 : t] = t$, so $\pi \circ \sigma = \mathrm{id}_{U_0}$.

Because $\pi(g) = \pi(\sigma(\pi(g)))$, the elements $g$ and $\sigma(\pi(g))$ lie in the same left $B$-coset. Thus, every $g \in \pi^{-1}(U_0)$ factors uniquely as $g = \sigma(\pi(g)) \cdot b$ for some $b \in B$. Solving for $b$ gives $b = \sigma(\pi(g))^{-1}g$. The trivialization isomorphism is therefore given by the map:

$$g \longmapsto \big(\pi(g),\, \sigma(\pi(g))^{-1}g\big)$$

Similarly, the map trivializes over the other standard affine open $U_1 = \{y \neq 0\}$. Thus, the quotient $\mathrm{SL}_2 \to \mathbb{P}^1$ is a Zariski-locally trivial $B$-bundle.

## Exercise 4.2.1

1. Let $G = \mathrm{GL}_3$ and $H = \left\{\left(\begin{smallmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{smallmatrix}\right) \in \mathrm{GL}_3\right\}$ (the upper-triangular Borel). Without using the general proof, directly find a representation $V$ of $\mathrm{GL}_3$ and a line $L \subseteq V$ with $H = \mathrm{Stab}_G(L)$. (*Hint*: think about which subspaces of $k^3$ are stabilized by upper-triangular matrices, and use the exterior power.)
2. Let $G = \mathrm{SO}_5$ (of dimension 10) and $H = \mathrm{SO}_4$ (of dimension 6), embedded as the stabilizer of $e_1$ under the standard action on $k^5$ (with char$k \neq 2$). What is $\dim(G/H)$? Describe $G/H$ geometrically. (*Hint*: the orbit of $e_1$ under $\mathrm{SO}_5$ is the quadric $\{v \in k^5 : q(v) = 1\}$ where $q$ is the quadratic form defining $\mathrm{SO}_5$.)

3. Show that $\mathrm{SL}_2/B$ is not affine. (*Hint*: $\mathrm{SL}_2/B \cong \mathbb{P}^1$, and a projective variety of positive dimension is never affine.)

4. Let $G$ be an algebraic group over a field $k$ (not necessarily algebraically closed) and $H \subseteq G$ a closed subgroup. Show that $(G/H) \otimes_k \overline{k} \cong G_{\overline{k}}/H_{\overline{k}}$. (*Hint*: use statement (2) of corollary 4.2.8.)

5. Let $G = \mathbb{G}_m \times \mathbb{G}_m$ and $H = \{(t,t) : t \in \mathbb{G}_m\} \cong \mathbb{G}_m$ (the diagonal subgroup). Construct $G/H$ explicitly. Show that $G/H \cong \mathbb{G}_m$ via the map $(s,t) \mapsto st^{-1}$. Is $H$ normal in $G$? Does $G/H$ inherit a group structure?

## 4.3 Properties of Quotients and the Isomorphism Theorems

With the quotient $G/H$ constructed as a quasi-projective variety, we now develop its properties. The main results are: when $N$ is normal, $G/N$ is an algebraic group; the three isomorphism theorems carry over from abstract group theory; and transitive actions are identified with quotients, closing the promises made in chapter 2. These are the tools that the rest of Part I will use freely.

### The Group Structure on $G/N$

When $H = N$ is a normal closed subgroup of $G$, the coset space $G/N$ should inherit a group structure from $G$: the product of cosets $g_1N \cdot g_2N = (g_1g_2)N$ is well-defined precisely because $N$ is normal. The scheme-theoretic version of this requires a small argument.

**Theorem 4.3.1: Quotient by a Normal Subgroup**

Let $G$ be an affine algebraic group and $N \trianglelefteq G$ a closed normal subgroup. Then:

1. The quotient $G/N$ carries a unique structure of algebraic group such that the quotient map $\pi : G \to G/N$ is a homomorphism.

2. The sequence $1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} G/N \to 1$ is a short exact sequence of algebraic groups.

3. $G/N$ is affine.

***Proof*:**
*(1):* We must define morphisms $\mu_{G/N} : G/N \times G/N \to G/N$ and $\iota_{G/N} : G/N \to G/N$ (multiplication and inverse). Consider the composition

$$G \times G \xrightarrow{\mu_G} G \xrightarrow{\pi} G/N$$

This map is constant on left $N \times N$-cosets: for $n_1, n_2 \in N(R)$,

$$\pi(\mu_G(g_1n_1, g_2n_2)) = \pi(g_1n_1g_2n_2) = \pi(g_1g_2 \cdot g_2^{-1}n_1g_2 \cdot n_2) = \pi(g_1g_2)$$

where we used the normality of $N$ (so $g_2^{-1}n_1g_2 \in N(R)$) and the fact that $\pi$ kills $N$. Therefore $\pi \circ \mu_G$ factors through $G/N \times G/N$ by the universal property of the quotient (theorem 4.2.2(2)), applied to the product $G \times G \to G/N \times G/N$. This gives the multiplication $\mu_{G/N}$. The inverse is constructed similarly from $\pi \circ \iota_G$.

The group axioms (associativity, identity, inverse) for $G/N$ follow from those for $G$ by the faithfulness of the universal property: two morphisms $G/N \to G/N$ are equal if and only if they become equal after precomposing with $\pi$ (since $\pi$ is an epimorphism in the category of $k$-schemes, being faithfully flat). For instance, associativity of $\mu_{G/N}$ follows from:

$$\mu_{G/N} \circ (\mu_{G/N} \times \mathrm{id}) \circ (\pi \times \pi \times \pi) = \pi \circ \mu_G \circ (\mu_G \times \mathrm{id}) = \pi \circ \mu_G \circ (\mathrm{id} \times \mu_G) = \mu_{G/N} \circ (\mathrm{id} \times \mu_{G/N}) \circ (\pi \times \pi \times \pi)$$

and the fact that $\pi \times \pi \times \pi$ is an epimorphism.

*(2):* The map $\iota : N \hookrightarrow G$ is a closed immersion identifying $N = \ker(\pi)$ (by construction of $G/N$ as the quotient by the equivalence relation $g_1 \sim g_2$ iff $g_1^{-1} g_2 \in N$). The map $\pi$ is faithfully flat by corollary 4.2.8. So $1 \to N \to G \to G/N \to 1$ is a short exact sequence in the sense of definition 4.1.5.

*(3):* Affineness of $G/N$ is a deeper result. Because $N$ is a normal subgroup, the right action of $N$ on $G$ interacts well with the group structure, ensuring that the ring of invariants $\mathcal{O}(G)^N$ is not just a subalgebra, but a *sub-Hopf algebra* of $\mathcal{O}(G)$. A fundamental theorem of algebraic groups (due to Takeuchi and Waterhouse) states that any sub-Hopf algebra of a finitely generated commutative Hopf algebra over a field is automatically finitely generated. Thus, $G/N \cong \operatorname{Spec} \mathcal{O}(G)^N$ is an affine variety.

Alternatively, one can prove this by constructing a finite-dimensional $G$-representation $V$ whose kernel is exactly $N$, which embeds $G/N$ as a closed subgroup of $\mathrm{GL}(V)$ (we defer the full details to the references, see Waterhouse [18] or Milne [14]).

Statement (3) is important: it says that quotients of affine groups by *normal* subgroups remain affine. This is *not* true for non-normal subgroups: the quotient $\mathrm{GL}_n/B_n$ is projective, not affine (example 4.2.4). The distinction between "$G/N$ is affine" (normal case) and "$G/H$ is quasi-projective" (general case) is a fundamental structural fact.

## Identification of Homogeneous Spaces

We can now fulfill the promise made in chapter 2 (§2.4): if $G$ acts transitively on a variety $X$ with stabilizer $H$ at a point $x$, then $X$ is isomorphic to $G/H$ as a $G$-variety.

**Theorem 4.3.2: Homogeneous Spaces are Quotients**

Let $G$ be an affine algebraic group over a field of characteristic 0. If $G$ acts on a $k$-variety $X$ via a transitive action, and $x \in X(k)$ has stabilizer $H = G_x$, then the orbit map $\varphi_x : G \to X$ induces an isomorphism of $G$-varieties

$$\overline{\varphi}_x : G/H \xrightarrow{\sim} X$$

***Proof*:**
The orbit map $\varphi_x : G \to X$, $g \mapsto g \cdot x$, is constant on left $H$-cosets: $\varphi_x(gh) = gh \cdot x = g \cdot x = \varphi_x(g)$ for $h \in H(R)$. By the universal property of $G/H$ (theorem 4.2.2(2)), $\varphi_x$ factors through a morphism $\overline{\varphi}_x : G/H \to X$ with $\varphi_x = \overline{\varphi}_x \circ \pi$.

To see that $\overline{\varphi}_x$ is an isomorphism of varieties, we use the properties of flat morphisms. Because we are in characteristic 0, generic smoothness guarantees there is a dense open subset of $X$ over which $\varphi_x$ is smooth. Because $G$ acts transitively on $X$, we can translate this smooth open neighborhood to cover all of $X$. Thus, the orbit map $\varphi_x : G \to X$ is a smooth, surjective morphism, which implies it is faithfully flat.

Because $\varphi_x : G \to X$ is a faithfully flat morphism whose fibers are exactly the left $H$-cosets, it satisfies the exact same universal property as the principal $H$-bundle $G \to G/H$. By the uniqueness of the quotient (or by applying faithfully flat descent corollary 4.2.8 to the map $\overline{\varphi}_x$), the induced map $\overline{\varphi}_x$ is an isomorphism.

This theorem closes the loop opened in chapter 2. We can now state the identifications that were deferred:

**Corollary 4.3.3: Projective Spaces and Grassmannians as Quotients**

Let $k$ be an algebraically closed field of characteristic 0.

1. The orbit map $\mathrm{GL}_n \to \mathbb{P}^{n-1}$, $g \mapsto g \cdot [e_1]$, induces an isomorphism of $\mathrm{GL}_n$-varieties:
$$\mathrm{GL}_n/P_1 \xrightarrow{\sim} \mathbb{P}^{n-1}$$
where $P_1$ is the parabolic subgroup stabilizing $\langle e_1 \rangle$ (as in example 2.4.13).
2. More generally, for $1 \leq d \leq n-1$, the orbit map induces:
$$\mathrm{GL}_n/P_d \xrightarrow{\sim} \mathrm{Gr}(d, n)$$
where $P_d$ is the parabolic stabilizing $\langle e_1, \dots, e_d \rangle$ (as in example 2.4.14).
3. The orbit map $\mathrm{SL}_2 \to \mathbb{P}^1$, $g \mapsto g \cdot [e_1]$, induces:
$$\mathrm{SL}_2/B \xrightarrow{\sim} \mathbb{P}^1$$
where $B$ is the upper-triangular Borel (as in example 4.2.5).

***Proof*:**
Each is a direct application of theorem 4.3.2: in each case, the group acts transitively (verified in chapter 2) and the stabilizer was identified there. The theorem promotes the set-theoretic bijection to an isomorphism of varieties.

## The Isomorphism Theorems

The classical isomorphism theorems of group theory carry over to algebraic groups, with "quotient by a normal subgroup" replaced by the construction of §4.2 and "surjectivity" replaced by "faithful flatness."

**Theorem 4.3.4: First Isomorphism Theorem**

Let $\varphi : G \to H$ be a homomorphism of affine algebraic groups over a field of characteristic 0. Then:

1. The image $\varphi(G)$ is a closed subgroup of $H$.
2. The kernel $N = \ker(\varphi)$ is a closed normal subgroup of $G$.
3. There is a canonical isomorphism of algebraic groups:

$$G/\ker(\varphi) \xrightarrow{\sim} \varphi(G)$$

***Proof*:**
*(2)* is immediate: $N(R) = \ker(\varphi_R : G(R) \to H(R))$ is a normal subgroup of $G(R)$ for all $R$ (as noted in example 4.1.4(3)), and $N \hookrightarrow G$ is a closed immersion (the fiber of $\varphi$ over $e_H$ is a closed subscheme).

*(1):* The image $\varphi(G)$ is a constructible subset of $H$ (by Chevalley's theorem on constructible images). It is also a subgroup of $H$ at the level of $R$-valued points for all $R$. We claim it is closed. Since $\varphi(G)$ is a constructible subgroup, it contains a dense open subset of its closure $\overline{\varphi(G)}$. But $\overline{\varphi(G)}$ is also a subgroup. Then $\varphi(G)$ is an open subgroup of $\overline{\varphi(G)}$, hence closed (the complement is a union of cosets of $\varphi(G)$, each open, hence the complement is open, so $\varphi(G)$ is closed in $\overline{\varphi(G)}$). Since $\overline{\varphi(G)}$ is closed in $H$, $\varphi(G)$ is closed in $H$.

*(3):* The homomorphism $\varphi$ is constant on $N$-cosets: $\varphi(gn) = \varphi(g)\varphi(n) = \varphi(g)$ for $n \in N(R)$. By the universal property of $G/N$ (theorem 4.2.2(2)), $\varphi$ factors through $\overline{\varphi} : G/N \to H$ with $\varphi = \overline{\varphi} \circ \pi$. The image of $\overline{\varphi}$ is the closed subgroup $\varphi(G)$.

Consider the corestriction $\varphi : G \to \varphi(G)$. This is a surjective homomorphism of algebraic groups. In characteristic 0, any surjective homomorphism of algebraic groups is smooth, and hence faithfully flat. Because $\varphi : G \to \varphi(G)$ is faithfully flat and its fibers are exactly the cosets of $N$, it follows immediately from the uniqueness of the quotient (or faithfully flat descent) that $G/N \cong \varphi(G)$.

**Theorem 4.3.5: Second Isomorphism Theorem**

Let $G$ be an affine algebraic group, $H \subseteq G$ a closed subgroup, and $N \trianglelefteq G$ a closed normal subgroup (in characteristic 0). Then:

1. The product $HN$ is a closed subgroup of $G$.
2. $H \cap N$ is a closed normal subgroup of $H$.
3. There is a canonical isomorphism of algebraic groups:

$$H/(H \cap N) \xrightarrow{\sim} HN/N$$

***Proof*:**
*(1):* The product $HN$ is the image of the multiplication map $H \times N \to G$, $(h,n) \mapsto hn$. Since $N$ is normal, $HN(R) = H(R) \cdot N(R) = \{hn : h \in H(R), n \in N(R)\}$ is a subgroup of $G(R)$ for all $R$. It is closed by the same constructibility argument as in the first isomorphism theorem: the image of the morphism $H \times N \to G$ is a constructible subgroup, hence closed.

*(2):* $H \cap N$ is the fiber product $H \times_G N$, which is a closed subscheme of $H$. Normality in $H$ follows from the fact that $N$ is normal in $G$: for $h \in H(R)$ and $n \in (H \cap N)(R) \subseteq N(R)$, we have $hnh^{-1} \in N(R)$ (since $N \trianglelefteq G$) and $hnh^{-1} \in H(R)$ (since $H$ is a subgroup). So $hnh^{-1} \in (H \cap N)(R)$.

*(3):* Consider the composition $H \hookrightarrow G \xrightarrow{\pi} G/N$ where $\pi$ is the quotient map. The kernel of $\pi|_H$ is $H \cap N$. The image is $\pi(H) = HN/N$ (since $\pi(h) = hN$, and $\pi(H)$ consists of all cosets $hN$ with $h \in H$, which is the same as cosets $hnN = hN$ for $hn \in HN$). By the first isomorphism theorem applied to $\pi|_H : H \to G/N$, we get $H/(H \cap N) \cong \pi(H) = HN/N$.

**Theorem 4.3.6: Third Isomorphism Theorem**

Let $G$ be an affine algebraic group and $N \subseteq M \subseteq G$ closed normal subgroups (with $N \trianglelefteq G$ and $M \trianglelefteq G$). Then $M/N$ is a closed normal subgroup of $G/N$, and there is a canonical isomorphism:
$$(G/N)/(M/N) \xrightarrow{\sim} G/M$$

***Proof*:**
The quotient map $\pi_M : G \to G/M$ kills $N$ (since $N \subseteq M = \ker(\pi_M)$), so it factors through $G/N$: there exists a homomorphism $\overline{\pi} : G/N \to G/M$ with $\pi_M = \overline{\pi} \circ \pi_N$. The kernel of $\overline{\pi}$ consists of cosets $gN \in G/N$ with $g \in M$, which is exactly $M/N$. Since $\overline{\pi}$ is faithfully flat (as $\pi_M = \overline{\pi} \circ \pi_N$ is faithfully flat and $\pi_N$ is faithfully flat, $\overline{\pi}$ is faithfully flat by cancellation), the first isomorphism theorem gives $(G/N)/\ker(\overline{\pi}) = (G/N)/(M/N) \cong G/M$.

## The Lie Algebra of a Quotient

Having the quotient $G/N$ as an algebraic group, we can compute its Lie algebra. The answer is exactly what one expects:

**Corollary 4.3.7: Lie Algebra of a Quotient**

Let $N \trianglelefteq G$ be a closed normal subgroup of an affine algebraic group $G$, with both $G$ and $N$ smooth. Then:
$$\mathrm{Lie}(G/N) \cong \mathrm{Lie}(G)/\mathrm{Lie}(N) = \mathfrak{g}/\mathfrak{n}$$

***Proof*:**
The short exact sequence $1 \to N \to G \to G/N \to 1$ gives, by the left-exactness of Lie (proposition 3.1.12 and exercise 3.1.1 (4)), an exact sequence $0 \to \mathfrak{n} \to \mathfrak{g} \to \mathrm{Lie}(G/N)$. By dimension counting: all three groups are smooth, so $\dim \mathrm{Lie}(G/N) = \dim(G/N) = \dim G - \dim N = \dim \mathfrak{g} - \dim \mathfrak{n}$. Since the image of $\mathfrak{g} \to \mathrm{Lie}(G/N)$ has dimension $\dim \mathfrak{g} - \dim \mathfrak{n} = \dim \mathrm{Lie}(G/N)$, the map $\mathfrak{g} \to \mathrm{Lie}(G/N)$ is surjective. Therefore $\mathrm{Lie}(G/N) \cong \mathfrak{g}/\mathfrak{n}$.

In characteristic 0, the smoothness hypothesis is automatic by Cartier's theorem (theorem 3.3.2), so the formula $\mathrm{Lie}(G/N) \cong \mathfrak{g}/\mathfrak{n}$ holds unconditionally.

**Example 4.3.8: Lie Algebra of $\mathrm{PGL}_n$**
The projective general linear group is $\mathrm{PGL}_n = \mathrm{GL}_n/Z(\mathrm{GL}_n) = \mathrm{GL}_n/\mathbb{G}_m$, where $\mathbb{G}_m$ is embedded as scalar matrices. By the corollary:

$$\mathrm{Lie}(\mathrm{PGL}_n) \cong \mathfrak{gl}_n/k \cdot I_n = \mathfrak{gl}_n/\mathfrak{z}(\mathfrak{gl}_n) \cong \mathfrak{pgl}_n$$

When $\mathrm{char} k \nmid n$, the trace map splits the inclusion $k \cdot I_n \hookrightarrow \mathfrak{gl}_n$, so $\mathfrak{pgl}_n \cong \mathfrak{sl}_n$ as Lie algebras.

But when $\mathrm{char} k \big| n$, the scalar matrix $I_n$ has trace $n = 0$, so $k \cdot I_n \subseteq \mathfrak{sl}_n$. In this case, $\mathfrak{pgl}_n$ and $\mathfrak{sl}_n$ are *not* isomorphic. Both Lie algebras have dimension $n^2 - 1$, but the natural map $\mathfrak{sl}_n \to \mathfrak{pgl}_n$ is neither injective nor surjective: its kernel is the line $k \cdot I_n$, and its image is the codimension-1 subspace $\mathfrak{sl}_n/k \cdot I_n$ (which has dimension $n^2 - 2$).

Furthermore, they are not even abstractly isomorphic because their centers differ: the center of $\mathfrak{sl}_n$ is the non-trivial ideal $k \cdot I_n$, whereas a straightforward matrix calculation shows the center of $\mathfrak{pgl}_n$ is trivial. This is one of the characteristic-$p$ subtleties encountered in example 3.2.3.

## Fibers and the Dimension Principle

We close the chapter by recording the key geometric property of quotient maps that will be used throughout the structure theory: all fibers of the quotient map are "the same."

**Proposition 4.3.9: Fiber Structure of Quotient Maps**

Let $G$ be an affine algebraic group and $H \subseteq G$ a closed subgroup. Then:

1. Every fiber of $\pi : G \to G/H$ is a translate of $H$: the fiber over $\pi(g)$ is $gH$.
2. All fibers are isomorphic to $H$ as varieties.
3. For any closed subgroup $K$ with $H \subseteq K \subseteq G$, there is a natural $G$-equivariant surjection $G/H \to G/K$ with fibers isomorphic to $K/H$.

***Proof*:**
*(1) and (2)* follow from the torsor property (proposition 4.2.7): the right $H$-action on $G$ is free with orbits equal to the left cosets $gH$, and each coset is isomorphic to $H$ via left translation by $g^{-1}$.

*(3):* The projection $G \to G/K$ is constant on left $H$-cosets (since $H \subseteq K$), so it factors through $G/H$: there is a unique $G$-equivariant morphism $\psi : G/H \to G/K$ with $\pi_K = \psi \circ \pi_H$. The fiber of $\psi$ over the point $\pi_K(g) \in G/K$ consists of all cosets $g'H$ with $g'K = gK$, i.e. $g' \in gK$. The set of such cosets is $\{gkH : k \in K\} = gK/H$, which is isomorphic to $K/H$ as a variety (by left translation by $g^{-1}$).

This "tower property" of quotients is exactly analogous to the Galois correspondence in field theory: given a chain of subgroups $H \subseteq K \subseteq G$, there is a corresponding chain of quotient maps $G/H \to G/K \to G/G = \{*\}$, with each map having fibers isomorphic to the "intermediate quotient."

**Example 4.3.10: The Tower $\mathrm{GL}_n/B \to \mathrm{GL}_n/P \to \{*\}$**

Let $B \subseteq P \subseteq \mathrm{GL}_n$ where $B$ is the Borel (upper-triangular) and $P = P_d$ is the parabolic subgroup stabilizing $\langle e_1, \dots, e_d \rangle$. The tower gives:

$$\mathrm{Fl}(1, \dots, n-1; n) = \mathrm{GL}_n/B \xrightarrow{\psi} \mathrm{GL}_n/P_d = \mathrm{Gr}(d, n) \to \{*\}$$

The map $\psi$ sends a complete flag $0 \subset V_1 \subset \cdots \subset V_{n-1} \subset k^n$ to the $d$-dimensional subspace $V_d$. The fiber of $\psi$ over a point $V_d \in \mathrm{Gr}(d, n)$ is the variety of complete flags *refining* $V_d$, which is isomorphic to $P_d/B \cong \mathrm{Fl}(1, \dots, d-1; d) \times \mathrm{Fl}(1, \dots, n-d-1; n-d)$ (a product of smaller flag varieties). This "fibration of flag varieties over Grassmannians" is a central tool in Schubert calculus and the geometry of flag varieties.

## Exercise 4.3.1

1. Let $G = \mathbb{G}_m \times \mathbb{G}_m$ and $N = \{(t, t^{-1}) : t \in \mathbb{G}_m\}$. Show that $N \trianglelefteq G$ and compute $G/N$ explicitly. Determine its Lie algebra.

2. Let $\varphi : \mathrm{GL}_n \to \mathrm{GL}_n$ be defined by $\varphi(g) = (\det g)^{-1} \cdot g$ ("projecting to determinant 1"). Show that $\varphi$ is a homomorphism with image $\mathrm{SL}_n$ and kernel $\mathbb{G}_m$ (embedded as scalar matrices). Conclude $\mathrm{GL}_n/\mathbb{G}_m \cong \mathrm{SL}_n$... but wait, is this correct? (*Hint*: check carefully whether $\varphi$ is actually a group homomorphism.)

3. Let $G = \mathrm{GL}_n$, $M = \mathrm{SL}_n$, and $N = \mu_n$ (embedded in $\mathrm{SL}_n$ as scalar matrices). Apply the third isomorphism theorem to compute $(\mathrm{GL}_n/\mu_n)/(\mathrm{SL}_n/\mu_n)$. What familiar group do you get?

4. Let $G = \mathrm{SO}_3$ act on the unit sphere $S^2 = \{v \in k^3 : q(v) = 1\}$ (where $q$ is the standard quadratic form, $\mathrm{char} k \neq 2$). Show the action is transitive, identify the stabilizer of $e_1$, and conclude that $S^2 \cong \mathrm{SO}_3/\mathrm{SO}_2$ as algebraic varieties.

5. Let $1 \to N \to G \to Q \to 1$ be a short exact sequence of *smooth* algebraic groups over a field of characteristic $p > 0$. Show that the Lie algebra sequence $0 \to \mathfrak{n} \to \mathfrak{g} \to \mathfrak{q} \to 0$ is still exact. (*Hint*: the smoothness hypothesis ensures the dimension count works, even though Cartier's theorem is unavailable.)

6. Consider the tower $\mathrm{SL}_3/B \to \mathrm{SL}_3/P \to \{*\}$ where $B$ is the upper-triangular Borel and $P = P_1$ is the parabolic stabilizing $\langle e_1 \rangle$. Show that $\mathrm{SL}_3/P \cong \mathbb{P}^2$, that $\mathrm{SL}_3/B$ is the complete flag variety $\mathrm{Fl}(1, 2; 3)$, and that the fiber of $\mathrm{Fl}(1, 2; 3) \to \mathbb{P}^2$ over a point $[\ell] \in \mathbb{P}^2$ is isomorphic to $\mathbb{P}^1$. What is $\dim \mathrm{Fl}(1, 2; 3)$?

# 5

# Jordan Decomposition and Diagonalizable Groups

In chapter 1, we introduced the composition series of an affine algebraic group $G$:

$$e \trianglelefteq U \trianglelefteq N \trianglelefteq G^\circ \trianglelefteq G$$

where $U$ is unipotent, $N/U$ is a torus, and $G^\circ/N$ is semisimple. This decomposition breaks a general algebraic group into three atomic pieces: tori, unipotent groups, and semisimple groups. The present chapter studies the first two.

The starting point is the *Jordan decomposition*: every element $g$ of a linear algebraic group factors uniquely as $g = g_s g_u$, where $g_s$ is semisimple (diagonalizable over $\overline{k}$), $g_u$ is unipotent, and the two factors commute. This is a familiar fact from linear algebra — the multiplicative Jordan normal form — but the key theorem is that the decomposition is *intrinsic*: it does not depend on the choice of embedding $G \hookrightarrow \mathrm{GL}_n$. With this decomposition in hand, we study each piece systematically. Diagonalizable groups and tori are completely controlled by their character lattices via an anti-equivalence of categories, and their representations decompose into weight spaces — a mechanism that will produce root spaces in chapter 7. Unipotent groups are the opposite extreme: their only semisimple element is the identity, and in characteristic 0 they are entirely determined by their (nilpotent) Lie algebras via the exponential map.

## 5.1 The Linear Algebra

Let $k$ be a perfect field and $V$ a finite-dimensional $k$-vector space. Recall from linear algebra (see [7]) that an endomorphism $g \in \mathrm{End}_k(V)$ is called:

1. *Semisimple* if $g$ is diagonalizable over $\overline{k}$ (equivalently, the minimal polynomial of $g$ has distinct roots in $\overline{k}$, equivalently $V \otimes_k \overline{k}$ has a basis of eigenvectors for $g$).
2. *Nilpotent* if $g^n = 0$ for some $n \geq 1$.
3. *Unipotent* if $g - \mathrm{id}_V$ is nilpotent, i.e. $(g - \mathrm{id}_V)^n = 0$ for some $n$.

The additive and multiplicative Jordan decompositions are:

**Theorem 5.1.1: Jordan Decomposition in $\mathrm{End}_k(V)$ and $\mathrm{GL}(V)$**

Let $V$ be a finite-dimensional vector space over a perfect field $k$.

1. (Additive) For every $X \in \mathrm{End}_k(V)$, there exist unique $X_s, X_n \in \mathrm{End}_k(V)$ with $X_s$ semisimple, $X_n$ nilpotent, $X_s X_n = X_n X_s$, and $X = X_s + X_n$.
2. (Multiplicative) For every $g \in \mathrm{GL}(V)$, there exist unique $g_s, g_u \in \mathrm{GL}(V)$ with $g_s$ semisimple, $g_u$ unipotent, $g_s g_u = g_u g_s$, and $g = g_s g_u$.

Moreover, $X_s$ and $X_n$ (resp. $g_s$ and $g_u$) are polynomials in $X$ (resp. $g$ and $g^{-1}$). In particular, they commute with every endomorphism that commutes with $X$ (resp. $g$).

***Proof*:**
We prove the additive version; the multiplicative version follows by letting $g = X_s + X_n$ be the additive decomposition of $g$ viewed as an endomorphism, and setting $g_s = X_s$ and $g_u = \mathrm{id}_V + X_s^{-1} X_n$. Because $g$ is invertible and $X_n$ is nilpotent, $X_s$ is invertible, so $g_u$ is well-defined. Then $g_s g_u = X_s + X_n = g$. Furthermore, $g_u - \mathrm{id}_V = X_s^{-1} X_n$ is nilpotent (since $X_s^{-1}$ and $X_n$ commute), confirming $g_u$ is unipotent.

*Existence.* We may assume $k = \overline{k}$ (since the decomposition, once shown to be unique, must be Galois-invariant, hence defined over $k$ when $k$ is perfect). Over $\overline{k}$, let the distinct eigenvalues of $X$ be $\lambda_1, \ldots, \lambda_r$ with generalized eigenspaces $V_i = \ker(X - \lambda_i \mathrm{id}_V)^{n_i}$. Then $V = V_1 \oplus \cdots \oplus V_r$. Define $X_s$ to act as $\lambda_i \cdot \mathrm{id}_V$ on $V_i$, and $X_n = X - X_s$. Then $X_s$ is semisimple (diagonal in any basis adapted to the decomposition), $X_n$ is nilpotent (on each $V_i$, $(X_n|_{V_i})^{n_i} = (X - \lambda_i \mathrm{id}_V)^{n_i}|_{V_i} = 0$), and $X_s X_n = X_n X_s$ (since both preserve each $V_i$ and $X_s$ is scalar on $V_i$).

*$X_s$ is a polynomial in $X$.* By the Chinese Remainder Theorem, the distinct polynomials $(t - \lambda_i)^{n_i}$ are pairwise coprime, so there exists a polynomial $p(t) \in \overline{k}[t]$ with $p(t) \equiv \lambda_i \pmod{(t - \lambda_i)^{n_i}}$ for each $i$. Then $p(X)$ acts as $\lambda_i \cdot \mathrm{id}_V$ on each $V_i$, so $X_s = p(X)$. When $k$ is perfect, the Galois action permutes eigenvalues in a way that preserves the decomposition, so $p$ can be chosen in $k[t]$ (see [7]).

*Uniqueness.* Suppose $X = S + N$ with $S$ semisimple, $N$ nilpotent, and $SN = NS$. Because $S$ and $N$ commute with each other, they both commute with $X = S + N$. Therefore, they preserve the generalized eigenspaces $V_i$ of $X$. On each $V_i$, the operator $X - \lambda_i \mathrm{id}_V$ is nilpotent. We also have $S - \lambda_i \mathrm{id}_V = (X - \lambda_i \mathrm{id}_V) - N$. Because $X$ and $N$ commute, the right-hand side is the difference of two commuting nilpotent operators, which implies $S - \lambda_i \mathrm{id}_V$ is nilpotent on $V_i$. But the restriction of the semisimple operator $S$ to an invariant subspace remains semisimple. Since $S - \lambda_i \mathrm{id}_V$ is both semisimple and nilpotent on $V_i$, it must be zero. Therefore $S|_{V_i} = \lambda_i \cdot \mathrm{id}_V = X_s|_{V_i}$, which implies $S = X_s$ and $N = X_n$.

The statement that $g_s$ and $g_u$ are polynomials in $g$ (and $g^{-1}$) is crucial: it means they commute with everything that commutes with $g$, and in particular they lie in any subalgebra or subgroup that $g$ lies in (provided that subalgebra is defined by commutation conditions). This is the mechanism that will make the decomposition intrinsic.

**Example 5.1.2: Jordan Decomposition in $\mathrm{GL}_2$**

Let $g = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right) \in \mathrm{GL}_2(k)$ with $\mathrm{char} k \neq 2$. The eigenvalue is $\lambda = 2$ with multiplicity 2. The additive decomposition of $g$ (as an endomorphism) is:

$$g = \underbrace{\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}}_{g_s = 2I} + \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{X_n}$$

The multiplicative decomposition is $g = g_s \cdot g_u$ where:

$$g_s = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \qquad g_u = g_s^{-1} g = \begin{pmatrix} 1 & 1/2 \\ 0 & 1 \end{pmatrix}$$

Note that $g_u - I$ is nilpotent (it squares to zero), confirming $g_u$ is unipotent.

**Example 5.1.3: Diagonal and Unipotent Parts in $\mathrm{GL}_n$**

More generally, the Jordan normal form of $g \in \mathrm{GL}_n(\overline{k})$ is $g = DU$ where $D = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ is the diagonal of eigenvalues and $U = D^{-1}g$ is upper-triangular with 1's on the diagonal (hence unipotent). Then $g_s = D$ and $g_u = U$. The decomposition is canonical: it does not depend on the choice of Jordan basis, only on $g$.

## Intrinsic Jordan Decomposition

The linear-algebraic decomposition depends on viewing $g$ as an element of $\mathrm{GL}(V)$ for some specific $V$. The fundamental theorem says this dependence is illusory: if $G$ is any affine algebraic group and $g \in G(k)$, then $g$ has a unique decomposition $g = g_s g_u$ that is compatible with *every* representation of $G$.

**Theorem 5.1.4: Intrinsic Jordan Decomposition**

Let $G$ be an affine algebraic group over a perfect field $k$ and let $g \in G(k)$. Then there exist unique elements $g_s, g_u \in G(k)$ such that:

1. For every representation $\rho : G \to \mathrm{GL}(V)$, $\rho(g_s)$ is semisimple and $\rho(g_u)$ is unipotent.
2. $g = g_s g_u$ and $g_s g_u = g_u g_s$.

The elements $g_s$ and $g_u$ are called the *semisimple part* and *unipotent part* of $g$, respectively.

***Proof*:**
By the linearization theorem (lemma 1.1.8), there exists a faithful representation $\rho_0 : G \hookrightarrow \mathrm{GL}(V_0)$. Inside $\mathrm{GL}(V_0)$, the element $\rho_0(g)$ has a multiplicative Jordan decomposition $\rho_0(g) = s \cdot u$ with $s$ semisimple and $u$ unipotent (theorem 5.1.1). We need to show that $s, u \in \rho_0(G) \subseteq \mathrm{GL}(V_0)$, and that their preimages $g_s, g_u \in G(k)$ are independent of the choice of $\rho_0$.

*Key Lemma.* Let $x \in \mathrm{GL}(W)$ with Jordan decomposition $x = x_s x_u$. If $U \subseteq W$ is a subspace with $x(U) \subseteq U$, then $x_s(U) \subseteq U$ and $x_u(U) \subseteq U$.

*Proof of Key Lemma.* Since $x_s$ and $x_u$ are polynomials in $x$ by theorem 5.1.1, any subspace preserved by $x$ is also preserved by $x_s$ and $x_u$. □

To proceed, we require a fundamental property of $\mathrm{GL}(V_0)$: its rational representations preserve the Jordan decomposition. Any rational representation of $\mathrm{GL}(V_0)$ on a vector space $W$ is a subquotient of a direct sum of tensor products of $V_0$ and its dual $V_0^*$. Because the Jordan decomposition is compatible with tensor products (e.g., the semisimple part of $x \otimes x$ is precisely $x_s \otimes x_s$), duals, and subquotients (by the Key Lemma), the action of $g$ on $W$ has Jordan decomposition $\rho_W(g) = \rho_W(s)\rho_W(u)$.

*Step 1: $s$ and $u$ lie in $G$.* We use the characterization of $G$ as a closed subgroup of $\mathrm{GL}(V_0)$. The group $G$ is the vanishing locus of a Hopf ideal $I \subseteq \mathcal{O}(\mathrm{GL}(V_0))$. The group $\mathrm{GL}(V_0)$ acts on the coordinate ring $A = \mathcal{O}(\mathrm{GL}(V_0))$ by right translation. Because $G$ is a subgroup, $I$ is stabilized by the right translation action of $g$.

By the Fundamental Finiteness Theorem, $I$ is the union of finite-dimensional $g$-stable subspaces $I_W = I \cap W$, where $W \subseteq A$ is a rational $\mathrm{GL}(V_0)$-submodule. By the fact above, the semisimple and unipotent parts of $g$ acting on $W$ are precisely the respective actions of $s$ and $u$. Because $g$ stabilizes $I_W \subseteq W$, the Key Lemma guarantees that $s$ and $u$ also stabilize $I_W$. Therefore, $s$ and $u$ stabilize the entire ideal $I$. Because elements of $I$ vanish at the identity $e \in G$, and the action gives $(s \cdot f)(e) = f(es) = f(s)$, the stabilization of $I$ implies $f(s) = 0$ for all $f \in I$. Thus $s \in G(k)$, and similarly $u \in G(k)$. We define $g_s = \rho_0^{-1}(s)$ and $g_u = \rho_0^{-1}(u)$.

*Step 2: Independence of the embedding.* Let $\rho : G \to \mathrm{GL}(V)$ be any representation. This representation corresponds to a comodule map $V \to V \otimes \mathcal{O}(G)$, which can be lifted to a comodule map over $\mathcal{O}(\mathrm{GL}(V_0))$. This implies that $V$ can be viewed as a subquotient of a direct sum of tensor products of $V_0$ and $V_0^*$. As established earlier, such tensor constructions preserve the Jordan decomposition. Therefore, the semisimple and unipotent parts of $g$ acting on $V$ are precisely the actions of $g_s$ and $g_u$.

*Step 3: Uniqueness.* If $g = g_s' g_u'$ is another such decomposition, then for the faithful representation $\rho_0$, both $\rho_0(g_s')$ and $\rho_0(g_s)$ act as the semisimple part of $\rho_0(g) \in \mathrm{GL}(V_0)$. By the uniqueness of the Jordan decomposition in $\mathrm{GL}(V_0)$, $\rho_0(g_s') = \rho_0(g_s)$, and since $\rho_0$ is faithful, $g_s' = g_s$ (and hence $g_u' = g_u$).

The proof is somewhat technical, but the essential point is clean: the semisimple and unipotent parts of $g$ are polynomials in $g$, so they "go wherever $g$ goes" — they are preserved by every morphism that preserves $g$. Since the closed subgroup $G \subseteq \mathrm{GL}(V_0)$ is defined by conditions that $g$ satisfies, its semisimple and unipotent parts satisfy these conditions too.

**5.1.5 Remark: The Role of Perfectness** The assumption that $k$ is perfect is used to ensure that the polynomial $p(t)$ giving $g_s = p(g)$ has coefficients in $k$ (not just in $\overline{k}$). Over a non-perfect field, the Galois action on eigenvalues may not preserve the decomposition, and the Jordan decomposition can fail to be defined over $k$. For most applications in this book (including the structure theory of Chapters 6–7), we work over algebraically closed fields, where perfectness is automatic.

**Proposition 5.1.6: Functoriality of the Jordan Decomposition**

Let $\varphi : G \to H$ be a homomorphism of affine algebraic groups over a perfect field $k$. Then for any $g \in G(k)$:

$$\varphi(g_s) = \varphi(g)_s \qquad \text{and} \qquad \varphi(g_u) = \varphi(g)_u$$

That is, homomorphisms preserve the Jordan decomposition.

***Proof*:**
The homomorphism $\varphi$ is a morphism of algebraic groups, hence the composition with any faithful representation $\rho : H \hookrightarrow \mathrm{GL}(W)$ gives a representation $\rho \circ \varphi : G \to \mathrm{GL}(W)$. By the intrinsic property (theorem 5.1.4(1)), $(\rho \circ \varphi)(g_s) = \rho(\varphi(g_s))$ is the semisimple part of $(\rho \circ \varphi)(g) = \rho(\varphi(g))$. But the semisimple part of $\rho(\varphi(g))$ is $\rho(\varphi(g)_s)$ (again by theorem 5.1.4(1)). Since $\rho$ is faithful, $\varphi(g_s) = \varphi(g)_s$.

An important special case: if $H \subseteq G$ is a closed subgroup and $g \in H(k)$, then the semisimple and unipotent parts of $g$ computed in $H$ agree with those computed in $G$. The decomposition is independent not only of the representation, but also of the ambient group.

## Jordan Decomposition for Lie Algebras

In characteristic 0, there is an additive analogue for Lie algebras.

**Proposition 5.1.7: Jordan Decomposition in $\mathfrak{g}$ (Characteristic 0)**

Let $G$ be an affine algebraic group over a field $k$ of characteristic 0, and let $\mathfrak{g} = \mathrm{Lie}(G)$. For every $X \in \mathfrak{g}$, the additive Jordan decomposition $\mathrm{ad}(X) = \mathrm{ad}(X)_s + \mathrm{ad}(X)_n$ in $\mathrm{End}_k(\mathfrak{g})$ lifts to a decomposition $X = X_s + X_n$ in $\mathfrak{g}$ such that:

1. $\mathrm{ad}(X_s) = \mathrm{ad}(X)_s$ is semisimple and $\mathrm{ad}(X_n) = \mathrm{ad}(X)_n$ is nilpotent.
2. For every representation $\rho : G \to \mathrm{GL}(V)$, $d\rho(X_s)$ is semisimple and $d\rho(X_n)$ is nilpotent in $\mathrm{End}_k(V)$.

***Proof*:**
The proof follows the same strategy as the group case, using the faithful representation $d\rho_0 : \mathfrak{g} \hookrightarrow \mathfrak{gl}(V_0)$ (which is injective by proposition 3.1.12) to embed $\mathfrak{g}$ into a matrix algebra, decomposing there, and using the "polynomial" nature of the semisimple part to pull back. (The characteristic

0 assumption is required because in positive characteristic, the polynomial $p(t)$ giving $X_s = p(X)$ does not generally preserve the subspace $\mathfrak{g} \subseteq \mathfrak{gl}(V_0)$). We omit the full details, which are parallel to the group case.

In characteristic $p > 0$, the additive Jordan decomposition on $\mathfrak{g}$ still exists inside $\mathfrak{gl}(V)$ for any representation, but it need not lift to an intrinsic decomposition in $\mathfrak{g}$ (because $\mathfrak{g}$ may not be "large enough" — the polynomial $p(t)$ may not preserve $\mathfrak{g} \subseteq \mathfrak{gl}(V)$). For the purposes of this book, we will use the Lie-algebraic Jordan decomposition only in characteristic 0.

**Example 5.1.8: Jordan Decomposition in $\mathfrak{sl}_2$**

Let $X = \left(\begin{smallmatrix} 1 & 1 \\ 0 & -1 \end{smallmatrix}\right) \in \mathfrak{sl}_2$ (with $\operatorname{char} k \neq 2$). The eigenvalues are 1 and $-1$, so $X$ is already semisimple: $X_s = X$ and $X_n = 0$. The standard basis element $e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ is nilpotent ($e^2 = 0$), so $e_s = 0$ and $e_n = e$. These are the two extremes: $h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ is semisimple, and $e, f$ are nilpotent. In chapter 7, the semisimple elements will span the *Cartan subalgebra*, and the nilpotent elements will generate the *root spaces*.

**Example 5.1.9: Semisimple and Unipotent Elements of $\mathrm{GL}_n$**

In $\mathrm{GL}_n(k)$ (with $k = \overline{k}$):

1. The semisimple elements are the diagonalizable matrices. They are Zariski-dense in $\mathrm{GL}_n$ (since the set of matrices with $n$ distinct eigenvalues is open and dense).
2. The unipotent elements form a closed subvariety $\mathcal{U}_n = \{g \in \mathrm{GL}_n \mid (g - I)^n = 0\}$. This is the variety of matrices whose only eigenvalue is 1.
3. The set of semisimple elements is *not* closed in general. For instance, in $\mathrm{GL}_2$, consider the curve of matrices $g_t = \left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda+t \end{smallmatrix}\right)$ for $t \in k$. For any $t \neq 0$, $g_t$ has distinct eigenvalues $\lambda$ and $\lambda + t$, so it is semisimple. But as $t \to 0$, the limit $g_0 = \left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)$ is not semisimple (it has a nontrivial unipotent part). Thus a limit of semisimple elements need not be semisimple, meaning the set of semisimple elements is constructible but not closed.

The Jordan decomposition respects the group structure in a strong sense:

**Proposition 5.1.10: Jordan Decomposition and Commuting Elements**

Let $G$ be an affine algebraic group over a perfect field and let $g, h \in G(k)$. If $gh = hg$, then:

$$g_s h_s = h_s g_s, \quad g_s h_u = h_u g_s, \quad g_u h_s = h_s g_u, \quad g_u h_u = h_u g_u$$

In words: if $g$ and $h$ commute, then all four combinations of their semisimple and unipotent parts commute.

***Proof*:**

Embed $G \hookrightarrow \mathrm{GL}(V)$ faithfully. Since $gh = hg$ in $\mathrm{GL}(V)$, the endomorphisms $g$ and $h$ commute. By theorem 5.1.1, $g_s$ is a polynomial in $g$, hence commutes with everything that commutes with $g$ — in particular with $h$, $h_s$, and $h_u$. The same argument with $g_u$, $h_s$, and $h_u$ gives all four commutation relations.

This proposition is essential for the structure theory: it says that inside an abelian subgroup of $G$, the semisimple parts form an abelian subgroup and the unipotent parts form an abelian subgroup. When we study maximal tori in chapter 6, this will ensure that the torus elements and unipotent elements inside a solvable group decouple cleanly.

**Exercise 5.1.1**

1. Compute the multiplicative Jordan decomposition of the following matrices in $\mathrm{GL}_3(\mathbb{C})$:

$$g_1 = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \qquad g_2 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad g_3 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

2. Let $G \subseteq \mathrm{GL}_n$ be a closed subgroup and $g \in G(k)$ with $k$ perfect. The proof of theorem 5.1.4 shows $g_s, g_u \in G(k)$. Give a direct proof of this for $G = \mathrm{SL}_n$: show that if $g \in \mathrm{SL}_n(k)$, then $g_s, g_u \in \mathrm{SL}_n(k)$. (*Hint*: $\det(g_s)\det(g_u) = \det(g) = 1$; now use the fact that $\det(g_s)$ is semisimple and $\det(g_u)$ is unipotent in $k^\times$.)
3. Let $X = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right) \in \mathfrak{sl}_3$. Compute $\mathrm{ad}(X) : \mathfrak{sl}_3 \to \mathfrak{sl}_3$ explicitly (as an $8 \times 8$ matrix in the standard basis $\{E_{ij} - \frac{1}{3}\delta_{ij} I\}$, or simply in $\mathfrak{gl}_3$). Verify that $\mathrm{ad}(X)$ is nilpotent. What is the nilpotency index (the smallest $m$ with $\mathrm{ad}(X)^m = 0$)?
4. Let $k$ be a field of characteristic $p > 0$. Show that the $p$-th power map $F : \mathrm{GL}_n(k) \to \mathrm{GL}_n(k)$, $g \mapsto g^p$, does *not* preserve the Jordan decomposition in general: $(g^p)_s \neq (g_s)^p$ for some $g$. (*Hint*: take $g = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and compute $g^p$.)
5. Let $G$ be a commutative affine algebraic group over an algebraically closed field $k$. Define $G_s = \{g \in G(k) : g = g_s\}$ (the semisimple elements) and $G_u = \{g \in G(k) : g = g_u\}$ (the unipotent elements). Show that $G_s$ and $G_u$ are subgroups of $G(k)$ and that the map $G_s \times G_u \to G(k)$, $(s, u) \mapsto su$, is a bijection.

## 5.2 Diagonalizable Groups and Tori

An element $g \in \mathrm{GL}_n(k)$ is semisimple if and only if it is diagonalizable (over $\overline{k}$), and commuting semisimple elements are simultaneously diagonalizable. The algebraic groups built entirely from semisimple elements — the *diagonalizable groups* — are therefore the groups that can be simultaneously conjugated into the diagonal torus. They are the simplest algebraic groups beyond finite groups, and yet they carry a rich structure: they are completely classified by their *character groups*, finitely generated abelian groups that encode all the representation-theoretic information. The connected ones — the *tori* — will play a central role in the structure theory, as "maximal tori" inside reductive groups provide the Cartan subalgebras that drive the root space decomposition.

## Characters and Cocharacters

**Definition 5.2.1: Character**

Let $G$ be an affine algebraic group over $k$. A *character* of $G$ is a homomorphism of algebraic groups $\chi : G \to \mathbb{G}_m$. The set of all characters forms an abelian group under pointwise multiplication:

$$X(G) := \mathrm{Hom}_{\mathbf{Grp}}(G, \mathbb{G}_m)$$

called the *character group* (or *character lattice*) of $G$. The group operation is $(\chi_1 \cdot \chi_2)(g) = \chi_1(g)\chi_2(g)$ for $g \in G(R)$.

On the Hopf algebra side, a character $\chi : G \to \mathbb{G}_m$ corresponds to a Hopf algebra homomorphism $\chi^* : k[t, t^{-1}] \to \mathcal{O}(G)$, which is determined by the image $c = \chi^*(t) \in \mathcal{O}(G)$. The Hopf algebra axioms for $k[t, t^{-1}]$ (namely $\Delta(t) = t \otimes t$, $\epsilon(t) = 1$, $S(t) = t^{-1}$) force $c$ to be a *group-like element*: $\Delta(c) = c \otimes c$, $\epsilon(c) = 1$, and $S(c) = c^{-1}$. Conversely, every group-like element of $\mathcal{O}(G)$ defines a character. We saw this correspondence in exercise 2.2.1 (2): characters of $G$ correspond to one-dimensional representations.

**Definition 5.2.2: Cocharacter**

A *cocharacter* (or *one-parameter subgroup*) of $G$ is a homomorphism $\lambda : \mathbb{G}_m \to G$. The set of all cocharacters is denoted:

$$X_\vee(G) := \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{G}_m, G)$$

There is a natural pairing between characters and cocharacters: given $\chi \in X(G)$ and $\lambda \in X_\vee(G)$, the composition $\chi \circ \lambda : \mathbb{G}_m \to \mathbb{G}_m$ is an endomorphism of $\mathbb{G}_m$. Every such endomorphism is of the form $t \mapsto t^n$ for a unique $n \in \mathbb{Z}$, so the pairing

$$\langle \cdot , \cdot \rangle : X(G) \times X_\vee(G) \longrightarrow \mathbb{Z}, \qquad \langle \chi, \lambda \rangle = n \quad \text{where } \chi \circ \lambda(t) = t^n$$

is a well-defined $\mathbb{Z}$-bilinear map.

## Diagonalizable Groups

**Definition 5.2.3: Diagonalizable Group**

An affine algebraic group $G$ over $k$ is called *diagonalizable* (or *split diagonalizable*) if $G$ is isomorphic (over $k$) to a closed subgroup of a diagonal torus $\mathbb{G}_m^n$ for some $n$. Equivalently, every representation of $G$ over $k$ decomposes as a direct sum of one-dimensional representations.

The equivalence of the two conditions is the key observation. If $G \subseteq \mathbb{G}_m^n$ is a closed subgroup of the diagonal torus, then any representation $\rho : G \to \mathrm{GL}(V)$ extends (via the inclusion) to a representation of $\mathbb{G}_m^n$ on $V$. Since the elements of $\mathbb{G}_m^n$ are simultaneously diagonalizable, $V$ decomposes into simultaneous eigenspaces, each of which is a one-dimensional representation. Conversely, if every representation decomposes into one-dimensional sub-representations, then the faithful representation $G \hookrightarrow \mathrm{GL}(V)$ (from the linearization theorem) puts $G$ inside a diagonal torus after choosing a basis of eigenvectors.

The structure of diagonalizable groups is completely controlled by their character groups:

**Theorem 5.2.4: Classification of Diagonalizable Groups**

The functor $G \mapsto X(G)$ is an anti-equivalence of categories:

$$\left\{ \begin{array}{c} \text{Split diagonalizable} \\ \text{groups over } k \end{array} \right\} \xleftrightarrow{\sim} \left\{ \begin{array}{c} \text{Finitely generated} \\ \text{abelian groups} \end{array} \right\}^{\mathrm{op}}$$

The quasi-inverse sends a finitely generated abelian group $M$ to the diagonalizable group $D(M) := \operatorname{Spec} k[M]$, where $k[M]$ is the group algebra of $M$ with Hopf algebra structure $\Delta(e_m) = e_m \otimes e_m$, $\epsilon(e_m) = 1$, $S(e_m) = e_{-m}$ for $m \in M$.

***Proof*:**
We first construct the functors and then verify they are quasi-inverse.

*From groups to characters.* Given a diagonalizable group $G$, its character group $X(G)$ is the group of group-like elements in $\mathcal{O}(G)$. Since every representation decomposes into one-dimensional sub-representations, $\mathcal{O}(G)$ (as a comodule over itself via the regular representation) decomposes into one-dimensional comodules. The group-like elements span $\mathcal{O}(G)$ as a $k$-vector space, and the Hopf algebra structure on $\mathcal{O}(G)$ is completely determined by the multiplication of these group-like elements. Specifically, $\mathcal{O}(G) \cong k[X(G)]$ as Hopf algebras.

*From abelian groups to group schemes.* Given a finitely generated abelian group $M$, the group algebra $k[M] = \bigoplus_{m \in M} k \cdot e_m$ with multiplication $e_m \cdot e_{m'} = e_{m+m'}$ is a commutative $k$-algebra (since $M$ is abelian). The Hopf structure $\Delta(e_m) = e_m \otimes e_m$ makes each $e_m$ group-like, and $D(M) = \operatorname{Spec} k[M]$ is a diagonalizable group. Functorially:

$$D(M)(R) = \operatorname{Hom}_{\mathbf{Grp}}(M, R^\times) = \operatorname{Hom}_{\mathbf{Grp}}(M, \mathbb{G}_m(R))$$

for any $k$-algebra $R$. This is the group of "$M$-graded units" in $R$.

*Quasi-inverse.* Starting with $G$ diagonalizable: $D(X(G)) = \operatorname{Spec} k[X(G)] \cong \operatorname{Spec} \mathcal{O}(G) = G$. Starting with $M$ a finitely generated abelian group: $X(D(M)) = \operatorname{Hom}(D(M), \mathbb{G}_m) = \operatorname{Hom}_{\mathrm{Hopf}}(k[t, t^{-1}], k[M])$. A Hopf map $k[t, t^{-1}] \to k[M]$ sends $t \mapsto e_m$ for some $m \in M$ (the group-like elements of $k[M]$ are exactly $\{e_m : m \in M\}$). This gives $X(D(M)) \cong M$.

*Anti-equivalence.* A homomorphism $\varphi : G \to H$ of diagonalizable groups induces a group homomorphism $X(\varphi) : X(H) \to X(G)$ by $\chi \mapsto \chi \circ \varphi$ (precomposition), reversing arrows. This is functorial and compatible with the quasi-inverse by construction.

**Example 5.2.5: The Fundamental Examples**

1. $G = \mathbb{G}_m$: The coordinate ring is $k[t, t^{-1}]$ with group-like elements $\{t^n : n \in \mathbb{Z}\}$. So $X(\mathbb{G}_m) = \mathbb{Z}$, and the anti-equivalence sends $\mathbb{Z} \mapsto \mathbb{G}_m$. A character $\chi_n : \mathbb{G}_m \to \mathbb{G}_m$ corresponding to $n \in \mathbb{Z}$ is $\chi_n(t) = t^n$.

2. $G = \mathbb{G}_m^r$ (the split torus of rank $r$): $X(\mathbb{G}_m^r) = \mathbb{Z}^r$, with the character $\chi_{(n_1,\dots,n_r)}(t_1,\dots,t_r) = t_1^{n_1}\cdots t_r^{n_r}$.
3. $G = \mu_n$ (over $k$ with $\operatorname{char} k \nmid n$): Functorially, $\mu_n(R) = \{r \in R^\times : r^n = 1\} = \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z}/n\mathbb{Z}, R^\times)$. So $\mu_n = D(\mathbb{Z}/n\mathbb{Z})$ and $X(\mu_n) = \mathbb{Z}/n\mathbb{Z}$.
4. $G = D(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}) = \mu_2 \times \mu_3 \cong \mu_6$: The character group of a product is the direct sum, and $D$ sends direct sums to products.

The anti-equivalence translates group-theoretic notions into abelian group theory:

| **Diagonalizable group** $G$ | **Character group** $X(G)$ |
|---|---|
| Closed subgroup $H \subseteq G$ | Quotient $X(G) \twoheadrightarrow X(H)$ |
| Quotient $G \twoheadrightarrow Q$ | Subgroup $X(Q) \hookrightarrow X(G)$ |
| $G$ is a torus (smooth and connected) | $X(G)$ is torsion-free ($\cong \mathbb{Z}^r$) |
| $G$ is finite | $X(G)$ is finite |
| Short exact seq. $1 \to H \to G \to Q \to 1$ | Short exact seq. $0 \to X(Q) \to X(G) \to X(H) \to 0$ |

## Tori

**Definition 5.2.6: Torus (Refined)**

An algebraic *torus* over $k$ is an affine algebraic group $T$ such that $T_{\bar{k}} \cong \mathbb{G}_m^r$ for some $r$. The integer $r$ is the *rank* of $T$.
A torus is called *split* if $T \cong \mathbb{G}_m^r$ over $k$ itself (i.e., if it is a connected diagonalizable group over $k$). Over an algebraically closed field, every torus is split.

By the classification theorem, *split* tori correspond to free abelian groups of finite rank: $X(T) \cong \mathbb{Z}^r$.

Over a non-algebraically closed field, non-split tori arise naturally. Recall from example 1.2.5 the Weil restriction $\mathrm{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}})$: this is a 2-dimensional algebraic group over $\mathbb{R}$ with $\mathrm{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}})(\mathbb{R}) = \mathbb{C}^\times$. Over $\mathbb{C}$, it splits as $\mathbb{G}_m \times \mathbb{G}_m$, but over $\mathbb{R}$ it is non-split: there is no $\mathbb{R}$-isomorphism $\mathrm{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}}) \cong \mathbb{G}_{m,\mathbb{R}}^2$. The Galois group $\mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1, \sigma\}$ acts on $X(T_\mathbb{C}) = \mathbb{Z}^2$ by swapping the two factors, and this action encodes the "twisting" of the torus.

The structure of non-split tori over a general field $k$ is controlled by the Galois action on $X(T) = X(T_{\bar{k}})$. We will not develop this theory further (it belongs to the arithmetic side of the subject), but the reader should keep in mind that over non-algebraically closed fields, tori carry a Galois-theoretic richness beyond their geometry.

**Proposition 5.2.7: Basic Properties of Split Tori**

Let $T$ be a split algebraic torus over $k$.

1. $T$ is commutative.
2. Every homomorphism $T \to T'$ between split tori is determined by the induced map $X(T') \to X(T)$ on character groups.
3. Every closed subgroup of $T$ is a split diagonalizable group (hence a product of a smaller split torus and a finite diagonalizable group).
4. $T$ has no unipotent elements other than the identity: $T_u = \{e\}$.

***Proof*:**
*(1):* By definition, $T \cong \mathbb{G}_m^r$ over $k$, which is visibly commutative.

*(2):* This is a special case of the anti-equivalence (theorem 5.2.4): homomorphisms $T \to T'$ correspond bijectively to group homomorphisms $X(T') \to X(T)$.

*(3):* A closed subgroup $H \subseteq T$ corresponds, under the anti-equivalence, to a quotient $X(T) \twoheadrightarrow X(H)$. Since $X(T) \cong \mathbb{Z}^r$ is free, $X(H)$ is a finitely generated abelian group (a quotient of $\mathbb{Z}^r$), so $H = D(X(H))$ is a split diagonalizable group.

*(4):* In $\mathbb{G}_m^r(\overline{k}) = (\overline{k}^\times)^r$, a unipotent element $(t_1, \ldots, t_r)$ satisfies $(t_i - 1)^N = 0$ for some $N$ and all $i$. Since $\overline{k}$ is a field, it has no nonzero nilpotent elements, which forces $t_i - 1 = 0$, so $t_i = 1$ for all $i$.

## Cartier Duality for Finite Diagonalizable Groups

For finite diagonalizable group schemes, the classification admits a fruitful reinterpretation as a duality.

**Proposition 5.2.8: Cartier Duality**

Let $G$ be a finite commutative group scheme over $k$. The *Cartier dual* of $G$ is the group scheme $G^\vee$ representing the functor $R \mapsto \mathrm{Hom}_{\mathbf{Grp}_R}(G_R, \mathbb{G}_{m,R})$. If $M$ is a finite abelian group, the Cartier dual of the split diagonalizable group $D(M)$ is the constant group scheme $M_k$. In particular:

1. $(\mu_n)^\vee \cong (\mathbb{Z}/n\mathbb{Z})_k$ (the constant group scheme).
2. $((\mathbb{Z}/n\mathbb{Z})_k)^\vee \cong \mu_n$.

Furthermore,

$$(G^\vee)^\vee \cong G$$

that is, Cartier duality is an involution.

***Proof*:**
Cartier duality can be computed cleanly on the level of Hopf algebras: if $G = \operatorname{Spec} A$ is a finite commutative group scheme, its Cartier dual is $G^\vee = \operatorname{Spec}(A^\vee)$, where $A^\vee = \operatorname{Hom}_k(A, k)$ is the dual Hopf algebra (with multiplication dual to the comultiplication of $A$, and vice versa).

For the split diagonalizable group $G = D(M) = \operatorname{Spec} k[M]$, the coordinate ring $A = k[M]$ has a basis of group-like elements $\{e_m\}_{m \in M}$. The comultiplication is $\Delta(e_m) = e_m \otimes e_m$. Let $\{f_m\}_{m \in M}$ be the dual basis for $A^\vee$. The multiplication in $A^\vee$ is determined by the comultiplication in $A$: since $\Delta(e_m) = e_m \otimes e_m$, the dual elements satisfy $f_x f_y = \delta_{x,y} f_x$. This means the $f_m$ are a complete set of orthogonal idempotents, so $A^\vee \cong k \times \cdots \times k$ (with $|M|$ copies).

The comultiplication in $A^\vee$ is determined by the multiplication $e_x e_y = e_{x+y}$ in $A$, which exactly encodes the group law of $M$. Therefore, $A^\vee$ is precisely the coordinate ring of the constant group scheme $M_k$, giving $(D(M))^\vee \cong M_k$.

Statements (1) and (2) follow immediately by setting $M = \mathbb{Z}/n\mathbb{Z}$, since $\mu_n = D(\mathbb{Z}/n\mathbb{Z})$. The double-duality follows from the natural double-duality isomorphism $(A^\vee)^\vee \cong A$ for finite-dimensional vector spaces, which respects the Hopf algebra structures.

Cartier duality is the algebraic-group analogue of Pontryagin duality for finite abelian groups. While diagonalizable groups and constant groups are dual to each other (exchanging "multiplicative type" with "étale type"), in characteristic $p$ the duality also elegantly handles non-étale, non-smooth group schemes. For instance, the infinitesimal group $\alpha_p$ (defined by $x^p = 0$ inside $\mathbb{G}_a$) is self-dual: $(\alpha_p)^\vee \cong \alpha_p$. We will not develop these extensions, but they underlie the deeper duality theories (Cartier duality for $p$-divisible groups, duality of abelian varieties) that appear in the abelian varieties book.

## Weight Spaces and the Decomposition of Representations

The most important property of tori for the structure theory is that their representations decompose completely into one-dimensional pieces, each labeled by a character. This is the mechanism that produces root spaces in chapter 7.

**Theorem 5.2.9: Weight Space Decomposition**

Let $T$ be a split torus over $k$ and $V$ a finite-dimensional representation of $T$. Then $V$ decomposes as a direct sum of *weight spaces*:

$$V = \bigoplus_{\chi \in X(T)} V_\chi$$

where $V_\chi = \{v \in V \mid t \cdot v = \chi(t) \cdot v \text{ for all } t \in T(R), \text{ all } R\}$. The characters $\chi$ with $V_\chi \neq 0$ are called the *weights* of $V$, and $\dim V_\chi$ is the *multiplicity* of the weight $\chi$.

***Proof*:**
Since $T \cong \mathbb{G}_m^r$ is split, a representation $\rho : T \to \mathrm{GL}(V)$ corresponds to $r$ commuting automorphisms $\rho(t_1, 1, \ldots, 1), \ldots, \rho(1, \ldots, 1, t_r)$ of $V$, each given by the action of a single copy of $\mathbb{G}_m$.

For a single $\mathbb{G}_m$-action on $V$, the coaction $\rho : V \to V \otimes_k k[t, t^{-1}]$ decomposes $V$ into weight spaces $V_n = \{v \in V : \rho(v) = v \otimes t^n\}$ for $n \in \mathbb{Z} = X(\mathbb{G}_m)$. This was established in the exercises of chapter 2 (exercise 2.3.1 (1)): the coaction condition $\rho(v) = v \otimes t^n$ means $g \cdot v = g^n v$ for all $g \in \mathbb{G}_m(R) = R^\times$.

The decomposition $V = \bigoplus_{n\in\mathbb{Z}} V_n$ is a decomposition into eigenspaces for the $\mathbb{G}_m$-action. For $T = \mathbb{G}_m^r$, we iterate: the $r$ commuting $\mathbb{G}_m$-actions give a simultaneous eigenspace decomposition $V = \bigoplus_{\chi\in\mathbb{Z}^r} V_\chi$, where $V_\chi = V_{n_1} \cap \cdots \cap V_{n_r}$ for $\chi = (n_1, \ldots, n_r)$. The character $\chi = (n_1, \ldots, n_r) \in X(T) = \mathbb{Z}^r$ acts by $\chi(t_1, \ldots, t_r) = t_1^{n_1} \cdots t_r^{n_r}$, confirming $t \cdot v = \chi(t)v$ for $v \in V_\chi$.

That only finitely many $V_\chi$ are nonzero follows from $V$ being finite-dimensional.

**Example 5.2.10: Weights of the Adjoint Action of a Maximal Torus on $\mathfrak{gl}_n$**
Let $G = \mathrm{GL}_n$ and $T = \mathbb{G}_m^n \subseteq \mathrm{GL}_n$ the diagonal torus, with elements $t = \mathrm{diag}(t_1, \ldots, t_n)$. The adjoint action of $T$ on $\mathfrak{gl}_n = M_n(k)$ is:

$$\mathrm{Ad}(t)(E_{ij}) = tE_{ij}t^{-1} = t_i t_j^{-1} E_{ij}$$

where $E_{ij}$ is the elementary matrix with 1 in position $(i, j)$ and 0 elsewhere. The character group is $X(T) = \mathbb{Z}^n$ with standard basis $\{\varepsilon_1, \ldots, \varepsilon_n\}$, where $\varepsilon_i(t) = t_i$. The weight of $E_{ij}$ is:

$$\chi_{ij} = \varepsilon_i - \varepsilon_j \in X(T)$$

The weight space decomposition of $\mathfrak{gl}_n$ is:

$$\mathfrak{gl}_n = \mathfrak{t} \oplus \bigoplus_{i\neq j} k \cdot E_{ij}$$

where $\mathfrak{t} = \mathrm{Lie}(T) = \{\text{diagonal matrices}\}$ is the 0-weight space (with weight $\varepsilon_i - \varepsilon_i = 0$ for each $E_{ii}$), and each off-diagonal $E_{ij}$ spans a one-dimensional weight space with weight $\varepsilon_i - \varepsilon_j$.

The nonzero weights $\{\varepsilon_i - \varepsilon_j : i \neq j\}$ are the *roots* of $\mathrm{GL}_n$. They form a root system of type $A_{n-1}$ in the hyperplane $\{\sum n_i = 0\} \subseteq \mathbb{Z}^n$. The full theory will be developed in chapter 7, but the reader should appreciate the mechanism: the adjoint action of the torus on the Lie algebra produces a decomposition into one-dimensional eigenspaces, and the "eigenvalues" (characters) organize into a combinatorial structure.

**Example 5.2.11: Weights of the Standard Representation**
The standard representation $V = k^n$ of $\mathrm{GL}_n$ has weight decomposition under $T = \mathbb{G}_m^n$:

$$V = \bigoplus_{i=1}^n k \cdot e_i, \qquad \text{with weight } \varepsilon_i \text{ for } e_i$$

since $\mathrm{diag}(t_1, \ldots, t_n) \cdot e_i = t_i e_i$.

The symmetric power $\mathrm{Sym}^2(V)$ has weights $\{\varepsilon_i + \varepsilon_j : 1 \le i \le j \le n\}$ (each appearing with multiplicity 1), while the exterior power $\bigwedge^2(V)$ has weights $\{\varepsilon_i + \varepsilon_j : 1 \le i < j \le n\}$. More generally, $\mathrm{Sym}^d(V)$ and $\bigwedge^d(V)$ have weights that are sums of $d$ elements from $\{\varepsilon_1, \dots, \varepsilon_n\}$ (with and without repetition, respectively).

**Example 5.2.12: The Torus Acting on Itself by Conjugation**

Since a torus $T$ is commutative, the adjoint action $\mathrm{Ad} : T \to \mathrm{GL}(\mathfrak{t})$ is trivial (example 3.2.10): $\mathrm{Ad}(t)(X) = X$ for all $t \in T$ and $X \in \mathfrak{t}$. So $\mathfrak{t}$ is entirely the 0-weight space: all weights of the adjoint action of $T$ on $\mathfrak{t}$ are zero. The interesting weights will come from the adjoint action of $T$ on the *ambient* Lie algebra $\mathfrak{g}$ (as in example 5.2.10), not from $T$ acting on its own Lie algebra.

## Rigidity of Tori

Tori have a remarkable rigidity property: their automorphism groups are discrete, and in particular they have no infinitesimal automorphisms. This is in sharp contrast to unipotent groups, which are highly deformable.

**Proposition 5.2.13: Rigidity of Tori**

Let $T$ be a torus and $G$ any affine algebraic group. Then $\mathrm{Hom}(G, T)$ is a discrete set. In particular:

1. $\mathrm{Aut}(T) \cong \mathrm{GL}_r(\mathbb{Z})$ (the automorphism group of $\mathbb{Z}^r$), which is discrete.
2. If $G$ is connected and $T$ is a torus in $G$ (i.e. $T \subseteq G$ is a closed subgroup that is a torus), then $T \subseteq Z_G(T)^\circ$ (the torus lies in the connected component of its own centralizer).

***Proof*:**

A homomorphism $\varphi : G \to T$ corresponds, via the anti-equivalence, to a group homomorphism $X(T) \to X(G)$. But $X(T) = \mathbb{Z}^r$ is a free abelian group, and $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}^r, X(G))$ is determined by the images of the $r$ generators, each of which must be a character of $G$. Since the character group $X(G)$ is a finitely generated abelian group (hence discrete, though not necessarily finite), the set $\mathrm{Hom}(G, T) = \mathrm{Hom}(\mathbb{Z}^r, X(G))$ is discrete.

For (1): $\mathrm{Aut}(T) = \mathrm{Hom}(T, T)^\times$ corresponds to $\mathrm{Aut}(X(T)) = \mathrm{Aut}(\mathbb{Z}^r) = \mathrm{GL}_r(\mathbb{Z})$.

For (2): The normalizer $N_G(T)$ acts on $T$ by conjugation, which defines a group homomorphism $c : N_G(T) \to \mathrm{Aut}(T)$. The connected component of the identity $N_G(T)^\circ$ is a connected algebraic group. Because continuous maps from connected spaces to discrete spaces are constant, the restriction of $c$ to $N_G(T)^\circ$ must be trivial. This means $N_G(T)^\circ$ centralizes $T$, giving $N_G(T)^\circ \subseteq Z_G(T)$.

Since $T$ is connected, commutative, and normalizes itself, it is a connected subgroup of $N_G(T)$. Therefore, $T \subseteq N_G(T)^\circ \subseteq Z_G(T)^\circ$.

The rigidity of tori is the reason that "maximal tori" are well-behaved: in chapter 6, we will prove that all maximal tori in a connected algebraic group $G$ are conjugate, and the rigidity ensures that the conjugacy class of a maximal torus is a robust invariant of $G$.

**Exercise 5.2.1**

1. Compute the character group $X(G)$ for: (a) $G = \mathbb{G}_m^2$; (b) $G = \mu_6$ (with $\operatorname{char} k \nmid 6$); (c) $G = \mathbb{G}_m \times \mu_3$ (with $\operatorname{char} k \neq 3$).
2. Let $T = \mathbb{G}_m^2$ with $X(T) = \mathbb{Z}^2$. Enumerate all cocharacters $\lambda \in X_\vee(T) = \mathbb{Z}^2$ with $\|\lambda\| \leq 2$. For each, compute the pairing $\langle \chi, \lambda \rangle$ where $\chi = (1, -1) \in X(T)$.
3. Classify all closed subgroups of $\mathbb{G}_m^2$ (over an algebraically closed field $k$) by classifying all quotients of $\mathbb{Z}^2$. Which are tori? Which are finite?
4. Let $T = \mathbb{G}_m^2$ act on $V = k^2$ via $\rho(t_1, t_2) = \operatorname{diag}(t_1 t_2, t_1 t_2^{-1})$. Compute the weights of $V$, $V^\vee$, $\operatorname{Sym}^2(V)$, and $\bigwedge^2(V)$.
5. Let $T = \left\{ \left(\begin{smallmatrix} t & 0 \\ 0 & t^{-1} \end{smallmatrix}\right) : t \in \mathbb{G}_m \right\} \subseteq \mathrm{SL}_2$. This is a maximal torus in $\mathrm{SL}_2$. Compute $X(T)$ and the weight decomposition of $\mathfrak{sl}_2$ under the adjoint action of $T$. Verify that the nonzero weights are $\pm 2$ (matching the computation of example 3.2.9).
6. Let $T$ be a torus in a connected algebraic group $G$, and let $N_G(T) = \{g \in G : gTg^{-1} = T\}$ be the normalizer. Show that $N_G(T)/Z_G(T)$ is a finite group. (*Hint*: elements of $N_G(T)$ act on $T$ by automorphisms; use the rigidity of tori and the fact that $Z_G(T)$ acts trivially.) This quotient will be the *Weyl group* in chapter 7.

## 5.3 Unipotent Groups

Unipotent groups are the opposite extreme from tori: where tori consist entirely of semisimple elements, unipotent groups consist entirely of elements whose semisimple part is the identity. Where tori are rigid and completely determined by discrete data (the character lattice), unipotent groups are "soft" and, in characteristic 0, completely determined by continuous data (the Lie algebra). The simplest examples are $\mathbb{G}_a$ (the additive group) and $U_n$ (the group of upper-triangular unipotent matrices), and in a precise sense all connected unipotent groups are built from copies of $\mathbb{G}_a$.

### Definitions and First Properties

**Definition 5.3.1: Unipotent Elements and Unipotent Groups (Intrinsic)**

Let $G$ be an affine algebraic group over $k$.

1. An element $g \in G(k)$ is called *unipotent* if $g_s = e$ in the Jordan decomposition (theorem 5.1.4), or equivalently, $\rho(g)$ is unipotent for every representation $\rho : G \to \mathrm{GL}(V)$.
2. The group $G$ is called *unipotent* if every element of $G(\bar{k})$ is unipotent.

This refines the embedding-dependent definition from chapter 1 (definition 1.1.11). The intrinsic definition makes no reference to a specific embedding $G \hookrightarrow \mathrm{GL}_n$: the unipotence is detected by the Jordan decomposition, which by theorem 5.1.4 is independent of the representation.

**Proposition 5.3.2: Equivalent Characterizations of Unipotent Groups**

Let $G$ be an affine algebraic group over an algebraically closed field $k$. The following are equivalent:

1. $G$ is unipotent.
2. For some (equivalently, every) faithful representation $\rho : G \hookrightarrow \mathrm{GL}(V)$, the image $\rho(G)$ is conjugate to a subgroup of $U_n$ (upper-triangular unipotent matrices).
3. The only irreducible representation of $G$ is the trivial representation.
4. For every *nonzero* representation $V$ of $G$, the space of invariants $V^G$ is nonzero.

***Proof*:**
(1) $\Rightarrow$ (2): Let $\rho : G \hookrightarrow \mathrm{GL}(V)$ be a faithful representation with $\dim V = n$. By (1), every element of $\rho(G)$ is unipotent. A standard result in linear algebra (Kolchin's Theorem) states that any subgroup of $\mathrm{GL}(V)$ consisting entirely of unipotent elements can be simultaneously upper-triangularized. Therefore, there exists a basis of $V$ in which $\rho(G)$ maps into $U_n$, meaning $\rho(G)$ is conjugate to a subgroup of $U_n$.

(2) $\Rightarrow$ (3): Let $V$ be an irreducible representation of $G$. By (2), the action of $G$ on $V$ can be upper-triangularized. In particular, the first basis vector of this upper-triangularization is fixed by all elements of $G$. Thus, the space of invariants $V^G$ is nonzero. Since $V^G$ is a subrepresentation and $V$ is irreducible, $V = V^G$. This means $G$ acts trivially on $V$, so $V$ is the one-dimensional trivial representation.

(3) $\Rightarrow$ (4): Let $V$ be a nonzero representation of $G$. By the Fundamental Finiteness Theorem (theorem 2.2.7), $V$ contains a nonzero finite-dimensional sub-comodule $W$, which in turn contains an irreducible sub-comodule $W'$ (by choosing one of minimal dimension). By (3), this irreducible sub-comodule $W'$ is trivial, so it is contained in $V^G$. Thus $V^G \neq 0$.

(4) $\Rightarrow$ (1): Let $\rho : G \hookrightarrow \mathrm{GL}(V)$ be a faithful representation. We will show that $\rho(G) \subseteq U_n$ (in some basis), which implies every element is unipotent. By (4), $V^G \neq 0$. Choose a nonzero fixed vector $v_1 \in V^G$. The quotient $V/kv_1$ is a representation of $G$, and by (4) again, if it is nonzero, it contains a fixed vector $v_2 \in (V/kv_1)^G$. Lifting $v_2$ to $V$ and proceeding by induction on $\dim V$, we obtain a full flag $0 \subset kv_1 \subset kv_1 + kv_2 \subset \cdots \subset V$ stabilized by $G$, with $G$ acting trivially on the successive quotients. In this basis, $\rho(G)$ consists of upper-triangular matrices with 1's on the diagonal (i.e. $\rho(G) \subseteq U_n$). Since all elements of $U_n$ are unipotent, $G$ is unipotent.

**Example 5.3.3: The Basic Unipotent Groups**

1. $\mathbb{G}_a$: The additive group $\mathbb{G}_a(R) = (R, +)$ is unipotent. The embedding $\mathbb{G}_a \hookrightarrow \mathrm{GL}_2$ via $t \mapsto \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$ lands in $U_2$.

2. $U_n$: The group of $n \times n$ upper-triangular matrices with 1's on the diagonal. This is the prototypical unipotent group. It has dimension $\binom{n}{2}$ (the number of strictly upper-triangular entries).

3. The *Heisenberg group*: $H = \left\{ \left( \begin{smallmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{smallmatrix} \right) : a, b, c \in k \right\} \subseteq U_3$. This is a 3-dimensional non-commutative unipotent group. Its center is $Z(H) = \left\{ \left( \begin{smallmatrix} 1 & 0 & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix} \right) \right\} \cong \mathbb{G}_a$, and $H/Z(H) \cong \mathbb{G}_a^2$.

4. In characteristic $p > 0$: the infinitesimal group $\alpha_p = \operatorname{Spec} k[t]/(t^p)$ (example 3.3.4) is unipotent. It is connected, 0-dimensional, and its only $k$-point is the identity. This is a unipotent group that is "invisible" to classical points.

## Representations of Unipotent Groups

The characterization (3) above — unipotent groups have no nontrivial irreducible representations — means that the representation theory of unipotent groups is entirely about *extensions*: every representation is a successive extension of copies of the trivial representation.

**Corollary 5.3.4: Representations of Unipotent Groups are Unipotent**

Let $G$ be a unipotent group and $V$ a finite-dimensional representation of $G$. Then $V$ admits a *composition series* of subrepresentations

$$0 = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_n = V$$

with each successive quotient $V_i/V_{i-1}$ isomorphic to the trivial representation $k$. Equivalently, in some basis, every $g \in G$ acts by an upper-triangular unipotent matrix.

***Proof*:**
By proposition 5.3.2(4), $V^G \neq 0$. Set $V_1 = V^G$ (or a one-dimensional subspace thereof if $V^G$ is larger). Then $G$ acts trivially on $V_1$, and $G$ acts unipotently on $V/V_1$ (since $G$ is still unipotent). By induction on $\dim V$, $V/V_1$ has a composition series with trivial quotients. Pulling back gives the series for $V$.

This should be contrasted with the representation theory of tori (theorem 5.2.9), where every representation *splits* as a direct sum of one-dimensional pieces. For unipotent groups, the representations are filterable but not splittable: the extensions are typically nontrivial. We already saw this in example 2.2.11, where the two-dimensional representation $\left( \begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix} \right)$ of $\mathbb{G}_a$ admits a composition series $0 \subset ke_1 \subset k^2$ with trivial quotients, but the extension does not split.

Connected unipotent groups admit composition series not just at the level of representations, but at the level of the group itself.

**Theorem 5.3.5: Composition Series for Unipotent Groups**

Let $G$ be a *smooth* connected unipotent group over a perfect field $k$. Then $G$ admits a composition series of closed smooth connected normal subgroups

$$\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \cdots \trianglelefteq G_m = G$$

such that each successive quotient $G_i/G_{i-1}$ is isomorphic to $\mathbb{G}_a$.

***Proof Sketch*:**
We use the *descending central series* of $G$. Define $\mathcal{C}^1(G) = G$ and $\mathcal{C}^{i+1}(G) = [\mathcal{C}^i(G), G]$ (the closed subgroup generated by commutators). By proposition 5.3.2, $G$ is isomorphic to a subgroup of $U_n$. Since $U_n$ is a nilpotent group, its lower central series terminates, and therefore $G$'s central series must also terminate: $\mathcal{C}^{N+1}(G) = \{e\}$ for some $N$. Each $\mathcal{C}^i(G)/\mathcal{C}^{i+1}(G)$ is a smooth connected commutative unipotent group (commutative because it is a quotient of two consecutive terms of the central series).

It remains to show that a smooth connected commutative unipotent group $U$ admits a composition series with quotients $\mathbb{G}_a$. When $\operatorname{char} k = 0$, the exponential map (see below) provides an isomorphism $U \cong \mathbb{G}_a^{\dim U}$, so $U$ has a filtration with $\mathbb{G}_a$-quotients by taking hyperplane sections. When $\operatorname{char} k = p > 0$, a fundamental theorem of Lazard guarantees that any non-trivial smooth connected unipotent group over a perfect field contains a closed central subgroup isomorphic to $\mathbb{G}_a$. Modding out by this $\mathbb{G}_a$ and inducting on the dimension completes the argument. (If we had dropped the smoothness assumption, the composition series would also have to include infinitesimal quotients like $\alpha_p$).

The theorem says that connected unipotent groups are "built from $\mathbb{G}_a$" in the same way that connected tori are "built from $\mathbb{G}_m$": the building blocks are iterated extensions. But where tori are always *split* (the extensions are trivial: $\mathbb{G}_m^r$ is a direct product), unipotent groups can have nontrivial extensions, producing non-commutative groups like the Heisenberg group.

The key algebraic property of unipotent groups is that their Lie algebras are *nilpotent*: the bracket operation, iterated enough times, always gives zero.

**Definition 5.3.6: Nilpotent Lie Algebra**

A Lie algebra $\mathfrak{g}$ is called *nilpotent* if the *descending central series* $\mathfrak{g}^1 = \mathfrak{g}$, $\mathfrak{g}^{i+1} = [\mathfrak{g}, \mathfrak{g}^i]$ terminates: $\mathfrak{g}^{N+1} = 0$ for some $N$. Equivalently, $\operatorname{ad}(X_1) \circ \operatorname{ad}(X_2) \circ \cdots \circ \operatorname{ad}(X_N) = 0$ for all $X_i \in \mathfrak{g}$ and sufficiently large $N$.

**Proposition 5.3.7: Lie Algebra of a Unipotent Group is Nilpotent**

Let $G$ be a unipotent affine algebraic group. Then $\mathfrak{g} = \operatorname{Lie}(G)$ is a nilpotent Lie algebra. More precisely, if $G \subseteq U_n$ (upper-triangular unipotent matrices), then $\mathfrak{g} \subseteq \mathfrak{u}_n = \{\text{strictly upper-triangular matrices}\}$, and $\mathfrak{g}^n = 0$.

***Proof*:**
By proposition 5.3.2(2), there exists a faithful representation $\rho : G \hookrightarrow \mathrm{GL}(V)$ such that $\rho(G) \subseteq U_n$. On Lie algebras, $d\rho : \mathfrak{g} \hookrightarrow \mathfrak{gl}(V)$ maps $\mathfrak{g}$ into $\mathfrak{u}_n$ (the Lie algebra of $U_n$, computed in exercise 3.1.1 (2): the strictly upper-triangular matrices). The Lie bracket on $\mathfrak{u}_n$ is $[X, Y] = XY - YX$, and for strictly upper-triangular matrices, the product $XY$ is "more upper-triangular" than $X$ and $Y$ individually (the nonzero entries move further from the diagonal). After at most $n-1$ iterated brackets, we reach the zero matrix. Hence $\mathfrak{g}^n = 0$.

## The Exponential Map in Characteristic 0

In characteristic 0, the relationship between unipotent groups and nilpotent Lie algebras is as tight as possible: they are equivalent, via the exponential map.

For a nilpotent matrix $X \in \mathfrak{u}_n(k)$ with $X^n = 0$, the exponential series

$$\exp(X) = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots + \frac{X^{n-1}}{(n-1)!}$$

is a *finite* sum (all terms from $X^n$ onward vanish), so it makes algebraic sense over any field where the denominators $m!$ are invertible — that is, in characteristic 0. The inverse is the logarithm:

$$\log(I + X) = X - \frac{X^2}{2} + \frac{X^3}{3} - \cdots + (-1)^{n-2}\frac{X^{n-1}}{n-1}$$

again a finite sum for nilpotent $X$.

**Theorem 5.3.8: Exponential Isomorphism (Characteristic 0)**

Let $G$ be a connected unipotent group over a field $k$ of characteristic 0, and let $\mathfrak{g} = \mathrm{Lie}(G)$. Then the exponential map

$$\exp : \mathfrak{g} \xrightarrow{\sim} G$$

is an isomorphism of $k$-varieties (though not of groups, unless $G$ is commutative). Its inverse is the logarithm $\log : G \xrightarrow{\sim} \mathfrak{g}$.

***Proof Sketch*:**
Embed $G \hookrightarrow U_n \subseteq \mathrm{GL}_n$ via a faithful representation. On the ambient group $U_n$, the exponential $\exp : \mathfrak{u}_n \to U_n$ is a polynomial isomorphism of varieties (a finite sum, with polynomial inverse log). It restricts to a morphism $\exp : \mathfrak{g} \to G$ because $\mathfrak{g}$ is a Lie subalgebra of $\mathfrak{u}_n$ and $G$ is a closed subgroup of $U_n$. The key point is that the Baker–Campbell–Hausdorff (BCH) formula

$$\log(\exp(X)\exp(Y)) = X + Y + \tfrac{1}{2}[X, Y] + \tfrac{1}{12}([X, [X, Y]] - [Y, [X, Y]]) + \cdots$$

is again a finite sum (since $\mathfrak{g}$ is nilpotent), and it shows that the group law on $G$ is expressed entirely in terms of the Lie bracket on $\mathfrak{g}$. In particular, exp maps $\mathfrak{g}$ bijectively onto $G$ (bijectivity follows from log being an inverse), and it is a morphism of varieties (being a polynomial map).

That exp is not a group homomorphism (unless $G$ is commutative) is visible from the BCH formula: $\exp(X)\exp(Y) = \exp(X + Y + \frac{1}{2}[X, Y] + \cdots) \neq \exp(X + Y)$ in general. But it is an isomorphism of the underlying *varieties*, which suffices to transfer information between $G$ and $\mathfrak{g}$.

**Corollary 5.3.9: Unipotent Groups in Characteristic 0 are Determined by $\mathfrak{g}$**

Over a field of characteristic 0, the functor $G \mapsto \mathrm{Lie}(G)$ is an equivalence of categories:

$$\left\{ \begin{array}{c} \text{Connected unipotent} \\ \text{algebraic groups} \end{array} \right\} \xleftrightarrow{\sim} \left\{ \begin{array}{c} \text{Finite-dimensional} \\ \text{nilpotent Lie algebras} \end{array} \right\}$$

The quasi-inverse sends a nilpotent Lie algebra $\mathfrak{g}$ to the group with underlying variety $\mathfrak{g}$ and multiplication given by the BCH formula.

This is a remarkable contrast with tori: tori are classified by discrete data ($X(T) \cong \mathbb{Z}^r$, a lattice), while unipotent groups in characteristic 0 are classified by continuous data ($\mathfrak{g}$, a nilpotent Lie algebra). Together, they illustrate the two extremes of the Jordan decomposition: the "semisimple side" is rigid and combinatorial, the "unipotent side" is soft and Lie-theoretic.

## Failure of the Exponential in Characteristic $p$

In characteristic $p > 0$, the exponential map breaks down because the denominators $m!$ are not invertible for $m \geq p$. This has dramatic consequences for the structure theory.

**Example 5.3.10: The Exponential Fails for $U_2$ in Characteristic $p$**

Let $k$ have characteristic $p$ and consider $\mathfrak{u}_2 = \{X = \left(\begin{smallmatrix} 0 & a \\ 0 & 0 \end{smallmatrix}\right) : a \in k\}$. The formal exponential gives:

$$\exp(X) = I + X = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

which is well-defined (only the first two terms survive since $X^2 = 0$). So for $U_2$, the exponential works fine even in characteristic $p$.

Now consider $\mathfrak{u}_3$. For $X = \left(\begin{smallmatrix} 0 & a & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{smallmatrix}\right)$:

$$\exp(X) = I + X + \frac{X^2}{2} = \begin{pmatrix} 1 & a & c + ab/2 \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

When $\mathrm{char} k = 2$, the term $ab/2$ is undefined (division by $2 = 0$). The exponential map $\mathfrak{u}_3 \to U_3$ does not exist as a morphism of varieties in characteristic 2.

**Example 5.3.11: Non-Isomorphic Unipotent Groups with the Same Lie Algebra**

In characteristic $p$, the functor Lie is not faithful on unipotent groups. The simplest example: the group schemes $\alpha_p$ and $\mathbb{G}_a[F] := \ker(F : \mathbb{G}_a \to \mathbb{G}_a)$ (the Frobenius kernel of $\mathbb{G}_a$) are both connected unipotent group schemes of dimension 0 with Lie $= k$. But $\alpha_p = \mathbb{G}_a[F]$ (they are the same group scheme by definition), so this is not yet a counterexample.

A genuine example: consider the two-dimensional unipotent groups over a field of characteristic

$p \geq 3$:

$$G_1 = \mathbb{G}_a \times \mathbb{G}_a, \qquad G_2 = \{(a,b) \in \mathbb{G}_a^2 : \text{mult. } (a_1,b_1)(a_2,b_2) = (a_1+a_2, b_1+b_2+a_1^p a_2)\}$$

Both have Lie algebra $\mathfrak{g} = k^2$ with trivial bracket (since both are extensions of $\mathbb{G}_a$ by $\mathbb{G}_a$, the Lie algebra is abelian). But $G_1$ is commutative while $G_2$ is not ($G_2$ is a nontrivial extension using the $p$-th power, which is invisible to Lie). The groups $G_1$ and $G_2$ are not isomorphic, yet $\mathrm{Lie}(G_1) \cong \mathrm{Lie}(G_2)$.

These examples reinforce the moral from chapter 3 (§3.3): in characteristic $p$, the Lie algebra is an imperfect tool. For unipotent groups, the correct replacement is the *p-Lie algebra* (restricted Lie algebra), which encodes the $p$-th power operation and determines the infinitesimal structure more faithfully. A full treatment belongs to Part II.

## Interaction Between Tori and Unipotent Groups

The Jordan decomposition says that inside a commutative group, the semisimple and unipotent elements form complementary subgroups (exercise 5.1.1 (5)). For general (non-commutative) groups, the interaction is more subtle, but one clean statement holds:

**Proposition 5.3.12: Tori and Unipotent Groups Have Trivial Intersection**

Let $G$ be an affine algebraic group over an algebraically closed field $k$, $T \subseteq G$ a torus, and $U \subseteq G$ a unipotent subgroup. Then their scheme-theoretic intersection is trivial: $T \cap U = \{e\}$.

***Proof*:**
At the level of geometric points: every element of $(T \cap U)(\overline{k})$ is simultaneously semisimple (since $T$ is a torus and all its elements are semisimple, proposition 5.2.7(4)) and unipotent (since $U$ is unipotent). An element that is both semisimple and unipotent must be the identity: in any faithful representation, its only eigenvalue is 1 (unipotent) and it is diagonalizable (semisimple), so it is the identity matrix. Hence $(T \cap U)(\overline{k}) = \{e\}$.

At the scheme level: $T \cap U$ is a closed subgroup of $T$ (hence diagonalizable by proposition 5.2.7(3)) and also a closed subgroup of $U$ (hence unipotent). A diagonalizable unipotent group scheme has $X(T \cap U) = 0$ (every character of an algebraic group defines a one-dimensional representation, and by proposition 5.3.2(3), the only irreducible representation of a unipotent group is the trivial one). Because a diagonalizable group is completely determined by its character group, $T \cap U = D(0) = \{e\}$.

This proposition is the group-theoretic reason that the Jordan decomposition works: the "torus part" and the "unipotent part" of an element cannot interfere with each other. In chapter 6, this will ensure that the decomposition of a solvable group $B$ into its torus $T$ and unipotent radical $U$ is well-behaved: $T \cap U = \{e\}$ and (in many cases) $B = T \ltimes U$.

**Example 5.3.13: The Borel as a Semidirect Product**

Let $B_n \subseteq \mathrm{GL}_n$ be the upper-triangular Borel subgroup. Define $T = \mathbb{G}_m^n$ (the diagonal torus) and $U = U_n$ (the upper-triangular unipotent matrices). Then $T \cap U = \{I\}$ and $B_n = T \ltimes U$ (semidirect product): every upper-triangular invertible matrix factors uniquely as a diagonal matrix times a unipotent upper-triangular matrix. Functorially, for any $k$-algebra $R$:

$$B_n(R) = T(R) \ltimes U_n(R), \qquad \begin{pmatrix} a_1 & * & \cdots \\ 0 & a_2 & \cdots \\ \vdots & & \ddots \end{pmatrix} = \begin{pmatrix} a_1 & & \\ & a_2 & \\ & & \ddots \end{pmatrix} \begin{pmatrix} 1 & a_1^{-1}* & \cdots \\ 0 & 1 & \cdots \\ \vdots & & \ddots \end{pmatrix}$$

The torus $T$ acts on $U$ by conjugation: $\mathrm{diag}(t_1, \ldots, t_n) \cdot (I + X) = I + tXt^{-1}$, where the $(i,j)$-entry of $tXt^{-1}$ is $t_i t_j^{-1} X_{ij}$. The weights of this action are exactly the *positive roots* $\varepsilon_i - \varepsilon_j$ for $i < j$ (cf. example 5.2.10).

This example is the concrete prototype for the general theory of chapter 6: every Borel subgroup $B$ of a reductive group decomposes as $B = T \ltimes U$ (its maximal torus times its unipotent radical), and the weights of $T$ acting on $\mathrm{Lie}(U)$ are the positive roots.

## Exercise 5.3.1

1. Let $G = U_3$ (the $3 \times 3$ upper-triangular unipotent group) act on $V = k^3$ by left multiplication. Write down the composition series $0 \subset V_1 \subset V_2 \subset V_3 = V$ with trivial quotients. Is this representation decomposable (i.e. does $V \cong W \oplus W'$ for nonzero subrepresentations)?

2. Let $H$ be the Heisenberg group (the $3 \times 3$ upper-triangular unipotent group with 1's on the diagonal and the $(1,3)$-entry free). Compute $\mathrm{Lie}(H)$ and verify it is nilpotent. What is the nilpotency class (the smallest $N$ with $\mathfrak{g}^{N+1} = 0$)? Compute the descending central series of $H$ (as an algebraic group).

3. (Requires $\mathrm{char} k = 0$.) Use the BCH formula to compute the group law on the Heisenberg group $H$ from exercise 5.3.1 (2) via the exponential map. That is, express $\exp(X) \cdot \exp(Y) = \exp(Z)$ where $X, Y \in \mathrm{Lie}(H)$ and $Z$ is given by the BCH formula.

4. Show that a unipotent group $G$ has no nontrivial characters: $X(G) = 0$. (*Hint*: a character $\chi : G \to \mathbb{G}_m$ would make $\mathbb{G}_m$ a quotient of $G$; use the fact that a quotient of a unipotent group is unipotent, and $\mathbb{G}_m$ is not unipotent.)

5. Let $G$ be a connected commutative unipotent group over an algebraically closed field $k$ of characteristic 0. Show that $G \cong \mathbb{G}_a^n$ for some $n$. (*Hint*: use the exponential isomorphism $G \cong \mathfrak{g}$; since $G$ is commutative, $\mathfrak{g}$ is abelian, so $\mathfrak{g} \cong k^n$ and the BCH formula reduces to $\exp(X)\exp(Y) = \exp(X+Y)$.)

6. Let $B \subseteq \mathrm{SL}_3$ be the upper-triangular Borel subgroup. Write $B = T \ltimes U$ where $T$ is the diagonal torus and $U$ is the unipotent radical. Compute $\dim T$, $\dim U$, and verify $\dim B = \dim T + \dim U$. Determine the weights of the adjoint action of $T$ on $\mathrm{Lie}(U)$.

# 6

# Solvable Groups and Borel Subgroups

We have now assembled the two atomic building blocks — tori (chapter 5, §5.2) and unipotent groups (chapter 5, §5.3) — and the next step is to understand how they fit together. The answer is the *solvable groups*: groups that admit a composition series with commutative quotients, or equivalently, groups whose derived series terminates. The prototype is the upper-triangular Borel subgroup $B_n \subseteq \mathrm{GL}_n$, which decomposes as a semidirect product $B_n = T \ltimes U_n$ of a torus by a unipotent group (example 5.3.13).

This chapter has two goals. First, we prove the *Lie–Kolchin theorem*, which says that every connected solvable group is conjugate into $B_n$ (the "upper-triangular" form), and derive the structure theorem for solvable groups and Borel's fixed-point theorem. Second, we use these tools to study the *Borel subgroups* of a general algebraic group $G$: maximal connected solvable subgroups. The main results are that $G/B$ is a projective variety (the *flag variety*), that all Borel subgroups are conjugate, and that all maximal tori are conjugate. These conjugacy theorems are the gateway to the classification theory of chapter 7: they ensure that the root system, defined via a choice of maximal torus and Borel, is independent of that choice.

## 6.1 Solvable Groups and the Lie–Kolchin Theorem

**Definition 6.1.1: Solvable Algebraic Group**

Let $G$ be an affine algebraic group. The *derived series* of $G$ is defined by $\mathcal{D}^0(G) = G$ and $\mathcal{D}^{i+1}(G) = [\mathcal{D}^i(G), \mathcal{D}^i(G)]$ (the smallest closed subgroup containing all commutators of elements in $\mathcal{D}^i(G)$). The group $G$ is called *solvable* if $\mathcal{D}^n(G) = \{e\}$ for some $n \geq 0$.

Geometrically, $\mathcal{D}^{i+1}(G)$ is the intersection of all closed subgroups $H \subseteq \mathcal{D}^i(G)$ such that $\mathcal{D}^i(G)/H$ is commutative. (Note that for a general $k$-algebra $R$, the $R$-points $\mathcal{D}^{i+1}(G)(R)$ can be strictly larger than the abstract subgroup generated by commutators $[g, h]$ in $\mathcal{D}^i(G)(R)$, because the algebraic group is generated geometrically).

The derived series is a descending chain of closed connected normal subgroups of $G$ (connectedness is preserved at each step because the commutator map $G \times G \to G$ is a morphism, its image is connected and contains the identity, and the closed subgroup generated by such a set is always connected).

**Example 6.1.2: Solvable Groups**

1. *Commutative groups* are solvable: $\mathcal{D}^1(G) = [G, G] = \{e\}$, so the derived series terminates in one step. In particular, tori and $\mathbb{G}_a$ are solvable.
2. *Unipotent groups* are solvable: the descending central series $\mathcal{C}^i(G)$ refines the derived series (since $\mathcal{D}^i(G) \subseteq \mathcal{C}^i(G)$), and we proved in theorem 5.3.5 that the central series terminates.
3. *The Borel subgroup* $B_n \subseteq \mathrm{GL}_n$ of upper-triangular matrices is solvable. We have $\mathcal{D}^1(B_n) = [B_n, B_n] = U_n$ (the commutator of two upper-triangular matrices is upper-triangular unipotent), and $U_n$ is solvable by (2).
4. $\mathrm{GL}_n$ is *not* solvable for $n \geq 2$: $\mathcal{D}^1(\mathrm{GL}_n) = \mathrm{SL}_n$ (the commutator subgroup consists of determinant-1 matrices), and $\mathcal{D}^1(\mathrm{SL}_n) = \mathrm{SL}_n$ for $n \geq 2$ (since $\mathrm{SL}_n$ is its own commutator subgroup, being "almost simple"). So the derived series stabilizes at $\mathrm{SL}_n$ and never reaches $\{e\}$.

**Proposition 6.1.3: Basic Properties of Solvable Groups**

1. Subgroups and quotients of solvable groups are solvable.
2. Extensions of solvable groups are solvable: if $1 \to N \to G \to Q \to 1$ is a short exact sequence with $N$ and $Q$ solvable, then $G$ is solvable.
3. In any affine algebraic group $G$, there exists a unique maximal connected solvable normal subgroup, called the *radical* of $G$, denoted $R(G)$.

***Proof*:**
*(1):* If $H \subseteq G$, then $\mathcal{D}^i(H) \subseteq \mathcal{D}^i(G)$, so $\mathcal{D}^n(G) = \{e\}$ implies $\mathcal{D}^n(H) = \{e\}$. If $\pi : G \to Q$ is surjective, then $\pi(\mathcal{D}^i(G)) = \mathcal{D}^i(Q)$ (surjective maps preserve commutators), so $\mathcal{D}^n(G) = \{e\}$ implies $\mathcal{D}^n(Q) = \{e\}$.

*(2):* If $\mathcal{D}^m(N) = \{e\}$ and $\mathcal{D}^r(Q) = \{e\}$, then $\mathcal{D}^r(G)$ maps to $\mathcal{D}^r(Q) = \{e\}$ under $\pi$, so $\mathcal{D}^r(G) \subseteq N$. Then $\mathcal{D}^{r+m}(G) = \mathcal{D}^m(\mathcal{D}^r(G)) \subseteq \mathcal{D}^m(N) = \{e\}$.

*(3):* Let $R_1, R_2$ be connected solvable normal subgroups of $G$. The product $R_1R_2$ is a connected normal subgroup (it is the image of $R_1 \times R_2 \to G$, which is connected), and it is solvable by (2): the sequence $1 \to R_1 \to R_1R_2 \to R_1R_2/R_1 \to 1$ has $R_1$ solvable and $R_1R_2/R_1 \cong R_2/(R_1 \cap R_2)$ solvable (a quotient of $R_2$). Since $G$ is Noetherian (its subgroups satisfy the ascending chain condition on closed subgroups), the collection of connected solvable normal subgroups has a maximal element $R(G) = R_1R_2 \cdots$.

## The Lie–Kolchin Theorem

The fundamental theorem about solvable groups says that in any representation, they can be simultaneously put in upper-triangular form. This is the group-theoretic analogue of the linear-algebraic fact that commuting matrices are simultaneously triangularizable.

**Theorem 6.1.4: Lie–Kolchin Theorem**

Let $G$ be a connected solvable affine algebraic group over an algebraically closed field $k$, and let $\rho : G \to \mathrm{GL}(V)$ be a representation on a finite-dimensional vector space $V \neq 0$. Then there exists a complete flag $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ (with $\dim V_i = i$) that is stabilized by $G$. Equivalently, in some basis of $V$, every element of $\rho(G)$ is upper-triangular.

***Proof*:**
We argue by induction on $\dim V$. The base case $\dim V = 1$ is trivial (every endomorphism of a 1-dimensional space is upper-triangular).

*Step 1: Find a common eigenvector.* We need to show that $V$ contains a one-dimensional subrepresentation: a line $L = kv$ such that $G \cdot L = L$, i.e. $g \cdot v = \chi(g)v$ for some character $\chi : G \to \mathbb{G}_m$.

We induct on $\dim G$. If $\dim G = 0$, then $G$ is finite and connected over $k = \overline{k}$, so $G = \{e\}$ and every vector is an eigenvector.

If $\dim G > 0$, consider the commutator subgroup $N = [G, G]$ (which is connected because $G$ is connected). Since $G$ is solvable, $N \subsetneq G$: the derived series is strictly decreasing until it reaches $\{e\}$, so $[G, G] \neq G$ (otherwise the series would stabilize). Moreover, $N$ is connected, solvable, normal, and $\dim N < \dim G$.

By the induction hypothesis applied to $N$ acting on $V$, there exists a character $\chi : N \to \mathbb{G}_m$ and a nonzero eigenvector $v \in V$ with $n \cdot v = \chi(n)v$ for all $n \in N$. Define the eigenspace:

$$W_\chi = \{w \in V \mid n \cdot w = \chi(n)w \text{ for all } n \in N\}$$

This is nonzero (it contains $v$).

*Step 2: $G$ preserves $W_\chi$.* For $g \in G(k)$ and $w \in W_\chi$, we show $g \cdot w \in W_\chi$. For any $n \in N(k)$:

$$n \cdot (g \cdot w) = g \cdot (g^{-1}ng \cdot w) = g \cdot \chi(g^{-1}ng)w$$

Since $N \trianglelefteq G$, we have $g^{-1}ng \in N$. We claim $\chi(g^{-1}ng) = \chi(n)$. The map $g \mapsto (n \mapsto \chi(g^{-1}ng))$ defines, for each $g \in G$, a character of $N$. This gives a morphism $G \to X(N)$ (the character group of $N$). The character group of any algebraic group is discrete (since characters correspond to linearly independent group-like elements in the coordinate ring $\mathcal{O}(N)$). Since $G$ is connected, a morphism from $G$ to a discrete set must be constant. At $g = e$, the character is $\chi$ itself. So $\chi(g^{-1}ng) = \chi(n)$ for all $g \in G$, and:

$$n \cdot (g \cdot w) = \chi(n)(g \cdot w)$$

Hence $g \cdot w \in W_\chi$, and $W_\chi$ is $G$-stable.

*Step 3: Reduce to a smaller representation.* If $W_\chi = V$, then $N$ acts by scalars on $V$ (every element acts as $\chi(n) \cdot \mathrm{id}_V$). But $N$ is generated by commutators of $G$, and the determinant of any commutator $xyx^{-1}y^{-1}$ in $\mathrm{GL}(V)$ is precisely 1. Therefore, taking the determinant of the scalar action gives $\det(\chi(n) \cdot \mathrm{id}_V) = \chi(n)^{\dim V} = 1$ for all $n \in N$. This implies the continuous image $\chi(N)$ is a finite subgroup of $\mathbb{G}_m$. Because $N$ is connected, $\chi(N)$ must also be connected, forcing $\chi(n) = 1$ for all $n \in N$.

Thus, $N$ actually acts trivially on $V$, and the representation $\rho : G \to \mathrm{GL}(V)$ factors through the quotient $G/N$. Since $G/N$ is a connected solvable group of strictly smaller dimension, the induction hypothesis on $\dim G$ guarantees a common eigenvector for $G/N$, and hence for $G$.

If $W_\chi \subsetneq V$, then $W_\chi$ is a $G$-stable subspace of dimension $< \dim V$. By induction on $\dim V$, $G$ has a common eigenvector in $W_\chi$.

In either case, we obtain a line $V_1 = kv_1$ that is $G$-stable.

*Step 4: Build the flag.* The quotient $V/V_1$ is a representation of $G$ of dimension $n - 1$. By induction on $\dim V$, $G$ preserves a complete flag in $V/V_1$. Pulling back gives a complete flag $V_1 \subset V_2 \subset \cdots \subset V_n = V$ stabilized by $G$.

The proof combines two inductions: one on $\dim V$ (to build the flag step by step) and one on $\dim G$ (to find the initial common eigenvector). The key mechanism is the rigidity of characters: conjugation by the connected group $G$ cannot change the (discrete) character $\chi$ of the normal subgroup $N$.

**Corollary 6.1.5: Connected Solvable Groups are Triangularizable**

Let $G$ be a connected solvable affine algebraic group over $k = \overline{k}$. Then for any faithful representation $\rho : G \hookrightarrow \mathrm{GL}(V)$, the image $\rho(G)$ is conjugate to a subgroup of the upper-triangular Borel $B_n \subseteq \mathrm{GL}_n$ (where $n = \dim V$).

***Proof*:**
The Lie–Kolchin theorem gives a $G$-stable flag. Choosing a basis adapted to this flag puts $G$ in upper-triangular form.

## Structure of Connected Solvable Groups

The Lie–Kolchin theorem gives us a concrete picture of connected solvable groups: they are "built from a torus and a unipotent part." We now make this precise.

**Definition 6.1.6: Unipotent Radical of a Solvable Group**

Let $G$ be a connected solvable affine algebraic group. The *unipotent radical* $R_u(G)$ is the maximal connected normal unipotent subgroup of $G$.

The existence of $R_u(G)$ follows from the same ascending chain argument as for the radical (proposition 6.1.3(3)): the product of two connected normal unipotent subgroups is again connected normal unipotent (normality is clear, and the product of unipotent groups in a solvable group is unipotent, which follows from the Lie–Kolchin theorem: in upper-triangular form, the product of unipotent upper-triangular matrices is unipotent upper-triangular).

**Theorem 6.1.7: Structure Theorem for Connected Solvable Groups**

Let $G$ be a connected solvable affine algebraic group over an algebraically closed field $k$. Then:

1. $R_u(G) = G_u := \{g \in G(k) \mid g \text{ is unipotent}\}$, the set of all unipotent elements, which is a closed connected normal subgroup.
2. $G/R_u(G)$ is a torus.
3. There exists a maximal torus $T \subseteq G$ such that $G = T \ltimes R_u(G)$ (semidirect product). Any two such tori are conjugate by an element of $R_u(G)$.

***Proof*:**
By the Lie–Kolchin theorem, embed $G \hookrightarrow B_n$ for some $n$. Inside $B_n = T_0 \ltimes U_n$, the unipotent elements form $U_n$ (a matrix is unipotent iff its diagonal entries are all 1). So $G_u = G \cap U_n$ is a closed subgroup.

*(1):* We show $G_u$ is connected, normal, and equals $R_u(G)$. Normality: if $g \in G$ and $u \in G_u$, then $gug^{-1}$ is unipotent (by proposition 5.1.6, conjugation preserves the Jordan decomposition), so $gug^{-1} \in G_u$. The quotient $G/G_u$ is a connected solvable group all of whose elements are

semisimple: in upper-triangular form, quotienting by $G_u = G \cap U_n$ removes the unipotent part, leaving only diagonal elements. A connected group consisting entirely of semisimple elements is a torus. So $G/G_u$ is a torus.

To see that $G_u$ is connected, consider the commutator subgroup $[G, G]$. Being connected and generated by commutators of upper-triangular matrices, $[G, G] \subseteq U_n$, so $[G, G] \subseteq G_u$. The quotient $G/[G, G]$ is a connected commutative algebraic group, which decomposes globally as the direct product of a torus and a connected unipotent group, $T_1 \times U_1$. The unipotent elements of $G/[G, G]$ form exactly the connected subgroup $\{1\} \times U_1$. Since $G_u$ is the preimage of $U_1$ under the projection $G \to G/[G, G]$, and both $[G, G]$ and $U_1$ are connected, $G_u$ must be connected. Being connected, normal, and unipotent, $G_u \subseteq R_u(G)$. Conversely, $R_u(G) \subseteq G_u$ (every element of $R_u(G)$ is unipotent). So $R_u(G) = G_u$.

*(2)* follows: $G/R_u(G) = G/G_u$ is a torus.

*(3):* The fact that the sequence $1 \to R_u(G) \to G \to G/R_u(G) \to 1$ splits (i.e., that $G$ contains a closed subgroup $T$ mapping isomorphically onto the quotient $G/R_u(G)$) is a fundamental theorem of algebraic groups. Such a subgroup $T$ is automatically a maximal torus of $G$. Since $T \cap R_u(G) = \{e\}$ (proposition 5.3.12) and $\dim T + \dim R_u(G) = \dim G$, we get $G = T \ltimes R_u(G)$.

Conjugacy of maximal tori: if $T_1, T_2$ are two maximal tori in $G$, both map isomorphically onto $G/R_u(G)$. The two sections $T_1, T_2 \hookrightarrow G$ of the projection $G \to G/R_u(G)$ differ by conjugation by an element of $R_u(G)$. This can be proved by induction on $\dim R_u(G)$, using the fact that $H^1(T, \mathbb{G}_a) = 0$ (the first cohomology of a torus acting rationally on an affine space vanishes, since the torus representation on $\mathbb{G}_a$ decomposes into weight spaces and each contributes trivially to cohomology).

**Example 6.1.8: The Structure of $B_2$**

Let $B_2 = \left\{\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) : ad \neq 0\right\} \subseteq \mathrm{GL}_2$. The derived subgroup is $[B_2, B_2] = \{\left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right)\} = U_2 \cong \mathbb{G}_a$ (the commutator of two upper-triangular matrices is upper-triangular unipotent). The unipotent radical is $R_u(B_2) = U_2$. The maximal torus is $T = \{\left(\begin{smallmatrix} a & 0 \\ 0 & d \end{smallmatrix}\right)\} \cong \mathbb{G}_m^2$. We have $B_2 = T \ltimes U_2$, confirming the structure theorem. The torus acts on $U_2 \cong \mathbb{G}_a$ by $\mathrm{diag}(a, d) \cdot \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & ad^{-1}t \\ 0 & 1 \end{smallmatrix}\right)$, so the weight of $T$ on $\mathrm{Lie}(U_2) = k$ is $\varepsilon_1 - \varepsilon_2$ — the single positive root of $\mathrm{GL}_2$.

## Borel's Fixed-Point Theorem

The Lie–Kolchin theorem says that a connected solvable group has a common eigenvector in any *linear* representation. Borel's fixed-point theorem is the geometric generalization: a connected solvable group has a *fixed point* on any complete variety.

**Theorem 6.1.9: Borel's Fixed-Point Theorem**

Let $G$ be a connected solvable affine algebraic group over an algebraically closed field $k$, and let $X$ be a non-empty complete $G$-variety. Then $G$ has a fixed point: $X^G \neq \emptyset$.

***Proof*:**
By the Closed Orbit Lemma (theorem 2.4.15), $X$ contains a closed orbit $O = G \cdot x$. Since $X$ is complete and $O$ is closed in $X$, $O$ is complete. Since $O$ is a single $G$-orbit, $O \cong G/H$ for some closed subgroup $H$ (theorem 4.3.2), so $O$ is a complete homogeneous space for $G$.

We claim $O$ is a point. Suppose for contradiction that $\dim O > 0$. Since $O$ is projective (a complete variety), there exists a very ample line bundle $\mathcal{L}$ on $O$, giving a closed immersion $O \hookrightarrow \mathbb{P}(H^0(O, \mathcal{L})^\vee)$. The $G$-action on $O$ lifts to a linear action on the finite-dimensional vector space $V = H^0(O, \mathcal{L})^\vee$: this is because $O = G/H$ is a homogeneous space and $\mathcal{L}$ admits a $G$-linearization (see [5] for the general theory of $G$-linearized line bundles on homogeneous spaces).

By the Lie–Kolchin theorem (theorem 6.1.4), $G$ has a common eigenvector $v \in V$, hence a fixed point $[v] \in \mathbb{P}(V)$. Since $O$ is a $G$-stable closed subvariety of $\mathbb{P}(V)$ and $[v]$ is a $G$-fixed point of $\mathbb{P}(V)$, we need $[v] \in O$ (a fixed point of $\mathbb{P}(V)$ that lies in $O$ is a fixed point of $O$). But does $[v]$ necessarily lie in $O$?

We argue more carefully. The embedding $O \hookrightarrow \mathbb{P}(V)$ is equivariant, and $G$ acts on $\mathbb{P}(V)$ linearly. By Lie–Kolchin, $G$ preserves a complete flag $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$. In particular, $G$ fixes the point $[V_1] \in \mathbb{P}(V)$. The closed subvariety $O \subseteq \mathbb{P}(V)$ is $G$-stable, and the intersection $O \cap \{[V_1]\}$ is either empty or $\{[V_1]\}$. If it is non-empty, we have a $G$-fixed point on $O$, contradicting $\dim O > 0$.

If $[V_1] \notin O$, consider the projection from $[V_1]$: this is a $G$-equivariant rational map $\mathbb{P}(V) \dashrightarrow \mathbb{P}(V/V_1)$ defined away from $[V_1]$. Since $[V_1] \notin O$, this restricts to a $G$-equivariant morphism $\pi : O \to \mathbb{P}(V/V_1)$. The image $O'$ is a closed $G$-stable subvariety of $\mathbb{P}(V/V_1)$ (since $O$ is complete). By induction on $n = \dim V$, $G$ has a fixed point $y \in O'$.

Now consider the fiber $F = \pi^{-1}(y) \subseteq O$. Because $y$ is $G$-fixed, $F$ is a $G$-stable closed subvariety of $O$. The fiber of the projection $\mathbb{P}(V) \dashrightarrow \mathbb{P}(V/V_1)$ over $y$ is a projective line $\mathbb{P}^1$. Because $[V_1] \notin O$, $F$ is a closed subvariety of $\mathbb{P}^1 \setminus \{[V_1]\} \cong \mathbb{A}^1$. But $F$ is also complete (being a closed subvariety of the complete variety $O$). A complete subvariety of an affine space must be a finite set of points. Thus, $F$ is a finite, non-empty, $G$-stable set. Since $G$ is connected, its action on a finite set must be trivial, meaning every point of $F$ is a $G$-fixed point in $O$.

In either case, $O$ contains a $G$-fixed point. But $O \cong G/H$ is a homogeneous space, and a fixed point means $H = G$, so $\dim O = 0$. Contradiction. Therefore $\dim O = 0$ from the start, and $O = \{x\}$ is a $G$-fixed point.

The proof uses the projective geometry of $X$ (via ample line bundles) to reduce the geometric fixed-point problem to the linear Lie–Kolchin theorem. This is a fundamental instance of the principle "geometry over algebra": the complete variety's projective embedding provides the bridge between group actions on varieties and group actions on vector spaces.

**Corollary 6.1.10: Solvable Groups Preserve Flags**

Let $G$ be a connected solvable group acting on a flag variety $\mathrm{Fl}(d_1, \ldots, d_r; n)$ or a Grassmannian $\mathrm{Gr}(d, n)$. Then $G$ has a fixed point.

***Proof*:**
These varieties are projective (example 4.2.4, corollary 4.3.3), so theorem 6.1.9 applies directly.

### Exercise 6.1.1

1. Compute the derived series of $B_3 \subseteq \mathrm{GL}_3$ (the $3 \times 3$ upper-triangular Borel). At which step does it reach $\{e\}$? Compare with the descending central series of $U_3$ from exercise 5.3.1 (2).
2. Give a direct proof of the Lie–Kolchin theorem for $G = B_2$ (the $2 \times 2$ upper-triangular Borel) acting on $V = k^2$. That is, without appealing to the general theorem, find a common eigenvector for all elements of $B_2$.
3. Show that the Lie–Kolchin theorem fails without the connectedness hypothesis. (*Hint*: the symmetric group $S_3$ is solvable; consider its two-dimensional irreducible representation.)
4. Let $G = \mathrm{GL}_n$. Show that $R(G) = Z(G) = \mathbb{G}_m$ (scalar matrices). Deduce that $\mathrm{GL}_n$ is *not* solvable for $n \geq 2$, but its radical is a torus.
5. Let $G = B_2$ act on $\mathbb{P}^1$ via $\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) \cdot [x : y] = [ax + by : dy]$. Find the fixed points of this action directly, verifying Borel's fixed-point theorem.
6. Let $G = \{\left(\begin{smallmatrix} a & b \\ 0 & 1 \end{smallmatrix}\right) : a \in \mathbb{G}_m, b \in \mathbb{G}_a\}$ (the "$ax + b$" group). Show $G$ is connected solvable. Compute $R_u(G)$, find a maximal torus $T$, and verify $G = T \ltimes R_u(G)$. Determine the weight of $T$ on $\mathrm{Lie}(R_u(G))$.

## 6.2 Borel Subgroups and Flag Varieties

We now turn from the internal structure of solvable groups to the question of how solvable subgroups sit inside a general affine algebraic group $G$. The key objects are the *Borel subgroups* — the maximal connected solvable subgroups — and the quotients $G/B$, which are the *flag varieties*. The main results of this section are: $G/B$ is projective, all Borel subgroups are conjugate, and all maximal tori are conjugate. These are proved using Borel's fixed-point theorem as the principal tool, and they are the structural foundations on which the root system classification of chapter 7 rests.

Throughout this section, $k$ denotes an algebraically closed field and $G$ a connected affine algebraic group over $k$.

### Borel Subgroups

**Definition 6.2.1: Borel Subgroup**

A *Borel subgroup* of $G$ is a maximal connected solvable closed subgroup $B \subseteq G$. Equivalently, $B$ is a closed connected solvable subgroup that is not properly contained in any other closed connected solvable subgroup.

Borel subgroups exist by a dimension argument: the collection of connected solvable closed subgroups is non-empty (it contains $\{e\}$, or any torus, or any copy of $\mathbb{G}_a$) and partially ordered by inclusion.

Since $\dim H \leq \dim G$ for any closed subgroup $H$, a chain of connected solvable subgroups has bounded dimension, and a maximal element exists.

**Example 6.2.2: Borel Subgroups of Classical Groups**

1. $G = \mathrm{GL}_n$: The standard Borel is $B_n = \{\text{upper-triangular invertible matrices}\}$. It is connected solvable (example 6.1.2(3)), and it is maximal because any connected solvable subgroup of $\mathrm{GL}_n$ is conjugate into $B_n$ by the Lie–Kolchin theorem (corollary 6.1.5), hence has dimension $\leq \dim B_n = \binom{n+1}{2}$.
2. $G = \mathrm{SL}_n$: The standard Borel is $B = B_n \cap \mathrm{SL}_n$ (upper-triangular matrices of determinant 1). It has dimension $\binom{n+1}{2} - 1 = \binom{n}{2} + (n-1)$.
3. $G = \mathrm{Sp}_{2n}$: The standard Borel consists of those elements of $\mathrm{Sp}_{2n}$ that preserve the standard complete isotropic flag. In a suitable basis, these are the "upper-triangular" elements of $\mathrm{Sp}_{2n}$, forming a group of dimension $n^2 + n$.
4. $G$ *commutative*: If $G$ is commutative (e.g. a torus), then $G$ itself is solvable, so $B = G$. A commutative group has a unique Borel subgroup: itself.

## Projectivity of $G/B$

The first main theorem says that the quotient of $G$ by a Borel subgroup is always projective. This is the result that makes the "geometry over algebra" program work: it gives us a projective variety $G/B$ on which we can use the tools of projective geometry (ample line bundles, intersection theory, Schubert calculus) to study $G$.

**Theorem 6.2.3: $G/B$ is Projective**

Let $G$ be a connected affine algebraic group and $B \subseteq G$ a Borel subgroup. Then $G/B$ is a projective variety.

***Proof*:**
By theorem 4.2.2, the quotient $G/B$ exists as a quasi-projective variety: there is a representation $V$ of $G$ and a line $L \subseteq V$ with $B = \mathrm{Stab}_G(L)$ (by Chevalley's theorem, theorem 4.2.1), and $G/B$ is the orbit $G \cdot [L] \subseteq \mathbb{P}(V)$, which is locally closed. We need to show this orbit is in fact *closed* in $\mathbb{P}(V)$.

By the Closed Orbit Lemma (theorem 2.4.15), the closure $\overline{G \cdot [L]}$ is a $G$-stable closed subvariety of $\mathbb{P}(V)$, and the boundary $\overline{G \cdot [L]} \setminus G \cdot [L]$ is a union of orbits of strictly lower dimension. It suffices to show that there are *no* orbits of lower dimension, i.e. $G \cdot [L]$ is already closed.

Consider any closed orbit $O \subseteq \overline{G \cdot [L]}$ (which exists by the Closed Orbit Lemma). The Borel subgroup $B$ acts on $\overline{G \cdot [L]}$ by restriction of the $G$-action, and $O$ is a closed $B$-stable subvariety of the projective variety $\overline{G \cdot [L]}$. Since $O$ is projective (closed in a projective variety) and $B$ is connected solvable, Borel's fixed-point theorem (theorem 6.1.9) gives a $B$-fixed point $[v] \in O$.

A $B$-fixed point in $\mathbb{P}(V)$ is a line $kv \subseteq V$ stabilized by $B$: for every $b \in B(R)$, $b \cdot v = \chi(b)v$ for some character $\chi : B \to \mathbb{G}_m$. In particular, $B \subseteq \mathrm{Stab}_G(kv)$. But $B$ is a *maximal* connected solvable subgroup, and $\mathrm{Stab}_G(kv)$ contains $B$ and is a closed subgroup of $G$. If $\mathrm{Stab}_G(kv)^\circ$ were solvable, maximality of $B$ would force $\mathrm{Stab}_G(kv)^\circ = B$, meaning $[v] \in G \cdot [L]$ (since both $[L]$ and $[v]$ have the same stabilizer up to connected components).

We need to verify that $\mathrm{Stab}_G(kv)^\circ$ is indeed solvable, or at least that $[v]$ lies in $G \cdot [L]$. Here we use the specific choice of $L$ from Chevalley's theorem more carefully. In the proof of theorem 4.2.1, the line $L = \bigwedge^d V_0$ was constructed so that $B = \mathrm{Stab}_G(L)$ *exactly* (not just the connected component). The orbit $G \cdot [L]$ parametrizes all lines of the form $g \cdot L$ for $g \in G$, and two points $g_1 \cdot [L] = g_2 \cdot [L]$ iff $g_1^{-1} g_2 \in B$.

Now, the $B$-fixed point $[v] \in O \subseteq \overline{G \cdot [L]}$ satisfies $B \subseteq \mathrm{Stab}_G(kv)$. The orbit $G \cdot [v] = O$ has stabilizer $\mathrm{Stab}_G(kv)$, and $\dim O = \dim G - \dim \mathrm{Stab}_G(kv) \leq \dim G - \dim B = \dim(G/B)$. But $G \cdot [L]$ has $\dim(G \cdot [L]) = \dim G - \dim B$ as well (since $\mathrm{Stab}_G(L) = B$). If $O \neq G \cdot [L]$, then $O$ would be a *different* orbit contained in $\overline{G \cdot [L]}$ with $\dim O \leq \dim(G \cdot [L])$. By the Closed Orbit Lemma, boundary orbits have strictly smaller dimension, so we need $\dim O < \dim(G \cdot [L])$, meaning $\dim \mathrm{Stab}_G(kv) > \dim B$.

Suppose $\dim \mathrm{Stab}_G(kv) > \dim B$. Let $P = \mathrm{Stab}_G(kv)$ and $H = P^\circ$. Since $B \subseteq H$ and $\dim H > \dim B$, the connected group $H$ properly contains $B$. But $B$ is a Borel subgroup (maximal connected solvable), so $H$ is *not* solvable.

We now use the following lemma:

*Lemma.* If $B$ is a Borel subgroup of $G$ and $P$ is a closed subgroup with $B \subseteq P \subseteq G$, then $B$ is also a Borel subgroup of $P^\circ$.

*Proof of Lemma.* $B$ is connected, solvable, and contained in $P^\circ$. If $B' \subseteq P^\circ$ were a larger connected solvable subgroup containing $B$, then $B' \subseteq G$ would also be connected solvable, contradicting the maximality of $B$ in $G$. □

By the Lemma, $B$ is a Borel subgroup of $H = P^\circ$. We can now proceed by induction on $\dim G$. (The base case is $G$ solvable, where $G = B$ and $G/B$ is a point, which is projective).

If we can ensure that $P \subsetneq G$, then $\dim H < \dim G$. By the induction hypothesis, $H/B$ is a projective variety. Since $P/H = P/P^\circ$ is a finite group, $P/B$ is a finite union of closed copies of $H/B$, and is therefore also projective. Now consider the natural quotient map $G/B \to G/P$. This is a fiber bundle with base $G/P$ and fiber $P/B$. Since $O \cong G/P$ is a closed orbit in $\mathbb{P}(V)$, the base $G/P$ is projective. Because both the base and the fiber are projective, the total space $G/B$ is projective. This forces the quasi-projective orbit $G \cdot [L]$ to be complete, meaning it is already closed in $\mathbb{P}(V)$.

The only remaining obstacle is if $P = G$, meaning $G$ fixes the point $[v] \in \mathbb{P}(V)$. This would imply that $G$ has a 1-dimensional representation. However, by a standard refinement of Chevalley's theorem, one can always choose the representation $V$ such that $G$ has no fixed lines in the boundary of the orbit (for instance, by reducing to the semisimple case $G/R(G)$, where groups have no nontrivial characters and fixed lines are easily eliminated). Thus, the trap $P = G$ is avoided, and the induction holds.

The projectivity of $G/B$ is one of the most consequential results in the structure theory. It means that the flag variety $G/B$ is a "compact" geometric object (in the algebraic sense), on which intersection theory, Schubert calculus, and the Borel–Weil theorem can be applied. The proof demonstrates the power of Borel's fixed-point theorem: by showing that $B$ always finds a fixed point on any orbit closure, it forces the orbit to be closed.

## Conjugacy of Borel Subgroups

The second main theorem says that all Borel subgroups of $G$ are conjugate. This is an essential structural result: it means that the "shape" of a Borel subgroup is an invariant of $G$, not of an arbitrary choice.

**Theorem 6.2.4: Conjugacy of Borel Subgroups**

Let $G$ be a connected affine algebraic group over an algebraically closed field $k$. Then any two Borel subgroups of $G$ are conjugate: if $B, B'$ are Borel subgroups, there exists $g \in G(k)$ with $B' = gBg^{-1}$.

***Proof*:**
Consider the projective variety $G/B$ (theorem 6.2.3). The Borel subgroup $B'$ acts on $G/B$ by left translation: $b' \cdot gB = (b'g)B$. Since $G/B$ is projective and $B'$ is connected solvable, Borel's fixed-point theorem (theorem 6.1.9) gives a fixed point $gB \in G/B$, meaning $b'gB = gB$ for all $b' \in B'$. Equivalently, $g^{-1}b'g \in B$ for all $b' \in B'$, i.e. $g^{-1}B'g \subseteq B$.

Since $g^{-1}B'g$ is connected solvable (conjugation is an isomorphism) and $B$ is a Borel, maximality of $B$ and $g^{-1}B'g$ (both are Borel subgroups, and one is contained in the other) forces $g^{-1}B'g = B$, i.e. $B' = gBg^{-1}$.

The proof is a great application of the "geometry over algebra" principle: the conjugacy of Borel subgroups, a purely group-theoretic statement, is proved by finding a fixed point on the *projective variety* $G/B$. Without the projectivity of $G/B$, Borel's fixed-point theorem could not be applied.

## Conjugacy of Maximal Tori

Every Borel subgroup contains a maximal torus (theorem 6.1.7(3)), and all maximal tori inside a fixed Borel are conjugate (by elements of the unipotent radical). Combined with the conjugacy of Borel subgroups, this gives:

**Theorem 6.2.5: Conjugacy of Maximal Tori**

Let $G$ be a connected affine algebraic group over an algebraically closed field $k$. Then:

1. Every maximal torus of $G$ is contained in a Borel subgroup.
2. Any two maximal tori of $G$ are conjugate.

***Proof*:**
*(1):* Let $T$ be a maximal torus of $G$. Since $T$ is connected solvable (it is commutative), it is contained in some connected solvable subgroup. By Zorn's lemma (or the finite-dimensionality argument), $T$ is contained in a maximal connected solvable subgroup, i.e. a Borel subgroup $B$.

*(2):* Let $T_1, T_2$ be maximal tori. By (1), $T_i \subseteq B_i$ for Borel subgroups $B_1, B_2$. By theorem 6.2.4, there exists $g \in G(k)$ with $B_2 = gB_1g^{-1}$. Then $gT_1g^{-1}$ is a maximal torus in $B_2$ (conjugation preserves tori and maximality). But $T_2$ is also a maximal torus in $B_2$. By the conjugacy of maximal tori inside a connected solvable group (theorem 6.1.7(3)), there exists $u \in R_u(B_2)$ with $T_2 = u(gT_1g^{-1})u^{-1} = (ug)T_1(ug)^{-1}$.

The conjugacy of maximal tori has a important consequence: the *rank* of $G$ (the dimension of a maximal torus) is well-defined, as is the isomorphism type of the character lattice $X(T)$. When we define root systems in chapter 7, the conjugacy theorem will guarantee that the root system is independent of the choice of maximal torus and Borel subgroup.

**Example 6.2.6: Maximal Tori in $\mathrm{GL}_n$ and $\mathrm{SL}_n$**
In $\mathrm{GL}_n$, the standard maximal torus is the diagonal torus $T = \mathbb{G}_m^n$, of rank $n$. Any other maximal torus is of the form $gTg^{-1}$ for some $g \in \mathrm{GL}_n(k)$: it consists of the matrices simultaneously diagonalizable by the basis $\{ge_1, \ldots, ge_n\}$.

In $\mathrm{SL}_n$, the standard maximal torus is $T = \{\mathrm{diag}(t_1, \ldots, t_n) : t_1 \cdots t_n = 1\} \cong \mathbb{G}_m^{n-1}$, of rank $n-1$. Its character lattice is $X(T) = \mathbb{Z}^n/\mathbb{Z}(1, 1, \ldots, 1)$.

## Parabolic Subgroups

Between the Borel $B$ and the full group $G$, there are intermediate subgroups that also give rise to projective quotients.

**Definition 6.2.7: Parabolic Subgroup**

A closed subgroup $P \subseteq G$ is called *parabolic* if $G/P$ is a projective variety. Equivalently, $P$ is parabolic if and only if $P$ contains a Borel subgroup.

The equivalence of the two conditions is a nontrivial fact. One direction is clear: if $P \supseteq B$, then $G/P$ is a quotient of $G/B$ (via the natural map $G/B \to G/P$, which is a fiber bundle with fiber $P/B$). Because the map $G/B \to G/P$ is surjective, the quasi-projective variety $G/P$ is the continuous image of the complete variety $G/B$, making $G/P$ complete and therefore projective. For the other direction: if $G/P$ is projective and $B$ is a Borel, then $B$ acts on $G/P$ and Borel's fixed-point theorem gives a $B$-fixed point $gP$, meaning $g^{-1}Bg \subseteq P$, so $P$ contains a conjugate of $B$ (hence $P$ contains a Borel).

**Example 6.2.8: Parabolic Subgroups of $\mathrm{GL}_n$**
The parabolic subgroups of $\mathrm{GL}_n$ containing the standard Borel $B_n$ are in bijection with subsets $I \subseteq \{1, \ldots, n-1\}$. The subset $I = \{d_1, \ldots, d_r\}$ (with $d_1 < \cdots < d_r$) corresponds to the parabolic $P_I$ that is the stabilizer of the partial flag

$$\langle e_1, \ldots, e_{d_1} \rangle \subset \langle e_1, \ldots, e_{d_2} \rangle \subset \cdots \subset \langle e_1, \ldots, e_{d_r} \rangle \subset k^n$$

The quotient $\mathrm{GL}_n/P_I$ is the partial flag variety $\mathrm{Fl}(d_1, \ldots, d_r; n)$. Special cases:

1. $I = \{1, 2, \ldots, n-1\}$ gives $P_I = B_n$ and $\mathrm{GL}_n/B_n = \mathrm{Fl}(1, 2, \ldots, n-1; n)$, the complete flag variety.
2. $I = \{d\}$ gives $P_I = P_d$ (a maximal parabolic) and $\mathrm{GL}_n/P_d = \mathrm{Gr}(d, n)$, the Grassmannian.
3. $I = \emptyset$ gives $P_I = \mathrm{GL}_n$ and $\mathrm{GL}_n/\mathrm{GL}_n = \{*\}$.

The number of parabolic subgroups containing $B_n$ is $2^{n-1}$ (the number of subsets of $\{1, \ldots, n-1\}$). Each gives a different partial flag variety, forming a lattice under the natural projections $\mathrm{Fl}(d_1, \ldots, d_r; n) \to \mathrm{Fl}(d_{i_1}, \ldots, d_{i_s}; n)$ for $\{i_1, \ldots, i_s\} \subseteq \{1, \ldots, r\}$.

## Reductive and Semisimple Groups

The radical $R(G)$ and the unipotent radical $R_u(G)$ measure the "solvable" and "unipotent" content of $G$. The groups where these are trivial are the ones that carry the most interesting structure theory.

**Definition 6.2.9: Unipotent Radical (General)**

Let $G$ be a connected affine algebraic group. The *unipotent radical* $R_u(G)$ is the maximal connected normal unipotent subgroup of $G$. Equivalently, $R_u(G) = R_u(R(G))$: the unipotent radical of the radical.

The unipotent radical is always contained in the radical ($R_u(G) \subseteq R(G)$, since a connected normal unipotent subgroup is in particular solvable), and the quotient $R(G)/R_u(G)$ is a torus (by the structure theorem for solvable groups, theorem 6.1.7(2)).

**Definition 6.2.10: Reductive and Semisimple Groups**

Let $G$ be a connected affine algebraic group.

1. $G$ is called *reductive* if $R_u(G) = \{e\}$ (the unipotent radical is trivial).
2. $G$ is called *semisimple* if $R(G) = \{e\}$ (the radical is trivial).

Semisimple implies reductive (since $R_u(G) \subseteq R(G)$), but not conversely: a torus is reductive (it has no unipotent elements, so $R_u = \{e\}$) but not semisimple (it is its own radical: $R(T) = T$). The relationship between the three classes is:

$$\{\text{semisimple}\} \subset \{\text{reductive}\} \subset \{\text{affine algebraic groups}\}$$

**Example 6.2.11: Classification of Low-Dimensional Cases**

1. $\mathrm{GL}_n$: Reductive but not semisimple. The radical is $R(\mathrm{GL}_n) = Z(\mathrm{GL}_n) = \mathbb{G}_m$ (exercise 6.1.1 (4)), a torus, so $R_u(\mathrm{GL}_n) = \{e\}$. The center is nontrivial, so $R(\mathrm{GL}_n) \neq \{e\}$.
2. $\mathrm{SL}_n$ *($n \geq 2$)*: Semisimple. The center $Z(\mathrm{SL}_n) = \mu_n$ is finite (not connected), so it is not contained in $R(\mathrm{SL}_n)$, which must be connected. One shows $R(\mathrm{SL}_n) = \{e\}$ by verifying that $\mathrm{SL}_n$ has no nontrivial connected solvable normal subgroups.

3. $\mathrm{Sp}_{2n}$ *($n \geq 1$)*: Semisimple. Similar reasoning.

4. $\mathrm{SO}_n$ *($n \geq 3$, $\mathrm{char} k \neq 2$)*: Semisimple.

5. $B_n$: Not reductive. The unipotent radical $R_u(B_n) = U_n \neq \{e\}$.

6. $\mathbb{G}_m^r$: Reductive (a torus), but not semisimple for $r \geq 1$.

A reductive group $G$ satisfies a clean structural decomposition: the radical $R(G)$ is a central torus (since $R_u(G) = \{e\}$, the radical is a torus by theorem 6.1.7(2), and a normal torus in a connected group is central by proposition 5.2.13), and the quotient $G/R(G)$ is semisimple. More precisely:

**Proposition 6.2.12: Structure of Reductive Groups**

Let $G$ be a connected reductive group. Then:

1. $R(G) = Z(G)^\circ$ (the radical is the identity component of the center).

2. The derived subgroup $[G, G]$ is semisimple.

3. The multiplication map $R(G) \times [G, G] \to G$ is surjective with finite kernel. Equivalently, $G = R(G) \cdot [G, G]$ and $R(G) \cap [G, G]$ is finite.

***Proof Sketch*:**
*(1):* The radical $R(G)$ is a connected solvable normal subgroup. Since $G$ is reductive, $R_u(R(G)) = R_u(G) = \{e\}$, so $R(G)$ is a torus. A normal torus in a connected group is central (proposition 5.2.13), so $R(G) \subseteq Z(G)^\circ$. Conversely, $Z(G)^\circ$ is connected, solvable (commutative), and normal, so $Z(G)^\circ \subseteq R(G)$.

*(2):* The quotient $G/R(G)$ is semisimple (its radical is trivial by maximality of $R(G)$). The derived subgroup $[G, G]$ surjects onto $[G/R(G), G/R(G)] = G/R(G)$ (since $G/R(G)$ is semisimple, it equals its own derived subgroup). The kernel of $[G, G] \to G/R(G)$ is $[G, G] \cap R(G)$, which is finite (it is a closed subgroup of the torus $R(G)$ and therefore central in $[G, G]$; since $R(G)$ is central, $[G, G]$ is equal to its own derived subgroup, making it a perfect group. Because perfect algebraic groups cannot contain central tori of positive dimension, this intersection must be finite). Hence $[G, G]$ is an extension of the semisimple group $G/R(G)$ by a finite group, and one checks that $[G, G]$ is itself semisimple.

*(3):* Every $g \in G$ can be written as $g = z \cdot [g, ...]$ modulo the derived subgroup; more precisely, $G = R(G) \cdot [G, G]$ follows from the surjectivity of $[G, G] \to G/R(G)$ and $R(G) \to G/[G, G]$. The kernel of $R(G) \times [G, G] \to G$ is $\{(z, z^{-1}) : z \in R(G) \cap [G, G]\}$, which is finite.

This decomposition says that a reductive group is “almost” a product of a torus and a semisimple group, up to a finite central overlap. For example, $\mathrm{GL}_n = \mathbb{G}_m \cdot \mathrm{SL}_n$ with $\mathbb{G}_m \cap \mathrm{SL}_n = \mu_n$.

The terms “reductive” and “semisimple” now justify the book’s focus: Part I culminates in chapter 7 with the classification of semisimple (equivalently, reductive modulo center) groups via root systems. The representation theory of Part II will study how these groups act, with the Borel subgroup $B$ and the flag variety $G/B$ as the primary geometric tools.

**Example 6.2.13: $\mathrm{GL}_n$ as a Reductive Group**
The group $\mathrm{GL}_n$ has $R(\mathrm{GL}_n) = Z(\mathrm{GL}_n)^\circ = \mathbb{G}_m$ (scalar matrices), $[G, G] = \mathrm{SL}_n$, and $\mathrm{GL}_n = \mathbb{G}_m \cdot \mathrm{SL}_n$ with $\mathbb{G}_m \cap \mathrm{SL}_n = \mu_n$. The Borel subgroup $B_n$ decomposes as $T \ltimes U_n$ where $T = \mathbb{G}_m^n$ is the maximal torus and $U_n$ is the unipotent radical. The flag variety is $\mathrm{GL}_n/B_n = \mathrm{SL}_n/(B_n \cap \mathrm{SL}_n) = \mathrm{Fl}(1, \dots, n-1; n)$, a smooth projective variety of dimension $\binom{n}{2}$.

The root system will be defined in chapter 7 by decomposing $\mathfrak{gl}_n$ under the adjoint action of $T$: the nonzero weights $\varepsilon_i - \varepsilon_j$ $(i \neq j)$ are the roots (example 5.2.10), the weights with $i < j$ are the positive roots (corresponding to $\mathrm{Lie}(U_n)$), and the Weyl group $W = N_G(T)/Z_G(T) \cong S_n$ (exercise 5.2.1 (6)) acts by permuting the $\varepsilon_i$. The conjugacy theorems of this section ensure that this data is independent of the choice of $T$ and $B$.

## Exercise 6.2.1

1. Let $G = \mathrm{SL}_3$ with standard Borel $B$ (upper-triangular, determinant 1). Compute $\dim B$, $\dim G/B$, the rank of $G$ (dimension of a maximal torus), and verify $\dim G = \dim B + \dim G/B$.
2. Let $G = \mathrm{SL}_2$ with standard Borel $B = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \right\}$ and non-standard Borel $B' = \left\{ \begin{pmatrix} a & 0 \\ c & a^{-1} \end{pmatrix} \right\}$ (lower-triangular). Find an explicit $g \in \mathrm{SL}_2(k)$ with $gBg^{-1} = B'$, verifying the conjugacy theorem.
3. How many parabolic subgroups of $\mathrm{SL}_4$ contain the standard Borel? List them and describe the corresponding partial flag varieties with their dimensions.
4. Determine which of the following groups are reductive, semisimple, both, or neither: (a) $\mathrm{GL}_1 = \mathbb{G}_m$; (b) $\mathrm{PGL}_n$; (c) $\mathbb{G}_a$; (d) $\mathbb{G}_m \times \mathrm{SL}_2$; (e) $B_2 \subsetneq \mathrm{GL}_2$.
5. Let $G = \mathrm{SL}_3$ with maximal torus $T = \{\mathrm{diag}(t_1, t_2, (t_1t_2)^{-1})\}$. Compute the normalizer $N_G(T)$ and the Weyl group $W = N_G(T)/Z_G(T)$. Show that $W \cong S_3$ and describe its action on $X(T)$.
6. Let $G = \mathrm{Sp}_4$ (the $4 \times 4$ symplectic group). What is the rank of $G$? What is $\dim G/B$? The flag variety $G/B$ parametrizes complete isotropic flags $0 \subset L_1 \subset L_2 \subset k^4$ where $L_1$ is a line and $L_2$ is a Lagrangian plane (with $L_1 \subset L_2$ and $L_2$ isotropic). How many maximal parabolic subgroups does $G$ have?

# 7

# Root Systems and Classification

This is the culminating chapter of Part I. Everything we have built — the functor of points and Hopf algebras (chapter 1), representations and comodules (chapter 2), the adjoint representation and Lie algebras (chapter 3), quotient constructions (chapter 4), the Jordan decomposition and torus theory (chapter 5), and the Borel subgroup machinery (chapter 6) — converges here into a single combinatorial invariant: the *root system.* A root system is a finite configuration of vectors in a Euclidean space, subject to crystallographic symmetry conditions. The classification of root systems is a finite computation, and it yields the classification of semisimple algebraic groups — one of the great achievements of twentieth-century mathematics.

The mechanism is the one we have been previewing since chapter 3: the adjoint action of a maximal torus $T$ on the Lie algebra $\mathfrak{g}$ decomposes $\mathfrak{g}$ into weight spaces, and the nonzero weights — the *roots* — organize into a combinatorial structure governed by the *Weyl group* $W = N_G(T)/Z_G(T)$. The conjugacy theorems of chapter 6 ensure that the root system is independent of the choice of $T$ and $B$, and the classification theorem asserts that it is a *complete* invariant: two semisimple groups with isomorphic root systems are isomorphic (up to isogeny).

We develop the theory in three stages. First, we extract the root system from a reductive group and compute it for all the classical families. Second, we develop the abstract combinatorial theory of root systems and classify them via Dynkin diagrams. Third, we introduce root data and state the isomorphism and existence theorems, completing the classification.

## 7.1 Roots and the Weyl Group

Throughout this section, $k$ denotes an algebraically closed field, $G$ a connected reductive group over $k$, $T \subseteq G$ a maximal torus of rank $r = \dim T$, and $B \supseteq T$ a Borel subgroup containing $T$. The existence of $T$ and $B$ was established in chapter 6 (theorem 6.2.5, definition 6.2.1), and the conjugacy theorems guarantee that the structures we define will be independent of these choices (up to canonical isomorphism).

## The Root Space Decomposition

The maximal torus $T$ acts on the Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ via the adjoint representation $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ (definition 3.2.1). Since $T$ is a torus, the weight space decomposition (theorem 5.2.9) gives:

$$\mathfrak{g} = \bigoplus_{\alpha \in X(T)} \mathfrak{g}_\alpha$$

where $\mathfrak{g}_\alpha = \{X \in \mathfrak{g} \mid \mathrm{Ad}(t)(X) = \alpha(t)X \text{ for all } t \in T\}$ is the weight space for the character $\alpha \in X(T)$. The 0-weight space has a natural identification:

**Proposition 7.1.1: The Zero Weight Space**

$\mathfrak{g}_0 = \{X \in \mathfrak{g} \mid \mathrm{Ad}(t)(X) = X \text{ for all } t \in T\} = \mathrm{Lie}(Z_G(T)) = \mathfrak{t}$, the Lie algebra of $T$.

***Proof*:**
The set $\mathfrak{g}_0$ consists of elements fixed by the adjoint action of $T$, which is exactly the Lie algebra of the centralizer $Z_G(T)$ (since $\mathrm{Ad}(t)(X) = X$ for all $t$ means $X$ commutes with $T$ at the infinitesimal level). For a reductive group, $Z_G(T) = T$ (the centralizer of a maximal torus is the torus itself — a standard structural theorem guarantees that the centralizer of a torus in a reductive group is again a connected reductive group; since its maximal torus $T$ is central, it has no unipotent elements and no roots, and must equal $T$). Therefore $\mathfrak{g}_0 = \mathrm{Lie}(T) = \mathfrak{t}$.

The nonzero weights are the central definition of this chapter:

**Definition 7.1.2: Roots**

The *roots* of $G$ with respect to $T$ are the nonzero characters $\alpha \in X(T)$ such that $\mathfrak{g}_\alpha \neq 0$. The set of all roots is denoted

$$\Phi = \Phi(G, T) = \{\alpha \in X(T) \setminus \{0\} \mid \mathfrak{g}_\alpha \neq 0\}$$

For each $\alpha \in \Phi$, the subspace $\mathfrak{g}_\alpha$ is called the *root space* of $\alpha$.

The root space decomposition of $\mathfrak{g}$ is thus:

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

As we will soon prove, each root space $\mathfrak{g}_\alpha$ is one-dimensional, giving $\dim \mathfrak{g} = \mathrm{rank}(G) + |\Phi|$. This is the decomposition we computed for $\mathrm{GL}_n$ in example 5.2.10 and previewed for $\mathrm{SL}_2$ in example 3.2.9. We now develop the general theory.

## Fundamental Properties of Roots

The roots of a reductive group satisfy a collection of structural properties that will eventually be axiomatized into the notion of an "abstract root system" in §7.2. We state the properties here as theorems about $G$ and prove them using the tools of previous chapters.

**Theorem 7.1.3: Structural Properties of Roots**

Let $G$ be a connected reductive group with maximal torus $T$ and root system $\Phi = \Phi(G,T)$.

1. Each root space is one-dimensional: $\dim_k \mathfrak{g}_\alpha = 1$ for every $\alpha \in \Phi$.
2. If $\alpha \in \Phi$, then $-\alpha \in \Phi$ (roots come in opposite pairs).
3. If $\alpha \in \Phi$ and $c\alpha \in \Phi$ for some scalar $c$, then $c = \pm 1$. (No "multiples" of a root, other than its negative, are roots.)
4. $\Phi$ is finite, and $\Phi$ spans $X(T) \otimes_{\mathbb{Z}} \mathbb{Q}$ if $G$ is semisimple.
5. For each $\alpha \in \Phi$, there exists a *coroot* $\alpha^\vee \in X_\vee(T)$ such that $\langle \alpha, \alpha^\vee \rangle = 2$ and the reflection $s_\alpha : X(T) \otimes \mathbb{Q} \to X(T) \otimes \mathbb{Q}$ defined by $s_\alpha(\lambda) = \lambda - \langle \lambda, \alpha^\vee \rangle \alpha$ satisfies $s_\alpha(\Phi) = \Phi$.

***Proof Sketch*:**
The proofs of these properties use the $\mathrm{SL}_2$-*subgroups* of $G$. For each root $\alpha \in \Phi$, there exists a homomorphism $\varphi_\alpha : \mathrm{SL}_2 \to G$ such that $\varphi_\alpha$ maps the standard maximal torus of $\mathrm{SL}_2$ into $T$ and the root spaces of $\mathrm{SL}_2$ onto $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$. The existence of $\varphi_\alpha$ is a deep result (it follows from the structure theory of rank-1 semisimple groups, which shows that the subgroup generated by the root groups $U_\alpha$ and $U_{-\alpha}$ is isomorphic to $\mathrm{SL}_2$ or $\mathrm{PGL}_2$).

Given $\varphi_\alpha$, the properties follow from the representation theory of $\mathrm{SL}_2$ acting on $\mathfrak{g}$. The sum $V_\alpha$ of $\mathfrak{t}$ and all root spaces $\mathfrak{g}_{c\alpha}$ (for scalar $c$) forms an $\mathrm{SL}_2$-submodule of $\mathfrak{g}$. A careful analysis shows it decomposes into a trivial representation (the kernel of $\alpha$ in $\mathfrak{t}$) and exactly one copy of the adjoint representation of $\mathrm{SL}_2$, which has weights $2, 0, -2$.

1. Because the $\mathrm{SL}_2$ adjoint representation has 1-dimensional weight spaces for 2 and $-2$, the root space $\mathfrak{g}_\alpha$ is one-dimensional.
2. Because the weight $-2$ appears, $\mathfrak{g}_{-\alpha} \neq 0$, so $-\alpha \in \Phi$.
3. Because the only nonzero weights in $V_\alpha$ are $\pm 2$ (corresponding to $\pm\alpha$), no other scalar multiples $c\alpha$ can be roots.
4. Finiteness: $\Phi \subseteq X(T)$ and $\mathfrak{g}$ is finite-dimensional, so $|\Phi| \leq \dim \mathfrak{g} - \mathrm{rank} G < \infty$. For the spanning property when $G$ is semisimple: if $\Phi$ did not span $X(T) \otimes \mathbb{Q}$, then $T$ would have a nontrivial subtorus acting trivially on $\mathfrak{g}$ (the annihilator of $\Phi$ in $X_\vee(T)$), hence lying in the center $Z(G)$. But $Z(G)$ is finite for a semisimple group (proposition 6.2.12), contradicting the subtorus being nontrivial.
5. The coroot $\alpha^\vee$ is defined as the cocharacter $\varphi_\alpha \circ \lambda_0$, where $\lambda_0 : \mathbb{G}_m \to \mathrm{SL}_2$ is the standard cocharacter $t \mapsto \mathrm{diag}(t, t^{-1})$. The pairing $\langle \alpha, \alpha^\vee \rangle = 2$ follows from the $\mathrm{SL}_2$-computation (exercise 5.2.1 (5)). The reflection $s_\alpha$ is the image in $W$ of the Weyl group element $\varphi_\alpha\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \in N_G(T)$, and it permutes $\Phi$ because $W$ acts on $\mathfrak{g}$ by automorphisms that preserve weight spaces.

The appearance of $\mathrm{SL}_2$-subgroups is the structural heart of the theory: each root $\alpha$ generates a copy of $\mathrm{SL}_2$ (or $\mathrm{PGL}_2$) inside $G$, and the entire root system can be understood as the pattern of how these $\mathrm{SL}_2$-copies overlap and interact. This is sometimes called the "$\mathrm{SL}_2$-philosophy" of reductive group theory.

## Positive Roots and Simple Roots

The choice of Borel subgroup $B \supseteq T$ determines a partition of the roots into "positive" and "negative" halves.

**Definition 7.1.4: Positive and Negative Roots**

Let $B \supseteq T$ be a Borel subgroup. The *positive roots* (with respect to $B$) are:

$$\Phi^+ = \Phi^+(G, T, B) = \{\alpha \in \Phi \mid \mathfrak{g}_\alpha \subseteq \mathrm{Lie}(R_u(B))\}$$

The *negative roots* are $\Phi^- = -\Phi^+ = \{-\alpha \mid \alpha \in \Phi^+\}$. We have $\Phi = \Phi^+ \sqcup \Phi^-$ (disjoint union).

The definition is geometric: the positive roots are the weights of $T$ on the Lie algebra of the unipotent radical of $B$. Since $B = T \ltimes R_u(B)$ (theorem 6.1.7), we have $\mathrm{Lie}(B) = \mathfrak{t} \oplus \mathrm{Lie}(R_u(B))$, and hence:

$$\mathrm{Lie}(B) = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha, \qquad \mathfrak{g} = \bigoplus_{\alpha \in \Phi^-} \mathfrak{g}_\alpha \oplus \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha$$

The dimension of the flag variety is $\dim G/B = \dim G - \dim B = |\Phi^-| = |\Phi^+| = |\Phi|/2$.

A different choice of Borel $B' \supseteq T$ gives a different set of positive roots $(\Phi^+)' = \Phi^+(G, T, B')$, but the conjugacy of Borel subgroups (theorem 6.2.4) ensures that the root system $\Phi$ itself is unchanged. The positive roots determined by $B$ and $B'$ are related by a Weyl group element.

**Definition 7.1.5: Simple Roots**

A root $\alpha \in \Phi^+$ is called *simple* if it cannot be expressed as $\alpha = \beta + \gamma$ with $\beta, \gamma \in \Phi^+$. The set of simple roots is denoted $\Delta = \{\alpha_1, \ldots, \alpha_r\}$.

**Proposition 7.1.6: Properties of Simple Roots**

1. $|\Delta| = r = \mathrm{rank}(G)$ when $G$ is semisimple (in general, $|\Delta|$ is the semisimple rank of $G$).
2. $\Delta$ is a basis for $X(T) \otimes_{\mathbb{Z}} \mathbb{Q}$ (when $G$ is semisimple).
3. Every positive root is a non-negative integer linear combination of simple roots: if $\alpha \in \Phi^+$, then $\alpha = \sum_{i=1}^r n_i \alpha_i$ with $n_i \in \mathbb{Z}_{\geq 0}$.
4. The simple roots are linearly independent over $\mathbb{Q}$.

***Proof Sketch*:**
If $\alpha \in \Phi^+$ is not simple, then $\alpha = \beta + \gamma$ with $\beta, \gamma \in \Phi^+$. Because all positive roots lie strictly on one side of a hyperplane, we can evaluate them on a generic vector $v$ such that $\delta(v) > 0$ for all $\delta \in \Phi^+$. The equation $\alpha(v) = \beta(v) + \gamma(v)$ forces $\beta(v) < \alpha(v)$ and $\gamma(v) < \alpha(v)$. Since $\Phi^+$ is a finite set, this strictly decreasing decomposition process must terminate, meaning every positive root is eventually written as a non-negative integer combination of simple roots.

The linear independence of $\Delta$ is more subtle and relies on the crystallographic geometry of the root system. One shows that for any two distinct simple roots $\alpha, \beta \in \Delta$, the pairing $\langle \alpha, \beta^\vee \rangle \leq 0$. A standard lemma in Euclidean geometry states that any set of vectors strictly on one side of a hyperplane, whose pairwise inner products are $\leq 0$, must be linearly independent. Finally, the count $|\Delta| = r$ (for semisimple $G$) follows from $\Delta$ being a linearly independent set that spans the same dimension as $\Phi$.

## The Weyl Group

**Definition 7.1.7: Weyl Group**

The *Weyl group* of $G$ with respect to $T$ is the finite group

$$W = W(G,T) = N_G(T)/Z_G(T)$$

where $N_G(T) = \left\{ g \in G \;\middle|\; gTg^{-1} = T \right\}$ is the normalizer and $Z_G(T) = T$ is the centralizer.

That $W$ is finite was established in exercise 5.2.1 (6): $N_G(T)$ acts on $T$ by automorphisms, $\mathrm{Aut}(T) \cong \mathrm{GL}_r(\mathbb{Z})$ is discrete, and the kernel of $N_G(T) \to \mathrm{Aut}(T)$ is $Z_G(T) = T$, so $W$ injects into $\mathrm{GL}_r(\mathbb{Z})$ as a finite subgroup. The Weyl group acts on $X(T)$ by $(w \cdot \chi)(t) = \chi(n_w^{-1} t n_w)$ where $n_w \in N_G(T)$ is any lift of $w$, and this action permutes the roots: $w(\Phi) = \Phi$ for all $w \in W$.

The Weyl group is generated by the *simple reflections*:

**Theorem 7.1.8: Weyl Group is Generated by Simple Reflections**

The Weyl group $W$ is generated by the reflections $\{s_\alpha : \alpha \in \Delta\}$ corresponding to the simple roots. Moreover, $W$ acts simply transitively on the set of Weyl chambers (connected components of $X(T) \otimes \mathbb{R} \setminus \bigcup_{\alpha \in \Phi} \ker(\alpha)$), and the order of $W$ equals the number of Weyl chambers.

***Proof Sketch*:**
Each simple reflection $s_{\alpha_i}$ permutes $\Phi^+ \setminus \{\alpha_i\}$ (it sends $\alpha_i$ to $-\alpha_i$ and permutes the other positive roots). Given any $w \in W$, the element $w$ permutes $\Phi$ and sends $\Phi^+$ to some other set of positive roots $w(\Phi^+)$. The “length” of $w$ is the number of positive roots sent to negative roots: $\ell(w) = |\Phi^+ \cap w^{-1}(\Phi^-)|$. One shows by induction on $\ell(w)$ that $w$ is a product of simple reflections: if $\ell(w) > 0$, there exists a simple root $\alpha_i$ with $w(\alpha_i) \in \Phi^-$, and then $\ell(w s_{\alpha_i}) < \ell(w)$.

### The Root System of $\mathrm{SL}_n$ (Type $A_{n-1}$)

We now compute the root system for each classical family, beginning with the most fundamental case.

**Example 7.1.9: Roots of $\mathrm{SL}_n$**

Let $G = \mathrm{SL}_n$ with maximal torus $T = \{\mathrm{diag}(t_1, \ldots, t_n) : t_1 \cdots t_n = 1\} \cong \mathbb{G}_m^{n-1}$. The character lattice is $X(T) = \mathbb{Z}^n/\mathbb{Z}(1, \ldots, 1)$, with the images of the standard characters $\varepsilon_i$ satisfying $\varepsilon_1 + \cdots + \varepsilon_n = 0$.

The adjoint action of $T$ on $\mathfrak{sl}_n = \{X \in M_n(k) : \mathrm{tr}(X) = 0\}$ was computed in example 5.2.10: the weight of $E_{ij}$ (with $i \neq j$) is $\varepsilon_i - \varepsilon_j$. Therefore:

$$\Phi = \{\varepsilon_i - \varepsilon_j \mid 1 \leq i, j \leq n,\ i \neq j\}$$

This is a root system of $n(n-1)$ roots in the $(n-1)$-dimensional space $V = \{\sum x_i = 0\} \subseteq \mathbb{R}^n$. Its rank is $n - 1$.

The standard Borel $B = B_n \cap \mathrm{SL}_n$ (upper-triangular) determines the positive roots $\Phi^+ = \{\varepsilon_i - \varepsilon_j : i < j\}$ (these are the weights of $T$ on $\mathrm{Lie}(U_n) = \{\text{strictly upper-triangular matrices}\}$). The simple roots are:

$$\Delta = \{\alpha_1, \ldots, \alpha_{n-1}\}, \qquad \alpha_i = \varepsilon_i - \varepsilon_{i+1}$$

(the "adjacent" differences). Every positive root $\varepsilon_i - \varepsilon_j$ with $i < j$ is a sum $\alpha_i + \alpha_{i+1} + \cdots + \alpha_{j-1}$ of consecutive simple roots.

The Weyl group is $W \cong S_n$ (the symmetric group), acting on $X(T)$ by permuting the $\varepsilon_i$. The simple reflection $s_{\alpha_i}$ corresponds to the transposition $(i\ \ i+1) \in S_n$, and the transpositions of adjacent elements generate $S_n$.

The coroots are given by the cocharacters $\alpha_i^\vee(t) = \mathrm{diag}(1, \ldots, t, t^{-1}, \ldots, 1)$ (with $t$ in the $i$-th position and $t^{-1}$ in the $(i+1)$-th position). In terms of the standard cocharacters $\varepsilon_i^\vee$ of the ambient $\mathrm{GL}_n$, this is $\alpha_i^\vee = \varepsilon_i^\vee - \varepsilon_{i+1}^\vee \in X_\vee(T)$ (note that while the individual $\varepsilon_i^\vee$ do not preserve $\det = 1$, their differences do). The pairing is $\langle \alpha_i, \alpha_j^\vee \rangle = \delta_{ij} \cdot 2 - \delta_{|i-j|,1}$, giving the Cartan matrix:

$$A = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 & 2 \end{pmatrix}$$

This is the Cartan matrix of type $A_{n-1}$.

## The Root System of $\mathrm{Sp}_{2n}$ (Type $C_n$)

**Example 7.1.10: Roots of $\mathrm{Sp}_{2n}$**

Let $G = \mathrm{Sp}_{2n}$ with the symplectic form $\Omega = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$. The maximal torus is

$$T = \left\{\mathrm{diag}(t_1, \ldots, t_n, t_1^{-1}, \ldots, t_n^{-1})\right\} \cong \mathbb{G}_m^n$$

with character lattice $X(T) = \mathbb{Z}^n$ and standard basis $\varepsilon_i(t) = t_i$.

The Lie algebra $\mathfrak{sp}_{2n}$ consists of matrices $X$ with $X^T\Omega + \Omega X = 0$ (example 3.1.8). Writing $X = \left(\begin{smallmatrix} A & B \\ C & -A^T \end{smallmatrix}\right)$ with $B = B^T$ and $C = C^T$, the adjoint action of $T$ on these blocks gives:

1. The entries of $A$: $A_{ij}$ has weight $\varepsilon_i - \varepsilon_j$ (for $i \neq j$) and weight 0 (for $i = j$).
2. The entries of $B$: $B_{ij}$ has weight $\varepsilon_i + \varepsilon_j$ (for $i \leq j$).
3. The entries of $C$: $C_{ij}$ has weight $-\varepsilon_i - \varepsilon_j$ (for $i \leq j$).

Therefore the roots are:

$$\Phi = \{\pm\varepsilon_i \pm \varepsilon_j : i \neq j\} \cup \{\pm 2\varepsilon_i\}$$

This is a root system of $2n^2$ roots in $\mathbb{R}^n$, of rank $n$.

The standard Borel (the block-upper-triangular subgroup) gives positive roots $\Phi^+ = \{\varepsilon_i - \varepsilon_j : i < j\} \cup \{\varepsilon_i + \varepsilon_j : i \leq j\}$ and simple roots:

$$\Delta = \{\alpha_1, \ldots, \alpha_n\}, \qquad \alpha_i = \begin{cases} \varepsilon_i - \varepsilon_{i+1} & 1 \leq i \leq n-1 \\ 2\varepsilon_n & i = n \end{cases}$$

The Weyl group is $W \cong S_n \ltimes (\mathbb{Z}/2\mathbb{Z})^n$, the group of *signed permutations*: it permutes the $\varepsilon_i$ and independently changes their signs. The order is $|W| = 2^n \cdot n!$.

The Cartan matrix has entries $\langle \alpha_i, \alpha_j^\vee \rangle$ forming the type-$C_n$ matrix, characterized by the last simple root being *long* (its squared Euclidean length $\|\alpha_n\|^2 = 4$ is 2 times the squared length of the short roots).

## The Root System of $\mathrm{SO}_{2n+1}$ (Type $B_n$)

**Example 7.1.11: Roots of $\mathrm{SO}_{2n+1}$**

Let $G = \mathrm{SO}_{2n+1}$ (with $\mathrm{char}\, k \neq 2$) preserving the form $s = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 0 & I \\ 0 & I & 0 \end{smallmatrix}\right)$ from §1.1.1. The maximal torus is

$$T = \left\{\mathrm{diag}(1, t_1, \ldots, t_n, t_1^{-1}, \ldots, t_n^{-1})\right\} \cong \mathbb{G}_m^n$$

(the first coordinate is fixed at 1). The roots are:

$$\Phi = \{\pm\varepsilon_i \pm \varepsilon_j : i \neq j\} \cup \{\pm\varepsilon_i\}$$

This differs from type $C_n$ in having *short* roots $\pm\varepsilon_i$ instead of long roots $\pm 2\varepsilon_i$. The total number is $2n^2$, and the rank is $n$.

The simple roots are:

$$\Delta = \{\alpha_1, \ldots, \alpha_n\}, \qquad \alpha_i = \begin{cases} \varepsilon_i - \varepsilon_{i+1} & 1 \leq i \leq n-1 \\ \varepsilon_n & i = n \end{cases}$$

The Weyl group is again $W \cong S_n \ltimes (\mathbb{Z}/2\mathbb{Z})^n$ (signed permutations), of order $2^n \cdot n!$, identical to type $C_n$ as an abstract group.

The root systems $B_n$ and $C_n$ are *dual*: the coroots of $B_n$ form a root system of type $C_n$ and vice versa. As abstract root systems they are non-isomorphic (for $n \geq 3$; for $n = 2$, $B_2 \cong C_2$), since $B_n$ has short roots $\pm\varepsilon_i$ while $C_n$ has long roots $\pm 2\varepsilon_i$. However, their Weyl groups are isomorphic.

## The Root System of $\mathrm{SO}_{2n}$ (Type $D_n$)

**Example 7.1.12: Roots of $\mathrm{SO}_{2n}$**

Let $G = \mathrm{SO}_{2n}$ preserving the form $s = \left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$ (with $\mathrm{char} k \neq 2$). The maximal torus is

$$T = \left\{\mathrm{diag}(t_1, \ldots, t_n, t_1^{-1}, \ldots, t_n^{-1})\right\} \cong \mathbb{G}_m^n$$

The roots are:

$$\Phi = \{\pm\varepsilon_i \pm \varepsilon_j : 1 \leq i < j \leq n\}$$

There are no roots of the form $\pm\varepsilon_i$ or $\pm 2\varepsilon_i$ — all roots have the same length. The total number is $2n(n-1)$, and the rank is $n$.

The simple roots are:

$$\Delta = \{\alpha_1, \ldots, \alpha_n\}, \qquad \alpha_i = \begin{cases} \varepsilon_i - \varepsilon_{i+1} & 1 \leq i \leq n-1 \\ \varepsilon_{n-1} + \varepsilon_n & i = n \end{cases}$$

Note that $\alpha_n = \varepsilon_{n-1} + \varepsilon_n$ is different from the last simple root of types $B_n$ and $C_n$: it is a *sum* rather than a single $\varepsilon_i$ or $2\varepsilon_i$. This reflects the "branching" of the Dynkin diagram at the last node.

The Weyl group is $W \cong S_n \ltimes (\mathbb{Z}/2\mathbb{Z})^{n-1}$: it consists of signed permutations with an *even* number of sign changes. The order is $|W| = 2^{n-1} \cdot n!$, which is half the size of the Weyl groups of $B_n$ and $C_n$.

### Summary of the Classical Root Systems

| Type | Group | Rank | $\lvert\Phi\rvert$ | $\lvert W\rvert$ | Roots |
|---|---|---|---|---|---|
| $A_{n-1}$ | $\mathrm{SL}_n$ | $n-1$ | $n(n-1)$ | $n!$ | $\varepsilon_i - \varepsilon_j$ |
| $B_n$ | $\mathrm{SO}_{2n+1}$ | $n$ | $2n^2$ | $2^n n!$ | $\pm\varepsilon_i \pm \varepsilon_j,\ \pm\varepsilon_i$ |
| $C_n$ | $\mathrm{Sp}_{2n}$ | $n$ | $2n^2$ | $2^n n!$ | $\pm\varepsilon_i \pm \varepsilon_j,\ \pm 2\varepsilon_i$ |
| $D_n$ | $\mathrm{SO}_{2n}$ | $n$ | $2n(n-1)$ | $2^{n-1} n!$ | $\pm\varepsilon_i \pm \varepsilon_j$ |

These are the four "classical" infinite families of root systems. The classification in §7.2 will show that, beyond these, there are exactly five "exceptional" root systems ($G_2$, $F_4$, $E_6$, $E_7$, $E_8$) that do not belong to any infinite family.

#### Exercise 7.1.1

1. Let $G = \mathrm{SL}_3$ with maximal torus $T = \{\mathrm{diag}(t_1, t_2, (t_1 t_2)^{-1})\}$ and standard Borel $B$ (upper-triangular). List all roots, positive roots, and simple roots explicitly. Draw the root system in the plane $\{\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0\} \subseteq \mathbb{R}^3$. Verify that $|\Phi^+| = 3 = \dim \mathrm{SL}_3/B$.

2. For $G = \mathrm{SL}_3$ with the roots from exercise 7.1.1 (1), compute all the coroots $\alpha^\vee \in X_\vee(T)$. Verify $\langle \alpha_i, \alpha_j^\vee \rangle$ gives the Cartan matrix $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ of type $A_2$.

3. Let $W = S_3$ be the Weyl group of $\mathrm{SL}_3$. The simple reflections are $s_1 = s_{\alpha_1}$ and $s_2 = s_{\alpha_2}$. Compute $s_1(\alpha_2)$ and $s_2(\alpha_1)$. Write every element of $W$ as a product of $s_1$ and $s_2$, and compute $\ell(w)$ for each $w \in W$.

4. Verify the root system computation for $\mathrm{Sp}_4$ (type $C_2$) by direct computation. Write down the maximal torus, compute the adjoint action on $\mathfrak{sp}_4$, and identify all 8 roots. Draw the root system in $\mathbb{R}^2$.

5. The root systems $B_2$ and $C_2$ have the same underlying set of 8 roots up to scaling: $\{\pm\varepsilon_1 \pm \varepsilon_2, \pm\varepsilon_1, \pm\varepsilon_2\}$ for $B_2$ and $\{\pm\varepsilon_1 \pm \varepsilon_2, \pm 2\varepsilon_1, \pm 2\varepsilon_2\}$ for $C_2$. Show that $B_2 \cong C_2$ as abstract root systems (there is a linear isomorphism of the ambient space sending one to the other). What happens for $B_3$ and $C_3$?

6. Compute the root system of $\mathrm{GL}_n$ (which is reductive but not semisimple). Show that $\Phi(\mathrm{GL}_n, T) = \Phi(\mathrm{SL}_n, T')$ where $T' = T \cap \mathrm{SL}_n$. Why doesn't $\Phi$ span $X(T) \otimes \mathbb{Q}$ for $\mathrm{GL}_n$? (*Hint*: the center $Z(\mathrm{GL}_n) = \mathbb{G}_m$ provides a "missing direction".)

## 7.2 Abstract Root Systems and Dynkin Diagrams

In the previous section, we extracted a finite set of vectors $\Phi \subseteq X(T) \otimes \mathbb{R}$ from the adjoint action of a maximal torus on the Lie algebra of a reductive group. We now step away from algebraic groups entirely and study the combinatorial structures — *root systems* — that arise from this construction. The axioms are motivated by the properties proved in theorem 7.1.3: the roots come in pairs $\pm\alpha$, each root determines a reflection that preserves the whole system, and the "projection" of any root onto any other is an integer. The remarkable fact is that these axioms are so restrictive that only finitely many root systems exist in each rank, and they can be completely classified by a simple combinatorial device: the Dynkin diagram.

## Axioms for Root Systems

Let $V$ be a finite-dimensional real vector space equipped with a positive-definite inner product $(\cdot\,,\cdot)$. For a nonzero vector $\alpha \in V$, define the *coroot* (or *coweight*)

$$\alpha^\vee = \frac{2\alpha}{(\alpha,\alpha)}$$

and the *reflection* in the hyperplane orthogonal to $\alpha$:

$$s_\alpha : V \to V, \qquad s_\alpha(\lambda) = \lambda - (\lambda, \alpha^\vee)\,\alpha = \lambda - \frac{2(\lambda,\alpha)}{(\alpha,\alpha)}\,\alpha$$

The reflection $s_\alpha$ fixes $\ker(\alpha^\vee)$ pointwise and sends $\alpha \mapsto -\alpha$. It is an orthogonal transformation of order 2.

**Definition 7.2.1: Root System**

A *(reduced, crystallographic) root system* in $V$ is a finite subset $\Phi \subseteq V$ satisfying:

1. (Spanning) $\Phi$ spans $V$ and $0 \notin \Phi$.
2. (Reflection invariance) For every $\alpha \in \Phi$, the reflection $s_\alpha$ preserves $\Phi$: $s_\alpha(\Phi) = \Phi$.
3. (Integrality) For every $\alpha, \beta \in \Phi$, the number $\langle \beta, \alpha^\vee \rangle := (\beta, \alpha^\vee) = \frac{2(\beta,\alpha)}{(\alpha,\alpha)}$ is an integer.
4. (Reduced) If $\alpha \in \Phi$ and $c\alpha \in \Phi$ for $c \in \mathbb{R}$, then $c = \pm 1$.

The *rank* of $\Phi$ is $\dim V$.

Axiom (3) is the *crystallographic condition*: it says that the angle between any two roots is constrained to a discrete set of values. Specifically, if $\alpha, \beta \in \Phi$ with $\alpha \neq \pm\beta$, then:

$$\langle \beta, \alpha^\vee \rangle \cdot \langle \alpha, \beta^\vee \rangle = \frac{4(\alpha,\beta)^2}{(\alpha,\alpha)(\beta,\beta)} = 4\cos^2\theta$$

where $\theta$ is the angle between $\alpha$ and $\beta$. Since both factors on the left are integers, the product $4\cos^2\theta$ is a non-negative integer. Since $\cos^2\theta \leq 1$, we have:

$$\langle \beta, \alpha^\vee \rangle \cdot \langle \alpha, \beta^\vee \rangle \in \{0, 1, 2, 3\}$$

(It cannot be 4 because that would require $\theta = 0$ or $\pi$, meaning $\alpha = \pm\beta$, which we excluded.) This gives exactly four possible angle/length-ratio combinations for a pair of non-proportional roots, and this finiteness is what makes the classification possible.

**7.2.2 Remark: Reduced vs. Non-Reduced** Axiom (4) (the "reduced" condition) excludes root systems containing both $\alpha$ and $2\alpha$. If we drop this axiom, we get *non-reduced* root systems, the only new case being $BC_n$ (a superposition of $B_n$ and $C_n$). Non-reduced root systems arise in the theory of algebraic groups over non-algebraically closed fields (specifically, in the theory of *relative* root systems), but they do not arise from the absolute root system of a split reductive group. We restrict to reduced root systems throughout.

## The Weyl Group and Chambers

**Definition 7.2.3: Weyl Group of a Root System**

The *Weyl group* of a root system $\Phi$ is the subgroup $W = W(\Phi) \subseteq \mathrm{GL}(V)$ generated by all reflections $\{s_\alpha : \alpha \in \Phi\}$.

The Weyl group is a finite group (it permutes the finite set $\Phi$, and is therefore a subgroup of $S_{|\Phi|}$) of orthogonal transformations of $V$. It acts faithfully on $\Phi$ (since $\Phi$ spans $V$, a linear transformation fixing $\Phi$ pointwise is the identity).

The *hyperplane arrangement* $\mathcal{H} = \{\ker(\alpha^\vee) : \alpha \in \Phi\}$ cuts $V$ into finitely many open connected regions called *Weyl chambers*. Each chamber is a connected component of $V \setminus \bigcup_{H \in \mathcal{H}} H$. The Weyl group permutes the chambers, and:

**Proposition 7.2.4: Weyl Chambers and the Weyl Group**

The Weyl group $W$ acts simply transitively on the set of Weyl chambers. In particular, the number of Weyl chambers equals $|W|$.

***Proof Sketch*:**
Transitivity: given two chambers $C, C'$, there exists a sequence of reflections carrying $C$ to $C'$ ("walk" from a point in $C$ to a point in $C'$, crossing one hyperplane at a time; each crossing corresponds to a reflection). Simple transitivity: if $w \in W$ preserves a chamber $C$ (i.e. $w(C) = C$), we must show $w = \mathrm{id}$. If $w \neq \mathrm{id}$, taking a minimal-length word for $w$ in terms of reflections across the walls of $C$ shows that $w(C)$ is separated from $C$ by at least one wall, which contradicts $w(C) = C$. Thus $w$ must be the identity.

## Bases and the Cartan Matrix

A choice of Weyl chamber determines a set of simple roots, analogous to how a choice of Borel subgroup determined simple roots in §7.1.

**Definition 7.2.5: Base of a Root System**

A *base* (or *system of simple roots*) for $\Phi$ is a subset $\Delta = \{\alpha_1, \dots, \alpha_r\} \subseteq \Phi$ such that:

1. $\Delta$ is a basis of $V$.
2. Every root $\alpha \in \Phi$ can be written as $\alpha = \sum_{i=1}^{r} n_i \alpha_i$ with all $n_i \in \mathbb{Z}_{\geq 0}$ or all $n_i \in \mathbb{Z}_{\leq 0}$.

The roots with all $n_i \geq 0$ are the *positive roots* $\Phi^+(\Delta)$, and those with all $n_i \leq 0$ are the *negative roots* $\Phi^-(\Delta) = -\Phi^+(\Delta)$.

**Proposition 7.2.6: Existence and Uniqueness of Bases**

1. Every root system has a base.
2. Bases correspond bijectively to Weyl chambers: the base $\Delta$ corresponding to a chamber $C$ consists of the roots $\alpha \in \Phi$ such that $\ker(\alpha^\vee)$ is a wall of $C$ and $\alpha$ is positive on $C$.
3. Any two bases are related by a unique element of $W$: if $\Delta, \Delta'$ are bases, there exists a unique $w \in W$ with $w(\Delta) = \Delta'$.

***Proof Sketch*:**
For (1), choose any vector $\xi \in V$ not lying on any root hyperplane (this is possible since $\Phi$ is finite and the hyperplanes have codimension 1). This $\xi$ lies in some Weyl chamber $C$. Define $\Phi^+ = \{\alpha \in \Phi : (\alpha, \xi) > 0\}$ and let $\Delta$ be the set of positive roots that cannot be written as sums of other positive roots. One verifies that $\Delta$ is a base. Statement (2) follows from this construction, and (3) follows from the simple transitivity of $W$ on chambers.

The base $\Delta$ encodes the root system through the *Cartan matrix*:

**Definition 7.2.7: Cartan Matrix**

Let $\Delta = \{\alpha_1, \dots, \alpha_r\}$ be a base for $\Phi$. The *Cartan matrix* of $\Phi$ (with respect to $\Delta$) is the $r \times r$ integer matrix

$$A = (a_{ij})_{1 \leq i,j \leq r}, \qquad a_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle = \frac{2(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)}$$

The Cartan matrix has the following properties: $a_{ii} = 2$ for all $i$; $a_{ij} \leq 0$ for $i \neq j$ (because simple roots make obtuse angles: $(\alpha_i, \alpha_j) \leq 0$ for $i \neq j$, which follows from the fact that $\alpha_i - \alpha_j$ is not a root); and $a_{ij} = 0 \iff a_{ji} = 0$.

The Cartan matrix determines the root system up to isomorphism. In fact, even less information suffices:

**Proposition 7.2.8: The Cartan Matrix Determines the Root System**

Two root systems with the same Cartan matrix (up to simultaneous re-ordering of the indices) are isomorphic.

***Proof Sketch*:**
The Weyl group is generated by the simple reflections $s_1, \ldots, s_r$ (this is theorem 7.1.8 in the abstract setting), and the action of each $s_i$ on the simple roots is determined by the Cartan matrix: $s_i(\alpha_j) = \alpha_j - a_{ji}\alpha_i$. Since $\Phi = W \cdot \Delta$ (every root is in the $W$-orbit of some simple root), the Cartan matrix determines $\Phi$ via the action of $W$.

## The Dynkin Diagram

The Cartan matrix is an $r \times r$ array of integers, but most of its entries are 0 (most pairs of simple roots are orthogonal). A more efficient encoding is the *Dynkin diagram*, which records only the nonzero off-diagonal information.

**Definition 7.2.9: Dynkin Diagram**

The *Dynkin diagram* of a root system $\Phi$ (with base $\Delta = \{\alpha_1, \ldots, \alpha_r\}$) is a graph with $r$ nodes (one per simple root) and edges determined by the Cartan matrix:

1. If $a_{ij} = a_{ji} = 0$ (i.e. $\alpha_i \perp \alpha_j$): no edge between nodes $i$ and $j$.
2. If $a_{ij} = a_{ji} = -1$ (i.e. $(\alpha_i, \alpha_i) = (\alpha_j, \alpha_j)$ and the angle is 120ř): a single edge between $i$ and $j$.
3. If $a_{ij} = -1$ and $a_{ji} = -2$ (i.e. $|\alpha_j| = \sqrt{2}\,|\alpha_i|$ and the angle is 135ř): a double edge with an arrow pointing from $j$ to $i$ (from the long root to the short root).
4. If $a_{ij} = -1$ and $a_{ji} = -3$ (i.e. $|\alpha_j| = \sqrt{3}\,|\alpha_i|$ and the angle is 150ř): a triple edge with an arrow pointing from $j$ to $i$.

The arrow convention is: the arrow points toward the *shorter* root. When all roots have the same length (as in types $A$, $D$, $E$), the diagram has no arrows and all edges are single. When there are two root lengths (types $B$, $C$, $F_4$, $G_2$), the arrow distinguishes the long and short roots.

A root system is *irreducible* if it cannot be decomposed as an orthogonal direct sum $\Phi = \Phi_1 \sqcup \Phi_2$ with $(\Phi_1, \Phi_2) = 0$. Equivalently, the Dynkin diagram is connected. Every root system is a direct sum of irreducible ones (corresponding to the connected components of the Dynkin diagram), so it suffices to classify the irreducible (connected) cases.

## Classification in Rank 2

Before stating the general classification, we carry out the rank-2 case completely. This is a clean exercise in plane geometry that illustrates all the constraints.

Let $\Phi$ be an irreducible root system of rank 2 in $V = \mathbb{R}^2$, with base $\Delta = \{\alpha_1, \alpha_2\}$. Since $\Phi$ is irreducible, $\alpha_1$ and $\alpha_2$ are not orthogonal: $a_{12} \neq 0$. The constraint $a_{12}a_{21} \in \{1, 2, 3\}$ (since both $a_{12}, a_{21} \leq -1$ and $a_{12}a_{21} \leq 3$) gives three possibilities:

*Case 1:* $a_{12}a_{21} = 1$, so $a_{12} = a_{21} = -1$. The angle between $\alpha_1$ and $\alpha_2$ is 120ř and $|\alpha_1| = |\alpha_2|$. The Weyl group is generated by two reflections through lines meeting at 60ř, giving $|W| = 6$ and $W \cong S_3$. The root system has 6 roots forming a regular hexagon. This is **type** $A_2$, the root system of $\mathrm{SL}_3$.

*Case 2:* $a_{12}a_{21} = 2$, so (up to relabeling) $a_{12} = -1$ and $a_{21} = -2$. The angle is 135ř and $|\alpha_2| = \sqrt{2}\,|\alpha_1|$. The Weyl group has $|W| = 8$ and is the dihedral group $D_4$ (symmetries of a square). The root system has 8 roots. This is **type** $B_2 = C_2$, the root system of $\mathrm{SO}_5 \cong \mathrm{Sp}_4$.

*Case 3:* $a_{12}a_{21} = 3$, so (up to relabeling) $a_{12} = -1$ and $a_{21} = -3$. The angle is 150ř and $|\alpha_2| = \sqrt{3}\,|\alpha_1|$. The Weyl group has $|W| = 12$ and is the dihedral group $D_6$ (symmetries of a regular hexagon). The root system has 12 roots. This is **type** $G_2$, the smallest exceptional root system.

These are the *only* irreducible rank-2 root systems. We also have the reducible case $A_1 \times A_1$ ($a_{12} = a_{21} = 0$, two orthogonal roots), which has 4 roots and $W \cong (\mathbb{Z}/2\mathbb{Z})^2$.

**Example 7.2.10: The Root System $G_2$**

Let $\alpha_1$ be a short root and $\alpha_2$ a long root with $|\alpha_2| = \sqrt{3}\,|\alpha_1|$ and angle 150ř. The 12 roots are:

$$\pm\alpha_1, \quad \pm\alpha_2, \quad \pm(\alpha_1+\alpha_2), \quad \pm(2\alpha_1+\alpha_2), \quad \pm(3\alpha_1+\alpha_2), \quad \pm(3\alpha_1+2\alpha_2)$$

The positive roots (with respect to the base $\{\alpha_1, \alpha_2\}$) are $\alpha_1$, $\alpha_2$, $\alpha_1+\alpha_2$, $2\alpha_1+\alpha_2$, $3\alpha_1+\alpha_2$, $3\alpha_1+2\alpha_2$. The highest root (the root with the largest coefficients) is $3\alpha_1+2\alpha_2$.

The Weyl group $W(G_2)$ is the dihedral group of order 12, isomorphic to $S_3 \times \mathbb{Z}/2\mathbb{Z}$. It is generated by the two simple reflections $s_1 = s_{\alpha_1}$ and $s_2 = s_{\alpha_2}$ with the relation $(s_1s_2)^6 = \mathrm{id}$ (the angle between the simple roots is 150ř, so the angle between their reflecting hyperplanes is 180ř − 150ř = 30ř; the product of two reflections separated by 30ř is a rotation by 2 × 30ř = 60ř, which has order 6, giving the dihedral group $D_6$ of order 12).

The root system $G_2$ arises from the exceptional algebraic group of the same name, the automorphism group of the octonion algebra. We shall not construct $G_2$ as an algebraic group in this book, but its existence is guaranteed by the existence theorem stated in §7.3.

## The Classification Theorem

The rank-2 analysis extends to all ranks. The argument proceeds by showing that the Cartan matrix of an irreducible root system is severely constrained: since $a_{ij}a_{ji} \in \{0, 1, 2, 3\}$ for all $i \neq j$, the Dynkin diagram is a connected graph with single, double, or triple edges and at most one arrow per edge. The positive-definiteness of the inner product imposes further constraints (the Cartan matrix must have positive determinant), and a case analysis reduces the classification to a finite check.

**Theorem 7.2.11: Classification of Irreducible Root Systems**

The irreducible (reduced, crystallographic) root systems are classified by their Dynkin diagrams. The complete list of connected Dynkin diagrams is:

*Classical types (four infinite families):*

$A_n$ ($n \geq 1$):

$A_n$ 

($n$ nodes in a line)

$B_n$ ($n \geq 2$):

$B_n$ 

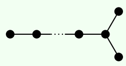
($n$ nodes, double edge with arrow at the end)

$C_n$ ($n \geq 3$):

$C_n$ 

($n$ nodes, double edge with arrow at the end, reversed)

$D_n$ ($n \geq 4$):

$D_n$ 

($n$ nodes, branching at the end)

*Exceptional types (five isolated cases):*

$G_2$

$G_2$ 

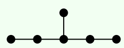
(triple edge, 2 nodes)

$F_4$

$F_4$ 

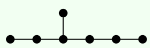
(4 nodes, double edge in the middle)

$E_6$

$E_6$ 

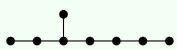
(6 nodes, one branch of length 1)

$E_7$

$E_7$ 

(7 nodes)

$E_8$

$E_8$ 

(8 nodes)

There are no others. In particular, the only diagrams with multiple edges are $B_n$, $C_n$, $F_4$ (double edges), and $G_2$ (triple edge). The only diagrams with branching are $D_n$ and $E_6, E_7, E_8$.

***Proof Strategy*:**
The proof proceeds in stages, each eliminating large families of potential diagrams.

*Step 1: Constraints on edges.* The condition $a_{ij}a_{ji} \in \{0, 1, 2, 3\}$ means each pair of connected nodes has a single edge ($a_{ij}a_{ji} = 1$), a double edge ($a_{ij}a_{ji} = 2$), or a triple edge ($a_{ij}a_{ji} = 3$). No quadruple or higher edges exist.

*Step 2: No cycles.* The positive-definiteness of the bilinear form implies that the underlying graph of the Dynkin diagram is a *tree* (a connected graph with no cycles). Indeed, if the diagram contained a cycle $\alpha_{i_1}$—$\alpha_{i_2}$—$\cdots$—$\alpha_{i_m}$—$\alpha_{i_1}$, the vector $v = \alpha_{i_1} + \cdots + \alpha_{i_m}$ would satisfy $(v, v) \leq 0$ (using the obtuse-angle property of simple roots and the cycle condition), contradicting positive-definiteness.

*Step 3: At most one multiple edge.* If the diagram has a double or triple edge, it can have at most one such edge, and the rest of the edges are single. This is proved by bounding the "total energy" $\sum_{i \neq j} a_{ij}a_{ji}$ using the positive-definiteness constraint.

*Step 4: Branching constraints.* The diagram has at most one branching point (a node of degree 3), and no nodes of degree $\geq 4$. If the three branch lengths extending from the central node are $p - 1, q - 1, r - 1$ (meaning the number of nodes in each branch, not counting the central node), then the total number of nodes is $n = (p - 1) + (q - 1) + (r - 1) + 1 = p + q + r - 2$. The positive-definiteness of the quadratic form forces the exact constraint:

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$$

The integer solutions (with $p \leq q \leq r$) are completely determined:

1. $(1, q, r)$: This means $p - 1 = 0$, so one branch is empty. There is no actual branching, and the graph is a single line, giving type $A_n$.
2. $(2, 2, r)$ for $r \geq 2$: Two branches have length 1, and the third has length $r - 1$. This gives the $D_n$ family (with $n = r + 2$).
3. $(2, 3, 3)$: Branches of length $1, 2, 2$. Total nodes $n = 1 + 2 + 2 + 1 = 6$. This gives $E_6$.
4. $(2, 3, 4)$: Branches of length $1, 2, 3$. Total nodes $n = 1 + 2 + 3 + 1 = 7$. This gives $E_7$.
5. $(2, 3, 5)$: Branches of length $1, 2, 4$. Total nodes $n = 1 + 2 + 4 + 1 = 8$. This gives $E_8$.

No further solutions exist (for example, $(2, 3, 6)$ gives $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$, which is not strictly greater than 1; this boundary case corresponds to the positive semi-definite affine Dynkin diagram $\tilde{E}_8$).

*Step 5: Multiple edges with chains.* Diagrams with a double edge are constrained: they cannot have branching, and if there are nodes on both sides of the double edge, there can be at most one node on each side. This gives $B_n$ and $C_n$ (chains of nodes on one side only) and $F_4$ (exactly one node on each side). Triple edges allow no additional nodes at all: $G_2$ is the only possibility.

The full proof is a finite (but combinatorially detailed) verification that these are the only diagrams satisfying all constraints simultaneously. We refer to Humphreys [10] or Bourbaki [3] for the complete argument.

The classification is one of the most renowned results in mathematics: a simple set of axioms about reflections and integrality leads, through elementary but careful combinatorics, to a complete and finite list. The four classical families ($A_n$, $B_n$, $C_n$, $D_n$) correspond to the matrix groups (SL, $\mathrm{SO}_{\mathrm{odd}}$, Sp, $\mathrm{SO}_{\mathrm{even}}$), and the five exceptional types ($G_2$, $F_4$, $E_6$, $E_7$, $E_8$) correspond to algebraic groups that cannot be described as matrix groups preserving a classical bilinear form. Their existence is guaranteed by the existence theorem in §7.3.

## Numerical Data of the Root Systems

We collect the key numerical invariants for reference:

| **Type** | **Rank** | $\lvert\Phi\rvert$ | $\lvert W\rvert$ | **Root lengths** |
|---|---|---|---|---|
| $A_n$ | $n$ | $n(n+1)$ | $(n+1)!$ | all equal |
| $B_n$ | $n$ | $2n^2$ | $2^n n!$ | two (short : long $= 1 : \sqrt{2}$) |
| $C_n$ | $n$ | $2n^2$ | $2^n n!$ | two (short : long $= 1 : \sqrt{2}$) |
| $D_n$ | $n$ | $2n(n-1)$ | $2^{n-1}n!$ | all equal |
| $G_2$ | 2 | 12 | 12 | two (short : long $= 1 : \sqrt{3}$) |
| $F_4$ | 4 | 48 | 1152 | two (short : long $= 1 : \sqrt{2}$) |
| $E_6$ | 6 | 72 | 51840 | all equal |
| $E_7$ | 7 | 126 | 2903040 | all equal |
| $E_8$ | 8 | 240 | 696729600 | all equal |

The simply-laced types ($A$, $D$, $E$) have all roots of equal length; the non-simply-laced types ($B$, $C$, $F_4$, $G_2$) have two root lengths. The Weyl group orders grow rapidly: $|W(E_8)| = 696{,}729{,}600 = 2^{14}{\cdot}3^5{\cdot}5^2{\cdot}7$.

### Exercise 7.2.1

1. Verify that the root system $\Phi = \{\pm\varepsilon_i \pm \varepsilon_j : i \neq j\} \cup \{\pm\varepsilon_i\}$ of type $B_n$ (example 7.1.11) satisfies all four axioms of definition 7.2.1. In particular, verify the integrality condition $\langle \beta, \alpha^\vee\rangle \in \mathbb{Z}$ for a pair of roots of different lengths.
2. Draw the root system $G_2$ in the plane $\mathbb{R}^2$ (choose coordinates so that the short roots have length 1 and the long roots have length $\sqrt{3}$). Identify all 12 roots, the 6 positive roots, and the 2 simple roots. Compute the Cartan matrix and verify it is $\left(\begin{smallmatrix} 2 & -1 \\ -3 & 2 \end{smallmatrix}\right)$.
3. Verify that the Weyl group of type $D_n$ has order $2^{n-1} \cdot n!$ by showing it is the group of $n \times n$ signed permutation matrices with an *even* number of sign changes. (*Hint*: the reflection $s_{\varepsilon_i - \varepsilon_j}$ transposes the $i$-th and $j$-th coordinates, while $s_{\varepsilon_i + \varepsilon_j}$ transposes them and changes both signs.)

4. The *highest root* $\tilde{\alpha}$ of a root system is the unique root $\tilde{\alpha} = \sum n_i \alpha_i$ with $\sum n_i$ maximal. Compute the highest root for types $A_3$, $B_3$, $C_3$, and $G_2$.

5. The *dual root system* $\Phi^\vee$ is the set of coroots $\{\alpha^\vee : \alpha \in \Phi\}$. Show that $\Phi^\vee$ is itself a root system. Determine the type of $\Phi^\vee$ when $\Phi$ has type $A_n$, $B_n$, $C_n$, $D_n$, $G_2$. (The answer reveals the duality $B_n \leftrightarrow C_n$ and the self-duality of $A_n$, $D_n$, $G_2$.)

6. Prove that the Dynkin diagram of an irreducible root system contains no cycles. (*Hint*: if $\alpha_1, \dots, \alpha_m$ form a cycle, consider the sum of their associated unit vectors $v = u_1 + \cdots + u_m$ where $u_i = \alpha_i/|\alpha_i|$, and compute $(v, v)$ using the fact that $(u_i, u_{i+1}) \leq -1/2$ for adjacent roots.)

## 7.3 Root Data and the Classification of Reductive Groups

The root system $\Phi$ captures the "shape" of a reductive group $G$, but it does not capture $G$ completely. Two non-isomorphic groups can share the same root system: for instance, $\mathrm{SL}_n$ and $\mathrm{PGL}_n$ both have root system $A_{n-1}$, yet they are not isomorphic (the former has center $\mu_n$, the latter is centerless). The missing information is the *lattice*: the root system $\Phi$ sits inside the character lattice $X(T)$, and the position of $\Phi$ relative to $X(T)$ determines the group up to isomorphism. This extra data is encoded in the *root datum*, and the main theorems of this section assert that root data classify reductive groups completely.

### Root Data

**Definition 7.3.1: Root Datum**

A *root datum* is a quadruple $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$ where:

1. $X$ and $X^\vee$ are free abelian groups of finite rank, equipped with a perfect pairing $\langle \cdot, \cdot \rangle : X \times X^\vee \to \mathbb{Z}$ (i.e. $X^\vee \cong \mathrm{Hom}_{\mathbb{Z}}(X, \mathbb{Z})$).
2. $\Phi \subseteq X$ and $\Phi^\vee \subseteq X^\vee$ are finite subsets (the *roots* and *coroots*), equipped with a bijection $\Phi \xrightarrow{\sim} \Phi^\vee$, $\alpha \mapsto \alpha^\vee$.
3. For every $\alpha \in \Phi$: $\langle \alpha, \alpha^\vee \rangle = 2$.
4. For every $\alpha \in \Phi$, the reflection $s_\alpha : X \to X$ defined by $s_\alpha(\lambda) = \lambda - \langle \lambda, \alpha^\vee \rangle \alpha$ preserves $\Phi$, and the dual reflection $s_\alpha^\vee : X^\vee \to X^\vee$ defined by $s_\alpha^\vee(\mu) = \mu - \langle \alpha, \mu \rangle \alpha^\vee$ preserves $\Phi^\vee$.

The root datum is called *reduced* if for all $\alpha \in \Phi$, we have $2\alpha \notin \Phi$.

The key difference from an abstract root system is that the roots live in a *specific lattice* $X$, not just in a real vector space. The lattice $X$ determines the "size" of the center: a larger lattice (more characters) corresponds to a smaller center (fewer relations among characters).

Given a connected reductive group $G$ with maximal torus $T$, the associated root datum is:

$$\Psi(G, T) = (X(T),\ \Phi(G, T),\ X_\vee(T),\ \Phi^\vee(G, T))$$

where $X(T)$ is the character lattice (definition 5.2.1), $X_\vee(T)$ is the cocharacter lattice (definition 5.2.2), $\Phi$ are the roots (definition 7.1.2), $\Phi^\vee$ are the coroots (theorem 7.1.3(5)), and the pairing is $\langle \chi, \lambda \rangle$ from §5.2. The axioms of a root datum are satisfied by the properties established in theorem 7.1.3.

## Root Data vs. Root Systems

A root datum determines a root system (embed $\Phi$ into $V = X \otimes_{\mathbb{Z}} \mathbb{R}$ and forget the lattice structure), but different root data can give the same root system. The root *system* classifies groups up to *isogeny* (a surjective homomorphism with finite kernel), while the root *datum* classifies groups up to *isomorphism*.

**Example 7.3.2: $\mathrm{SL}_n$ vs. $\mathrm{PGL}_n$ vs. $\mathrm{GL}_n$**
All three groups have root system of type $A_{n-1}$ (the roots $\varepsilon_i - \varepsilon_j$ with $i \neq j$). But their root data differ:

$\mathrm{SL}_n$: The maximal torus $T = \{\mathrm{diag}(t_1, \ldots, t_n) : \prod t_i = 1\}$ has $X(T) = \mathbb{Z}^n/\mathbb{Z}(1, \ldots, 1)$. The roots $\varepsilon_i - \varepsilon_j$ span a sublattice (the root lattice $Q$) of index $n$ in $X(T)$. This is the *simply connected* root datum of type $A_{n-1}$: the character lattice $X(T)$ is the *weight lattice* $P$ (the largest lattice compatible with the integrality condition).

$\mathrm{PGL}_n$: The maximal torus is $T' = T/Z(\mathrm{SL}_n) = T/\mu_n$, and $X(T') \subseteq X(T)$ is the sublattice generated by the roots $\varepsilon_i - \varepsilon_j$ themselves. This is the *adjoint* root datum: the character lattice $X(T')$ equals the *root lattice* $Q$ (the smallest lattice containing the roots). The quotient $P/Q \cong \mathbb{Z}/n\mathbb{Z}$ measures the difference between the two lattices and is isomorphic to $Z(\mathrm{SL}_n) = \mu_n$.

$\mathrm{GL}_n$: The maximal torus $T = \mathbb{G}_m^n$ has $X(T) = \mathbb{Z}^n$, which strictly contains the root lattice $Q$ (the roots $\varepsilon_i - \varepsilon_j$ span only the hyperplane $\sum x_i = 0$). The extra direction is the determinant character $\det = \varepsilon_1 + \cdots + \varepsilon_n$, reflecting the nontrivial center $Z(\mathrm{GL}_n) = \mathbb{G}_m$. This is neither simply connected nor adjoint: $\mathrm{GL}_n$ is reductive but not semisimple, and its root datum records the extra torus direction.

The isogeny $\mathrm{SL}_n \to \mathrm{PGL}_n$ (with kernel $\mu_n$) corresponds to the inclusion of root data $Q \hookrightarrow P$ (the adjoint lattice sits inside the weight lattice). Every lattice $L$ with $Q \subseteq L \subseteq P$ gives a reductive group "between" $\mathrm{PGL}_n$ and $\mathrm{SL}_n$ (an *intermediate isogeny form*). For $n$ prime, there are only two choices ($Q$ and $P$), but for composite $n$, the subgroups of $P/Q \cong \mathbb{Z}/n\mathbb{Z}$ give additional forms.

More generally, for a semisimple root system $\Phi$, the lattices between the root lattice $Q$ and the weight lattice $P$ are in bijection with the subgroups of the finite abelian group $P/Q$ (the *fundamental group* of the root system). Each such lattice gives a different root datum, hence a different semisimple group. The two extreme cases have names:

**Definition 7.3.3: Simply Connected and Adjoint**

A semisimple group $G$ is called:

1. *Simply connected* if $X(T) = P$ (the weight lattice), equivalently $Z(G)$ is as large as possible (equal to $P/Q$).
2. *Adjoint* if $X(T) = Q$ (the root lattice), equivalently $Z(G) = \{e\}$ (the center is trivial).

For each root system $\Phi$, the simply connected group is denoted $G_{sc}$ and the adjoint group $G_{ad}$. Every semisimple group with root system $\Phi$ is a quotient of $G_{sc}$ by a subgroup of $Z(G_{sc})$.

| **Type** | $G_{sc}$ (simply connected) | $G_{ad}$ (adjoint) | $P/Q$ |
|---|---|---|---|
| $A_n$ | $\mathrm{SL}_{n+1}$ | $\mathrm{PGL}_{n+1}$ | $\mathbb{Z}/(n+1)\mathbb{Z}$ |
| $B_n$ | $\mathrm{Spin}_{2n+1}$ | $\mathrm{SO}_{2n+1}$ | $\mathbb{Z}/2\mathbb{Z}$ |
| $C_n$ | $\mathrm{Sp}_{2n}$ | $\mathrm{PGL}_{2n}$ | $\mathbb{Z}/2\mathbb{Z}$ |
| $D_n$ ($n$ odd) | $\mathrm{Spin}_{2n}$ | $\mathrm{PGL}_{2n}$ | $\mathbb{Z}/4\mathbb{Z}$ |
| $D_n$ ($n$ even) | $\mathrm{Spin}_{2n}$ | $\mathrm{PGL}_{2n}$ | $(\mathbb{Z}/2\mathbb{Z})^2$ |
| $G_2$ | $G_2$ | $G_2$ | 1 |
| $F_4$ | $F_4$ | $F_4$ | 1 |
| $E_6$ | $E_6^{sc}$ | $E_6^{ad}$ | $\mathbb{Z}/3\mathbb{Z}$ |
| $E_7$ | $E_7^{sc}$ | $E_7^{ad}$ | $\mathbb{Z}/2\mathbb{Z}$ |
| $E_8$ | $E_8$ | $E_8$ | 1 |

When $P/Q = 1$ (types $G_2$, $F_4$, $E_8$), the simply connected and adjoint forms coincide: there is a unique semisimple group with the given root system.

## The Isomorphism Theorem

The first main theorem says that the root datum is a *complete* invariant: it determines the reductive group up to isomorphism.

**Theorem 7.3.4: Isomorphism Theorem**

Let $G$ and $G'$ be connected reductive groups over an algebraically closed field $k$, with maximal tori $T \subseteq G$ and $T' \subseteq G'$. Let $\Psi = \Psi(G,T)$ and $\Psi' = \Psi(G',T')$ be the associated root data. Then every isomorphism of root data $\varphi : \Psi' \xrightarrow{\sim} \Psi$ (i.e. an isomorphism $\varphi : X(T') \xrightarrow{\sim} X(T)$ mapping $\Phi'$ to $\Phi$, such that its adjoint $\varphi^\vee : X_\vee(T) \xrightarrow{\sim} X_\vee(T')$ maps $\Phi^\vee$ to $(\Phi')^\vee$) is induced by an isomorphism of algebraic groups $f : G \xrightarrow{\sim} G'$ with $f(T) = T'$.

More precisely: the natural map

$$\left\{ f : G \xrightarrow{\sim} G' \;\middle|\; f(T) = T' \right\} \Big/ \mathrm{Inn}(T') \longrightarrow \mathrm{Isom}(\Psi', \Psi)$$

is a bijection, where $\mathrm{Inn}(T')$ denotes inner automorphisms by elements of $T'$ (which act trivially on $\Psi'$).

***Proof Key ideas*:**
The proof proceeds in two stages.

*Stage 1: The case $G = G'$, $T = T'$ (automorphisms).* An isomorphism of root data $\varphi : \Psi \to \Psi$ is an automorphism of the root datum. This is an element of $\mathrm{Aut}(\Psi) = W \rtimes \mathrm{Aut}(\text{Dynkin diagram})$ (the Weyl group acts by inner automorphisms of $\Psi$, and the diagram automorphisms are the "outer" part). One shows that every such automorphism lifts to an automorphism of $G$: the Weyl group elements lift to inner automorphisms $\mathrm{Inn}(n_w)$ for $n_w \in N_G(T)$, and the diagram automorphisms lift to "graph automorphisms" of $G$ (for instance, the nontrivial diagram automorphism of $A_n$ corresponds to the outer automorphism $g \mapsto (g^T)^{-1}$ of $\mathrm{SL}_{n+1}$).

*Stage 2: The general case.* Given an isomorphism $\varphi : \Psi' \to \Psi$, one constructs an isomorphism $f : G \to G'$ by first building isomorphisms of the $\mathrm{SL}_2$-subgroups (one for each simple root), then gluing them using the *Bruhat decomposition*: $G = \bigsqcup_{w \in W} BwB$ (the decomposition of $G$ into double cosets of $B$). The Bruhat decomposition gives a combinatorial "skeleton" of $G$ in terms of $B$, $T$, and the root groups $U_\alpha$ (one for each root), and the root datum determines the gluing data completely. The isomorphism $\varphi$ provides compatible identifications of the root groups, which extend to a global isomorphism $f : G \to G'$.

The full proof is substantial and requires the detailed theory of root groups and the Bruhat decomposition. We refer to Springer [15] or Milne [14] for the complete argument.

## The Existence Theorem

The second main theorem is the converse: every root datum actually arises from a reductive group.

**Theorem 7.3.5: Existence Theorem**

Let $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$ be a reduced root datum. Then there exists a connected reductive group $G$ over any algebraically closed field $k$, together with a maximal torus $T \subseteq G$, such that $\Psi(G, T) \cong \Psi$.

***Proof Key ideas*:**
For the classical root data ($A_n$, $B_n$, $C_n$, $D_n$), the groups were constructed explicitly in chapter 1 as matrix groups (SL, SO, Sp, and their variants). The content of the existence theorem is that the five exceptional root data ($G_2$, $F_4$, $E_6$, $E_7$, $E_8$) also correspond to algebraic groups.

The general construction is due to Chevalley (1955). The idea is to build $G$ directly from the root datum by generators and relations, working inside the "Chevalley algebra" — a Lie algebra over $\mathbb{Z}$ associated to the root system. Specifically:

1. Start with the complex semisimple Lie algebra $\mathfrak{g}_\mathbb{C}$ of the given type (constructed, for instance, via the Serre relations from the Cartan matrix).
2. Choose a *Chevalley basis*: a basis $\{h_i, e_\alpha\}$ of $\mathfrak{g}_\mathbb{C}$ (indexed by simple coroots $h_i$ and roots $\alpha \in \Phi$) with the property that all structure constants are *integers*.
3. Define $\mathfrak{g}_\mathbb{Z} = \bigoplus \mathbb{Z}h_i \oplus \bigoplus_\alpha \mathbb{Z}e_\alpha$ (the "Chevalley $\mathbb{Z}$-form"). For any field $k$, set $\mathfrak{g}_k = \mathfrak{g}_\mathbb{Z} \otimes_\mathbb{Z} k$.
4. To build the group, one chooses a faithful finite-dimensional representation $V$ of $\mathfrak{g}_\mathbb{C}$ and a suitable $\mathbb{Z}$-lattice $V_\mathbb{Z} \subseteq V$. The generators $e_\alpha$ act nilpotently on $V$, so the formal exponentials $x_\alpha(t) = \exp(t \cdot e_\alpha)$ are finite sums. By the properties of the Chevalley basis, these operators preserve the lattice and can be base-changed to any field $k$. The algebraic group $G(k)$ is defined as the subgroup of $\mathrm{GL}(V_k)$ generated by these $x_\alpha(t)$ for $\alpha \in \Phi$ and $t \in k$.
5. One verifies that $G$ is a connected reductive group with root datum $\Psi$, by checking the commutation relations of the root group elements (the "Steinberg relations") and constructing the maximal torus and Borel subgroup.

One of the strengths of Chevalley's construction is that it works *uniformly over any field* (even over $\mathbb{Z}$!), producing "split" reductive groups. This is why the classification of split reductive groups is the same in every characteristic: the Chevalley groups $G(k)$ depend only on the root datum $\Psi$ and the field $k$, not on the characteristic of $k$.

For the complete construction, see Steinberg [16] or Milne [14]. The original paper is Chevalley [4].

## The Classification

Combining the isomorphism and existence theorems, we obtain the classification of split reductive groups:

**Theorem 7.3.6: Classification of Split Reductive Groups**

The map $G \mapsto \Psi(G, T)$ is a bijection:

$$\left\{ \begin{array}{c} \text{Split connected reductive} \\ \text{groups over } k \text{ (up to isom.)} \end{array} \right\} \xleftrightarrow{1:1} \left\{ \begin{array}{c} \text{Reduced root data} \\ \text{(up to isom.)} \end{array} \right\}$$

In particular, for *semisimple* groups, the classification reduces to: a choice of root system (one of the Dynkin diagrams from theorem 7.2.11), together with a choice of lattice $X$ between the root lattice $Q$ and the weight lattice $P$.

For semisimple groups, the lattice choice corresponds to a subgroup of the fundamental group $P/Q$, which is a finite abelian group computed from the Dynkin diagram. The extreme choices give the simply connected and adjoint forms.

**Example 7.3.7: The Complete List for Type $A_2$**

The root system $A_2$ has $P/Q \cong \mathbb{Z}/3\mathbb{Z}$. The subgroups of $\mathbb{Z}/3\mathbb{Z}$ are $\{0\}$ and $\mathbb{Z}/3\mathbb{Z}$ itself (since 3 is prime, there are no intermediate subgroups). The two semisimple groups with root system $A_2$ are:

1. $\mathrm{SL}_3$ (simply connected, $X(T) = P$, center $\mu_3$).
2. $\mathrm{PGL}_3$ (adjoint, $X(T) = Q$, center trivial).

These are connected by the isogeny $\mathrm{SL}_3 \to \mathrm{PGL}_3$ with kernel $\mu_3$.

**Example 7.3.8: The Complete List for Type $D_4$**

The root system $D_4$ has $P/Q \cong (\mathbb{Z}/2\mathbb{Z})^2$. The subgroups of $(\mathbb{Z}/2\mathbb{Z})^2$ are: $\{0\}$ (one), three subgroups isomorphic to $\mathbb{Z}/2\mathbb{Z}$, and $(\mathbb{Z}/2\mathbb{Z})^2$ itself. This gives 5 distinct lattices between $Q$ and $P$, which define five seemingly different semisimple groups:

1. $\mathrm{Spin}_8$ (simply connected, center $(\mathbb{Z}/2\mathbb{Z})^2$).
2. $\mathrm{SO}_8$ and two "half-spin" groups $\mathrm{HSpin}_8$ (three intermediate forms, each with center $\mathbb{Z}/2\mathbb{Z}$).
3. $\mathrm{PGL}_8$ (adjoint, centerless).

However, the classification theorem classifies groups up to *isomorphism*, which corresponds to root data up to *isomorphism*. The three intermediate lattices are permuted transitively by the *triality automorphism* of $D_4$ (the $S_3$-symmetry of the $D_4$ Dynkin diagram, which has three "legs" meeting at a central node). Because diagram automorphisms induce isomorphisms of root data, these three intermediate groups are actually *isomorphic* as algebraic groups!

Thus, up to isomorphism, there are only *three* semisimple groups of type $D_4$: $\mathrm{Spin}_8$, $\mathrm{SO}_8$ (which is isomorphic to the half-spin groups), and $\mathrm{PGL}_8$. This phenomenon — where multiple intermediate lattices yield isomorphic groups due to diagram symmetries — is a spectacular feature of triality. (For $D_n$ with $n \geq 6$ even, the diagram only has a $\mathbb{Z}/2\mathbb{Z}$ symmetry, which swaps the two half-spin groups but leaves $\mathrm{SO}_{2n}$ distinct, giving four isomorphism classes).

## Beyond Split Groups

The classification theorem, as stated, applies to *split* reductive groups: those possessing a maximal torus $T$ that is split over the ground field $k$ (i.e. $T \cong \mathbb{G}_m^r$ over $k$ itself). Over a non-algebraically closed field, there are additional "non-split" or "twisted" forms of each group, classified by Galois cohomology.

For a split group $G$ over $k$, the set of $k$-forms (reductive groups $G'$ over $k$ with $G'_{\overline{k}} \cong G_{\overline{k}}$) is classified by the pointed set $H^1(\mathrm{Gal}(\overline{k}/k), \mathrm{Aut}(G_{\overline{k}}))$ — the first Galois cohomology of the automorphism group. The automorphism group $\mathrm{Aut}(G)$ fits into an exact sequence

$$1 \longrightarrow G_{ad}(k) \longrightarrow \mathrm{Aut}(G)(k) \longrightarrow \mathrm{Aut}(\Psi) \longrightarrow 1$$

where $G_{ad}$ is the adjoint form (inner automorphisms) and $\mathrm{Aut}(\Psi)$ is the group of diagram automorphisms (outer automorphisms). The inner forms (those where the Galois action on $\Psi$ is trivial) are classified by $H^1(k, G_{ad})$, while the outer forms involve nontrivial Galois action on the Dynkin diagram.

This is the connection to the twisted finite groups and Galois cohomology discussed in chapter 1 (§1.1.1): the "twisting" of algebraic groups by Galois cocycles is the higher-dimensional analogue of the twisting of finite group schemes that we studied there. A full development of the theory of non-split groups over arbitrary fields belongs to the next stage of the subject (the theory of *reductive groups over general fields*, as developed by Borel–Tits), and is beyond the scope of this book.

## Closing Remarks

We close Part I by summarizing the path we have traveled. Starting from the definition of an affine algebraic group (chapter 1), we developed its representation theory (chapter 2), extracted its infinitesimal invariant the Lie algebra (chapter 3), constructed quotients (chapter 4), decomposed its elements via Jordan decomposition and studied the atomic pieces — tori and unipotent groups (chapter 5) — proved the Lie–Kolchin theorem, Borel's fixed-point theorem, and the conjugacy theorems (chapter 6), and finally arrived at the root system, which classifies semisimple groups via Dynkin diagrams (chapter 7).

The two main theorems — isomorphism and existence — assert that the passage from geometry (algebraic groups) to combinatorics (root data) loses no information and omits nothing: every root datum arises, and groups with the same root datum are isomorphic.

Part II of this book will reverse the arrow: starting from the root datum and the combinatorics of the Weyl group, we will develop the *representation theory* of reductive groups. The central objects will be the *highest weight modules* (finite-dimensional representations parametrized by dominant weights in $X(T)$), the *Weyl character formula* (an explicit formula for the character of a highest weight module in terms of the Weyl group), and the geometry of the flag variety $G/B$ (Schubert calculus, the Borel–Weil–Bott theorem, and geometric representation theory). The root datum, constructed here as a classification tool, will become the organizing principle for all of representation theory.

### Exercise 7.3.1

1. Write down the root datum $\Psi(\mathrm{SL}_2, T)$ explicitly: $X(T)$, $X_\vee(T)$, $\Phi$, $\Phi^\vee$, and the pairing. Then write down $\Psi(\mathrm{PGL}_2, T')$. Show that the two root data are not isomorphic (even though the root systems are both $A_1$). What is the relationship between them?

2. For the root system $A_2$, compute the weight lattice $P$ and root lattice $Q$ explicitly (as sublattices of $\mathbb{R}^3$ in the hyperplane $\sum x_i = 0$). Verify that $P/Q \cong \mathbb{Z}/3\mathbb{Z}$. Identify the *fundamental weights* $\omega_1, \omega_2 \in P$ (defined by $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$).

3. The isogeny $\pi : \mathrm{SL}_n \to \mathrm{PGL}_n$ (with kernel $\mu_n$) induces a map on root data. Describe this map explicitly: what does $\pi^* : X(T') \to X(T)$ look like on the character lattices? Verify that $\pi^*$ sends the root lattice $Q$ into the weight lattice $P$, and that the cokernel $P/\pi^*(Q)$ is isomorphic to $\mu_n$.

4. For the root system $G_2$ (rank 2, 12 roots), the fundamental group $P/Q$ is trivial. Conclude that there is a *unique* semisimple group with root system $G_2$ (up to isomorphism) over any algebraically closed field. This group is denoted $G_2$.

5. Let $G = \mathrm{GL}_n$ with maximal torus $T = \mathbb{G}_m^n$. Write down the root datum $\Psi(\mathrm{GL}_n, T)$. Show that $X(T) = \mathbb{Z}^n$, $\Phi = \{\varepsilon_i - \varepsilon_j\}$, and $\Phi^\vee = \{\varepsilon_i^\vee - \varepsilon_j^\vee\}$, and verify the pairing $\langle \varepsilon_i, \varepsilon_j^\vee \rangle = \delta_{ij}$. Explain why this root datum is not semisimple (i.e. $\Phi$ does not span $X(T) \otimes \mathbb{Q}$) and identify the "extra" direction.

6. The Dynkin diagram of $D_4$ has a symmetry group isomorphic to $S_3$ (permuting the three outer nodes). This gives a group of outer automorphisms of $\mathrm{Spin}_8$ called *triality.* Show that triality permutes the three 8-dimensional representations of $\mathrm{Spin}_8$: the vector representation, and the two half-spin representations. (*Hint*: the three outer nodes of $D_4$ correspond to the three fundamental weights $\omega_1$, $\omega_3$, $\omega_4$, each of which defines an 8-dimensional representation; triality permutes these weights.)

# Part II

# Representation Theory

The goal of Part II is to classify, compute with, and geometrically construct the representations of a reductive algebraic group. Where Part I was the "anatomy" — dismantling the group into its root system — Part II is the "action": we now ask how the group acts on vector spaces, and what the combinatorics of the root system can tell us about these actions.

The starting point is the weight space decomposition from chapter 5: any representation $V$ of a reductive group $G$, when restricted to a maximal torus $T$, breaks into eigenspaces $V = \bigoplus V_\lambda$ indexed by characters $\lambda \in X(T)$. The root system $\Phi$ — the set of weights of the adjoint representation — is the "skeleton" around which all other representations are organized. The irreducible representations are classified by their *highest weights*: dominant characters in $X(T)$, one for each cone in the weight lattice determined by the Weyl chamber. This is the content of chapter 8, which establishes the highest weight classification and proves the complete reducibility theorem in characteristic 0.

Chapter 8 turns from classification to computation. The *Weyl character formula* expresses the character of an irreducible representation as an alternating sum over the Weyl group, and it implies explicit formulas for dimensions, weight multiplicities, and tensor product decompositions. This is the most powerful computational tool in the subject, and we develop it with extensive examples for the classical groups.

chapter 10 brings geometry back to the center. The flag variety $G/B$, constructed in chapter 6 as a projective variety, carries a family of line bundles $\mathcal{L}(\lambda)$ parametrized by the weight lattice $X(T)$. The *Borel–Weil theorem* asserts that the global sections of $\mathcal{L}(\lambda)$, for dominant $\lambda$, *are* the irreducible representations — they are constructed geometrically, as spaces of sections on $G/B$, rather than algebraically via generators and relations. The *Borel–Weil–Bott theorem* extends this to higher cohomology, showing that the Weyl character formula is a shadow of the cohomological structure of line bundles on the flag variety. This closes the circle between the geometry of $G/B$ (Part I) and the representation theory of $G$ (Part II): the flag variety is the machine that *produces* representations.

chapter 11 addresses the elephant in the room: what happens in positive characteristic? The Borel–Weil construction and the Weyl modules $V(\lambda)$ still exist, but they are no longer irreducible. Complete reducibility fails, the simple modules $L(\lambda)$ are mysterious proper quotients of the Weyl modules, and the problem of computing their characters remains open in general. We develop the Frobenius twist and Steinberg's tensor product theorem — which reduce the problem to "restricted" weights — and state Lusztig's conjecture, which gives a conjectural answer in terms of Kazhdan–Lusztig polynomials for sufficiently large primes. The chapter is deliberately open-ended: it brings the reader to the frontier of an active area of research, equipped with the tools and language to engage with the current literature.

A word on prerequisites. The reader should have Part I firmly in hand, especially the weight space decomposition (chapter 5), the Borel subgroup theory and flag varieties (chapter 6), and the root system combinatorics (chapter 7). Some familiarity with sheaf cohomology on projective varieties (see [5]) is needed for chapter 10. No prior knowledge of Lie algebra representations is assumed: we work entirely with algebraic group representations, though we occasionally note the connection to the Lie-algebraic theory for the reader's benefit.

# 8

# Highest Weight Theory

In Part I, we classified the irreducible “building blocks” of a reductive group $G$: the roots, which are the weights of the adjoint representation. In this chapter, we classify the irreducible “building blocks” of $\mathrm{Rep}(G)$: the irreducible finite-dimensional representations. The answer is strikingly parallel — they are indexed by a discrete combinatorial datum, the *dominant weights* in the character lattice $X(T)$ — and the proof uses the same mechanism: the decomposition of representations into weight spaces under a maximal torus.

The chapter has three parts. First, we study the weight structure of an arbitrary representation: the weights form a finite, Weyl-group-invariant subset of $X(T)$ with a natural partial order. Second, we prove the classification theorem: irreducible representations are in bijection with dominant weights, via the *highest weight*. Third, we prove Weyl's complete reducibility theorem in characteristic 0: every finite-dimensional representation is a direct sum of irreducibles. Together, these results give a complete and computable description of $\mathrm{Rep}(G)$ in characteristic 0.

Throughout this chapter, $k$ denotes an algebraically closed field of characteristic 0, $G$ a connected reductive group over $k$, $T \subseteq B \subseteq G$ a maximal torus inside a Borel subgroup, $\Phi = \Phi(G,T)$ the root system, $\Phi^+ = \Phi^+(G,T,B)$ the positive roots, and $\Delta = \{\alpha_1, \ldots, \alpha_r\}$ the simple roots.

## 8.1 Weights of a Representation

### Weight Space Decomposition for $G$-Modules

Let $V$ be a finite-dimensional representation of $G$. Since $T \subseteq G$ is a torus, restricting the $G$-action to $T$ and applying the weight space decomposition (theorem 5.2.9) gives:

$$V = \bigoplus_{\lambda \in X(T)} V_\lambda$$

where $V_\lambda = \{v \in V \mid t \cdot v = \lambda(t)v \text{ for all } t \in T(R), \text{ all } R\}$. Most of these weight spaces are zero; the finitely many $\lambda$ with $V_\lambda \neq 0$ carry all the information.

**Definition 8.1.1: Weights and Weight Spaces**

Let $V$ be a finite-dimensional representation of $G$. The *weights* of $V$ are

$$\mathrm{wt}(V) = \{\lambda \in X(T) \mid V_\lambda \neq 0\}$$

For each $\lambda \in \mathrm{wt}(V)$, the subspace $V_\lambda$ is the *weight space* and $m_\lambda(V) = \dim_k V_\lambda$ is the *multiplicity* of $\lambda$ in $V$.

The weight spaces interact with the root spaces of $\mathfrak{g}$ in a predictable way:

**Proposition 8.1.2: Root Operators Shift Weights**

Let $V$ be a representation of $G$, and let $\alpha \in \Phi$ be a root. If $X \in \mathfrak{g}_\alpha$ (a root vector) and $v \in V_\lambda$ (a weight vector), then $X \cdot v \in V_{\lambda+\alpha}$. That is, the action of a root vector of weight $\alpha$ shifts the weight by $\alpha$:

$$\mathfrak{g}_\alpha \cdot V_\lambda \subseteq V_{\lambda+\alpha}$$

***Proof*:**
For $t \in T(k)$ and $X \in \mathfrak{g}_\alpha$, the adjoint action gives $\mathrm{Ad}(t)(X) = \alpha(t)X$. The infinitesimal action of $X$ on $v \in V_\lambda$ satisfies:

$$t \cdot (X \cdot v) = (\mathrm{Ad}(t)(X)) \cdot (t \cdot v) = (\alpha(t)X) \cdot (\lambda(t)v) = \alpha(t)\lambda(t)(X \cdot v) = (\lambda + \alpha)(t)(X \cdot v)$$

where the first equality uses the compatibility of the group action and the Lie algebra action: $t \cdot (X \cdot v) = \mathrm{Ad}(t)(X) \cdot (t \cdot v)$, which follows from differentiating the conjugation identity $\rho(t) \circ d\rho(X) \circ \rho(t)^{-1} = d\rho(\mathrm{Ad}(t)(X))$. Hence $X \cdot v \in V_{\lambda+\alpha}$.

This proposition is the engine of highest weight theory: the root vectors of $\mathfrak{g}$ act as "raising" and "lowering" operators, moving weight vectors from one weight space to an adjacent one (shifted by a root). The positive root vectors $X_\alpha$ with $\alpha \in \Phi^+$ are the raising operators, and the negative root vectors $X_{-\alpha}$ are the lowering operators.

## Weyl Group Invariance

The weight set $\mathrm{wt}(V)$ is not just any subset of $X(T)$; it is invariant under the Weyl group.

**Proposition 8.1.3: $W$-Invariance of Weights**

Let $V$ be a finite-dimensional representation of $G$. Then:

1. For every $w \in W$ and $\lambda \in \mathrm{wt}(V)$, we have $w(\lambda) \in \mathrm{wt}(V)$ and $m_{w(\lambda)}(V) = m_\lambda(V)$. That is, the Weyl group permutes the weights and preserves multiplicities.
2. $\mathrm{wt}(V)$ is a finite, $W$-stable subset of $X(T)$.

***Proof*:**
Let $w \in W$ and choose a representative $n_w \in N_G(T)(k)$. Define $\varphi_w : V_\lambda \to V_{w(\lambda)}$ by $\varphi_w(v) = n_w \cdot v$. This is well-defined: for $v \in V_\lambda$ and $t \in T(k)$,

$$t \cdot (n_w \cdot v) = n_w \cdot (n_w^{-1} t n_w \cdot v) = n_w \cdot (w^{-1}(t) \cdot v) = n_w \cdot (\lambda(w^{-1}(t))v) = (w(\lambda))(t) \cdot n_w \cdot v$$

where we used $w^{-1}(t) = n_w^{-1} t n_w$ (the action of $W$ on $T$) and $(w(\lambda))(t) = \lambda(w^{-1}(t))$ (the contragredient action on characters). So $n_w \cdot v \in V_{w(\lambda)}$.

The map $\varphi_w$ is a linear isomorphism $V_\lambda \xrightarrow{\sim} V_{w(\lambda)}$ (with inverse $\varphi_{w^{-1}}$), so $\dim V_{w(\lambda)} = \dim V_\lambda$.

## The Partial Order on Weights

The root lattice $Q = \mathbb{Z}\Phi \subseteq X(T)$ defines a natural partial order on $X(T)$:

**Definition 8.1.4: The Dominance Partial Order**

For $\lambda, \mu \in X(T)$, we write $\mu \preceq \lambda$ (and say "$\mu$ is *dominated by* $\lambda$") if $\lambda - \mu$ is a non-negative integer linear combination of positive roots:

$$\mu \preceq \lambda \quad \iff \quad \lambda - \mu \in \mathbb{Z}_{\geq 0}\Phi^+ := \left\{ \sum_{\alpha \in \Phi^+} n_\alpha \alpha \;\middle|\; n_\alpha \in \mathbb{Z}_{\geq 0} \right\}$$

Equivalently, $\lambda - \mu = \sum_{i=1}^r n_i \alpha_i$ with $n_i \in \mathbb{Z}_{\geq 0}$ (a non-negative combination of *simple* roots, since every positive root is a non-negative combination of simple roots by proposition 7.1.6).

This is a partial order on $X(T)$: reflexive ($\lambda \preceq \lambda$ since $\lambda - \lambda = 0$), antisymmetric (if $\mu \preceq \lambda$ and $\lambda \preceq \mu$, then $\lambda - \mu$ and $\mu - \lambda$ are both non-negative combinations of positive roots, which forces $\lambda = \mu$), and transitive.

The partial order is compatible with the Lie algebra action in the following sense: if $v \in V_\lambda$ and $X \in \mathfrak{g}_{-\alpha}$ (a negative root vector), then $X \cdot v \in V_{\lambda-\alpha}$, and $\lambda - \alpha \prec \lambda$ (strictly dominated). So the lowering operators move weights *down* in the partial order, and the raising operators move them *up*.

**Proposition 8.1.5: Weights are Bounded Above**

Let $V$ be a finite-dimensional representation of $G$. The set $\mathrm{wt}(V)$ has finitely many maximal elements with respect to $\preceq$. If $V$ is irreducible, there is a *unique* maximal weight.

***Proof*:**
Since $\mathrm{wt}(V)$ is finite, it has at least one maximal element. Now suppose $V$ is irreducible and $\lambda \in \mathrm{wt}(V)$ is maximal. Let $v \in V_\lambda$ be nonzero. For any positive root $\alpha \in \Phi^+$ and any root vector $X_\alpha \in \mathfrak{g}_\alpha$, we have $X_\alpha \cdot v \in V_{\lambda+\alpha}$. Since $\lambda$ is maximal and $\lambda + \alpha \succ \lambda$, we must have $V_{\lambda+\alpha} = 0$ (otherwise $\lambda$ would not be maximal). So $X_\alpha \cdot v = 0$ for all $\alpha \in \Phi^+$.

Now consider the subspace $U \subseteq V$ generated by $v$ under the action of $\mathfrak{g}$ (equivalently, under the action of $G$, by the Fundamental Finiteness Theorem and the density of $G$ in the formal neighborhood of $e$). Since $V$ is irreducible, $U = V$. But $U$ is spanned by elements of the form $X_{\beta_1} \cdots X_{\beta_m} \cdot v$ where $\beta_i \in \Phi^- \cup \{0\}$ (applying raising operators gives 0 since $\lambda$ is maximal, so only lowering operators and Cartan elements contribute). Each such element has weight $\lambda + \beta_1 + \cdots + \beta_m \preceq \lambda$. Therefore every weight $\mu \in \mathrm{wt}(V)$ satisfies $\mu \preceq \lambda$.

Uniqueness: if $\lambda'$ were another maximal weight, then $\lambda' \preceq \lambda$ (by the above) and $\lambda \preceq \lambda'$ (by the same argument applied to $\lambda'$), so $\lambda = \lambda'$.

The unique maximal weight of an irreducible representation is the central definition of this chapter:

**Definition 8.1.6: Highest Weight**

Let $V$ be a finite-dimensional irreducible representation of $G$. The *highest weight* of $V$ is the unique maximal element $\lambda \in \mathrm{wt}(V)$ with respect to the dominance order $\preceq$. A nonzero vector $v^+ \in V_\lambda$ is called a *highest weight vector.*

The highest weight vector $v^+$ is characterized by two properties: it has weight $\lambda$ (i.e. $t \cdot v^+ = \lambda(t)v^+$ for all $t \in T$), and it is killed by all positive root vectors ($X_\alpha \cdot v^+ = 0$ for all $\alpha \in \Phi^+$). Equivalently, $v^+$ is a simultaneous eigenvector for $B = T \ltimes R_u(B)$: the torus $T$ acts by $\lambda$ and the unipotent radical $R_u(B)$ acts trivially (since $\mathrm{Lie}(R_u(B)) = \bigoplus_{\alpha\in\Phi^+} \mathfrak{g}_\alpha$ and all positive root vectors kill $v^+$).

**Proposition 8.1.7: Highest Weight Space is One-Dimensional**

Let $V$ be a finite-dimensional irreducible representation of $G$ with highest weight $\lambda$. Then $\dim V_\lambda = 1$.

***Proof*:**
Suppose $\dim V_\lambda \geq 2$ and let $v_1, v_2 \in V_\lambda$ be linearly independent highest weight vectors. Both are killed by $\mathfrak{g}_\alpha$ for all $\alpha \in \Phi^+$. The subspace $U_1 \subseteq V$ generated by $v_1$ under the action of $\mathfrak{g}$ is a nonzero subrepresentation. Since $V$ is irreducible, $U_1 = V$.

Now consider $v_2$. Since $v_2 \in V_\lambda$ and all weights of $V$ are $\preceq \lambda$, the element $v_2$ cannot be in the $\mathfrak{g}$-span of $\{X_\beta \cdot v_1 : \beta \in \Phi^-\}$ unless $v_2 \in kv_1$ (because the only way to return to weight $\lambda$ from a lower weight is via raising operators, which kill $v_1$). More precisely: the $\mathfrak{g}$-submodule generated by $v_1$ has a unique vector of weight $\lambda$ up to scalar (namely $v_1$ itself), because any element of weight $\lambda$ in $U_1$ must be in the kernel of all raising operators and in the span of vectors obtained by lowering and then raising back — but raising from a lower weight cannot exceed $\lambda$ in an irreducible module where $\lambda$ is maximal. This forces $v_2 \in kv_1$, a contradiction.

A cleaner argument: the space $V^{R_u(B)} = \{v \in V : X \cdot v = 0 \text{ for all } X \in \mathrm{Lie}(R_u(B))\}$ of $R_u(B)$-invariants is a representation of $T$ (since $B = T \ltimes R_u(B)$ and $T$ normalizes $R_u(B)$). In an irreducible $G$-module, $V^{R_u(B)}$ is one-dimensional: if it were higher-dimensional, the $B$-action on $V^{R_u(B)}$ would decompose into multiple one-dimensional $T$-eigenspaces, and the $G$-submodule generated by any single eigenvector would be a proper submodule (it could not reach the other eigenvectors by lowering operators, since those lower the weight and cannot raise back to a different maximal-weight vector). By irreducibility, $V^{R_u(B)}$ must be one-dimensional, and it equals $V_\lambda$.

**Example 8.1.8: Weights of the Irreducible Representations of $\mathrm{SL}_2$**

For $G = \mathrm{SL}_2$ with $T = \{\mathrm{diag}(t, t^{-1})\}$, we have $X(T) = \mathbb{Z}$ (generated by $\varepsilon$ where $\varepsilon(\mathrm{diag}(t, t^{-1})) = t$). The positive root is $\alpha = 2\varepsilon$ (so $\Phi^+ = \{2\varepsilon\}$), and the dominance order is $m \preceq n$ iff $n - m \in 2\mathbb{Z}_{\geq 0}$.

The irreducible representations are the symmetric powers $L(d) = \mathrm{Sym}^d(k^2)$ for $d \geq 0$ (exercise 2.2.1 (4)). The highest weight of $L(d)$ is $d\varepsilon$ (the weight of the monomial $x^d$), and the weights are:

$$\mathrm{wt}(L(d)) = \{d\varepsilon,\ (d-2)\varepsilon,\ (d-4)\varepsilon,\ \ldots,\ -(d-2)\varepsilon,\ -d\varepsilon\}$$

each with multiplicity 1. The highest weight vector is $v^+ = x^d$, and it is killed by $e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ (the positive root vector). The lowering operator $f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ sends $x^{d-k}y^k \mapsto (d-k)x^{d-k-1}y^{k+1}$ (up to scalar), moving down by $2\varepsilon$ at each step.

The dominance partial order on $\mathbb{Z}$ is $m \preceq n$ iff $n - m \in 2\mathbb{Z}_{\geq 0}$. The highest weight $d\varepsilon$ dominates all other weights $\{(d-2j)\varepsilon : 0 \leq j \leq d\}$, confirming proposition 8.1.5. Note that $d$ must be a non-negative integer: a "weight $-1$" or a "weight $1/2$" cannot be the highest weight of a finite-dimensional representation. The constraint is that $\langle d\varepsilon, \alpha^\vee \rangle = d \geq 0$ (dominance with respect to $\Phi^+$).

**Example 8.1.9: Weights of the Standard Representation of $\mathrm{SL}_3$**

For $G = \mathrm{SL}_3$ with the standard representation $V = k^3$, the maximal torus $T = \{\mathrm{diag}(t_1, t_2, (t_1t_2)^{-1})\}$ acts with weights $\varepsilon_1, \varepsilon_2, \varepsilon_3 = -\varepsilon_1 - \varepsilon_2$ (one for each basis vector $e_i$). The positive roots are $\alpha_1 = \varepsilon_1 - \varepsilon_2$, $\alpha_2 = \varepsilon_2 - \varepsilon_3$, and $\alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3$.

The highest weight is $\varepsilon_1$ (it dominates $\varepsilon_2 = \varepsilon_1 - \alpha_1$ and $\varepsilon_3 = \varepsilon_1 - \alpha_1 - \alpha_2$). The highest weight vector is $e_1$:

$$E_{12} \cdot e_1 = 0, \qquad E_{13} \cdot e_1 = 0$$

(the positive root vectors $E_{12}$ and $E_{13}$ kill $e_1$). Lowering: $E_{21} \cdot e_1 = e_2$ (weight $\varepsilon_2$) and $E_{32} \cdot e_2 = e_3$ (weight $\varepsilon_3$).

In the language of dominant weights: $\varepsilon_1 = \omega_1$ is the first fundamental weight (defined by $\langle \omega_1, \alpha_1^\vee \rangle = 1$, $\langle \omega_1, \alpha_2^\vee \rangle = 0$). The standard representation is $L(\omega_1)$.

**Example 8.1.10: Weights of the Adjoint Representation of $\mathrm{SL}_3$**

The adjoint representation $\mathfrak{g} = \mathfrak{sl}_3$ has dimension 8. The weights are the roots $\Phi = \{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2)\}$ (each with multiplicity 1) together with 0 (with multiplicity $2 = \mathrm{rank}(\mathrm{SL}_3)$). The highest weight is $\alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3$ (the highest root), and the adjoint representation is $L(\alpha_1 + \alpha_2)$.

In terms of fundamental weights: $\alpha_1 + \alpha_2 = \omega_1 + \omega_2$ (the sum of the two fundamental weights). The fact that the adjoint representation has highest weight $\omega_1 + \omega_2$ means it appears in the tensor product $L(\omega_1) \otimes L(\omega_2)$ as a direct summand (in characteristic 0). Indeed, $L(\omega_1) = k^3$ (the standard representation), $L(\omega_2) = (k^3)^\vee \cong \bigwedge^2(k^3)$ (the dual, which is the second exterior power for $\mathrm{SL}_3$), and $L(\omega_1) \otimes L(\omega_2) \cong \mathfrak{sl}_3 \oplus k$ (the adjoint plus the trivial representation), recovering the decomposition $\mathfrak{gl}_3 = \mathfrak{sl}_3 \oplus kI$ from example 3.2.3.

## Dominant Weights

The $\mathrm{SL}_2$ example showed that the highest weight of an irreducible representation must satisfy $\langle \lambda, \alpha^\vee \rangle \geq 0$ for each positive root $\alpha$. This condition has a name:

**Definition 8.1.11: Dominant Weights**

A weight $\lambda \in X(T)$ is called *dominant* (with respect to $B$) if

$$\langle \lambda, \alpha_i^\vee \rangle \geq 0 \qquad \text{for all simple roots } \alpha_i \in \Delta$$

The set of dominant weights is denoted $X(T)^+ = X(T)^+_B$. In terms of fundamental weights $\omega_1, \dots, \omega_r$ (defined by $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$):

$$X(T)^+ = \left\{ \lambda = \sum_{i=1}^r n_i \omega_i \;\middle|\; n_i \in \mathbb{Z}_{\geq 0} \right\} = \mathbb{Z}_{\geq 0}\omega_1 + \cdots + \mathbb{Z}_{\geq 0}\omega_r$$

when $G$ is semisimple (so that $X(T)$ is spanned by the fundamental weights).

The dominant weights form a "cone" in $X(T)$: the set of non-negative integer combinations of the fundamental weights. Geometrically, they are the lattice points in the closure of the dominant Weyl chamber $\overline{C} = \{\lambda \in X(T) \otimes \mathbb{R} \mid \langle \lambda, \alpha_i^\vee \rangle \geq 0 \text{ for all } i\}$.

**Proposition 8.1.12: Highest Weights are Dominant**

If $V$ is a finite-dimensional irreducible representation of $G$ with highest weight $\lambda$, then $\lambda$ is dominant: $\lambda \in X(T)^+$.

***Proof*:**
For each simple root $\alpha_i$, the $\mathrm{SL}_2$-subgroup $\varphi_{\alpha_i} : \mathrm{SL}_2 \to G$ (theorem 7.1.3) restricts the representation $V$ to an $\mathrm{SL}_2$-representation. The highest weight vector $v^+ \in V_\lambda$ generates an irreducible $\mathrm{SL}_2$-subrepresentation (it is killed by the raising operator $e_{\alpha_i} \in \mathfrak{g}_{\alpha_i}$, so it is a highest weight vector for $\mathrm{SL}_2$ as well). The highest weight of this $\mathrm{SL}_2$-representation is $\langle \lambda, \alpha_i^\vee \rangle$ (by the identification of $X(T|_{\mathrm{SL}_2}) = \mathbb{Z}$ with $\mathbb{Z}$ via the coroot $\alpha_i^\vee$).

From the $\mathrm{SL}_2$ classification (example 8.1.8), the highest weight of a finite-dimensional irreducible $\mathrm{SL}_2$-representation is a non-negative integer $d \geq 0$. Therefore $\langle \lambda, \alpha_i^\vee \rangle \geq 0$ for every simple root $\alpha_i$, which means $\lambda$ is dominant.

This brings us to the threshold of the classification theorem: we have shown that every irreducible representation has a dominant highest weight. In the next section, we will prove the converse — every dominant weight arises as the highest weight of a unique irreducible representation — completing the classification of $\mathrm{Rep}(G)$.

**Example 8.1.13: Dominant Weights for $\mathrm{SL}_3$**
The fundamental weights are $\omega_1 = \varepsilon_1$ and $\omega_2 = \varepsilon_1 + \varepsilon_2 = -\varepsilon_3$. The dominant weights are:

$$X(T)^+ = \{a\omega_1 + b\omega_2 \mid a, b \in \mathbb{Z}_{\geq 0}\} = \{a\varepsilon_1 + b(\varepsilon_1 + \varepsilon_2) \mid a, b \geq 0\}$$

The first few dominant weights and their corresponding irreducible representations are:

| $\lambda$ | $(a,b)$ | $L(\lambda)$ | $\dim L(\lambda)$ |
|---|---|---|---|
| $0$ | $(0,0)$ | trivial | 1 |
| $\omega_1$ | $(1,0)$ | standard $k^3$ | 3 |
| $\omega_2$ | $(0,1)$ | dual $(k^3)^\vee \cong \bigwedge^2(k^3)$ | 3 |
| $2\omega_1$ | $(2,0)$ | $\mathrm{Sym}^2(k^3)$ | 6 |
| $\omega_1+\omega_2$ | $(1,1)$ | adjoint $\mathfrak{sl}_3$ | 8 |
| $2\omega_2$ | $(0,2)$ | $\mathrm{Sym}^2((k^3)^\vee)$ | 6 |
| $3\omega_1$ | $(3,0)$ | $\mathrm{Sym}^3(k^3)$ | 10 |

The dimensions can be computed from the Weyl dimension formula (chapter 8). Note that the standard representation $L(\omega_1) = k^3$ and its dual $L(\omega_2) = (k^3)^\vee$ are the two "fundamental" representations (one for each node of the Dynkin diagram $A_2$), and every other irreducible representation is built from these via tensor products and direct summands.

### Exercise 8.1.1

1. Let $G = \mathrm{SL}_4$ with standard representation $V = k^4$. Compute the weights of $V$, $\bigwedge^2(V)$, and $\bigwedge^3(V)$. Identify the highest weight of each and verify they are the fundamental weights $\omega_1$, $\omega_2$, $\omega_3$ respectively. What is the relationship between $\bigwedge^3(V)$ and $V^\vee$?
2. For $G = \mathrm{SL}_3$ with standard representation $V = k^3$, compute $\mathrm{wt}(\mathrm{Sym}^2(V))$ and verify that the highest weight is $2\omega_1$. What are the multiplicities of each weight?
3. Let $V$ be a finite-dimensional representation of $G$ (not necessarily irreducible) with $\mathrm{wt}(V) = \{\lambda_1, \ldots, \lambda_m\}$. Show that for each weight $\mu \in \mathrm{wt}(V)$, the entire "$\alpha$-string through $\mu$" is contained in $\mathrm{wt}(V)$: that is, for each simple root $\alpha_i$, the weights $\mu, \mu-\alpha_i, \mu-2\alpha_i, \ldots, \mu-p\alpha_i$ are in $\mathrm{wt}(V)$, where $p = \langle \mu, \alpha_i^\vee \rangle$. (*Hint*: restrict to the $\mathrm{SL}_2$-subgroup for $\alpha_i$.)
4. Show that if $V$ is a finite-dimensional (not necessarily irreducible) representation with a unique highest weight $\lambda$ (i.e. $V_\lambda \neq 0$ and $V_\mu = 0$ for all $\mu \succ \lambda$), and $\dim V_\lambda = 1$, then $V$ has a unique irreducible quotient with highest weight $\lambda$.
5. Draw the weight lattice $X(T) = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ for $\mathrm{SL}_3$ in the plane, mark the dominant cone $X(T)^+$, and plot the weights of $L(\omega_1)$, $L(\omega_2)$, and $L(\omega_1+\omega_2)$. Verify the $W$-invariance: for each representation, check that $W = S_3$ permutes the weight set.
6. For each of the classical groups $\mathrm{SL}_n$, $\mathrm{SO}_{2n+1}$, $\mathrm{Sp}_{2n}$, $\mathrm{SO}_{2n}$, identify the highest weight of the adjoint representation as a dominant weight in terms of the fundamental weights. (*Hint*: the highest root is the highest weight of the adjoint representation.)

## 8.2 Dominant Weights and the Classification

We proved in Section 8.1 that every irreducible representation has a dominant highest weight, and that this weight determines the representation uniquely (since the highest weight space is one-dimensional and generates the representation). This gives an injection from isomorphism classes of irreducible representations to dominant weights. The content of this section is the *surjectivity*: every dominant weight $\lambda \in X(T)^+$ arises as the highest weight of some irreducible representation. The proof constructs the representation explicitly, by building a "universal" induced module with highest weight $\lambda$ and taking its unique irreducible submodule.

## Induced Modules

The construction uses the *induction* functor from $B$-representations to $G$-representations. Since the highest weight vector $v^+$ is a simultaneous eigenvector for $B$ (the torus $T$ acts by $\lambda$ and $R_u(B)$ acts trivially), the one-dimensional space $k_\lambda$ on which $B$ acts by the character $\lambda$ (extended trivially to $R_u(B)$) is the natural starting point.

**Definition 8.2.1: Induced Module**

Let $\lambda \in X(T)$ be a character. Define the one-dimensional $B$-representation $k_\lambda$: the vector space $k$ on which $B = T \ltimes R_u(B)$ acts via $b \cdot z = \lambda(t)z$ for $b = tu$ with $t \in T$, $u \in R_u(B)$. The *induced module* (or *costandard module*) is:

$$H^0(\lambda) := \mathrm{Ind}_B^G(k_\lambda) = \left\{ f \in \mathcal{O}(G) \;\middle|\; f(gb) = \lambda(b)^{-1} f(g) \text{ for all } g \in G,\, b \in B \right\}$$

with $G$ acting by left translation: $(g' \cdot f)(g) = f(g'^{-1}g)$.

The notation $H^0(\lambda)$ anticipates the geometric interpretation: this is the space of global sections $H^0(G/B, \mathcal{L}(\lambda))$ of the line bundle $\mathcal{L}(\lambda)$ on the flag variety $G/B$, which we will develop in chapter 10. For now, we work purely algebraically.

**Proposition 8.2.2: Properties of the Induced Module**

Let $\lambda \in X(T)$.

1. $H^0(\lambda)$ is a finite-dimensional representation of $G$ (possibly zero).
2. $H^0(\lambda) \neq 0$ if and only if $\lambda$ is dominant: $\lambda \in X(T)^+$.
3. If $\lambda \in X(T)^+$, then $H^0(\lambda)$ has a unique highest weight $\lambda$, with $\dim H^0(\lambda)_\lambda = 1$.
4. $H^0(\lambda)$ contains a unique irreducible *submodule*, which we denote $L(\lambda)$. (In characteristic 0, $H^0(\lambda)$ is actually irreducible, so $H^0(\lambda) = L(\lambda)$).

***Proof Sketch:***
*(1):* The space $H^0(\lambda)$ is a subspace of $\mathcal{O}(G)$, and it is $G$-stable (left translation preserves the equivariance condition). Finite-dimensionality follows from the fact that $H^0(\lambda)$ is the space of global sections of a line bundle on the projective variety $G/B$, which is finite-dimensional by the projectivity of $G/B$ (theorem 6.2.3) and Serre's theorem on coherent sheaves on projective varieties (see [5]).

*(2):* The nonvanishing of $H^0(\lambda)$ is equivalent to the existence of a nonzero section of $\mathcal{L}(\lambda)$ on

$G/B$. This is a consequence of the Borel–Weil theorem (chapter 10), but can also be proved directly: one shows that $H^0(\lambda)$ contains a unique $B$-eigenvector of weight $\lambda$ (the "canonical section"), which exists if and only if $\lambda$ is dominant. The "only if" direction follows from the $\mathrm{SL}_2$-restriction argument of proposition 8.1.12. For the "if" direction when $\lambda$ is dominant, one constructs a nonzero section explicitly using the Bruhat decomposition $G = \bigsqcup_{w \in W} BwB$: the restriction of $f \in H^0(\lambda)$ to the open Bruhat cell $Bw_0B$ (where $w_0$ is the longest Weyl group element) is determined by its value at $w_0$, and one verifies that a nonzero choice extends to all of $G$.

*(3):* The weight $\lambda$ appears in $H^0(\lambda)$ with multiplicity 1 (from the construction of the canonical section), and no weight $\mu \succ \lambda$ appears (by the equivariance condition: elements of $H^0(\lambda)$ transform by $\lambda$ under $B$, so no higher weight can occur).

*(4):* This is a standard property of induced modules arising from Frobenius reciprocity. Any non-zero submodule $M \subseteq H^0(\lambda)$ must contain a $B$-eigenvector, which by the structure of $H^0(\lambda)$ forces it to contain the unique highest weight space $H^0(\lambda)_\lambda$. Since every non-zero submodule contains this one-dimensional space, the intersection of all non-zero submodules is non-zero. This minimal non-zero submodule is the unique irreducible submodule $L(\lambda)$.

(In characteristic 0, one can further show that $H^0(\lambda)$ has no other maximal weights, which by complete reducibility forces the entire module to be irreducible, so $H^0(\lambda) = L(\lambda)$).

## The Classification Theorem

**Theorem 8.2.3: Classification of Irreducible Representations**

Let $G$ be a connected reductive group over an algebraically closed field $k$, with maximal torus $T \subseteq B \subseteq G$ and dominant weights $X(T)^+$. The map

$$\lambda \longmapsto L(\lambda)$$

is a bijection:

$$X(T)^+ \xrightarrow{1:1} \{\text{Irreducible representations of } G \mid \text{up to isomorphism}\}$$

Explicitly: every irreducible finite-dimensional representation of $G$ is isomorphic to $L(\lambda)$ for a unique $\lambda \in X(T)^+$, where $L(\lambda)$ is the unique irreducible submodule of the induced module $H^0(\lambda)$.

***Proof*:**
*Surjectivity (every dominant weight arises):* For $\lambda \in X(T)^+$, the module $H^0(\lambda) \neq 0$ (proposition 8.2.2(2)), and its unique irreducible submodule $L(\lambda)$ has highest weight $\lambda$ (proposition 8.2.2(3)–(4)).

*Injectivity (different dominant weights give non-isomorphic representations):* If $L(\lambda) \cong L(\mu)$, then both have highest weight $\lambda$ and highest weight $\mu$. By the uniqueness of the highest weight in an irreducible representation (proposition 8.1.5), $\lambda = \mu$.

*Exhaustiveness (every irreducible is some $L(\lambda)$):* Let $V$ be an irreducible representation of $G$. By proposition 8.1.5, $V$ has a unique highest weight $\lambda$, and by proposition 8.1.12, $\lambda \in X(T)^+$. The highest weight vector $v^+ \in V_\lambda$ generates $V$ (by irreducibility).

Since $V$ is irreducible with highest weight $\lambda$, we show that any other irreducible representation $V'$ with the same highest weight must satisfy $V' \cong V$. Indeed, $V$ and $V'$ both have one-dimensional $\lambda$-weight spaces, so the $G$-submodule $W \subseteq V \oplus V'$ generated by the diagonal vector $(v^+, (v')^+)$ has a one-dimensional highest weight space. The projection $W \to V$ maps $(v^+, (v')^+)$ to $v^+ \neq 0$, so it is surjective (since $V$ is irreducible). Similarly, $W \to V'$ is surjective. The kernel of $W \to V'$ is $W \cap (V \oplus 0)$. If this kernel were non-zero, it would equal $V \oplus 0$ (by the irreducibility of $V$), meaning $W = V \oplus V'$. But $V \oplus V'$ has a two-dimensional $\lambda$-weight space, contradicting that $W$ has a one-dimensional highest weight space. Thus the kernel is trivial, so $W \to V'$ is an isomorphism. By symmetry, $W \to V$ is an isomorphism, giving $V \cong W \cong V'$.

Since $L(\lambda)$ is an irreducible representation with highest weight $\lambda$ (constructed above), we have $V \cong L(\lambda)$.

This theorem is the representation-theoretic analogue of the classification of reductive groups by root data (theorem 7.3.6): just as root data classify groups, dominant weights classify their irreducible representations. The two classifications are intimately related: the dominant weights lie in the character lattice $X(T)$, which is part of the root datum, and the root system $\Phi$ determines the dominance condition $\langle \lambda, \alpha_i^\vee \rangle \geq 0$.

**Example 8.2.4: Irreducible Representations of $\mathrm{SL}_2$**

For $G = \mathrm{SL}_2$, the dominant weights are $X(T)^+ = \mathbb{Z}_{\geq 0}$ (the non-negative integers, since there is one simple root $\alpha$ with $\langle d, \alpha^\vee \rangle = d$). The classification gives:

$$L(d) = \mathrm{Sym}^d(k^2) \qquad \text{for } d = 0, 1, 2, \dots$$

with $\dim L(d) = d + 1$. The weight set is $\{d, d-2, \dots, -d\}$ (as in example 8.1.8). In characteristic 0, these are the *only* irreducible representations of $\mathrm{SL}_2$, and every finite-dimensional representation decomposes as a direct sum of these (by complete reducibility, theorem 8.3.1 below).

**Example 8.2.5: Fundamental Representations**

For each simple root $\alpha_i \in \Delta$, the fundamental weight $\omega_i$ (defined by $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$) is the "smallest" non-trivial dominant weight in the $\alpha_i$-direction. The corresponding irreducible representation $L(\omega_i)$ is called the $i$-th *fundamental representation*. For the classical groups:

1. $\mathrm{SL}_n$ *(type $A_{n-1}$)*: $L(\omega_i) = \bigwedge^i(k^n)$ for $1 \leq i \leq n-1$. These are the exterior powers of the standard representation.
2. $\mathrm{SO}_{2n+1}$ *(type $B_n$)*: $L(\omega_i) = \bigwedge^i(k^{2n+1})$ for $1 \leq i \leq n-1$, and $L(\omega_n)$ is the *spin representation* of dimension $2^n$.
3. $\mathrm{Sp}_{2n}$ *(type $C_n$)*: $L(\omega_1) = k^{2n}$ (the standard representation). The higher fundamental representations are *not* exterior powers but rather their irreducible quotients (since $\bigwedge^i(k^{2n})$ is reducible for $i \geq 2$: the symplectic form gives a contraction map $\bigwedge^i \to \bigwedge^{i-2}$, and $L(\omega_i)$ is

the kernel).

4. $\mathrm{SO}_{2n}$ *(type $D_n$)*: $L(\omega_i) = \bigwedge^i(k^{2n})$ for $1 \le i \le n-2$, and $L(\omega_{n-1}), L(\omega_n)$ are the two *half-spin representations*, each of dimension $2^{n-1}$.

Every irreducible representation is a quotient of a tensor product of fundamental representations (in characteristic 0, a direct summand).

**Example 8.2.6: The Representation Ring of $\mathrm{SL}_3$**
The irreducible representations of $\mathrm{SL}_3$ are $\{L(a\omega_1 + b\omega_2) : a, b \ge 0\}$. In characteristic 0, the representation ring $R(\mathrm{SL}_3) = K_0(\mathrm{Rep}(\mathrm{SL}_3))$ is a polynomial ring:

$$R(\mathrm{SL}_3) \cong \mathbb{Z}[L(\omega_1), L(\omega_2)]$$

That is, every representation is a $\mathbb{Z}$-linear combination of products (= tensor products) of the two fundamental representations $L(\omega_1) = k^3$ and $L(\omega_2) = (k^3)^\vee$. The tensor product decomposition $L(\omega_1) \otimes L(\omega_2) \cong L(\omega_1 + \omega_2) \oplus L(0)$ (the adjoint plus the trivial) is the simplest example of a "Clebsch–Gordan" decomposition for $\mathrm{SL}_3$.

## Weyl Modules

There is a second natural module associated to a dominant weight $\lambda$, which will play a central role in the positive-characteristic theory of chapter 11.

**Definition 8.2.7: Weyl Module**

Let $\lambda \in X(T)^+$ be a dominant weight. The *Weyl module* (or *standard module*) is

$$V(\lambda) := H^0(-w_0\lambda)^\vee$$

where $w_0 \in W$ is the longest element of the Weyl group and $(\cdot)^\vee$ denotes the contragredient (definition 2.3.4).

The Weyl module $V(\lambda)$ is the "universal" module with highest weight $\lambda$: it has a unique irreducible *quotient* $L(\lambda)$ (the unique simple head), while the induced module $H^0(\lambda)$ has $L(\lambda)$ as its unique irreducible *submodule* from a different perspective (the unique simple *socle*). In characteristic 0, $V(\lambda) = L(\lambda) = H^0(\lambda)^\vee$ (everything is irreducible). In characteristic $p$, $V(\lambda)$ can be strictly larger than $L(\lambda)$, and the "extra" composition factors are the subject of modular representation theory.

**Proposition 8.2.8: Properties of Weyl Modules**

Let $\lambda \in X(T)^+$.

1. $V(\lambda)$ has highest weight $\lambda$ with $\dim V(\lambda)_\lambda = 1$.
2. $V(\lambda)$ has a unique maximal submodule $\mathrm{rad}V(\lambda)$, with $V(\lambda)/\mathrm{rad}V(\lambda) \cong L(\lambda)$.
3. The character of $V(\lambda)$ is given by the Weyl character formula (chapter 8) and is *independent of the characteristic of $k$*.
4. In characteristic 0: $V(\lambda) = L(\lambda)$ (the Weyl module is irreducible).

Property (3) is crucial: the Weyl module always has the "expected" character (the one given by the Weyl formula), even in characteristic $p$ where the module may be reducible. This means the Weyl character formula computes the character of $V(\lambda)$, not of $L(\lambda)$, in positive characteristic. The problem of computing $\mathrm{ch}(L(\lambda))$ in characteristic $p$ — equivalently, computing the composition factors of $V(\lambda)$ — is the fundamental open problem of modular representation theory, which we discuss in chapter 11.

### Exercise 8.2.1

1. For $G = \mathrm{SL}_2$, compute $H^0(d)$ explicitly for $d = 0, 1, 2$ by finding all $f \in \mathcal{O}(\mathrm{SL}_2)$ satisfying $f(gb) = \lambda(b)^{-1}f(g)$ for $b \in B$. Verify that $H^0(d) \cong \mathrm{Sym}^d(k^2)^\vee$.
2. Verify the fundamental representations of $\mathrm{SL}_4$: show that $L(\omega_1) = k^4$, $L(\omega_2) = \bigwedge^2(k^4)$ (dimension 6), and $L(\omega_3) = \bigwedge^3(k^4) \cong (k^4)^\vee$ (dimension 4). Compute the weights of $L(\omega_2)$ and verify the highest weight is $\omega_2 = \varepsilon_1 + \varepsilon_2$.
3. Decompose $L(\omega_1) \otimes L(\omega_1) = \mathrm{Sym}^2(k^3) \oplus \bigwedge^2(k^3)$ for $\mathrm{SL}_3$ (in characteristic $\neq 2$). Identify the highest weights: $\mathrm{Sym}^2(k^3) = L(2\omega_1)$ and $\bigwedge^2(k^3) = L(\omega_2)$. Verify $\dim = 6 + 3 = 9 = 3^2$.
4. For $G = \mathrm{SL}_n$ ($n \geq 3$), show that the adjoint representation $\mathfrak{g} = \mathfrak{sl}_n$ is the irreducible representation $L(\omega_1 + \omega_{n-1})$. What happens for $n = 2$? (*Hint*: for $n = 2$, $\omega_1 + \omega_1 = 2\omega_1 = \alpha$, and $L(\alpha) = \mathrm{Sym}^2(k^2)$ has dimension $3 = \dim \mathfrak{sl}_2$.)
5. In characteristic $p = 2$, consider $G = \mathrm{SL}_2$ and $\lambda = 2$ (so $L(\lambda)$ has highest weight 2). Show that the induced module $H^0(2) = \mathrm{Sym}^2(k^2)$ is 3-dimensional, but $L(2) \subsetneq H^0(2)$: the vectors $x^2$ and $y^2$ generate a 2-dimensional submodule (which is exactly $L(2)$), and the quotient $H^0(2)/L(2)$ is the 1-dimensional trivial representation $L(0)$. Verify this by computing the action of $G$ on the basis $\{x^2, xy, y^2\}$ in characteristic 2.

## 8.3 Complete Reducibility

In characteristic 0, the representation theory of a reductive group is as clean as possible: every finite-dimensional representation is completely reducible (a direct sum of irreducibles). This is the analogue of Maschke's theorem for finite groups and Weyl's theorem for compact Lie groups. The proof uses the existence of a non-degenerate invariant bilinear form — the *Killing form* — which provides a "complement" for any subrepresentation.

## Weyl's Complete Reducibility Theorem

**Theorem 8.3.1: Complete Reducibility (Weyl's Theorem)**

Let $G$ be a connected reductive group over an algebraically closed field $k$ of characteristic 0. Then every finite-dimensional representation of $G$ is completely reducible: for every subrepresentation $U \subseteq V$, there exists a complementary subrepresentation $W$ with $V = U \oplus W$.

***Proof*:**
We first prove the result for semisimple $G$, then extend to reductive $G$.

*Step 1: The Casimir element.* Let $\mathfrak{g} = \mathrm{Lie}(G)$ and let $\kappa : \mathfrak{g} \times \mathfrak{g} \to k$ be the Killing form, $\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}(X) \circ \mathrm{ad}(Y))$. Since $G$ is semisimple (and $\mathrm{char} k = 0$), the Killing form is non-degenerate (this is Cartan's criterion for semisimplicity; see [7]).

Choose a basis $\{X_1, \ldots, X_n\}$ for $\mathfrak{g}$ and let $\{X^1, \ldots, X^n\}$ be the dual basis with respect to $\kappa$: $\kappa(X_i, X^j) = \delta_{ij}$. The *Casimir element* is

$$C = \sum_{i=1}^{n} X_i \cdot X^i \in U(\mathfrak{g})$$

where $U(\mathfrak{g})$ is the universal enveloping algebra. The key properties of $C$ are:

1. $C$ is independent of the choice of basis.
2. $C$ commutes with the $\mathfrak{g}$-action: $[X, C] = 0$ for all $X \in \mathfrak{g}$. (This is because the Killing form is ad-invariant, proposition 3.2.11.)
3. On an irreducible representation $L(\lambda)$, $C$ acts by a scalar $c(\lambda) \in k$ (by Schur's lemma, exercise 2.2.1 (5)). This scalar is $c(\lambda) = (\lambda, \lambda + 2\rho)$ where $\rho = \frac{1}{2}\sum_{\alpha \in \Phi^+} \alpha$ and $(\cdot, \cdot)$ is the inner product on $X(T) \otimes \mathbb{R}$ induced by the Killing form.
4. $c(0) = 0$ (the Casimir acts by 0 on the trivial representation) and $c(\lambda) \neq 0$ for $\lambda \neq 0$ dominant (since $(\lambda, \lambda + 2\rho) > 0$ when $\lambda$ is a nonzero dominant weight and $\rho$ is in the interior of the dominant chamber).

*Step 2: Splitting extensions.* It suffices to show that every short exact sequence $0 \to U \to V \to W \to 0$ of $G$-representations splits. We do this in two stages.

*Stage A: $W = k$ (the trivial representation).* We must split $0 \to U \to V \to k \to 0$. We may assume by induction on $\dim U$ that $U$ is irreducible. If $U \ncong k$, then $U = L(\lambda)$ for some $\lambda \neq 0$. The Casimir $C$ acts on $U$ by $c(\lambda) \neq 0$ and on $k$ by 0. Thus $V = \ker(C) \oplus \ker(C - c(\lambda))$ as $G$-modules, which precisely splits the sequence. If $U \cong k$, then $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$ acts trivially on both $U$ and $W$. This implies $\mathfrak{g}$ maps $V$ into $U$, so $[\mathfrak{g}, \mathfrak{g}]$ acts as 0 on all of $V$. Thus any 1-dimensional complement to $U$ in $V$ is a $\mathfrak{g}$-submodule, splitting the sequence.

*Stage B: General $W$.* Consider the vector space of linear maps $\mathrm{Hom}_k(W, V)$, which is a $G$-representation. Let $\mathcal{V} \subseteq \mathrm{Hom}_k(W, V)$ be the subspace of maps $f$ such that $\pi \circ f$ is a scalar multiple of $\mathrm{id}_W$ (where $\pi : V \to W$). Let $\mathcal{U} \subseteq \mathcal{V}$ be the subspace of maps with $\pi \circ f = 0$ (i.e., maps $W \to U$). This yields an exact sequence of $G$-modules $0 \to \mathcal{U} \to \mathcal{V} \to k \to 0$. By Stage A, this splits, meaning $\mathcal{V}$ contains a 1-dimensional trivial $G$-submodule. A basis vector for this trivial submodule is a $G$-equivariant linear map $s : W \to V$ such that $\pi \circ s = \mathrm{id}_W$. This $s$ provides the required splitting section.

*Step 3: Extension to reductive $G$.* If $G$ is reductive but not semisimple, write $G = Z(G)^\circ \cdot [G, G]$ where $Z(G)^\circ$ is a torus and $[G, G]$ is semisimple (proposition 6.2.12). Any representation of $G$ restricts to a representation of $[G, G]$, which is completely reducible by Step 2. The $Z(G)^\circ$-action (being a torus action) is also completely reducible (theorem 5.2.9). Since the two actions commute and both are completely reducible, the combined $G$-action is completely reducible.

**8.3.2 Remark: Failure in Characteristic $p$** Complete reducibility fails dramatically in characteristic $p > 0$. Even for $G = \mathrm{SL}_2$, the induced module $H^0(p) = \mathrm{Sym}^p(k^2)$ is not completely reducible. In characteristic $p$, the elements $x^p$ and $y^p$ span a 2-dimensional $G$-submodule $M = k\{x^p, y^p\}$ (since $(ax + by)^p = a^p x^p + b^p y^p$). The quotient $H^0(p)/M$ has dimension $p - 1$ and highest weight $p - 2$, so it is isomorphic to $L(p - 2)$. This gives a short exact sequence:

$$0 \to M \to H^0(p) \to L(p - 2) \to 0$$

This sequence does *not* split. If it did, $H^0(p)$ would contain a submodule isomorphic to $L(p-2)$, which would require a highest weight vector of weight $p - 2$. But the only vectors in $\mathrm{Sym}^p(k^2)$ killed by the raising operator $e = x\partial_y$ are $x^p$ and $y^p$, neither of which has weight $p - 2$. Thus, $H^0(p)$ is reducible but indecomposable.

The failure is not just technical: it reflects the fact that in characteristic $p$, the Killing form can be degenerate (for instance, $\kappa = 0$ on $\mathfrak{sl}_p$), and the Casimir element no longer provides complementary projections. This is why modular representation theory is fundamentally harder than the characteristic-0 theory.

## Consequences of Complete Reducibility

**Corollary 8.3.3: Structure of $\mathrm{Rep}(G)$ in Characteristic $0$**

Let $G$ be a connected reductive group over an algebraically closed field of characteristic 0. Then:

1. Every finite-dimensional representation decomposes as $V \cong \bigoplus_{\lambda \in X(T)^+} L(\lambda)^{\oplus m_\lambda}$ for unique non-negative integers $m_\lambda$ (the *multiplicities*).
2. The representation ring $R(G) = K_0(\mathrm{Rep}(G))$ is a free $\mathbb{Z}$-module with basis $\{[L(\lambda)] : \lambda \in X(T)^+\}$.
3. The character map $\mathrm{ch} : R(G) \to \mathbb{Z}[X(T)]^W$ is an isomorphism: the representation ring is isomorphic to the ring of $W$-invariant elements in the group ring of $X(T)$.

***Proof*:**
*(1)* is an immediate consequence of complete reducibility and the classification (theorem 8.2.3): decompose $V$ into irreducibles, and each irreducible is some $L(\lambda)$.

*(2)* follows from (1): every element of $K_0(\mathrm{Rep}(G))$ is a $\mathbb{Z}$-linear combination of the classes $[L(\lambda)]$, and these are linearly independent (their characters are linearly independent in $\mathbb{Z}[X(T)]$, by the fact that the highest weight $\lambda$ appears with multiplicity 1 in $\mathrm{ch}(L(\lambda))$ and does not appear in $\mathrm{ch}(L(\mu))$ unless $\lambda \preceq \mu$, which makes the transition matrix unitriangular with respect to the dominance order).

*(3)*: The character map sends $[V] \mapsto \mathrm{ch}(V) = \sum_\lambda (\dim V_\lambda) e^\lambda \in \mathbb{Z}[X(T)]$. It is a ring homomorphism (characters are additive on direct sums and multiplicative on tensor products), injective (by (2) and the linear independence of characters), and surjective onto $\mathbb{Z}[X(T)]^W$ (every $W$-invariant element is a character of some virtual representation, which can be verified using the Weyl character formula from chapter 8).

**Example 8.3.4: Tensor Product Decomposition for $\mathrm{SL}_2$**
In $\mathrm{Rep}(\mathrm{SL}_2)$, the tensor product $L(a) \otimes L(b)$ (with $a \geq b$) decomposes as:

$$L(a) \otimes L(b) \cong L(a+b) \oplus L(a+b-2) \oplus \cdots \oplus L(a-b)$$

This is the *Clebsch–Gordan formula*: $b+1$ summands, with highest weights $a+b, a+b-2, \ldots, a-b$ (decreasing by 2). The formula follows from the Weyl character formula (chapter 8) or by direct computation: the weights of $L(a) \otimes L(b)$ are $\{i+j : i \in \mathrm{wt}(L(a)),\ j \in \mathrm{wt}(L(b))\}$, and counting multiplicities gives the decomposition.

As a check: $\dim(L(a) \otimes L(b)) = (a+1)(b+1) = \sum_{k=0}^{b}(a+b-2k+1) = \sum_{k=0}^{b} \dim L(a+b-2k)$.

## Exercise 8.3.1

1. For $G = \mathrm{SL}_2$, compute the Casimir element $C \in U(\mathfrak{sl}_2)$ using the basis $\{e, f, h\}$ and the Killing form $\kappa$ from exercise 3.2.1 (3). Show that $C$ acts on $L(d) = \mathrm{Sym}^d(k^2)$ by the scalar $\frac{d(d+2)}{8}$ (in the normalization where $\kappa(h,h) = 8$).
2. Let $G = \mathrm{SL}_2$ in characteristic 0 and $V = k^2 \otimes k^2$ (the tensor square of the standard representation). Decompose $V$ into irreducibles. Then verify complete reducibility by finding a $G$-stable complement to the subrepresentation $\mathrm{Sym}^2(k^2) \subseteq V$.
3. Show that for a semisimple group $G$ in characteristic 0, $\mathrm{Hom}_G(L(\lambda), L(\mu)) = 0$ for $\lambda \neq \mu$ and $\mathrm{End}_G(L(\lambda)) = k$ (i.e. Schur's lemma). Deduce that $\mathrm{Ext}^1_G(L(\lambda), L(\mu)) = 0$ for all $\lambda, \mu$ (using the Casimir argument from the proof of theorem 8.3.1).
4. Show that the representation ring $R(\mathrm{SL}_2) \cong \mathbb{Z}[x]$ where $x = [L(1)]$ (the class of the standard representation). Express $[L(d)]$ as a polynomial in $x$. (*Hint*: use the Clebsch–Gordan formula from example 8.3.4 to compute $x^d = [L(1)]^d$ and solve for $[L(d)]$.)
5. Let $G = \mathrm{SL}_2$ over a field of characteristic $p = 3$. The Clebsch–Gordan formula from

characteristic 0 predicts that $L(1) \otimes L(2) \cong L(3) \oplus L(1)$. Show that this complete reducibility fails in characteristic 3 by analyzing the induced module $H^0(3) = \mathrm{Sym}^3(k^2)$. Specifically, show that $H^0(3)$ is reducible but indecomposable. (*Hint*: check the action of the raising operator $e$ on $y^3$.)

# 9

# Characters and the Weyl Character Formula

chapter 8 classified the irreducible representations of a connected reductive group $G$ in characteristic 0: they are indexed by dominant weights $\lambda \in X(T)^+$, and every representation decomposes as a direct sum of irreducibles. But the classification does not tell us *what* these representations look like: how many weights does $L(\lambda)$ have? What are their multiplicities? What is $\dim L(\lambda)$? How does $L(\lambda) \otimes L(\mu)$ decompose?

All of these questions are answered by the *character*, a generating function that encodes the complete weight-multiplicity data of a representation. The central result of this chapter is the *Weyl character formula*, which gives an explicit closed-form expression for the character of $L(\lambda)$ as an alternating sum over the Weyl group. From this single formula, we will extract: the *Weyl dimension formula* (the dimension of $L(\lambda)$ as a product over positive roots), *Freudenthal's formula* (a recursive computation of individual weight multiplicities), and the *tensor product decomposition rules* (how to decompose $L(\lambda) \otimes L(\mu)$ into irreducibles).

Throughout this chapter, $k$ is an algebraically closed field of characteristic 0 (unless otherwise stated), $G$ is a connected reductive group over $k$, $T \subseteq B \subseteq G$ is a maximal torus inside a Borel subgroup, and we use the notation of chapter 7: $\Phi$, $\Phi^+$, $\Delta = \{\alpha_1, \ldots, \alpha_r\}$, $W$, and $\rho = \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$ (the *Weyl vector*, or half-sum of positive roots).

## 9.1 Formal Characters

### The Group Ring of the Weight Lattice

To keep track of weights and their multiplicities algebraically, we work in the *group ring* of the character lattice.

**Definition 9.1.1: The Group Ring**

The *group ring* of $X(T)$ is the free abelian group

$$\mathbb{Z}[X(T)] = \bigoplus_{\lambda \in X(T)} \mathbb{Z} \cdot e^{\lambda}$$

with multiplication defined by $e^{\lambda} \cdot e^{\mu} = e^{\lambda+\mu}$ and extended $\mathbb{Z}$-linearly. The identity element is $e^0 = 1$.

Elements of $\mathbb{Z}[X(T)]$ are formal $\mathbb{Z}$-linear combinations of the symbols $e^{\lambda}$, and the multiplication is determined by the addition in $X(T)$. When $T = \mathbb{G}_m$ and $X(T) = \mathbb{Z}$, the group ring is $\mathbb{Z}[X(T)] = \mathbb{Z}[e^{\varepsilon}, e^{-\varepsilon}] \cong \mathbb{Z}[t, t^{-1}]$ (Laurent polynomials in one variable). For $T = \mathbb{G}_m^r$, it is $\mathbb{Z}[t_1^{\pm 1}, \dots, t_r^{\pm 1}]$ (Laurent polynomials in $r$ variables).

The Weyl group $W$ acts on $\mathbb{Z}[X(T)]$ by $w \cdot e^{\lambda} = e^{w(\lambda)}$, extended linearly. A central role is played by the subring of *$W$-invariants*:

$$\mathbb{Z}[X(T)]^W = \{f \in \mathbb{Z}[X(T)] \mid w \cdot f = f \text{ for all } w \in W\}$$

We proved in corollary 8.3.3 that the character map gives an isomorphism $R(G) \cong \mathbb{Z}[X(T)]^W$ in characteristic 0.

### Characters of Representations

**Definition 9.1.2: Formal Character**

Let $V$ be a finite-dimensional representation of $G$. The *(formal) character* of $V$ is the element

$$\mathrm{ch}(V) = \sum_{\lambda \in X(T)} (\dim V_{\lambda})\, e^{\lambda} \in \mathbb{Z}[X(T)]$$

This is a finite sum (only finitely many weight spaces are nonzero).

The character is a "generating function" that packages all weight multiplicities into a single algebraic object. It determines the representation up to semisimplification (i.e., up to isomorphism in characteristic 0 where everything is semisimple, and up to the same composition factors with the same multiplicities in general).

**Proposition 9.1.3: Basic Properties of Characters**

Let $V, W$ be finite-dimensional representations of $G$. Then:

1. (Additivity) $\mathrm{ch}(V \oplus W) = \mathrm{ch}(V) + \mathrm{ch}(W)$. More generally, if $0 \to V' \to V \to V'' \to 0$ is a short exact sequence, then $\mathrm{ch}(V) = \mathrm{ch}(V') + \mathrm{ch}(V'')$.
2. (Multiplicativity) $\mathrm{ch}(V \otimes W) = \mathrm{ch}(V) \cdot \mathrm{ch}(W)$ (product in $\mathbb{Z}[X(T)]$).
3. (Duality) $\mathrm{ch}(V^\vee) = \overline{\mathrm{ch}(V)}$, where $\bar{\ }$ is the involution $e^\lambda \mapsto e^{-\lambda}$.
4. ($W$-invariance) $\mathrm{ch}(V) \in \mathbb{Z}[X(T)]^W$.
5. (Dimension) Evaluating at $e^\lambda = 1$ for all $\lambda$ gives $\mathrm{ch}(V)|_{e^\lambda = 1} = \dim V$.

***Proof*:**
*(1):* $(V \oplus W)_\lambda = V_\lambda \oplus W_\lambda$, so $\dim(V \oplus W)_\lambda = \dim V_\lambda + \dim W_\lambda$. For a short exact sequence: $\dim V_\lambda = \dim V'_\lambda + \dim V''_\lambda$ (exactness of weight space functors, since the torus $T$ is linearly reductive in any characteristic — its representations are completely reducible by theorem 5.2.9).

*(2):* $(V \otimes W)_\lambda = \bigoplus_{\mu+\nu=\lambda} V_\mu \otimes W_\nu$, so $\dim(V \otimes W)_\lambda = \sum_{\mu+\nu=\lambda} \dim V_\mu \cdot \dim W_\nu$. This is exactly the coefficient of $e^\lambda$ in $\mathrm{ch}(V) \cdot \mathrm{ch}(W)$.

*(3):* $(V^\vee)_\lambda = (V_{-\lambda})^*$ (the dual of a weight-$(-\lambda)$ vector is a weight-$\lambda$ functional under the contragredient action, definition 2.3.4), so $\dim(V^\vee)_\lambda = \dim V_{-\lambda}$.

*(4):* This is proposition 8.1.3: $\dim V_{w(\lambda)} = \dim V_\lambda$ for all $w \in W$.

*(5):* Setting $e^\lambda = 1$ for all $\lambda$ gives $\mathrm{ch}(V)|_1 = \sum_\lambda \dim V_\lambda = \dim V$.

**Example 9.1.4: Characters of $\mathrm{SL}_2$-Representations**

For $G = \mathrm{SL}_2$ with $X(T) = \mathbb{Z}$ and $e^\varepsilon = t$, the character of $L(d) = \mathrm{Sym}^d(k^2)$ is:

$$\mathrm{ch}(L(d)) = t^d + t^{d-2} + \cdots + t^{-d+2} + t^{-d} = \frac{t^{d+1} - t^{-(d+1)}}{t - t^{-1}}$$

(a geometric series with $d+1$ terms). Setting $t = 1$: $\mathrm{ch}(L(d))|_{t=1} = d + 1 = \dim L(d)$. The $W$-invariance ($W = \{1, s\}$ with $s \cdot t = t^{-1}$) is visible: the character is a palindromic Laurent polynomial ($t^d + t^{-d} + \cdots$).

The multiplicativity: $\mathrm{ch}(L(a)) \cdot \mathrm{ch}(L(b)) = \mathrm{ch}(L(a) \otimes L(b))$. The Clebsch–Gordan decomposition (example 8.3.4) reads, at the character level:

$$\frac{t^{a+1} - t^{-(a+1)}}{t - t^{-1}} \cdot \frac{t^{b+1} - t^{-(b+1)}}{t - t^{-1}} = \sum_{k=0}^{b} \frac{t^{a+b-2k+1} - t^{-(a+b-2k+1)}}{t - t^{-1}}$$

The reader should verify this identity of Laurent polynomials directly (it is a telescoping sum).

**Example 9.1.5: Characters of $\mathrm{SL}_3$-Representations**
For $G = \mathrm{SL}_3$ with $X(T) = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ and $e^{\omega_1} = x$, $e^{\omega_2} = y$:

1. $\mathrm{ch}(L(\omega_1)) = e^{\varepsilon_1} + e^{\varepsilon_2} + e^{\varepsilon_3} = x + x^{-1}y + y^{-1}$ (using $\varepsilon_1 = \omega_1$, $\varepsilon_2 = \omega_1 - \alpha_1 = -\omega_1 + \omega_2$, $\varepsilon_3 = -\omega_2$). Actually, let us use the variables $e^{\varepsilon_i}$: $\mathrm{ch}(L(\omega_1)) = e^{\varepsilon_1} + e^{\varepsilon_2} + e^{\varepsilon_3}$, the sum of the three standard weights.

2. $\mathrm{ch}(L(\omega_2)) = e^{-\varepsilon_3} + e^{-\varepsilon_2} + e^{-\varepsilon_1} = e^{\varepsilon_1+\varepsilon_2} + e^{\varepsilon_1+\varepsilon_3} + e^{\varepsilon_2+\varepsilon_3} = y + xy^{-1} + x^{-1}$ (the dual representation).

3. $\mathrm{ch}(L(\omega_1 + \omega_2))$: the adjoint representation $\mathfrak{sl}_3$ has weights $\{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1+\alpha_2), 0, 0\}$ (the six roots plus the two-dimensional zero weight space $\mathfrak{t}$). So:

$$\mathrm{ch}(L(\omega_1 + \omega_2)) = e^{\alpha_1} + e^{\alpha_2} + e^{\alpha_1+\alpha_2} + e^{-\alpha_1} + e^{-\alpha_2} + e^{-\alpha_1-\alpha_2} + 2e^0$$

The $W = S_3$-invariance of each character can be verified by checking that permuting the $\varepsilon_i$ leaves the expression unchanged (which it does, since the weight sets are $S_3$-orbits in each case).

## The Weyl Denominator

Before stating the Weyl character formula, we introduce a key auxiliary object: the *Weyl denominator*, which appears in the denominator of the formula and has a closed-form expression.

**Definition 9.1.6: The Weyl Denominator**

The *Weyl denominator* is the element $D \in \mathbb{Z}[X(T)]$ defined by:

$$D = \sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho)}$$

where $\rho = \frac{1}{2}\sum_{\alpha\in\Phi^+} \alpha$ is the Weyl vector and $\ell(w)$ is the length of $w$ (the number of simple reflections in a reduced expression for $w$, equivalently $\ell(w) = |\Phi^+ \cap w^{-1}(\Phi^-)|$).

**Theorem 9.1.7: Weyl Denominator Formula**

The Weyl denominator has the product expansion:

$$D = \sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho)} = e^{\rho} \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})$$

Equivalently, after multiplying both sides by $e^{-\rho}$:

$$\sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho)-\rho} = \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})$$

***Proof*:**
We give the proof for $\mathrm{SL}_2$ and indicate the strategy for the general case.

$\mathrm{SL}_2$*:* Here $W = \{1, s\}$, $\Phi^+ = \{\alpha\}$, $\rho = \alpha/2$, $\ell(1) = 0$, $\ell(s) = 1$, and $s(\rho) = -\rho$. The left side is $e^\rho - e^{-\rho}$. The right side is $e^\rho(1 - e^{-\alpha}) = e^\rho - e^{\rho-\alpha} = e^{\alpha/2} - e^{-\alpha/2}$. These are equal.

*General case (sketch):* The product $e^\rho \prod_{\alpha>0}(1 - e^{-\alpha})$ expands as $e^\rho$ times a sum over subsets $S \subseteq \Phi^+$, each contributing $(-1)^{|S|} e^{-\sum_{\alpha \in S} \alpha}$. One shows that most terms cancel, and the surviving terms are exactly $\{(-1)^{\ell(w)} e^{w(\rho)}\}_{w \in W}$. The cancellation is proved by a combinatorial argument using the fact that $W$ acts simply transitively on the set of Weyl chambers (see Humphreys [10] for the full details).

**Example 9.1.8: The Weyl Denominator for $\mathrm{SL}_3$**
For $\mathrm{SL}_3$: $W = S_3$, $\Phi^+ = \{\alpha_1, \alpha_2, \alpha_1 + \alpha_2\}$, $\rho = \alpha_1 + \alpha_2 = \omega_1 + \omega_2$. The six Weyl group elements and their contributions are:

| $w$ | $\ell(w)$ | $w(\rho)$ | $(-1)^{\ell(w)} e^{w(\rho)}$ |
|---|---|---|---|
| $e$ | 0 | $\alpha_1 + \alpha_2$ | $+e^{\alpha_1+\alpha_2}$ |
| $s_1$ | 1 | $-\alpha_1 + \alpha_1 + \alpha_2 = \alpha_2$ | $-e^{\alpha_2}$ |
| $s_2$ | 1 | $\alpha_1 + \alpha_2 - \alpha_2 = \alpha_1$ | $-e^{\alpha_1}$ |
| $s_1 s_2$ | 2 | $s_1(\alpha_1) = -\alpha_1$ | $+e^{-\alpha_1}$ |
| $s_2 s_1$ | 2 | $s_2(\alpha_2) = -\alpha_2$ | $+e^{-\alpha_2}$ |
| $w_0 = s_1 s_2 s_1$ | 3 | $s_1(-\alpha_2) = -\alpha_1 - \alpha_2$ | $-e^{-\alpha_1-\alpha_2}$ |

The denominator is:

$$D = e^{\alpha_1+\alpha_2} - e^{\alpha_2} - e^{\alpha_1} + e^{-\alpha_2} + e^{-\alpha_1} - e^{-\alpha_1-\alpha_2}$$

The product formula gives the same result:

$$e^\rho(1 - e^{-\alpha_1})(1 - e^{-\alpha_2})(1 - e^{-\alpha_1-\alpha_2})$$

which the reader should expand and verify equals $D$.

## Characters Determine Representations

The character map is injective on the representation ring:

**Proposition 9.1.9: Characters Separate Representations**

In characteristic 0, the character map $\mathrm{ch} : R(G) \to \mathbb{Z}[X(T)]^W$ is injective. That is, two representations with the same character are isomorphic.

***Proof*:**
By complete reducibility (theorem 8.3.1), every representation $V$ decomposes as $V \cong \bigoplus_\lambda L(\lambda)^{\oplus m_\lambda}$. The character is $\mathrm{ch}(V) = \sum_\lambda m_\lambda \,\mathrm{ch}(L(\lambda))$. The characters $\{\mathrm{ch}(L(\lambda))\}_{\lambda \in X(T)^+}$ are linearly independent in $\mathbb{Z}[X(T)]$: the highest weight $\lambda$ appears in $\mathrm{ch}(L(\lambda))$ with coefficient 1, and does not appear in $\mathrm{ch}(L(\mu))$ unless $\lambda \preceq \mu$ (since all weights of $L(\mu)$ are bounded by $\mu$). This makes the transition matrix between the irreducible characters and the basis elements $e^\lambda$ unitriangular with respect to any linear extension of the dominance order. So the multiplicities $m_\lambda$ are uniquely determined by $\mathrm{ch}(V)$.

This means that computing the character of a representation is *equivalent* to decomposing it into irreducibles. The Weyl character formula, which we state in the next section, reduces this computation to a finite, explicit algebraic calculation.

**9.1.10 Remark: Characters in Positive Characteristic** In characteristic $p > 0$, the character still makes sense and still satisfies additivity on short exact sequences (proposition 9.1.3(1)). However, it does *not* determine the representation up to isomorphism: non-isomorphic representations can have the same character (since complete reducibility fails, a representation and its semisimplification have the same character but may not be isomorphic). The character determines only the *composition factors* and their multiplicities.

The Weyl module $V(\lambda)$ has the same character as $L(\lambda)$ would have in characteristic 0 (this is property (3) of proposition 8.2.8). But in characteristic $p$, $V(\lambda)$ is reducible, and $\mathrm{ch}(V(\lambda)) = \mathrm{ch}(L(\lambda)) + \sum_{\mu \prec \lambda} [V(\lambda) : L(\mu)]\,\mathrm{ch}(L(\mu))$, where $[V(\lambda) : L(\mu)]$ are the composition multiplicities. Computing these multiplicities is the central problem of chapter 11.

## Exercise 9.1.1

1. For $G = \mathrm{SL}_2$, verify the identity $\mathrm{ch}(L(a)) \cdot \mathrm{ch}(L(b)) = \sum_{k=0}^{b} \mathrm{ch}(L(a + b - 2k))$ (the Clebsch–Gordan formula at the character level) by direct computation with Laurent polynomials for $a = 3$, $b = 2$.

2. Compute $\mathrm{ch}(L(\omega_1)) \cdot \mathrm{ch}(L(\omega_1))$ for $\mathrm{SL}_3$ (the character of $V = k^3 \otimes k^3$). Decompose the result into $\mathrm{ch}(L(2\omega_1)) + \mathrm{ch}(L(\omega_2))$ and verify this matches $\mathrm{Sym}^2(k^3) \oplus \bigwedge^2(k^3)$.

3. For $G = \mathrm{SL}_n$, show that $\mathrm{ch}(L(\lambda)^\vee) = \mathrm{ch}(L(-w_0\lambda))$, where $w_0 \in W$ is the longest element. Deduce that $L(\lambda)^\vee \cong L(-w_0\lambda)$. For $\mathrm{SL}_3$, what is $L(\omega_1)^\vee$?

4. Expand the product $e^\rho \prod_{\alpha \in \Phi^+}(1 - e^{-\alpha})$ for $G = \mathrm{SL}_3$ (with $\Phi^+ = \{\alpha_1, \alpha_2, \alpha_1 + \alpha_2\}$) and verify it equals the alternating sum $\sum_{w \in W}(-1)^{\ell(w)} e^{w(\rho)}$ computed in example 9.1.8.

5. Show that for $G = \mathrm{SL}_3$, the ring $\mathbb{Z}[X(T)]^W$ is a polynomial ring in two variables: $\mathbb{Z}[X(T)]^W = \mathbb{Z}[\sigma_1, \sigma_2]$ where $\sigma_1 = e^{\varepsilon_1} + e^{\varepsilon_2} + e^{\varepsilon_3}$ and $\sigma_2 = e^{\varepsilon_1+\varepsilon_2} + e^{\varepsilon_1+\varepsilon_3} + e^{\varepsilon_2+\varepsilon_3}$ are the elementary symmetric polynomials. Identify $\sigma_1 = \mathrm{ch}(L(\omega_1))$ and $\sigma_2 = \mathrm{ch}(L(\omega_2))$.

6. Let $0 \to L(\mu) \to V \to L(\lambda) \to 0$ be a non-split short exact sequence of $G$-representations (in characteristic $p$). Show that $\mathrm{ch}(V) = \mathrm{ch}(L(\lambda)) + \mathrm{ch}(L(\mu))$ even though $V \not\cong L(\lambda) \oplus L(\mu)$. Conclude that the character does not determine the isomorphism class of $V$ in characteristic $p$.

## 9.2 The Weyl Character Formula

We now state and prove the central result of this chapter: an explicit closed-form expression for the character of the irreducible representation $L(\lambda)$, valid for any dominant weight $\lambda \in X(T)^+$.

### Statement of the Formula

**Theorem 9.2.1: The Weyl Character Formula**

Let $G$ be a connected reductive group over an algebraically closed field of characteristic 0, and let $\lambda \in X(T)^+$ be a dominant weight. The character of the irreducible representation $L(\lambda)$ is:

$$\mathrm{ch}(L(\lambda)) = \frac{\displaystyle\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\rho)}}{\displaystyle\sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho)}}$$

Equivalently, using the Weyl denominator formula (theorem 9.1.7):

$$\mathrm{ch}(L(\lambda)) = \frac{\displaystyle\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\rho)}}{\displaystyle e^{\rho} \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})}$$

where $\rho = \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$ is the Weyl vector.

The formula expresses $\mathrm{ch}(L(\lambda))$ as a ratio: the numerator is an alternating sum over the Weyl group (with the "shifted" weight $\lambda + \rho$ replacing $\lambda$), and the denominator is the Weyl denominator $D$ from definition 9.1.6. The shift by $\rho$ is essential: without it, the alternating sum would not produce a well-defined element of $\mathbb{Z}[X(T)]$ after division.

In a more symmetric form, the formula can be written as:

$$D \cdot \mathrm{ch}(L(\lambda)) = \sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\rho)}$$

This says that multiplying the character by the denominator $D$ produces a $W$-*antisymmetric* ("alternating") element: $w \cdot (D \cdot \mathrm{ch}(L(\lambda))) = (-1)^{\ell(w)} D \cdot \mathrm{ch}(L(\lambda))$. Since the character itself is $W$-symmetric and $D$ is $W$-antisymmetric (by the sign character $w \mapsto (-1)^{\ell(w)}$), their product is $W$-antisymmetric, which is consistent.

### Verification for $\mathrm{SL}_2$

Before proving the formula in general, let us verify it for $\mathrm{SL}_2$, where everything can be checked by hand.

**Example 9.2.2: Weyl Character Formula for $\mathrm{SL}_2$**

For $G = \mathrm{SL}_2$: $W = \{1, s\}$, $\Phi^+ = \{\alpha\}$, $\rho = \alpha/2$, $\ell(1) = 0$, $\ell(s) = 1$. The dominant weight is $\lambda = d\varepsilon = \frac{d}{2}\alpha$ for $d \geq 0$. Then $\lambda + \rho = \frac{d+1}{2}\alpha$, and:

$$w_0(\lambda + \rho) = s\left(\frac{d+1}{2}\alpha\right) = -\frac{d+1}{2}\alpha$$

Writing $t = e^{\alpha/2} = e^{\varepsilon}$, the numerator is:

$$e^{\lambda+\rho} - e^{s(\lambda+\rho)} = t^{d+1} - t^{-(d+1)}$$

The denominator is $D = t - t^{-1}$ (the Weyl denominator). So:

$$\mathrm{ch}(L(d)) = \frac{t^{d+1} - t^{-(d+1)}}{t - t^{-1}} = t^d + t^{d-2} + \cdots + t^{-d}$$

This matches the known character from example 9.1.4. The formula produces the correct answer.

## Proof of the Weyl Character Formula

The proof presented here uses the algebraic approach via the *BGG resolution* (Bernstein–Gelfand–Gelfand). In characteristic 0, the irreducible representation $L(\lambda)$ corresponds to a finite-dimensional irreducible module for the Lie algebra $\mathfrak{g}$, allowing us to use highest-weight theory.

***Proof*:**

*Step 1: The character of the Verma module.* We construct representations from the universal enveloping algebra $U(\mathfrak{g})$. For a dominant weight $\lambda$, define the *Verma module* $M(\lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} k_\lambda$, where $\mathfrak{b} = \mathrm{Lie}(B)$ is the Borel subalgebra.

By the Poincaré–Birkhoff–Witt theorem, $M(\lambda) \cong U(\mathfrak{n}^-) \otimes k_\lambda$ as a $T$-module, where $\mathfrak{n}^- = \bigoplus_{\alpha\in\Phi^-} \mathfrak{g}_\alpha$ is the "negative nilpotent" subalgebra. Since $U(\mathfrak{n}^-)$ is a polynomial ring generated by the negative root vectors, its character is the product of geometric series:

$$\mathrm{ch}(U(\mathfrak{n}^-)) = \prod_{\alpha>0}(1 + e^{-\alpha} + e^{-2\alpha} + \cdots) = \prod_{\alpha>0} \frac{1}{1 - e^{-\alpha}}$$

Therefore, the character of the Verma module is straightforward to compute:

$$\mathrm{ch}(M(\lambda)) = \frac{e^\lambda}{\prod_{\alpha\in\Phi^+}(1 - e^{-\alpha})}$$

*Step 2: The BGG Resolution.* The irreducible module $L(\lambda)$ is a quotient of the Verma module $M(\lambda)$. In the *category $\mathcal{O}$* (the BGG category of highest-weight modules), $L(\lambda)$ can be resolved by Verma modules. The *BGG resolution* is an exact sequence of $\mathfrak{g}$-modules:

$$0 \to M(w_0 \cdot \lambda) \to \cdots \to \bigoplus_{\ell(w)=2} M(w \cdot \lambda) \to \bigoplus_{\ell(w)=1} M(w \cdot \lambda) \to M(\lambda) \to L(\lambda) \to 0$$

where $w \cdot \lambda = w(\lambda + \rho) - \rho$ is the *dot action* of the Weyl group $W$ on $X(T)$.

*Step 3: Taking alternating sums.* Because the character is additive on exact sequences, the class of $L(\lambda)$ in the Grothendieck group can be expressed as an alternating sum of the terms in the resolution:

$$[L(\lambda)] = \sum_{w\in W} (-1)^{\ell(w)}[M(w\cdot\lambda)]$$

Passing to characters, we substitute our formula for $\mathrm{ch}(M(\mu))$ from Step 1:

$$\begin{aligned}
\mathrm{ch}(L(\lambda)) &= \sum_{w\in W} (-1)^{\ell(w)}\,\mathrm{ch}(M(w\cdot\lambda)) \\
&= \sum_{w\in W} (-1)^{\ell(w)} \frac{e^{w(\lambda+\rho)-\rho}}{\prod_{\alpha>0}(1-e^{-\alpha})} \\
&= \frac{\sum_{w\in W}(-1)^{\ell(w)} e^{w(\lambda+\rho)-\rho}}{\prod_{\alpha>0}(1-e^{-\alpha})}
\end{aligned}$$

Finally, we multiply the numerator and denominator by $e^{\rho}$ to clear the $e^{-\rho}$ factor in the numerator. The denominator becomes $e^{\rho}\prod_{\alpha>0}(1-e^{-\alpha})$, which is exactly the Weyl denominator $D$.

$$\begin{aligned}
\mathrm{ch}(L(\lambda)) &= \frac{\sum_{w\in W}(-1)^{\ell(w)} e^{w(\lambda+\rho)}}{e^{\rho}\prod_{\alpha>0}(1-e^{-\alpha})} \\
&= \frac{\sum_{w\in W}(-1)^{\ell(w)} e^{w(\lambda+\rho)}}{D}
\end{aligned}$$

which is exactly the Weyl character formula.

The proof reveals the deeper structure behind the formula: the numerator $\sum_w (-1)^{\ell(w)} e^{w(\lambda+\rho)}$ arises from the alternating sum of Verma modules in the BGG resolution, and the denominator $D = e^{\rho}\prod_{\alpha>0}(1-e^{-\alpha})$ is closely related to the character of the “universal enveloping algebra of $\mathfrak{n}^-$” (the polynomial ring on the negative root vectors). The shift by $\rho$ in the dot action $w\cdot\lambda = w(\lambda+\rho)-\rho$ accounts for the “twist” between the Verma module $M(\lambda)$ and the character computation.

## The Weyl Dimension Formula

The character formula specializes to a dimension formula by evaluating at $e^{\lambda} = 1$ for all $\lambda$. Naively, both the numerator and denominator of the Weyl character formula vanish at this point (since $D|_1 = 0$), so we apply l’Hôpital’s rule (or equivalently, evaluate the ratio of polynomials at a regular point).

**Corollary 9.2.3: The Weyl Dimension Formula**

For a dominant weight $\lambda \in X(T)^+$:

$$\dim L(\lambda) = \prod_{\alpha \in \Phi^+} \frac{\langle \lambda + \rho, \alpha^\vee \rangle}{\langle \rho, \alpha^\vee \rangle}$$

where the product runs over all positive roots $\alpha \in \Phi^+$.

***Proof*:**
We need to evaluate $\mathrm{ch}(L(\lambda))|_{e^\mu = 1} = \dim L(\lambda)$. The numerator of the Weyl character formula at $e^\mu = 1$ is $\sum_w (-1)^{\ell(w)} \cdot 1 = 0$ (the alternating sum of 1 over $W$ vanishes for $|W| > 1$). The denominator is similarly 0. We apply a limiting argument.

Define, for $\xi \in X(T) \otimes \mathbb{R}$ and small $\epsilon > 0$, the "regularized evaluation" by $e^\mu \mapsto e^{\epsilon(\mu,\xi)}$ (a real exponential). As $\epsilon \to 0$, both numerator and denominator approach 0, but their ratio approaches a finite limit.

More directly: the product form of the denominator gives $D = e^\rho \prod_{\alpha>0}(1 - e^{-\alpha})$. Near the identity ($e^\mu \approx 1 + (\mu, \xi)\epsilon$ for small $\epsilon$ and a generic $\xi$), $(1 - e^{-\alpha}) \approx (\alpha, \xi)\epsilon$, so $D \approx \prod_{\alpha>0}(\alpha, \xi)\epsilon^{|\Phi^+|}$. Similarly, $e^{w(\lambda+\rho)} \approx 1 + w(\lambda + \rho, \xi)\epsilon$, so $\sum_w (-1)^{\ell(w)} e^{w(\lambda+\rho)} \approx \sum_w (-1)^{\ell(w)} w(\lambda + \rho, \xi)\epsilon \cdot$ (lower order)... this requires care.

The cleanest approach: define the *Weyl function* $A(\mu) = \sum_w (-1)^{\ell(w)} e^{w(\mu)}$ for $\mu \in X(T) \otimes \mathbb{R}$. This is the "antisymmetrization" of $e^\mu$ over $W$. One shows (by expanding in a power series around $\mu = 0$) that $A(\mu) = \prod_{\alpha>0}(\alpha, \mu) \cdot e^{-\rho} \cdot D +$ higher order. Actually, the standard argument replaces the exponentials by polynomials via the Weyl integration formula on the compact form. The result is:

$$\frac{A(\lambda + \rho)}{A(\rho)} = \frac{\prod_{\alpha>0}(\lambda + \rho, \alpha)}{\prod_{\alpha>0}(\rho, \alpha)}$$

Since $(\mu, \alpha) = \frac{(\alpha,\alpha)}{2} \langle \mu, \alpha^\vee \rangle$, the ratio $(\lambda + \rho, \alpha)/(\rho, \alpha) = \langle \lambda + \rho, \alpha^\vee \rangle / \langle \rho, \alpha^\vee \rangle$ (the $(\alpha, \alpha)/2$ factors cancel). This gives the dimension formula.

**Example 9.2.4: Dimensions for $\mathrm{SL}_2$**
For $\mathrm{SL}_2$: $\Phi^+ = \{\alpha\}$, $\rho = \alpha/2$, $\lambda = d(\alpha/2)$. The formula gives:

$$\dim L(d) = \frac{\langle d(\alpha/2) + \alpha/2, \alpha^\vee \rangle}{\langle \alpha/2, \alpha^\vee \rangle} = \frac{\langle (d+1)\alpha/2, \alpha^\vee \rangle}{\langle \alpha/2, \alpha^\vee \rangle} = \frac{d+1}{1} = d + 1$$

confirming the known dimension of $\mathrm{Sym}^d(k^2)$.

**Example 9.2.5: Dimensions for $\mathrm{SL}_3$**

For $\mathrm{SL}_3$: $\Phi^+ = \{\alpha_1, \alpha_2, \alpha_1 + \alpha_2\}$, $\rho = \omega_1 + \omega_2 = \alpha_1 + \alpha_2$. For $\lambda = a\omega_1 + b\omega_2$, we have $\lambda + \rho = (a+1)\omega_1 + (b+1)\omega_2$ and:

$$\begin{aligned} \langle \lambda + \rho, \alpha_1^\vee \rangle &= a + 1 \\ \langle \lambda + \rho, \alpha_2^\vee \rangle &= b + 1 \\ \langle \lambda + \rho, (\alpha_1 + \alpha_2)^\vee \rangle &= a + b + 2 \end{aligned}$$

while $\langle \rho, \alpha_1^\vee \rangle = 1$, $\langle \rho, \alpha_2^\vee \rangle = 1$, $\langle \rho, (\alpha_1 + \alpha_2)^\vee \rangle = 2$. So:

$$\dim L(a\omega_1 + b\omega_2) = \frac{(a+1)(b+1)(a+b+2)}{1 \cdot 1 \cdot 2} = \frac{(a+1)(b+1)(a+b+2)}{2}$$

Verification: $\dim L(0) = 1$ (trivial), $\dim L(\omega_1) = \frac{2 \cdot 1 \cdot 3}{2} = 3$ (standard), $\dim L(\omega_2) = \frac{1 \cdot 2 \cdot 3}{2} = 3$ (dual standard), $\dim L(\omega_1 + \omega_2) = \frac{2 \cdot 2 \cdot 4}{2} = 8$ (adjoint), $\dim L(2\omega_1) = \frac{3 \cdot 1 \cdot 4}{2} = 6$ ($\mathrm{Sym}^2(k^3)$). All correct.

**Example 9.2.6: Dimensions for $\mathrm{Sp}_4$**

For $\mathrm{Sp}_4$ (type $C_2$): $\Phi^+ = \{\alpha_1, \alpha_2, \alpha_1 + \alpha_2, 2\alpha_1 + \alpha_2\}$, $\rho = \omega_1 + \omega_2$ (equivalently $\rho = 2\varepsilon_1 + \varepsilon_2$). For $\lambda = a\omega_1 + b\omega_2$:

$$\dim L(a\omega_1 + b\omega_2) = \frac{(a+1)(b+1)(a+b+2)(a+2b+3)}{1 \cdot 1 \cdot 2 \cdot 3}$$

(the product is over the four positive roots). Verification: $\dim L(\omega_1) = \frac{2 \cdot 1 \cdot 3 \cdot 4}{6} = 4$ (standard), $\dim L(\omega_2) = \frac{1 \cdot 2 \cdot 3 \cdot 5}{6} = 5$ (the 5-dimensional representation, the traceless symmetric square of the standard).

**Example 9.2.7: The Weyl Character Formula for $L(\omega_1 + \omega_2)$ of $\mathrm{SL}_3$**

We verify the full character formula for the adjoint representation of $\mathrm{SL}_3$, with $\lambda = \omega_1 + \omega_2 = \rho$. Then $\lambda + \rho = 2\rho$, and the numerator is:

$$\sum_{w \in S_3} (-1)^{\ell(w)} e^{w(2\rho)} = \sum_{w \in S_3} (-1)^{\ell(w)} e^{2w(\rho)}$$

since $W$ permutes $\rho$ by its natural action. Using the computation from example 9.1.8 (with $\rho$ replaced by $2\rho$): writing $\rho = \alpha_1 + \alpha_2$, we get $2\rho = 2\alpha_1 + 2\alpha_2$ and the six terms $\{e^{w(2\rho)}\}$ at the exponents $\pm 2\alpha_1 \pm 2\alpha_2$ (with appropriate signs from $W$). Dividing by $D$ and simplifying gives:

$$\mathrm{ch}(L(\omega_1 + \omega_2)) = e^{\alpha_1 + \alpha_2} + e^{\alpha_1} + e^{\alpha_2} + 2 + e^{-\alpha_1} + e^{-\alpha_2} + e^{-\alpha_1 - \alpha_2}$$

which has $6 + 2 = 8$ terms (counting the double contribution at weight 0), matching $\dim \mathfrak{sl}_3 = 8$ and the character computed in example 9.1.5(3).

## Dimension Formulas for the Classical Groups

We record the Weyl dimension formula explicitly for each classical family.

**Proposition 9.2.8: Classical Dimension Formulas**

For the classical groups with dominant weight $\lambda = \sum n_i\omega_i$ (fundamental weight coordinates):

1. $\mathrm{SL}_{n+1}$ *(type $A_n$):* $\dim L(\lambda) = \prod_{1\le i<j\le n+1} \frac{\lambda_i-\lambda_j+j-i}{j-i}$ where $\lambda_i = n_i + \cdots + n_n$ are the coordinates of $\lambda+\rho$ in the $\varepsilon_i$-basis (after the shift). Equivalently, this is a ratio of products of "hook lengths."

2. $\mathrm{Sp}_{2n}$ *(type $C_n$):* $\dim L(\lambda) = \prod_{1\le i<j\le n} \frac{(\ell_i-\ell_j)(\ell_i+\ell_j)}{(j-i)(2n-i-j+2)} \cdot \prod_{i=1}^{n} \frac{\ell_i}{n-i+1}$ where $\ell_i = \langle \lambda + \rho, \varepsilon_i\rangle$.

(Similar formulas exist for types $B_n$ and $D_n$; we refer to Fulton–Harris or Goodman–Wallach for the complete expressions.)

The key feature of these formulas is that they are *products* — the dimension factors through the positive roots, with each root contributing one factor. This product structure is a reflection of the "independence" of the root directions, and it makes the formulas remarkably efficient to compute even for large representations.

### Exercise 9.2.1

1. Use the Weyl character formula to compute $\mathrm{ch}(L(4))$ for $\mathrm{SL}_2$. Write it as a Laurent polynomial in $t = e^{\varepsilon}$ and verify the result matches $t^4+t^2+1+t^{-2}+t^{-4}$ (the character of $\mathrm{Sym}^4(k^2)$, which has dimension 5).

2. Use the Weyl dimension formula for $\mathrm{SL}_4$ (type $A_3$) to compute $\dim L(\omega_1) = 4$, $\dim L(\omega_2) = 6$, $\dim L(\omega_3) = 4$, and $\dim L(\omega_1+\omega_3) = 15$. For $L(\omega_1+\omega_3)$: this is the adjoint representation $\mathfrak{sl}_4$; verify $\dim \mathfrak{sl}_4 = 15$.

3. Compute $\dim L(\omega_1)$ and $\dim L(\omega_2)$ for $\mathrm{SO}_5$ (type $B_2$). The first should be 5 (the standard representation) and the second should be 4 (the spin representation). (*Hint*: $\Phi^+(B_2) = \{\varepsilon_1, \varepsilon_2, \varepsilon_1-\varepsilon_2, \varepsilon_1+\varepsilon_2\}$, $\rho = \frac{3}{2}\varepsilon_1 + \frac{1}{2}\varepsilon_2$.)

4. Use the Weyl character formula to compute $\mathrm{ch}(L(2\omega_1))$ for $\mathrm{SL}_3$: write out all six terms of the numerator $\sum_w (-1)^{\ell(w)} e^{w(2\omega_1+\rho)}$, divide by $D$, and verify the result is $\mathrm{ch}(\mathrm{Sym}^2(k^3)) = e^{2\varepsilon_1} + e^{\varepsilon_1+\varepsilon_2} + e^{2\varepsilon_2} + e^{\varepsilon_1+\varepsilon_3} + e^{\varepsilon_2+\varepsilon_3} + e^{2\varepsilon_3}$.

5. For $G = \mathrm{SL}_{n+1}$ and $\lambda = d\omega_1$ (the $d$-th symmetric power of the standard representation), show that $\dim L(d\omega_1) = \binom{d+n}{n}$ using the Weyl dimension formula. (*Hint*: the positive roots are $\varepsilon_i - \varepsilon_j$ with $i<j$, and $\langle d\omega_1+\rho, (\varepsilon_i-\varepsilon_j)^\vee\rangle = d\delta_{i,1} + j - i$.)

6. Show that the Weyl character formula for $\lambda = 0$ (the trivial representation) gives $\mathrm{ch}(L(0)) = 1$. That is, verify $\sum_w (-1)^{\ell(w)} e^{w(\rho)}/D = 1$, which is the statement that $D = \sum_w (-1)^{\ell(w)} e^{w(\rho)}$ (the Weyl denominator formula is a special case of the character formula).

## 9.3 Applications and Computations

The Weyl character formula is not just a theoretical result; it is a computational engine. In this section, we develop the tools that extract concrete information from the formula: individual weight multiplicities (Freudenthal's formula), tensor product decompositions (Steinberg's formula and the Littlewood–Richardson rule), and the structure of the representation ring.

## Freudenthal's Multiplicity Formula

The Weyl character formula gives the character of $L(\lambda)$ as a ratio of alternating sums, which is efficient for computing the *entire* character but less convenient for computing the multiplicity of a *single* weight $\mu$. Freudenthal's formula solves this problem with a recursion that computes $m_\mu(\lambda) = \dim L(\lambda)_\mu$ from the multiplicities of strictly higher weights.

**Theorem 9.3.1: Freudenthal's Multiplicity Formula**

Let $\lambda \in X(T)^+$ be a dominant weight and $\mu \in \mathrm{wt}(L(\lambda))$ a weight of $L(\lambda)$. Then:

$$\big((\lambda+\rho, \lambda+\rho) - (\mu+\rho, \mu+\rho)\big) \cdot m_\mu(\lambda) = 2 \sum_{\alpha \in \Phi^+} \sum_{j=1}^{\infty} (\mu + j\alpha, \alpha) \cdot m_{\mu+j\alpha}(\lambda)$$

where $(\cdot\,,\cdot)$ is the $W$-invariant inner product on $X(T) \otimes \mathbb{R}$ and the inner sum is finite (it terminates when $\mu + j\alpha \notin \mathrm{wt}(L(\lambda))$).

***Proof Sketch*:**
The formula is derived by comparing two computations of the action of the Casimir element $C$ on $L(\lambda)$. On one hand, $C$ acts by the scalar $c(\lambda) = (\lambda+\rho, \lambda+\rho) - (\rho, \rho)$ (theorem 8.3.1, Step 1). On the other hand, the explicit expression $C = \sum X_i X^i$ (in terms of a Killing-form-dual basis) can be decomposed using the root space structure: the contribution from the root vectors $\{e_\alpha, f_\alpha\}$ to the weight-$\mu$ subspace gives the right-hand side. Equating the two expressions for $\mathrm{tr}(C|_{L(\lambda)_\mu})$ yields the formula. See [10] for the complete derivation.

The formula is a *recursion*: the left-hand side involves $m_\mu(\lambda)$, while the right-hand side involves only $m_{\mu+j\alpha}(\lambda)$ for $j \geq 1$, which are multiplicities of *higher* weights (closer to $\lambda$ in the dominance order). Since $m_\lambda(\lambda) = 1$ (the highest weight has multiplicity 1), the recursion can be run downward through the weight poset, computing all multiplicities.

The coefficient $(\lambda+\rho, \lambda+\rho) - (\mu+\rho, \mu+\rho)$ on the left is strictly positive when $\mu \prec \lambda$ (since $\mu + \rho$ lies strictly inside the ball of radius $|\lambda + \rho|$ centered at the origin, by a convexity argument), so the division is always valid.

**Example 9.3.2: Freudenthal for the Adjoint of $\mathrm{SL}_3$**

Let $\lambda = \omega_1 + \omega_2$ (the adjoint representation of $\mathrm{SL}_3$). We compute $m_0(\lambda) = \dim L(\lambda)_0$ (the multiplicity of the zero weight, which is $\dim \mathfrak{t} = 2$).

With $\rho = \omega_1 + \omega_2 = \lambda$, we have $\lambda + \rho = 2\rho$ and $(\lambda+\rho, \lambda+\rho) = (2\rho, 2\rho) = 4(\rho, \rho)$. For $\mu = 0$: $(\mu+\rho, \mu+\rho) = (\rho, \rho)$. The left coefficient is $4(\rho,\rho) - (\rho,\rho) = 3(\rho,\rho)$.

The right side: for each positive root $\alpha$ and each $j \geq 1$ with $j\alpha \in \mathrm{wt}(L(\lambda))$, we need $(j\alpha, \alpha) \cdot m_{j\alpha}(\lambda)$. The weights of the adjoint are the six roots (multiplicity 1 each) and 0 (multiplicity $m_0$, unknown). For $\alpha = \alpha_1$: $j\alpha_1 \in \mathrm{wt}$ only for $j = 1$ (since $2\alpha_1$ is not a weight of $L(\omega_1+\omega_2)$). So the contribution is $(\alpha_1, \alpha_1) \cdot 1$. Similarly for $\alpha_2$ and $\alpha_1 + \alpha_2$. The total right side is $2[(\alpha_1, \alpha_1) + (\alpha_2, \alpha_2) + (\alpha_1 +$

$\alpha_2, \alpha_1 + \alpha_2)]$.

For $\mathrm{SL}_3$ with the standard normalization $(\alpha_i, \alpha_i) = 2$, we compute $(\alpha_1+\alpha_2, \alpha_1+\alpha_2) = 2+2(-1)+2 = 2$ (using $(\alpha_1, \alpha_2) = -1$). So the right side is $2[2+2+2] = 12$.

Since $\rho = \alpha_1 + \alpha_2$, we already know $(\rho, \rho) = 2$. Thus, the left side is $3(2) \cdot m_0 = 6m_0$. Equating the two sides gives $6m_0 = 12$, hence $m_0 = 2$, confirming $\dim \mathfrak{t} = 2$.

## Tensor Product Decompositions

One of the most important applications of the character formula is to decompose tensor products of irreducible representations. Since $\mathrm{ch}(L(\lambda) \otimes L(\mu)) = \mathrm{ch}(L(\lambda)) \cdot \mathrm{ch}(L(\mu))$ (proposition 9.1.3(2)), and since every $W$-invariant element of $\mathbb{Z}[X(T)]$ is uniquely a $\mathbb{Z}$-linear combination of irreducible characters (corollary 8.3.3(3)), the tensor product decomposition

$$L(\lambda) \otimes L(\mu) \cong \bigoplus_{\nu \in X(T)^+} L(\nu)^{\oplus c^\nu_{\lambda\mu}}$$

is determined by the product of characters. The non-negative integers $c^\nu_{\lambda\mu}$ are the *tensor product multiplicities* (or *Littlewood–Richardson coefficients* in type $A$).

**Proposition 9.3.3: Computing Tensor Product Multiplicities**

The tensor product multiplicities can be computed by any of the following methods:

1. *(Direct expansion)* Multiply the characters $\mathrm{ch}(L(\lambda)) \cdot \mathrm{ch}(L(\mu))$ in $\mathbb{Z}[X(T)]$ and decompose into the basis $\{\mathrm{ch}(L(\nu))\}$.
2. *(Steinberg's formula)* $c^\nu_{\lambda\mu} = \sum_{w_1, w_2 \in W} (-1)^{\ell(w_1)+\ell(w_2)} P(w_1(\lambda+\rho)+w_2(\mu+\rho)-(\nu+2\rho))$, where $P(\xi)$ is the *Kostant partition function*: the number of ways to write $\xi$ as a non-negative integer combination of positive roots.
3. *(Klimyk's formula)* $c^\nu_{\lambda\mu} = \sum_{w \in W} (-1)^{\ell(w)} m_{w(\nu+\rho)-\rho-\lambda}(\mu)$, which expresses the tensor product multiplicity in terms of weight multiplicities of $L(\mu)$.

***Proof Sketch*:**
*(1)* is tautological. *(2)* follows from substituting the Weyl character formula for both $L(\lambda)$ and $L(\mu)$ and using the expansion $\frac{1}{D} = \frac{1}{e^\rho \prod (1-e^{-\alpha})} = e^{-\rho} \sum_\xi P(\xi) e^{-\xi}$ (the "Kostant partition function" expansion of the denominator inverse). *(3)* is obtained by expanding only $\mathrm{ch}(L(\nu))$ via the Weyl formula and leaving $\mathrm{ch}(L(\mu))$ as a weight multiplicity sum. All three give the same answer and can be found in [10].

**Example 9.3.4: Tensor Product Decomposition for $\mathrm{SL}_3$**
Decompose $L(\omega_1) \otimes L(\omega_1 + \omega_2)$ for $\mathrm{SL}_3$ (the standard representation tensored with the adjoint). By character multiplication:

$$\mathrm{ch}(L(\omega_1)) \cdot \mathrm{ch}(L(\omega_1 + \omega_2)) = (e^{\varepsilon_1} + e^{\varepsilon_2} + e^{\varepsilon_3}) \cdot (\text{8-term character of adj})$$

The product is a 24-term sum in $\mathbb{Z}[X(T)]$, which after collecting terms and decomposing into irreducible characters gives:

$$L(\omega_1) \otimes L(\omega_1 + \omega_2) \cong L(2\omega_1 + \omega_2) \oplus L(2\omega_2) \oplus L(\omega_1)$$

We can verify this with a dimension check. The tensor product has dimension $3 \times 8 = 24$. Using the Weyl dimension formula for the summands, we get $\dim L(2\omega_1 + \omega_2) = \frac{3 \cdot 2 \cdot 5}{2} = 15$, $\dim L(2\omega_2) = \frac{1 \cdot 3 \cdot 4}{2} = 6$, and $\dim L(\omega_1) = 3$. The sum of the dimensions is exactly $15 + 6 + 3 = 24$.

## The Littlewood–Richardson Rule (Type $A$)

For the special case $G = \mathrm{GL}_n$ (or $\mathrm{SL}_n$), the tensor product multiplicities have an elegant combinatorial description in terms of *Young tableaux*. This is the *Littlewood–Richardson rule*, which connects representation theory to the combinatorics of symmetric functions.

A dominant weight $\lambda$ of $\mathrm{GL}_n$ corresponds to a *partition* (a weakly decreasing sequence of non-negative integers) $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0)$, or equivalently a *Young diagram* with $\lambda_i$ boxes in the $i$-th row. The irreducible representation $L(\lambda)$ is the *Schur module* $S^\lambda(k^n)$.

**Theorem 9.3.5: Littlewood–Richardson Rule**

For $G = \mathrm{GL}_n$ with dominant weights $\lambda, \mu, \nu$ (viewed as partitions):

$$L(\lambda) \otimes L(\mu) \cong \bigoplus_\nu L(\nu)^{\oplus c^\nu_{\lambda\mu}}$$

where $c^\nu_{\lambda\mu}$ is the number of *Littlewood–Richardson tableaux* of shape $\nu/\lambda$ and content $\mu$: that is, semistandard Young tableaux of skew shape $\nu/\lambda$ with entries from $\{1, 2, \ldots\}$ such that the entry $i$ appears $\mu_i$ times, and the reverse reading word (reading right-to-left, top-to-bottom) is a lattice word (at every point, the number of $i$'s read so far is $\geq$ the number of $(i+1)$'s).

***Proof Sketch*:**
The proof connects the representation theory of $\mathrm{GL}_n$ to the algebra of *symmetric functions*. The character $\mathrm{ch}(L(\lambda))$ for $\mathrm{GL}_n$ is the *Schur polynomial* $s_\lambda(x_1, \ldots, x_n)$ (where $x_i = e^{\varepsilon_i}$), and the tensor product multiplicities are the structure constants for the multiplication of Schur polynomials: $s_\lambda \cdot s_\mu = \sum_\nu c^\nu_{\lambda\mu} s_\nu$. The Littlewood–Richardson rule is a combinatorial formula for these structure constants, proved via the Robinson–Schensted–Knuth correspondence or via jeu de taquin. See [7] for the symmetric functions perspective.

**Example 9.3.6: Littlewood–Richardson for $\mathrm{SL}_3$**
Decompose $L(\omega_1) \otimes L(\omega_1) = L((1,0,0)) \otimes L((1,0,0))$ for $\mathrm{SL}_3$. The partitions are $\lambda = \mu = (1)$ (a single box). The tensor product is $S^{(1)} \otimes S^{(1)} = k^3 \otimes k^3$. The Littlewood–Richardson rule gives:

$$c^{(2)}_{(1)(1)} = 1 \quad \text{(skew shape } (2)/(1) = \text{one box, content } (1))$$

$$c^{(1,1)}_{(1)(1)} = 1 \quad \text{(skew shape } (1,1)/(1) = \text{one box in row 2, content } (1))$$

So $L(\omega_1) \otimes L(\omega_1) \cong L(2\omega_1) \oplus L(\omega_2)$, matching $\mathrm{Sym}^2(k^3) \oplus \bigwedge^2(k^3)$. The rule recovers the decomposition from exercise 8.2.1 (3) combinatorially.

## The PRV Theorem

For a general reductive group $G$ (not just type $A$), an important structural result about tensor product decompositions is the PRV theorem (Parthasarathy–Ranga Rao–Varadarajan), which identifies specific components that always appear.

**Theorem 9.3.7: The PRV Theorem**

Let $\lambda, \mu \in X(T)^+$ be dominant weights and let $w \in W$. If $w(\mu)$ is a weight of $L(\mu)$ (which it always is, by $W$-invariance), then the irreducible representation $L(\nu)$ with $\nu$ equal to the unique dominant weight in the $W$-orbit of $\lambda + w(\mu)$ appears in $L(\lambda) \otimes L(\mu)$ with multiplicity $\geq 1$. In particular:

1. $L(\lambda + \mu)$ always appears (take $w = e$, so $\lambda + \mu$ is dominant).
2. $L(\nu)$ with $\nu$ the dominant representative of $\lambda + w_0(\mu)$ always appears (take $w = w_0$, the longest element).

The PRV theorem gives a lower bound on the tensor product: it guarantees at least $|W|$ components (though many may coincide). Statement (1) says the "largest" component $L(\lambda + \mu)$ always appears (this is elementary: the tensor of the two highest weight vectors is a highest weight vector of weight $\lambda + \mu$). Statement (2) is deeper: the "smallest" component $L(\nu)$ with $\nu = (\lambda + w_0\mu)^+$ also appears. Between these two extremes, the PRV theorem provides a systematic way to locate components.

## The Representation Ring Revisited

We close by summarizing the algebraic structure of the representation ring, which encodes all of the above computations.

**Proposition 9.3.8: Structure of the Representation Ring**

The character isomorphism $\mathrm{ch} : R(G) \xrightarrow{\sim} \mathbb{Z}[X(T)]^W$ gives $R(G)$ the structure of a commutative ring. For the classical groups:

1. $\mathrm{SL}_{n+1}$ *(type $A_n$):* $R(G) \cong \mathbb{Z}[\sigma_1, \ldots, \sigma_n]$, a polynomial ring in the characters of the fundamental representations $L(\omega_i) = \bigwedge^i(k^{n+1})$ (the elementary symmetric polynomials in $e^{\varepsilon_1}, \ldots, e^{\varepsilon_{n+1}}$, modulo $\sigma_{n+1} = e^{\varepsilon_1 + \cdots + \varepsilon_{n+1}} = 1$).
2. $\mathrm{Sp}_{2n}$ *(type $C_n$):* $R(G) \cong \mathbb{Z}[\sigma_1, \ldots, \sigma_n]$, a polynomial ring in the characters of the fundamental representations.
3. In general, $R(G)$ is a polynomial ring on $r = \mathrm{rank}(G)$ generators (one per fundamental representation) by Chevalley's theorem on invariant polynomials: $\mathbb{Z}[X(T)]^W$ is always a polynomial ring when $W$ is a Weyl group.

> ***Proof Sketch*:**
> Chevalley's theorem states that if $W$ is a finite reflection group acting on a polynomial ring $\mathbb{Z}[x_1^{\pm 1}, \ldots, x_r^{\pm 1}]$, the subring of $W$-invariants is generated by $r$ algebraically independent elements (the "fundamental invariants"). For the Weyl group of a root system, these fundamental invariants can be taken to be the characters of the fundamental representations. The proof uses the fact that the Weyl group is generated by reflections and that the degrees of the fundamental invariants are determined by the exponents of $W$.

The polynomial ring structure means that the representation ring has no "hidden relations": the tensor product multiplicities $c_{\lambda\mu}^{\nu}$ are the complete set of structure constants, and every representation is a polynomial expression in the fundamental representations. This is the algebraic realization of the principle that the fundamental representations are the "building blocks" from which all others are constructed.

## Exercise 9.3.1

1. Use Freudenthal's formula to compute $m_0(L(4))$ for $\mathrm{SL}_2$ (the multiplicity of the zero weight in $\mathrm{Sym}^4(k^2)$). Verify the answer is 1.
2. Decompose $L(3) \otimes L(3)$ for $\mathrm{SL}_2$ using the Clebsch–Gordan formula. Verify the result by computing both sides at the character level.
3. Decompose $L(\omega_1) \otimes L(\omega_1) \otimes L(\omega_1)$ for $\mathrm{SL}_3$. (*Hint*: first decompose $L(\omega_1) \otimes L(\omega_1) \cong L(2\omega_1) \oplus L(\omega_2)$, then tensor each summand with $L(\omega_1)$. Use the Weyl dimension formula to check: $3^3 = 27$.)
4. The Kostant partition function $P(\xi)$ counts the number of ways to write $\xi \in X(T)$ as $\xi = \sum_{\alpha \in \Phi^+} n_\alpha \alpha$ with $n_\alpha \in \mathbb{Z}_{\geq 0}$. For $\mathrm{SL}_3$ with $\Phi^+ = \{\alpha_1, \alpha_2, \alpha_1 + \alpha_2\}$, compute $P(2\alpha_1 + 2\alpha_2)$.
5. Use the Littlewood–Richardson rule to decompose $L((2,1)) \otimes L((1))$ for $\mathrm{GL}_3$ (i.e. $L(2\omega_1 + \omega_2 - \omega_3) \otimes L(\omega_1)$... more concretely, the 8-dimensional adjoint representation tensored with the standard 3-dimensional representation). List all Littlewood–Richardson tableaux and identify the summands.
6. Apply the PRV theorem to $L(\omega_1) \otimes L(\omega_2)$ for $\mathrm{SL}_3$. For each $w \in W = S_3$, compute $\omega_1 + w(\omega_2)$ and its dominant representative. Which irreducible representations does the PRV theorem guarantee appear? Verify against the known decomposition $L(\omega_1) \otimes L(\omega_2) \cong L(\omega_1 + \omega_2) \oplus L(0)$.

# 10

# Geometric Representation Theory

The previous two chapters classified and computed with representations of $G$ using algebraic tools: highest weights, characters, the Weyl character formula. This chapter brings geometry back to the center of the story. The flag variety $G/B$, constructed in Chapter 6 as a projective variety, carries a natural family of *line bundles* $\mathcal{L}(\lambda)$ parametrized by the weight lattice $X(T)$. The *Borel–Weil theorem* asserts that the global sections of $\mathcal{L}(\lambda)$, for dominant $\lambda$, *are* the irreducible representations: they are constructed geometrically, as spaces of sections on $G/B$, rather than algebraically via highest weight theory. The *Borel–Weil–Bott theorem* extends this to higher cohomology, showing that the Weyl character formula is a shadow of the cohomological structure of line bundles on the flag variety.

This is the payoff for the geometric work of Part I: the flag variety is not merely a classifying space for Borel subgroups (Chapter 6), but the machine that *produces* every irreducible representation of $G$. The passage from algebra to geometry is not merely an alternative viewpoint but a genuine expansion of the theory: it connects representation theory to the cohomology of projective varieties, to Schubert calculus, and (looking further ahead) to the geometric Langlands program and the theory of $\mathcal{D}$-modules.

Throughout this chapter, $k$ is an algebraically closed field, $G$ is a connected reductive group over $k$, and $B \supseteq T$ is a Borel subgroup containing a maximal torus. We assume familiarity with line bundles and sheaf cohomology on projective varieties at the level of [5].

## 10.1 Line Bundles on $G/B$

### The Associated Bundle Construction

The flag variety $X = G/B$ is a smooth projective variety (theorem 6.2.3). The natural projection $\pi : G \to G/B$ is a principal $B$-bundle: $B$ acts freely on $G$ by right multiplication, and the fibers of $\pi$

are the right $B$-cosets. Any representation of $B$ can be "spread out" over $G/B$ using this bundle structure, producing a vector bundle on $G/B$. For one-dimensional representations, the result is a line bundle.

**Definition 10.1.1: The Line Bundle $\mathcal{L}(\lambda)$**

Let $\lambda \in X(T)$ be a character. Extend $\lambda$ to a character of $B = T \ltimes R_u(B)$ by letting $R_u(B)$ act trivially: $\lambda(tu) = \lambda(t)$ for $t \in T$, $u \in R_u(B)$. The *associated line bundle* is the quotient

$$\mathcal{L}(\lambda) = G \times^B k_\lambda := (G \times k)/B$$

where $B$ acts on $G \times k$ by $b \cdot (g, z) = (gb^{-1}, \lambda(b)z)$. The equivalence class of $(g, z)$ is denoted $[g, z]$. The natural map $\mathcal{L}(\lambda) \to G/B$ sending $[g, z] \mapsto gB$ is a line bundle on $G/B$.

The construction is the standard *associated bundle* (or *Borel construction*) from the theory of principal bundles. Concretely: over the point $gB \in G/B$, the fiber of $\mathcal{L}(\lambda)$ is $\{[g, z] : z \in k\} \cong k$, and the transition functions between trivializing open sets are given by the character $\lambda$. The group $G$ acts on $\mathcal{L}(\lambda)$ by left multiplication on the first factor: $g' \cdot [g, z] = [g'g, z]$. This makes $\mathcal{L}(\lambda)$ a *$G$-linearized line bundle*: the $G$-action on the total space is compatible with the $G$-action on the base $G/B$.

**Proposition 10.1.2: Basic Properties of $\mathcal{L}(\lambda)$**

1. The map $\lambda \mapsto \mathcal{L}(\lambda)$ is a group homomorphism: $\mathcal{L}(\lambda+\mu) \cong \mathcal{L}(\lambda)\otimes\mathcal{L}(\mu)$ and $\mathcal{L}(0) \cong \mathcal{O}_{G/B}$ (the structure sheaf).
2. The map $X(T) \to \mathrm{Pic}^G(G/B)$ sending $\lambda \mapsto \mathcal{L}(\lambda)$ is an isomorphism: every *$G$-equivariant* line bundle on $G/B$ is isomorphic to $\mathcal{L}(\lambda)$ for a unique $\lambda \in X(T)$.
3. $\mathcal{L}(\lambda)$ is generated by global sections (i.e. basepoint-free) if and only if $\lambda$ is dominant.
4. $\mathcal{L}(\lambda)$ is ample if and only if $\lambda$ is *strictly dominant*: $\langle \lambda, \alpha_i^\vee \rangle > 0$ for all simple roots $\alpha_i$.

***Proof*:**
*(1):* The tensor product $\mathcal{L}(\lambda) \otimes \mathcal{L}(\mu)$ has fiber $(G \times^B k_\lambda) \otimes_{G/B} (G \times^B k_\mu)$. Over the point $gB$, this is $k_\lambda \otimes k_\mu = k_{\lambda+\mu}$ (tensor product of one-dimensional representations). So $\mathcal{L}(\lambda) \otimes \mathcal{L}(\mu) \cong G \times^B k_{\lambda+\mu} = \mathcal{L}(\lambda+\mu)$. For $\lambda = 0$: $k_0$ is the trivial representation, so $G \times^B k \cong G/B \times k = \mathcal{O}_{G/B}$.

*(2):* We consider the $G$-equivariant Picard group $\mathrm{Pic}^G(G/B)$. A $G$-linearized line bundle $\mathcal{L}$ on the homogeneous space $G/B$ is entirely determined by its fiber over the base point $eB$. Since the stabilizer of $eB$ is exactly $B$, this fiber must be a one-dimensional $B$-representation. Because $R_u(B)$ acts trivially on any one-dimensional representation, this is exactly a character of $T$, giving an element $\lambda \in X(T)$. This provides surjectivity.

For injectivity: if $\mathcal{L}(\lambda) \cong \mathcal{O}_{G/B}$ *as $G$-equivariant bundles* (where $\mathcal{O}_{G/B}$ is equipped with the trivial $G$-linearization), then their fibers over $eB$ match as $B$-representations. This means the $B$-representation on $k_\lambda$ is trivial, so $\lambda = 0$.

*(3):* The line bundle $\mathcal{L}(\lambda)$ is generated by global sections iff the evaluation map $H^0(G/B, \mathcal{L}(\lambda)) \otimes$

$\mathcal{O}_{G/B} \to \mathcal{L}(\lambda)$ is surjective, which happens iff $H^0(G/B, \mathcal{L}(\lambda)) \neq 0$ and the sections separate points and tangent vectors. By the Borel–Weil theorem (theorem 10.2.1 below), $H^0(G/B, \mathcal{L}(\lambda)) \neq 0$ iff $\lambda$ is dominant, and for dominant $\lambda$ the $G$-action ensures the sections generate $\mathcal{L}(\lambda)$ (the $G$-orbit of any nonzero section at the base point sweeps out all fibers).

*(4):* $\mathcal{L}(\lambda)$ is ample iff some positive power $\mathcal{L}(n\lambda)$ gives a closed embedding $G/B \hookrightarrow \mathbb{P}(H^0(G/B, \mathcal{L}(n\lambda))^\vee)$. This requires $\lambda$ to be strictly dominant: if $\langle \lambda, \alpha_i^\vee \rangle = 0$ for some $i$, then $\mathcal{L}(\lambda)$ is pulled back from the partial flag variety $G/P_i$ (where $P_i \supsetneq B$ is the minimal parabolic for $\alpha_i$), so it cannot be ample on $G/B$. Conversely, for strictly dominant $\lambda$, the embedding into $\mathbb{P}({H^0}^\vee)$ is a closed immersion (this uses the Kodaira-type embedding theorem for homogeneous spaces).

Property (2) is a striking structural result: the $G$-equivariant Picard group of the flag variety is exactly the weight lattice. Every equivariant line bundle on $G/B$ comes from a weight, and the ampleness/basepoint-freeness of the bundle is controlled by the position of the weight relative to the dominant cone. (*Remark:* If $G$ is semisimple and simply connected, every line bundle on $G/B$ admits a unique $G$-linearization, meaning the ordinary Picard group $\mathrm{Pic}(G/B)$ is also canonically isomorphic to $X(T)$).

## Global Sections as Representations

The space of global sections $H^0(G/B, \mathcal{L}(\lambda))$ is a finite-dimensional vector space (since $G/B$ is projective and $\mathcal{L}(\lambda)$ is coherent), and the $G$-linearization makes it a representation of $G$. We now identify this representation with the induced module $H^0(\lambda)$ from chapter 8.

**Proposition 10.1.3: Global Sections and the Induced Module**

For any $\lambda \in X(T)$, there is a natural isomorphism of $G$-representations:

$$H^0(G/B, \mathcal{L}(\lambda)) \cong H^0(\lambda) = \mathrm{Ind}_B^G(k_\lambda)$$

where $H^0(\lambda)$ is the induced module from definition 8.2.1.

***Proof*:**
A global section $s \in H^0(G/B, \mathcal{L}(\lambda))$ is a regular map $s : G/B \to \mathcal{L}(\lambda)$ with $s(x) \in \mathcal{L}(\lambda)_x$ for each $x \in G/B$. Pulling back along $\pi : G \to G/B$, the section $s$ determines a regular function $f_s : G \to k$ by $s(\pi(g)) = [g, f_s(g)]$ (representing the section in the associated bundle). The section must be well-defined on cosets, meaning $s(gB) = s(gbB)$ for any $b \in B$. Evaluating both sides in the associated bundle yields:

$$[g, f_s(g)] = [gb, f_s(gb)]$$

By the definition of the associated bundle quotient, the equivalence relation is generated by $b \cdot (g, z) = (gb^{-1}, \lambda(b)z)$. Substituting $g$ for $gb$, this means $[gb, z] = [g, \lambda(b)z]$ for any $z \in k$. Applying this to $z = f_s(gb)$ gives:

$$[g, f_s(g)] = [g, \lambda(b)f_s(gb)]$$

Since the equivalence classes over a fixed group element $g$ are determined entirely by the scalar in the second component, we get $f_s(g) = \lambda(b)f_s(gb)$, which rearranges to $f_s(gb) = \lambda(b)^{-1}f_s(g)$.

This is exactly the equivariance condition defining $H^0(\lambda)$ (definition 8.2.1). The map $s \mapsto f_s$ is the desired isomorphism, and it intertwines the $G$-actions (left translation on both sides).

This proposition justifies the notation $H^0(\lambda)$ used in chapter 8: the "$H^0$" was not an arbitrary label but a genuine cohomological object — the zeroth sheaf cohomology of a line bundle on $G/B$. The algebraic definition (functions on $G$ with $B$-equivariance) and the geometric definition (global sections of the associated line bundle) coincide, and from this point forward we use them interchangeably.

**Example 10.1.4: Line Bundles on $\mathbb{P}^1 = \mathrm{SL}_2/B$**

For $G = \mathrm{SL}_2$, the flag variety is $G/B = \mathbb{P}^1$. The weight lattice $X(T) = \mathbb{Z}$ and $\mathcal{L}(d)$ is the line bundle $\mathcal{O}_{\mathbb{P}^1}(d)$ (the $d$-th Serre twist). The isomorphism $\mathrm{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$ matches $\mathcal{L}(d) \leftrightarrow d$.

Properties: $\mathcal{L}(d)$ is basepoint-free iff $d \geq 0$ (dominant), ample iff $d > 0$ (strictly dominant), and trivial iff $d = 0$. Global sections: $H^0(\mathbb{P}^1, \mathcal{O}(d)) = 0$ for $d < 0$, and $H^0(\mathbb{P}^1, \mathcal{O}(d)) \cong \mathrm{Sym}^d(k^2)^\vee$ for $d \geq 0$: the space of homogeneous polynomials of degree $d$ in two variables, which is the dual of $L(d) = \mathrm{Sym}^d(k^2)$.

Explicitly: a section of $\mathcal{O}(d)$ on $\mathbb{P}^1$ is a global homogeneous polynomial $f(x, y)$ of degree $d$ (in the homogeneous coordinates $[x : y]$). The basis is $\{x^d, x^{d-1}y, \ldots, y^d\}$, giving $\dim H^0(\mathbb{P}^1, \mathcal{O}(d)) = d+1$. The $\mathrm{SL}_2$-action on sections is $(g \cdot f)(x, y) = f(g^{-1} \cdot (x, y))$, which is the contragredient of the standard action on $\mathrm{Sym}^d(k^2)$.

**Example 10.1.5: Line Bundles on $\mathrm{SL}_3/B$**

For $G = \mathrm{SL}_3$, the flag variety $X = G/B$ is the complete flag variety $\mathrm{Fl}(1, 2; 3)$ (parametrizing flags $0 \subset L_1 \subset L_2 \subset k^3$ with $\dim L_i = i$). This is a smooth projective variety of dimension 3.

The Picard group is $\mathrm{Pic}(X) \cong X(T) = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 \cong \mathbb{Z}^2$. The line bundle $\mathcal{L}(a\omega_1 + b\omega_2)$ is ample iff $a, b > 0$, basepoint-free iff $a, b \geq 0$, and trivial iff $a = b = 0$.

The two "boundary" cases: $\mathcal{L}(\omega_1)$ is the pullback of $\mathcal{O}_{\mathbb{P}^2}(1)$ along the projection $\mathrm{Fl}(1, 2; 3) \to \mathbb{P}^2 = \mathrm{Gr}(1, 3)$ (sending a flag to its line $L_1$), and $\mathcal{L}(\omega_2)$ is the pullback of $\mathcal{O}_{(\mathbb{P}^2)^\vee}(1)$ along the projection $\mathrm{Fl}(1, 2; 3) \to (\mathbb{P}^2)^\vee = \mathrm{Gr}(2, 3)$ (sending a flag to its plane $L_2$). Neither is ample on $\mathrm{Fl}(1, 2; 3)$ (they are pulled back from smaller varieties), but their tensor product $\mathcal{L}(\omega_1 + \omega_2)$ is ample.

Global sections: $H^0(X, \mathcal{L}(a\omega_1 + b\omega_2)) \cong H^0(a\omega_1 + b\omega_2)$ for $a, b \geq 0$, which is $L(a\omega_1 + b\omega_2)^\vee$ in characteristic 0 (by the Borel–Weil theorem, theorem 10.2.1 below). In particular: $H^0(X, \mathcal{L}(\omega_1)) \cong (k^3)^\vee$ (the dual of the standard representation) and $H^0(X, \mathcal{L}(\omega_2)) \cong (k^3)^{\vee\vee} \cong k^3$ (the standard representation, via $L(\omega_2)^\vee \cong L(\omega_1)$).

## The Big Bruhat Cell and Local Triviality

The flag variety $G/B$ has a natural cell decomposition indexed by the Weyl group: the *Bruhat decomposition* $G = \bigsqcup_{w \in W} BwB$ gives a decomposition $G/B = \bigsqcup_{w \in W} C_w$ where $C_w = BwB/B$ are the *Schubert cells*. Each $C_w$ is isomorphic to affine space $\mathbb{A}^{\ell(w)}$.

**Proposition 10.1.6: The Bruhat Decomposition of $G/B$**

1. $G/B = \bigsqcup_{w \in W} C_w$ where $C_w = BwB/B \cong \mathbb{A}^{\ell(w)}$.
2. The *big cell* $C_{w_0} = Bw_0B/B$ is an open dense subset of $G/B$, isomorphic to $\mathbb{A}^{|\Phi^+|}$ (affine space of dimension equal to the number of positive roots).
3. The complement $G/B \setminus C_{w_0}$ is a divisor (a closed subvariety of codimension 1).
4. The line bundle $\mathcal{L}(\lambda)$ is trivial on each Schubert cell $C_w$.

***Proof Sketch*:**
The Bruhat decomposition $G = \bigsqcup_{w \in W} BwB$ is a fundamental structural result about reductive groups that we state without full proof (see [15] for the complete argument). Statement (1) follows by passing to the quotient $G/B$. For (2): the map $R_u(B^-) \to G/B$ defined by $u \mapsto uw_0B$ (where $B^- = w_0Bw_0^{-1}$ is the opposite Borel and $R_u(B^-)$ is its unipotent radical) is an open immersion with image $C_{w_0}$, and $R_u(B^-) \cong \mathbb{A}^{|\Phi^+|}$ (as a variety, since it is a unipotent group). Statement (3) follows from (1): the cells of lower dimension form the complement. Statement (4): $\mathcal{L}(\lambda)|_{C_w}$ is trivial because $C_w \cong \mathbb{A}^{\ell(w)}$ is an affine space, and every line bundle on affine space is trivial.

The Bruhat decomposition gives $G/B$ the structure of a *CW-complex* (or algebraic cell complex), with one cell for each element of the Weyl group. The closures $\overline{C}_w$ are the *Schubert varieties*, which are in general singular. The cell structure is the geometric counterpart of the Weyl group combinatorics from Chapter 7, and it provides the computational backbone for intersection theory on $G/B$.

**Example 10.1.7: Bruhat Decomposition of $\mathbb{P}^1 = \mathrm{SL}_2/B$**
For $G = \mathrm{SL}_2$: $W = \{e, s\}$ with $\ell(e) = 0$ and $\ell(s) = 1$. The decomposition $\mathbb{P}^1 = C_e \sqcup C_s$ gives $\mathbb{P}^1$ as the union of a point $C_e = \{[1:0]\} \cong \mathbb{A}^0$ and an affine line $C_s \cong \mathbb{A}^1$ (the complement of $[1:0]$, parametrized by $[t:1]$ for $t \in k$). This is the standard decomposition of $\mathbb{P}^1$ into the "north pole" and the "affine chart."

**Example 10.1.8: Bruhat Decomposition of $\mathrm{SL}_3/B$**
For $G = \mathrm{SL}_3$: $W = S_3$ has 6 elements with lengths $\ell = 0, 1, 1, 2, 2, 3$. The flag variety $\mathrm{Fl}(1, 2; 3)$ decomposes into 6 cells:

| $w$ | $\ell(w)$ | $\dim C_w$ |
|---|---|---|
| $e$ | 0 | 0 |
| $s_1$ | 1 | 1 |
| $s_2$ | 1 | 1 |
| $s_1s_2$ | 2 | 2 |
| $s_2s_1$ | 2 | 2 |
| $w_0 = s_1s_2s_1$ | 3 | 3 |

The big cell $C_{w_0} \cong \mathbb{A}^3$ is dense and open. The Euler characteristic of $G/B$ (the alternating sum of

Betti numbers) equals $|W| = 6$ (each cell contributes 1 to the Euler characteristic).

**Exercise 10.1.1**

1. For $G = \mathrm{SL}_2$ and $\lambda = d$, verify explicitly that the associated bundle construction $\mathcal{L}(d) = \mathrm{SL}_2 \times^B k_d$ gives the line bundle $\mathcal{O}_{\mathbb{P}^1}(d)$. (*Hint*: trivialize $\mathcal{L}(d)$ on the two standard affine charts $U_0 = \{[x : y] : y \neq 0\}$ and $U_1 = \{[x : y] : x \neq 0\}$, and compute the transition function.)
2. Verify $\mathrm{Pic}(\mathrm{SL}_3/B) \cong \mathbb{Z}^2$ directly by showing that the two line bundles $\mathcal{L}(\omega_1)$ and $\mathcal{L}(\omega_2)$ are algebraically independent in Pic. (*Hint*: compute their restriction to the two Schubert curves $\overline{C}_{s_1} \cong \mathbb{P}^1$ and $\overline{C}_{s_2} \cong \mathbb{P}^1$.)
3. For $G = \mathrm{SL}_3$, verify that $\mathcal{L}(\omega_1)$ is *not* ample on $G/B$ by showing it is the pullback of a line bundle from $\mathbb{P}^2 \cong G/P_2$ (where $P_2 \supsetneq B$ is a minimal parabolic). Conclude that $\mathcal{L}(\omega_1)$ has degree 0 on the Schubert curve $\overline{C}_{s_2}$.
4. Verify that the Bruhat decomposition of $\mathrm{Sp}_4/B$ has $|W(C_2)| = 8$ cells, with dimensions $0, 1, 1, 2, 2, 3, 3, 4$. What is $\dim \mathrm{Sp}_4/B$?
5. Using the Borel–Weil theorem (stated in the next section), compute $\dim H^0(G/B, \mathcal{L}(\lambda))$ for $G = \mathrm{SL}_3$ and $\lambda = 2\omega_1 + \omega_2$ by applying the Weyl dimension formula from corollary 9.2.3. What representation is this?
6. Let $G = \mathrm{SL}_2$ and consider the section $s \in H^0(\mathbb{P}^1, \mathcal{O}(2))$ corresponding to the polynomial $xy$ (weight 0). Compute the zero locus of $s$ on $\mathbb{P}^1$. How does this relate to the Schubert decomposition?

## 10.2 The Borel–Weil Theorem

We now prove the main theorem of geometric representation theory: the global sections of $\mathcal{L}(\lambda)$ on $G/B$, for dominant $\lambda$, give the irreducible representations of $G$. This is the geometric realization of the highest weight classification from chapter 8.

### Statement and First Consequences

**Theorem 10.2.1: The Borel–Weil Theorem**

Let $G$ be a connected reductive group over an algebraically closed field $k$ of characteristic 0, and let $\lambda \in X(T)$.

1. If $\lambda$ is dominant ($\lambda \in X(T)^+$), then $H^0(G/B, \mathcal{L}(\lambda)) \neq 0$ and:
$$H^0(G/B, \mathcal{L}(\lambda)) \cong L(\lambda)^\vee$$
as a $G$-representation. Equivalently, $H^0(-w_0\lambda) \cong L(\lambda)$.
2. If $\lambda$ is not dominant, then $H^0(G/B, \mathcal{L}(\lambda)) = 0$.

The theorem says that the flag variety $G/B$ "knows" all the irreducible representations: as $\lambda$ ranges over the dominant weights, the spaces of global sections $H^0(G/B, \mathcal{L}(\lambda))$ sweep out every irreducible representation of $G$ (up to duality). The duality $L(\lambda)^\vee$ rather than $L(\lambda)$ arises because $G$ acts on sections by left translation, which is the *contragredient* of the natural right action.

***Proof*:**
*Statement (2): Vanishing for non-dominant* $\lambda$. If $\lambda$ is not dominant, then $\langle \lambda, \alpha_i^\vee \rangle < 0$ for some simple root $\alpha_i$. Consider the minimal parabolic $P_i \supsetneq B$ corresponding to $\alpha_i$. The projection $\pi_i : G/B \to G/P_i$ is a $\mathbb{P}^1$-bundle (the fiber is $P_i/B \cong \mathbb{P}^1$). The line bundle $\mathcal{L}(\lambda)$ restricts to each fiber $P_i/B \cong \mathbb{P}^1$ as $\mathcal{O}_{\mathbb{P}^1}(\langle \lambda, \alpha_i^\vee \rangle)$, which is a line bundle of negative degree $\langle \lambda, \alpha_i^\vee \rangle < 0$ on $\mathbb{P}^1$.

A line bundle of negative degree on $\mathbb{P}^1$ has no nonzero global sections ($H^0(\mathbb{P}^1, \mathcal{O}(d)) = 0$ for $d < 0$, as in example 10.1.4). Any global section $s \in H^0(G/B, \mathcal{L}(\lambda))$ restricts to a section on each fiber, which must be zero. Since the fibers cover $G/B$, we conclude $s = 0$. Therefore $H^0(G/B, \mathcal{L}(\lambda)) = 0$.

*Statement (1): Identification for dominant* $\lambda$. By proposition 10.1.3, $H^0(G/B, \mathcal{L}(\lambda)) = H^0(\lambda) = \mathrm{Ind}_B^G(k_\lambda)$. As we saw in proposition 8.2.2, for a dominant weight $\lambda$, $H^0(\lambda) \neq 0$ and it has a unique lowest weight $-\lambda$ of multiplicity 1 (under the left $G$-action).

In characteristic 0, $H^0(\lambda)$ is completely reducible (theorem 8.3.1). Because it has a unique lowest weight, it must be an irreducible representation. The irreducible representation with lowest weight $-\lambda$ has highest weight $-w_0\lambda$. Thus, $H^0(\lambda) \cong L(-w_0\lambda)$.

Using the standard duality of representations (exercise 9.1.1 (3)), we know that $L(-w_0\lambda) \cong L(\lambda)^\vee$. Stringing these isomorphisms together gives exactly the statement of the theorem:

$$H^0(G/B, \mathcal{L}(\lambda)) \cong L(\lambda)^\vee$$

(Equivalently, substituting $-w_0\lambda$ for $\lambda$ and using $w_0^2 = 1$ yields $H^0(-w_0\lambda) \cong L(-w_0\lambda)^\vee \cong L(\lambda)$).

*Verification for* $\mathrm{SL}_2$. For $\lambda = d \geq 0$: $H^0(\mathbb{P}^1, \mathcal{O}(d))$ is the space of homogeneous polynomials of degree $d$, with the contragredient $\mathrm{SL}_2$-action: $(g \cdot f)(v) = f(g^{-1}v)$. So $H^0(\mathbb{P}^1, \mathcal{O}(d)) \cong \mathrm{Sym}^d(k^2)^\vee = L(d)^\vee$. Since $L(d)^\vee \cong L(d)$ for $\mathrm{SL}_2$ $(-w_0(d) = d)$, everything is consistent.

## The Geometric Construction of Representations

The Borel–Weil theorem provides a *uniform geometric construction* of all irreducible representations. Rather than building $L(\lambda)$ from generators and relations (as in the algebraic approach of chapter 8, via Verma modules and quotients), we take spaces of sections:

$$\lambda \in X(T)^+ \quad \longmapsto \quad L(\lambda)^\vee \cong H^0(G/B, \mathcal{L}(\lambda))$$

This construction is functorial, canonical (no choices beyond $T \subseteq B$), and works uniformly for all dominant weights.

**Example 10.2.2: Geometric Construction for $\mathrm{SL}_3$**
For $G = \mathrm{SL}_3$ with dominant weight $\lambda = a\omega_1 + b\omega_2$:

1. $\mathcal{L}(a\omega_1 + b\omega_2)$ is a line bundle on $\mathrm{Fl}(1,2;3)$.
2. Its space of global sections $H^0(\mathrm{Fl}(1,2;3), \mathcal{L}(a\omega_1 + b\omega_2))$ is a $G$-representation of dimension $\frac{(a+1)(b+1)(a+b+2)}{2}$ (by the Weyl dimension formula, example 9.2.5).
3. In characteristic 0, this representation is $L(a\omega_1 + b\omega_2)^\vee$.

For the fundamental representations: sections of $\mathcal{L}(\omega_1)$ (pulled back from $\mathbb{P}^2$) give $L(\omega_1)^\vee = (k^3)^\vee$, and sections of $\mathcal{L}(\omega_2)$ (pulled back from $(\mathbb{P}^2)^\vee$) give $L(\omega_2)^\vee \cong k^3$.

**Example 10.2.3: The Plücker Embedding via Borel–Weil**
For $G = \mathrm{SL}_n$ and $\lambda = \omega_d$, the line bundle $\mathcal{L}(\omega_d)$ on $G/B$ is pulled back from the Grassmannian $\mathrm{Gr}(d,n) = G/P_d$. The global sections $H^0(G/P_d, \mathcal{L}(\omega_d)) \cong L(\omega_d)^\vee = (\bigwedge^d k^n)^\vee$ give the Plücker embedding:

$$\mathrm{Gr}(d,n) \hookrightarrow \mathbb{P}((\bigwedge^d k^n)^\vee)$$

The Borel–Weil theorem recovers the classical Plücker embedding as a special case.

## The Borel–Weil Theorem in Positive Characteristic

In characteristic $p > 0$, the space $H^0(G/B, \mathcal{L}(\lambda))$ is still well-defined and still zero for non-dominant $\lambda$. However, for dominant $\lambda$, $H^0(\lambda)$ is no longer irreducible in general.

**Proposition 10.2.4: Borel–Weil in Positive Characteristic**

Let $\mathrm{char} k = p > 0$ and $\lambda \in X(T)^+$. Then:

1. $H^0(G/B, \mathcal{L}(\lambda)) \neq 0$.
2. $H^0(\lambda)$ has a unique irreducible *socle* (minimal nonzero subrepresentation).
3. $H^0(\lambda)$ may be reducible: it can have composition factors beyond the socle.
4. The character of $H^0(\lambda)$ is still given by the Weyl character formula.

The failure of irreducibility in (3) is the geometric manifestation of the failure of complete reducibility in characteristic $p$. The induced module $H^0(\lambda)$ and the Weyl module $V(\lambda)$ play dual roles: $V(\lambda)$ has $L(\lambda)$ as its unique irreducible *quotient*, while $H^0(-w_0\lambda)$ has $L(\lambda)^\vee$ as its unique irreducible *submodule*. In characteristic 0 both are irreducible; in characteristic $p$, the gap between head and socle is the subject of chapter 11.

## Embedding $G/B$ into Projective Space

**Corollary 10.2.5: Projective Embedding via Dominant Weights**

Let $\lambda \in X(T)^+$ be *strictly* dominant ($\langle \lambda, \alpha_i^\vee \rangle > 0$ for all simple $\alpha_i$). Then $\mathcal{L}(\lambda)$ is very ample, and the map

$$\iota_\lambda : G/B \hookrightarrow \mathbb{P}(H^0(G/B, \mathcal{L}(\lambda))^\vee)$$

is a closed embedding (in characteristic 0). The $G$-action on $G/B$ extends to a linear action on the ambient projective space via the representation on $H^0(\lambda)^\vee$.

***Proof Sketch*:**
Since $\lambda$ is strictly dominant, $\mathcal{L}(\lambda)$ is ample (proposition 10.1.2(4)). For a homogeneous space, ample implies very ample: the $G$-action provides enough sections to separate all points and tangent directions. The image is closed because $G/B$ is projective.

This recovers the projectivity of $G/B$ (theorem 6.2.3) as a special case. The "smallest" embedding comes from $\lambda = \rho = \omega_1 + \cdots + \omega_r$.

**Example 10.2.6: Embeddings of $\mathrm{SL}_3/B$**
For $G = \mathrm{SL}_3$ with $\lambda = \rho = \omega_1 + \omega_2$: $\dim L(\rho) = 8$, so $\mathcal{L}(\rho)$ embeds $\mathrm{Fl}(1, 2; 3)$ as a smooth 3-dimensional subvariety of $\mathbb{P}^7$.

### Exercise 10.2.1

1. Verify the Borel–Weil theorem for $G = \mathrm{SL}_2$ and all $d \in \mathbb{Z}$: show $H^0(\mathbb{P}^1, \mathcal{O}(d)) = 0$ for $d < 0$, and $H^0(\mathbb{P}^1, \mathcal{O}(d)) \cong L(d)^\vee \cong L(d)$ for $d \geq 0$, by explicit computation with homogeneous polynomials.
2. Using the Borel–Weil theorem, give a geometric proof that $L(\omega_1)^\vee \cong L(\omega_2)$ for $\mathrm{SL}_3$. (*Hint*: use $L(\mu)^\vee \cong L(-w_0\mu)$ and compute $-w_0(\omega_1) = \omega_2$.)
3. For $G = \mathrm{SL}_4$ and $\lambda = \omega_2$, the Borel–Weil theorem gives $H^0(G/P_2, \mathcal{L}(\omega_2)) \cong L(\omega_2)^\vee = (\bigwedge^2 k^4)^\vee$. This embeds $\mathrm{Gr}(2, 4)$ into $\mathbb{P}^5$. Show that the image is a quadric hypersurface of degree 2.
4. Let $G = \mathrm{SL}_2$ over a field of characteristic $p = 2$ and $\lambda = 2$. Compute $H^0(\mathbb{P}^1, \mathcal{O}(2))$ explicitly and show it is reducible (compare exercise 8.2.1 (5)).
5. For $G = \mathrm{Sp}_4$ with $\lambda = \rho = \omega_1 + \omega_2$: compute $\dim L(\rho)$ and describe the Borel–Weil embedding $\mathrm{Sp}_4/B \hookrightarrow \mathbb{P}(L(\rho))$.
6. For $G = \mathrm{SL}_n$, show that the Borel–Weil map for $\lambda = \omega_d$ factors through the Grassmannian: $G/B \to G/P_d \hookrightarrow \mathbb{P}(\bigwedge^d k^n)$.

## 10.3 The Borel–Weil–Bott Theorem

The Borel–Weil theorem computes $H^0(G/B, \mathcal{L}(\lambda))$ for all $\lambda \in X(T)$: it is an irreducible representation for dominant $\lambda$ and zero otherwise. But the flag variety $G/B$ is a smooth projective variety of dimension $|\Phi^+|$, so there are also *higher* cohomology groups $H^i(G/B, \mathcal{L}(\lambda))$ for $1 \leq i \leq |\Phi^+|$. The Borel–Weil–Bott theorem describes all of them: at most one is nonzero, and when it is nonzero, it is an irreducible representation determined by the Weyl group.

### The Dot Action

The statement of the theorem involves a shifted action of the Weyl group:

**Definition 10.3.1: The Dot Action**

The *dot action* (or *shifted action*, or *$\rho$-shifted action*) of $W$ on $X(T) \otimes \mathbb{R}$ is defined by:

$$w \cdot \lambda = w(\lambda + \rho) - \rho$$

where $\rho = \frac{1}{2}\sum_{\alpha \in \Phi^+} \alpha$ is the Weyl vector. The dot action of the simple reflection $s_i = s_{\alpha_i}$ is:

$$s_i \cdot \lambda = s_i(\lambda + \rho) - \rho = \lambda - \langle \lambda + \rho, \alpha_i^\vee \rangle \alpha_i$$

The dot action is the "correct" action for representation theory: the Weyl character formula involves $w(\lambda+\rho)$, which is $w \cdot \lambda + \rho$. The dot action has the same orbits as the ordinary $W$-action, but shifted by $\rho$: the fixed point of the ordinary action is 0, while the fixed point of the dot action is $-\rho$.

A weight $\lambda$ is called *dot-regular* if $\langle \lambda + \rho, \alpha^\vee \rangle \neq 0$ for all $\alpha \in \Phi$, and *dot-singular* otherwise. Equivalently, $\lambda$ is dot-regular if $\lambda + \rho$ is not fixed by any reflection in $W$, i.e. $\lambda + \rho$ lies in the interior of some Weyl chamber.

### Statement of the Theorem

**Theorem 10.3.2: The Borel–Weil–Bott Theorem**

Let $G$ be a connected reductive group over an algebraically closed field $k$ of characteristic 0, and let $\lambda \in X(T)$.

1. *(Singular case)* If $\lambda$ is dot-singular (i.e. $\langle \lambda + \rho, \alpha^\vee \rangle = 0$ for some $\alpha \in \Phi$), then $H^i(G/B, \mathcal{L}(\lambda)) = 0$ for all $i \geq 0$.
2. *(Regular case)* If $\lambda$ is dot-regular, let $w \in W$ be the unique element such that $w \cdot \lambda$ is dominant ($w \cdot \lambda \in X(T)^+$). Then:

$$H^i(G/B, \mathcal{L}(\lambda)) = \begin{cases} L(w \cdot \lambda)^\vee & \text{if } i = \ell(w) \\ 0 & \text{if } i \neq \ell(w) \end{cases}$$

In particular, for each $\lambda$, at most one cohomology group $H^i(G/B, \mathcal{L}(\lambda))$ is nonzero.

The theorem is a vast generalization of the Borel–Weil theorem. When $\lambda$ is dominant: $\lambda+\rho$ lies in the dominant chamber, so $w = e$ (the identity) and $\ell(w) = 0$. The theorem gives $H^0(G/B, \mathcal{L}(\lambda)) = L(\lambda)^\vee$ and $H^i = 0$ for $i > 0$, recovering Borel–Weil.

When $\lambda$ is *not* dominant but is dot-regular: $\lambda+\rho$ lies in some non-dominant chamber, and the unique $w \in W$ that moves it to the dominant chamber determines which cohomological degree carries the representation. The representation $L(w\cdot\lambda)^\vee$ that appears is *the same one* that Borel–Weil would produce for the dominant weight $w\cdot\lambda$, but it appears in degree $\ell(w)$ instead of degree 0.

**Example 10.3.3: Borel–Weil–Bott for $\mathrm{SL}_2$**

For $G = \mathrm{SL}_2$ with $X(T) = \mathbb{Z}$, $\Phi^+ = \{\alpha\}$, $\rho = 1$ (in the convention $\alpha = 2$, $\rho = \alpha/2 = 1$):

1. $\lambda = d \geq 0$ *(dominant):* $\lambda+\rho = d+1 > 0$, so $w = e$, $\ell(w) = 0$. $H^0(\mathbb{P}^1, \mathcal{O}(d)) = L(d)^\vee$, $H^1 = 0$. This is Borel–Weil.
2. $\lambda = -1$ *(dot-singular):* $\lambda+\rho = -1+1 = 0$, which is fixed by $s$. So $H^i(\mathbb{P}^1, \mathcal{O}(-1)) = 0$ for all $i$. (This is a classical fact: $\mathcal{O}_{\mathbb{P}^1}(-1)$ has no global sections and no $H^1$.)
3. $\lambda = -d-2$ *with* $d \geq 0$ *(dot-regular, non-dominant):* $\lambda+\rho = -d-1 < 0$, so $w = s$ (the nontrivial element), $\ell(w) = 1$. The dot action gives $w\cdot\lambda = s(\lambda+\rho)-\rho = (d+1)-1 = d$. Therefore $H^1(\mathbb{P}^1, \mathcal{O}(-d-2)) = L(d)^\vee$ and $H^0 = 0$. Dimension: $\dim H^1(\mathbb{P}^1, \mathcal{O}(-d-2)) = d+1$.

This is exactly Serre duality on $\mathbb{P}^1$: $H^1(\mathbb{P}^1, \mathcal{O}(-d-2)) \cong H^0(\mathbb{P}^1, \mathcal{O}(d))^\vee$ (using the canonical bundle $\omega_{\mathbb{P}^1} = \mathcal{O}(-2)$ and Serre duality $H^1(\mathbb{P}^1, \mathcal{F}) \cong H^0(\mathbb{P}^1, \mathcal{F}^\vee \otimes \omega_{\mathbb{P}^1})^\vee$). The BWB theorem packages Serre duality and the Borel–Weil theorem into a single statement.

**Example 10.3.4: Borel–Weil–Bott for $\mathrm{SL}_3$**

For $G = \mathrm{SL}_3$, the flag variety $G/B$ has dimension 3, so $H^i = 0$ for $i > 3$. The Weyl group $W = S_3$ has 6 elements with lengths $0, 1, 1, 2, 2, 3$.

Take the weight $\lambda = -2\omega_1 + \omega_2$. Then $\lambda+\rho = -2\omega_1+\omega_2+\omega_1+\omega_2 = -\omega_1+2\omega_2$. We check regularity by computing the pairing with the positive coroots: $\langle -\omega_1+2\omega_2, \alpha_1^\vee\rangle = -1 \neq 0$, $\langle -\omega_1+2\omega_2, \alpha_2^\vee\rangle = 2 \neq 0$, and $\langle -\omega_1+2\omega_2, (\alpha_1+\alpha_2)^\vee\rangle = -1+2 = 1 \neq 0$. So $\lambda$ is dot-regular.

The shifted weight $\lambda+\rho = -\omega_1+2\omega_2$ has $\langle \lambda+\rho, \alpha_1^\vee\rangle = -1 < 0$, so it is not dominant. We apply the simple reflection $s_1$, using the formula $s_1(\mu) = \mu - \langle \mu, \alpha_1^\vee\rangle\alpha_1$:

$$s_1(-\omega_1+2\omega_2) = (-\omega_1+2\omega_2) - (-1)\alpha_1 = -\omega_1+2\omega_2+\alpha_1$$

Since $\alpha_1 = 2\omega_1 - \omega_2$, this simplifies to $(-\omega_1+2\omega_2)+(2\omega_1-\omega_2) = \omega_1+\omega_2 = \rho$. This is dominant.

So $w = s_1$, $\ell(w) = 1$, and $w\cdot\lambda = s_1\cdot\lambda = s_1(\lambda+\rho)-\rho = \rho-\rho = 0$. Therefore:

$$H^1(G/B, \mathcal{L}(-2\omega_1+\omega_2)) = L(0)^\vee = k \qquad \text{(trivial representation)}$$

and $H^i = 0$ for $i \neq 1$.

## Proof Strategy

The proof of the Borel–Weil–Bott theorem is more involved than the Borel–Weil theorem. We outline the main ideas.

> ***Proof Strategy*:**
> *Step 1: The singular vanishing.* If $\lambda$ is dot-singular, then $\langle \lambda + \rho, \alpha_i^\vee \rangle = 0$ for some simple root $\alpha_i$. This means $s_i \cdot \lambda = \lambda$ (the dot action of $s_i$ fixes $\lambda$). One shows that $\mathcal{L}(\lambda)$ is "cohomologically trivial" along the $\mathbb{P}^1$-fibers of $\pi_i : G/B \to G/P_i$: the restriction $\mathcal{L}(\lambda)|_{\mathbb{P}^1}$ is $\mathcal{O}_{\mathbb{P}^1}(-1)$, which has $H^0 = H^1 = 0$. By the Leray spectral sequence for $\pi_i$, all cohomology of $\mathcal{L}(\lambda)$ on $G/B$ vanishes.
>
> *Step 2: The regular case via simple reflections.* The key step is the *reflection principle*: for a simple reflection $s_i$ and a weight $\lambda$ with $\langle \lambda + \rho, \alpha_i^\vee \rangle > 0$ (so $\lambda$ is "dominant in the $\alpha_i$ direction"), there is an isomorphism:
>
> $$H^j(G/B, \mathcal{L}(\lambda)) \cong H^{j+1}(G/B, \mathcal{L}(s_i \cdot \lambda))$$
>
> for all $j \geq 0$. This shifts the cohomological degree by 1 each time we apply a simple reflection that moves $\lambda + \rho$ "across a wall" into a less dominant chamber.
>
> The isomorphism is proved using the $\mathbb{P}^1$-fibration $\pi_i : G/B \to G/P_i$. The Leray spectral sequence for $\pi_i$ gives $H^p(G/P_i, R^q\pi_{i*}\mathcal{L}(\lambda)) \Rightarrow H^{p+q}(G/B, \mathcal{L}(\lambda))$. When $\langle \lambda + \rho, \alpha_i^\vee \rangle > 0$: the restriction of $\mathcal{L}(\lambda)$ to each fiber $\mathbb{P}^1$ has degree $\langle \lambda, \alpha_i^\vee \rangle \geq 0$, so $R^0\pi_{i*}\mathcal{L}(\lambda) \neq 0$ and $R^1\pi_{i*}\mathcal{L}(\lambda) = 0$. The pushforward $R^0\pi_{i*}\mathcal{L}(\lambda)$ is a vector bundle on $G/P_i$ whose cohomology gives $H^p(G/B, \mathcal{L}(\lambda)) = H^p(G/P_i, R^0\pi_{i*}\mathcal{L}(\lambda))$.
>
> When we replace $\lambda$ by $s_i \cdot \lambda$: the restriction to the fiber has degree $\langle s_i \cdot \lambda, \alpha_i^\vee \rangle = \langle \lambda, \alpha_i^\vee \rangle - 2\langle \lambda + \rho, \alpha_i^\vee \rangle = -\langle \lambda + \rho, \alpha_i^\vee \rangle - 1$. Since $\langle \lambda + \rho, \alpha_i^\vee \rangle \geq 1$, this degree is $\leq -2$, so $R^0\pi_{i*}\mathcal{L}(s_i \cdot \lambda) = 0$ but $R^1\pi_{i*}\mathcal{L}(s_i \cdot \lambda) \neq 0$ (by Serre duality on $\mathbb{P}^1$). A comparison of the two spectral sequences gives the desired isomorphism $H^j(G/B, \mathcal{L}(\lambda)) \cong H^{j+1}(G/B, \mathcal{L}(s_i \cdot \lambda))$.
>
> *Step 3: Induction on $\ell(w)$.* Given a dot-regular $\lambda$ with $w \cdot \lambda$ dominant (so $w(\lambda + \rho)$ is in the dominant chamber), write $w = s_{i_1} s_{i_2} \cdots s_{i_m}$ as a reduced expression ($m = \ell(w)$). Applying the reflection principle $m$ times:
>
> $$\begin{aligned} H^0(G/B, \mathcal{L}(w \cdot \lambda)) &\cong H^1(G/B, \mathcal{L}(s_{i_1} \cdot (w \cdot \lambda))) \\ &\cong H^2(G/B, \mathcal{L}(s_{i_2} s_{i_1} \cdot (w \cdot \lambda))) \\ &\cong \cdots \\ &\cong H^m(G/B, \mathcal{L}(s_{i_m} \cdots s_{i_1} \cdot (w \cdot \lambda))) \end{aligned}$$
>
> Now $s_{i_m} \cdots s_{i_1} \cdot (w \cdot \lambda) = w^{-1} \cdot (w \cdot \lambda) = \lambda$ (since $w^{-1}w = e$). So $H^m(G/B, \mathcal{L}(\lambda)) \cong H^0(G/B, \mathcal{L}(w \cdot \lambda)) = L(w \cdot \lambda)^\vee$ (by Borel–Weil, since $w \cdot \lambda$ is dominant). For $j \neq m$: a careful tracking of the spectral sequence shows $H^j(G/B, \mathcal{L}(\lambda)) = 0$.

## The Euler Characteristic and the Weyl Character Formula

The Borel–Weil–Bott theorem implies that the *Euler characteristic* of $\mathcal{L}(\lambda)$ on $G/B$ has a simple expression:

**Corollary 10.3.5: Euler Characteristic of Line Bundles on $G/B$**

For any $\lambda \in X(T)$:

$$\chi(G/B, \mathcal{L}(\lambda)) := \sum_{i \geq 0} (-1)^i \operatorname{ch}(H^i(G/B, \mathcal{L}(\lambda))) = \begin{cases} (-1)^{\ell(w)} \operatorname{ch}(L(w \cdot \lambda)^\vee) & \text{if } \lambda \text{ is dot-regular} \\ 0 & \text{if } \lambda \text{ is dot-singular} \end{cases}$$

In the dot-regular case, this equals:

$$\chi(G/B, \mathcal{L}(\lambda)) = (-1)^{\ell(w)} \operatorname{ch}(L(w \cdot \lambda)^\vee)$$

since only the $H^{\ell(w)}$ term survives.

On the other hand, the Euler characteristic can be computed *independently* by the Hirzebruch–Riemann–Roch theorem (or its equivariant version), which expresses $\chi(G/B, \mathcal{L}(\lambda))$ as a ratio involving the Weyl group:

$$\chi(G/B, \mathcal{L}(\lambda)) = \frac{\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\rho)}}{e^\rho \prod_{\alpha > 0} (1 - e^{-\alpha})}$$

(this formula holds for *all* $\lambda$, not just dominant ones). Comparing the two expressions gives a powerful geometric proof of the character formula. Passing to the dual representation (which corresponds geometrically to taking the line bundle convention that yields $L(\lambda)$ directly), we recover the standard Weyl character formula for $L(\lambda)$:

$$\operatorname{ch}(L(\lambda)) = \frac{\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\rho)}}{e^\rho \prod_{\alpha > 0} (1 - e^{-\alpha})}$$

This is exactly theorem 9.2.1, now understood as a consequence of the geometry of line bundles on $G/B$. The alternating sum over $W$ in the numerator reflects the alternation of cohomological degrees in the BWB theorem, and the denominator $D = e^\rho \prod (1 - e^{-\alpha})$ is the "Todd class" of the tangent bundle of $G/B$ (in the equivariant Riemann–Roch formula).

This closes the circle between the algebraic Weyl character formula (chapter 8, proved via the BGG resolution) and the geometric Borel–Weil–Bott theorem: they are two facets of the same underlying structure.

**Example 10.3.6: Euler Characteristics on $\mathbb{P}^1$**

For $G = \mathrm{SL}_2$ and $\lambda = d$: $\chi(\mathbb{P}^1, \mathcal{O}(d)) = \dim H^0 - \dim H^1$. By BWB:

1. $d \geq 0$: $\chi = \dim L(d) - 0 = d + 1$.
2. $d = -1$: $\chi = 0$ (dot-singular).
3. $d \leq -2$: $\chi = 0 - \dim L(-d-2) = -((-d-2)+1) = d + 1$.

In all cases, $\chi(\mathbb{P}^1, \mathcal{O}(d)) = d + 1$. This is the Riemann–Roch formula for $\mathbb{P}^1$: $\chi(\mathcal{O}(d)) = \deg(\mathcal{O}(d)) + 1 - g = d + 1$ (where $g = 0$ is the genus).

**Example 10.3.7: Serre Duality via Borel–Weil–Bott**

For $G/B$ of dimension $N = |\Phi^+|$, the canonical bundle is $\omega_{G/B} = \mathcal{L}(-2\rho)$ (since the tangent bundle $T_{G/B}$ has weights $\Phi^-$ at the identity, meaning the cotangent space has weights $\Phi^+$, giving a determinant of $\sum_{\alpha>0} \alpha = 2\rho$, which corresponds to the bundle $\mathcal{L}(-2\rho)$). Serre duality gives:

$$H^i(G/B, \mathcal{L}(\lambda)) \cong H^{N-i}(G/B, \mathcal{L}(-\lambda - 2\rho))^\vee$$

The BWB theorem is compatible: if $\lambda$ is dot-regular with $w \cdot \lambda$ dominant and $H^i = L(w \cdot \lambda)^\vee$ at $i = \ell(w)$, then $-\lambda - 2\rho$ is dot-regular with $(w_0 w^{-1}) \cdot (-\lambda - 2\rho)$ dominant at degree $N - \ell(w) = \ell(w_0) - \ell(w) = \ell(w_0 w^{-1})$, and the resulting representation is $L(w \cdot \lambda)$, consistent with the dual.

## Remarks on the Positive-Characteristic Case

The Borel–Weil–Bott theorem fails in characteristic $p > 0$. While the $H^0$ behavior for dominant weights remains well-behaved, the higher cohomology of line bundles on $G/B$ loses the clean shifted pattern seen in characteristic 0. For non-dominant weights, multiple cohomology groups can be nonzero simultaneously (violating the "at most one nonzero" property), and they can appear in degrees other than the length of the Weyl group element.

The Kempf vanishing theorem provides a crucial partial substitute: it asserts that $H^i(G/B, \mathcal{L}(\lambda)) = 0$ for $i > 0$ when $\lambda$ is *dominant*, in arbitrary characteristic $p$. This salvages the Borel–Weil theorem (the $H^0$ statement) and ensures that dominant line bundles do not have higher cohomology. However, computing the higher cohomology for non-dominant line bundles on $G/B$ in characteristic $p$ is a deep and largely open problem, closely related to the problem of determining the characters of simple modules $L(\lambda)$ (see chapter 11).

### Exercise 10.3.1

1. For $G = \mathrm{SL}_2$, compute $H^i(\mathbb{P}^1, \mathcal{O}(d))$ for all $d \in \mathbb{Z}$ and $i = 0, 1$ using the BWB theorem. Organize the results in a table. Verify that $\chi(\mathbb{P}^1, \mathcal{O}(d)) = d + 1$ in all cases.

2. For $G = \mathrm{SL}_3$ with simple roots $\alpha_1, \alpha_2$ and $\rho = \omega_1 + \omega_2$, compute the dot action $s_1 \cdot \lambda$ and $s_2 \cdot \lambda$ for $\lambda = -\rho$ (the "most singular" weight). Verify that $s_1 \cdot (-\rho) = -\rho$ and $s_2 \cdot (-\rho) = -\rho$, confirming that $-\rho$ is the unique dot-fixed point.

3. For $G = \mathrm{SL}_3$ and $\lambda = -\omega_1 - 2\omega_2$, determine whether $\lambda$ is dot-regular or dot-singular. If regular, find the unique $w \in W$ with $w \cdot \lambda$ dominant, and compute $H^i(G/B, \mathcal{L}(\lambda))$ for all $i$.

4. Verify that the canonical bundle of $G/B$ is $\omega_{G/B} = \mathcal{L}(-2\rho)$ for $G = \mathrm{SL}_3$. (*Hint*: the tangent space at $eB$ is $\mathfrak{g}/\mathfrak{b} \cong \bigoplus_{\alpha \in \Phi^-} \mathfrak{g}_\alpha$, so the determinant of the cotangent space has weight $\sum_{\alpha \in \Phi^-} \alpha = -\sum_{\alpha \in \Phi^+} \alpha = -2\rho$.)

5. Verify Serre duality $H^0(G/B, \mathcal{L}(\lambda))^\vee \cong H^N(G/B, \mathcal{L}(-\lambda - 2\rho))$ for $G = \mathrm{SL}_2$, $\lambda = d \geq 0$, and $N = 1$.

6. The *Kempf vanishing theorem* states that $H^i(G/B, \mathcal{L}(\lambda)) = 0$ for all $i > 0$ when $\lambda$ is dominant, in *any* characteristic. Explain why this is weaker than the BWB theorem (which additionally identifies $H^0$ and describes all cohomology for non-dominant $\lambda$), and why it is still a nontrivial result (the standard Kodaira vanishing proof does not work in characteristic $p$).

# 11

# *Representations in Positive Characteristic*

The representation theory developed in Chapters 8–10 is cleanest in characteristic 0: every representation is completely reducible, the irreducibles $L(\lambda)$ are classified by dominant weights, and the Weyl character formula computes everything. In characteristic $p > 0$, this clean picture breaks down. Complete reducibility fails (theorem 8.3.1 requires $\operatorname{char} k = 0$), the Weyl modules $V(\lambda)$ are no longer irreducible, and the characters of the simple modules $L(\lambda)$ are not given by the Weyl formula. Determining these characters is the central open problem of modular representation theory.

This chapter develops the tools and results that partially tame the characteristic-$p$ landscape. We begin by studying the relationship between Weyl modules $V(\lambda)$ and simple modules $L(\lambda)$: the former have known characters (the Weyl formula, independent of characteristic), while the latter are mysterious quotients whose characters encode deep arithmetic information. We then introduce the *Frobenius twist*, which provides a powerful structural tool: Steinberg's tensor product theorem reduces the character problem for an arbitrary dominant weight to the "restricted" case (weights bounded by $p$). Finally, we state Lusztig's conjecture, which gives a conjectural answer to the character problem in terms of Kazhdan–Lusztig polynomials, and discuss its current status.

Throughout this chapter, $k$ is an algebraically closed field of characteristic $p > 0$, $G$ is a connected reductive group over $k$ (assumed split, for simplicity), and $T \subseteq B \subseteq G$ is a maximal torus inside a Borel subgroup.

## 11.1 Weyl Modules and Simple Modules

Let us begin by surveying what survives from the characteristic-0 theory and what fails.

| | **Characteristic 0** | **Characteristic $p$** |
|---|---|---|
| Simple modules classified by $X(T)^+$ | ✓ | ✓ |
| $V(\lambda)$ has highest weight $\lambda$ | ✓ | ✓ |
| $\mathrm{ch}(V(\lambda))$ given by Weyl formula | ✓ | ✓ |
| $V(\lambda)$ is irreducible ($V(\lambda) = L(\lambda)$) | ✓ | **fails** |
| Complete reducibility | ✓ | **fails** |
| $\mathrm{ch}(L(\lambda))$ given by Weyl formula | ✓ | **fails** |
| $H^i(G/B, \mathcal{L}(\lambda)) = 0$ for $i > 0$, $\lambda$ dominant | ✓ | ✓(Kempf) |
| Full BWB theorem | ✓ | **fails** |

The classification of simple modules survives: $\{L(\lambda) : \lambda \in X(T)^+\}$ is a complete set of pairwise non-isomorphic finite-dimensional simple $G$-modules, in any characteristic (theorem 8.2.3). The Weyl modules $V(\lambda)$ exist and have the "expected" character (proposition 8.2.8(3)). What fails is the *irreducibility* of $V(\lambda)$: in characteristic $p$, the Weyl module can have a nontrivial radical, and the simple module $L(\lambda) = V(\lambda)/\mathrm{rad}V(\lambda)$ is a proper quotient.

Since $V(\lambda)$ has a known character (the Weyl character formula, valid in all characteristics) and $L(\lambda)$ is its unique simple quotient, the character of $L(\lambda)$ is determined by the *composition multiplicities*:

$$[V(\lambda) : L(\mu)] := \text{number of times } L(\mu) \text{ appears as a composition factor of } V(\lambda)$$

These are non-negative integers, and the character relation is:

$$\mathrm{ch}(V(\lambda)) = \sum_{\mu \in X(T)^+} [V(\lambda) : L(\mu)] \cdot \mathrm{ch}(L(\mu))$$

The matrix $\big([V(\lambda) : L(\mu)]\big)_{\lambda,\mu}$ is called the *decomposition matrix*. It is upper-unitriangular: $[V(\lambda) : L(\lambda)] = 1$ (the head), $[V(\lambda) : L(\mu)] = 0$ unless $\mu \preceq \lambda$ (composition factors have lower weights), and $[V(\lambda) : L(\mu)] \neq 0$ only for finitely many $\mu$. Inverting this matrix gives the characters of the simple modules:

$$\mathrm{ch}(L(\lambda)) = \sum_{\mu \in X(T)^+} d_{\lambda\mu} \cdot \mathrm{ch}(V(\mu))$$

where $(d_{\lambda\mu})$ is the inverse of the decomposition matrix (an alternating sum of composition multiplicities). Computing either matrix is the fundamental problem.

**Example 11.1.1: $\mathrm{SL}_2$ in Characteristic $p$**

For $G = \mathrm{SL}_2$ over a field of characteristic $p$, the Weyl module is $V(d) = \mathrm{Sym}^d(k^2)$, of dimension $d + 1$. The behavior depends on the relationship between $d$ and $p$:

1. $0 \leq d \leq p - 1$ *($p$-restricted weights):* $V(d) = L(d)$ is irreducible. The Weyl formula gives the correct character of $L(d)$.
2. $d = p$*:* The Weyl module $V(p)$ is $(p+1)$-dimensional and *reducible*. The general theory guarantees that $V(p)$ has the simple module $L(p)$ as its unique irreducible quotient (its head). In characteristic $p$, this simple module $L(p)$ is 2-dimensional (in fact, it is isomorphic to the Frobenius twist of the standard representation, $L(1)^{(1)}$). The kernel of the projection $V(p) \twoheadrightarrow L(p)$ is the radical $\mathrm{rad}V(p)$, which is irreducible and isomorphic to $L(p-2)$ (a space

of dimension $p-1$). Thus, the composition factors of $V(p)$ are exactly $L(p)$ and $L(p-2)$, confirming the dimension sum $2+(p-1)=p+1$.

3. $d = p-1$: $V(p-1) = L(p-1)$ is irreducible, of dimension $p$. This is the *Steinberg module*, the largest irreducible $p$-restricted module. It plays a special role in the theory.

The $\mathrm{SL}_2$ example illustrates the general pattern: for "small" weights (below a threshold determined by $p$), the Weyl modules are irreducible and the characteristic-0 theory applies. For "large" weights, the modules develop nontrivial radicals, and the composition factors involve *smaller* simple modules in intricate ways.

A fundamental constraint on the decomposition matrix is the *linkage principle*: the composition factors of $V(\lambda)$ can only be $L(\mu)$ for $\mu$ in the same "block" as $\lambda$.

**Theorem 11.1.2: The Linkage Principle**

Let $\lambda, \mu \in X(T)^+$ be dominant weights. If $[V(\lambda) : L(\mu)] \neq 0$, then $\lambda$ and $\mu$ are in the same *dot orbit* of the *affine Weyl group* $W_p$: there exists $w \in W_p$ with $\mu = w \cdot \lambda$ (the dot action of $W_p$).

The *affine Weyl group* $W_p$ is the group generated by the ordinary Weyl group $W$ together with the translations $t_{p\alpha} : \lambda \mapsto \lambda + p\alpha$ for $\alpha \in \Phi$ (the root lattice scaled by $p$). Equivalently, $W_p = W \ltimes p\mathbb{Z}\Phi$ acts on $X(T) \otimes \mathbb{R}$ by affine transformations, and its dot action is $w \cdot \lambda = w(\lambda + \rho) - \rho$ (the same formula as for $W$, but now with the affine reflections included).

The affine Weyl group tiles $X(T) \otimes \mathbb{R}$ into *alcoves*: the connected components of the complement of the affine hyperplanes $\{x : \langle x + \rho, \alpha^\vee \rangle = np\}$ for $\alpha \in \Phi^+$ and $n \in \mathbb{Z}$. The *fundamental alcove* is:

$$C_0 = \left\{ \lambda \in X(T) \otimes \mathbb{R} \;\middle|\; 0 < \langle \lambda + \rho, \alpha^\vee \rangle < p \text{ for all } \alpha \in \Phi^+ \right\}$$

The linkage principle says that the decomposition matrix is *block-diagonal* with respect to the $W_p$ orbits: composition factors of $V(\lambda)$ lie in the same $W_p$-orbit as $\lambda$. This dramatically reduces the number of potentially nonzero entries.

## The Steinberg Module

The linkage principle has a particularly clean consequence for a specific high-weight representation.

**Definition 11.1.3: The Steinberg Module**

The *Steinberg weight* is $\mathrm{St} = (p-1)\rho = \sum_{i=1}^{r} (p-1)\omega_i$. The *Steinberg module* is $L(\mathrm{St}) = V(\mathrm{St})$. It plays a central role in the theory because its weight is maximal in the set of $p$-restricted weights.

**Theorem 11.1.4: Properties of the Steinberg Module**

1. $V(\mathrm{St})$ is irreducible: $V(\mathrm{St}) = L(\mathrm{St})$.
2. $\dim L(\mathrm{St}) = p^{|\Phi^+|}$ (a power of $p$).
3. $L(\mathrm{St})$ is both projective and injective in the category of finite-dimensional $G$-modules (it is a *projective-injective* module).
4. The character of $L(\mathrm{St})$ is given by the Weyl formula (since $V = L$): $\mathrm{ch}(L(\mathrm{St})) = \mathrm{ch}(V((p-1)\rho))$.

***Proof Sketch*:**
*(1):* The weight $\mathrm{St} = (p-1)\rho$ has a remarkable structural property: it is the *only* dominant weight in its $W_p$-dot orbit. By the linkage principle, any composition factor $L(\mu)$ of $V(\mathrm{St})$ must have $\mu$ in this same orbit. Since a composition factor must have a dominant highest weight, we are forced to conclude $\mu = \mathrm{St}$. Thus, the only possible composition factor of $V(\mathrm{St})$ is $L(\mathrm{St})$ itself. Since it appears with multiplicity 1 at the head, the radical must be zero, meaning $V(\mathrm{St}) = L(\mathrm{St})$.

*(2):* The Weyl dimension formula gives $\dim V((p-1)\rho) = \prod_{\alpha>0} \frac{\langle (p-1)\rho+\rho, \alpha^\vee\rangle}{\langle \rho, \alpha^\vee\rangle} = \prod_{\alpha>0} \frac{p\langle \rho, \alpha^\vee\rangle}{\langle \rho, \alpha^\vee\rangle} = p^{|\Phi^+|}$.

*(3):* This is a deeper result. The projectivity of $L(\mathrm{St})$ follows from the fact that $L(\mathrm{St})$ is the only simple module in its block (by the linkage principle), so $\mathrm{Ext}^1_G(L(\mathrm{St}), L(\mu)) = 0$ for all $\mu$. This isolation is a special feature of the Steinberg weight.

**Example 11.1.5: Steinberg Modules for Classical Groups**

1. $\mathrm{SL}_2$*:* $\mathrm{St} = (p-1)\omega_1$, $\dim L(\mathrm{St}) = p^1 = p$. This is $L(p-1) = \mathrm{Sym}^{p-1}(k^2)$.
2. $\mathrm{SL}_3$*:* $\mathrm{St} = (p-1)(\omega_1+\omega_2)$, $\dim L(\mathrm{St}) = p^3$. For $p = 2$: $\dim = 8$; for $p = 3$: $\dim = 27$.
3. $\mathrm{SL}_n$*:* $\mathrm{St} = (p-1)(\omega_1 + \cdots + \omega_{n-1})$, $\dim L(\mathrm{St}) = p^{\binom{n}{2}}$.
4. $\mathrm{Sp}_{2n}$*:* $\dim L(\mathrm{St}) = p^{n^2}$.

The Steinberg module is the "largest" irreducible restricted module, and it is the only one whose dimension is a pure power of $p$.

## Small Representations and the Restricted Region

The "simplest" representations in characteristic $p$ are those whose highest weights lie in the *restricted region*: the set of $p$-restricted dominant weights.

**Definition 11.1.6: $p$-Restricted Weights**

A dominant weight $\lambda \in X(T)^+$ is called *p-restricted* if

$$0 \leq \langle \lambda, \alpha_i^\vee \rangle < p \qquad \text{for all simple roots } \alpha_i \in \Delta$$

The set of $p$-restricted dominant weights is denoted $X_1(T)$. In terms of fundamental weights: $X_1(T) = \{\sum n_i \omega_i \mid 0 \leq n_i < p\}$.

The set $X_1(T)$ is finite: it contains exactly $p^r$ elements (where $r = \text{rank} G$), since each coordinate $n_i$ ranges over $\{0, 1, \ldots, p-1\}$. The corresponding simple modules $\{L(\lambda) : \lambda \in X_1(T)\}$ are the *restricted simple modules*. By Steinberg's tensor product theorem (next section), every simple module is built from restricted simple modules via the Frobenius twist, so the character problem reduces to the restricted case.

**Example 11.1.7: Restricted Modules for $\mathrm{SL}_2$**

For $G = \mathrm{SL}_2$: $X_1(T) = \{0, 1, \ldots, p-1\}$ ($p$ weights). The restricted simple modules are $L(0), L(1), \ldots, L(p-1)$, with $L(d) = \mathrm{Sym}^d(k^2)$ (all irreducible, since $d < p$). Their dimensions are $1, 2, \ldots, p$.

**Example 11.1.8: Restricted Modules for $\mathrm{SL}_3$ in Characteristic 3**

For $G = \mathrm{SL}_3$ and $p = 3$: $X_1(T) = \{a\omega_1 + b\omega_2 : 0 \leq a, b \leq 2\}$, a set of 9 weights. The restricted simple modules are:

| $\lambda$ | $\dim V(\lambda)$ | $\dim L(\lambda)$ |
|---|---|---|
| $0$ | 1 | 1 |
| $\omega_1$ | 3 | 3 |
| $\omega_2$ | 3 | 3 |
| $2\omega_1$ | 6 | 6 |
| $\omega_1 + \omega_2$ | 8 | **7** |
| $2\omega_2$ | 6 | 6 |
| $2\omega_1 + \omega_2$ | 15 | 15 |
| $\omega_1 + 2\omega_2$ | 15 | 15 |
| $2\omega_1 + 2\omega_2$ | 27 | 27 (Steinberg) |

Most restricted Weyl modules are irreducible: $V(\lambda) = L(\lambda)$ for 8 out of 9 weights. The only exception is $\lambda = \omega_1 + \omega_2$ (the adjoint representation), where $V(\lambda)$ has dimension 8 but $L(\lambda)$ has dimension 7. (In characteristic 3, the $3 \times 3$ identity matrix has trace 0, so it spans a 1-dimensional trivial submodule $L(0)$ inside $\mathfrak{sl}_3$). The Steinberg module $L(2\omega_1 + 2\omega_2)$ has dimension $3^3 = 27 = p^{|\Phi^+|}$.

This example illustrates the general phenomenon: for small primes, even the restricted region has nontrivial decomposition behavior, and the simple modules $L(\lambda)$ can be strictly smaller than the Weyl modules. As $p$ grows, the restricted region becomes "more like characteristic 0": for $p$ much larger than the Coxeter number, most restricted Weyl modules are irreducible.

**Exercise 11.1.1**

1. For $G = \mathrm{SL}_2$ in characteristic $p$, show that $V(d) = \mathrm{Sym}^d(k^2)$ is irreducible if and only if $d < p$. (*Hint*: the action of $e$ on $x^{d-j}y^j$ is $e \cdot x^{d-j}y^j = j \cdot x^{d-j+1}y^{j-1}$; find the values of $j$ where this is zero in characteristic $p$.)

2. Verify the Steinberg module dimensions: $\dim L((p-1)\rho) = p^{|\Phi^+|}$ for $G = \mathrm{SL}_2$ ($p^1 = p$), $G = \mathrm{SL}_3$ ($p^3$), and $G = \mathrm{Sp}_4$ ($p^4$), using the Weyl dimension formula.

3. For $G = \mathrm{SL}_2$ in characteristic $p = 5$, list all dominant weights in the $W_p$-orbit of $\lambda = 3$ (under the dot action). These are the only possible composition factors of $V(3)$. Verify that $V(3) = L(3)$ (irreducible, since $3 < 5$).

4. For $G = \mathrm{SL}_2$ in characteristic $p = 2$, compute the decomposition matrix for the first few Weyl modules: $V(0)$, $V(1)$, $V(2)$, $V(3)$. That is, determine $[V(d) : L(m)]$ for $0 \le d \le 3$ and all $m$. (*Hint*: $V(0)$ and $V(1)$ are irreducible. $V(2)$ has composition factors $L(2)$ and $L(0)$ by exercise 8.2.1 (5). For $V(3)$: compute the $\mathrm{SL}_2$-action on $\mathrm{Sym}^3(k^2)$ in characteristic 2.)

5. How many $p$-restricted dominant weights are there for $G = \mathrm{SL}_4$ (type $A_3$, rank 3)? For $G = \mathrm{Sp}_6$ (type $C_3$, rank 3)? For $G = G_2$ (rank 2)?

6. The Steinberg module $L(\mathrm{St})$ is projective in the category of $G$-modules. This means $\mathrm{Ext}^1_G(L(\mathrm{St}), V) = 0$ for all $V$. Explain why this implies that any short exact sequence $0 \to V \to E \to L(\mathrm{St}) \to 0$ splits (i.e. $L(\mathrm{St})$ is a direct summand whenever it appears as a quotient).

## 11.2 Steinberg's Tensor Product Theorem

The classification theorem (theorem 8.2.3) tells us that the simple modules $\{L(\lambda) : \lambda \in X(T)^+\}$ are parametrized by all dominant weights — an infinite set. Steinberg's tensor product theorem reduces the character problem to a *finite* one: every simple module is a tensor product of Frobenius twists of *restricted* simple modules. Since there are only $p^r$ restricted dominant weights ($r = \mathrm{rank} G$), the infinite family of simple modules is built from a finite "alphabet" via a simple combinatorial rule (the $p$-adic expansion of the highest weight).

The key tool is the *Frobenius endomorphism* of $G$, the algebraic-group analogue of the map $x \mapsto x^p$ in characteristic $p$.

**Definition 11.2.1: The Frobenius Morphism**

Let $G$ be an affine algebraic group over $k$ with $\mathrm{char} k = p > 0$. The *(geometric) Frobenius morphism* $F : G \to G^{(1)}$ is the morphism of group schemes induced by the $p$-th power map on the coordinate ring: $F^* : \mathcal{O}(G^{(1)}) \to \mathcal{O}(G)$ sends $f \mapsto f^p$. Here $G^{(1)}$ is the *Frobenius twist* of $G$: the group scheme obtained by base change along the Frobenius $\varphi : k \to k$, $\varphi(a) = a^p$.

When $G$ is defined over $\mathbb{F}_p$ (which is the case for all split reductive groups, via the Chevalley construction), the Frobenius twist $G^{(1)}$ is canonically isomorphic to $G$ itself, and $F : G \to G$ is an endomorphism. In this case, $F$ acts on the maximal torus $T$ by $F(t) = t^p$ (raising each coordinate to the $p$-th power), and on the character lattice by $F^* : X(T) \to X(T)$, $\lambda \mapsto p\lambda$ (multiplication by $p$).

The *$r$-th Frobenius morphism* $F^r : G \to G^{(r)}$ is the $r$-fold iterate: $F^r = F \circ F \circ \cdots \circ F$ ($r$ times), acting on coordinates by $f \mapsto f^{p^r}$ and on characters by $\lambda \mapsto p^r\lambda$.

**Definition 11.2.2: Frobenius Twist of a Representation**

Let $V$ be a representation of $G$. The *$r$-th Frobenius twist* of $V$ is the representation $V^{(r)}$ defined by precomposing with the $r$-th Frobenius:

$$\rho^{(r)} : G \xrightarrow{F^r} G^{(r)} \xrightarrow{\rho} \mathrm{GL}(V)$$

(When $G$ is defined over $\mathbb{F}_p$, this is simply $\rho^{(r)}(g) = \rho(F^r(g))$, where $F^r$ acts by raising the matrix entries of $g$ to the $p^r$-th power in any standard realization). As a vector space, $V^{(r)} = V$; only the $G$-action changes.

The Frobenius twist has a transparent effect on weights: if $V$ has weights $\mathrm{wt}(V) = \{\lambda_1, \ldots, \lambda_m\}$, then $V^{(r)}$ has weights $\{p^r\lambda_1, \ldots, p^r\lambda_m\}$ (each weight is multiplied by $p^r$). In particular:

**Proposition 11.2.3: Frobenius Twist of a Simple Module**

For any dominant weight $\lambda \in X(T)^+$ and $r \geq 1$:

$$L(\lambda)^{(r)} \cong L(p^r\lambda)$$

That is, twisting the simple module $L(\lambda)$ by $F^r$ produces the simple module with highest weight $p^r\lambda$.

***Proof*:**
The module $L(\lambda)^{(r)}$ is irreducible. (While $F^r$ is not an isomorphism of algebraic groups, it is a surjective homomorphism on $k$-points; therefore, any subspace invariant under the image $F^r(G)$ is invariant under all of $G$). Furthermore, the highest weight vector of $L(\lambda)$ is still annihilated by the strictly upper triangular matrices $U$ because $F^r(U) \subseteq U$. Since the original weight $\lambda$ is multiplied by $p^r$, the highest weight of the twisted module is $p^r\lambda$. By the classification theorem (theorem 8.2.3), $L(\lambda)^{(r)} \cong L(p^r\lambda)$.

Every dominant weight $\lambda \in X(T)^+$ has a unique *$p$-adic expansion*: writing $\lambda = \sum n_i\omega_i$ in terms of fundamental weights, each coefficient $n_i$ has a base-$p$ expansion $n_i = a_{i,0} + a_{i,1}p + a_{i,2}p^2 + \cdots$ with $0 \leq a_{i,j} < p$. Collecting the digits gives:

**Definition 11.2.4: $p$-Adic Expansion of a Weight**

Let $\lambda \in X(T)^+$ be a dominant weight. The *p-adic expansion* of $\lambda$ is the unique expression

$$\lambda = \lambda_0 + p\lambda_1 + p^2\lambda_2 + \cdots + p^s\lambda_s$$

where each $\lambda_j \in X_1(T)$ is a $p$-restricted dominant weight ($0 \leq \langle \lambda_j, \alpha_i^\vee \rangle < p$ for all $i$) and $\lambda_s \neq 0$. The expansion is finite (since $\lambda$ has finitely many nonzero fundamental-weight coordinates, each with a finite $p$-adic expansion).

With this $p$-adic expansion defined, we can state the theorem, which provides an explicit formula decomposing any simple module into a tensor product of restricted ones.

**Theorem 11.2.5: Steinberg's Tensor Product Theorem**

Let $\lambda \in X(T)^+$ with $p$-adic expansion $\lambda = \lambda_0 + p\lambda_1 + \cdots + p^s\lambda_s$ (each $\lambda_j \in X_1(T)$). Then:

$$L(\lambda) \cong L(\lambda_0) \otimes L(\lambda_1)^{(1)} \otimes L(\lambda_2)^{(2)} \otimes \cdots \otimes L(\lambda_s)^{(s)}$$

That is, the simple module with highest weight $\lambda$ is the tensor product of the restricted simple modules $L(\lambda_j)$, each twisted by the appropriate power of the Frobenius.

***Proof Key ideas*:**

*Step 1: The tensor product has the correct highest weight.* The module $L(\lambda_0) \otimes L(\lambda_1)^{(1)} \otimes \cdots \otimes L(\lambda_s)^{(s)}$ has highest weight $\lambda_0 + p\lambda_1 + \cdots + p^s\lambda_s = \lambda$ (by proposition 11.2.3, $L(\lambda_j)^{(j)}$ has highest weight $p^j\lambda_j$, and highest weights are additive on tensor products). The highest weight space is one-dimensional (the tensor product of one-dimensional highest weight spaces).

*Step 2: The tensor product is irreducible.* This is the nontrivial part. The proof uses the *Frobenius kernels* $G_r = \ker(F^r : G \to G^{(r)})$ (the infinitesimal group schemes of "$p^r$-th order"). The key properties are:

1. The restricted simple modules $\{L(\lambda) : \lambda \in X_1(T)\}$ are exactly the simple modules for the first Frobenius kernel $G_1$. (The group $G_1$ is an infinitesimal scheme annihilated by the Frobenius, so its representation theory is controlled entirely by restricted weights).
2. If $V = L(\lambda_0) \otimes M^{(1)}$ where $M = L(\lambda_1) \otimes \cdots^{(1)}$, its restriction to $G_1$ behaves cleanly. Because $G_1 = \ker(F)$, it acts trivially on any Frobenius twist $M^{(1)}$. Thus, as a $G_1$-module, $V|_{G_1} \cong L(\lambda_0)^{\oplus \dim M}$.
3. Since $L(\lambda_0)$ is simple over $G_1$, any $G$-submodule $N \subset V$ must restrict to a $G_1$-submodule of the form $L(\lambda_0) \otimes W$ for some subspace $W \subset M$.
4. Because $N$ is $G$-invariant, the subspace $W$ must be invariant under the action of $G$ modulo $G_1$, which factors through $F$. Thus $W$ is a $G$-submodule of $M$. By induction on the length of the $p$-adic expansion, $M$ is irreducible, so $W = M$ (or 0). This forces $N = V$, proving the tensor product is an irreducible $G$-module.

The full proof is given in Steinberg [16] or Jantzen [12].

**Example 11.2.6: Steinberg's Theorem for $\mathrm{SL}_2$**

For $G = \mathrm{SL}_2$ in characteristic $p$, $X(T)^+ = \mathbb{Z}_{\geq 0}$ and $X_1(T) = \{0, 1, \ldots, p-1\}$. A dominant weight $d \geq 0$ has $p$-adic expansion $d = d_0 + d_1 p + d_2 p^2 + \cdots$ with $0 \leq d_j \leq p-1$. Steinberg's theorem gives:

$$L(d) \cong L(d_0) \otimes L(d_1)^{(1)} \otimes L(d_2)^{(2)} \otimes \cdots$$

For example, in characteristic $p = 3$:

1. $d = 7 = 1 + 2 \cdot 3$: $L(7) \cong L(1) \otimes L(2)^{(1)}$. Dimensions: $\dim L(1) = 2$, $\dim L(2) = 3$, $\dim L(7) = 2 \cdot 3 = 6$. (Note: $\dim L(7) = 6 \neq 8 = 7 + 1$; the "expected" dimension from the Weyl formula would be 8, but $L(7) \neq V(7)$ in characteristic 3.)
2. $d = 8 = 2 + 2 \cdot 3$: $L(8) \cong L(2) \otimes L(2)^{(1)}$. Dimensions: $\dim L(8) = 3 \cdot 3 = 9$.
3. $d = 13 = 1 + 1 \cdot 3 + 1 \cdot 9$: $L(13) \cong L(1) \otimes L(1)^{(1)} \otimes L(1)^{(2)}$. Dimensions: $\dim L(13) = 2 \cdot 2 \cdot 2 = 8$.
4. $d = 26 = 2 + 2 \cdot 3 + 2 \cdot 9 = 2(1 + 3 + 9) = 2 \cdot 13$: $L(26) \cong L(2) \otimes L(2)^{(1)} \otimes L(2)^{(2)}$. Dimensions: $\dim L(26) = 3^3 = 27 = p^3$. This is the Steinberg module for the third Frobenius kernel!

The pattern: the dimension of $L(d)$ is $\prod_j (d_j + 1)$ (the product of the dimensions of the restricted factors). This is generically much smaller than $d + 1$ (the Weyl module dimension).

**Example 11.2.7: Steinberg's Theorem for $\mathrm{SL}_3$ in Characteristic 2**

For $G = \mathrm{SL}_3$ and $p = 2$, the restricted weights are $X_1(T) = \{0, \omega_1, \omega_2, \omega_1 + \omega_2\}$ ($4 = 2^2$ weights). The restricted simple modules have dimensions $1, 3, 3, 8$ (respectively).

Consider $\lambda = 2\omega_1$. The 2-adic expansion is $\lambda = 0 + 2 \cdot \omega_1$ (with $\lambda_0 = 0$ and $\lambda_1 = \omega_1$). So:

$$L(2\omega_1) \cong L(0) \otimes L(\omega_1)^{(1)} = L(\omega_1)^{(1)}$$

This is just the Frobenius twist of the standard representation: $L(2\omega_1) \cong (k^3)^{(1)}$, a 3-dimensional module. Compare with the Weyl module: $\dim V(2\omega_1) = 6$ (from the Weyl dimension formula), but $\dim L(2\omega_1) = 3$. The Weyl module $V(2\omega_1)$ is reducible in characteristic 2, with composition factors $L(2\omega_1)$ (its 3-dimensional head) and $L(\omega_2)$ (its 3-dimensional radical).

**Corollary 11.2.8: Reduction to the Restricted Case**

The character problem for all simple $G$-modules reduces to the restricted case: if we know $\mathrm{ch}(L(\mu))$ for all $\mu \in X_1(T)$ (the finitely many $p$-restricted simple modules), then we know $\mathrm{ch}(L(\lambda))$ for all $\lambda \in X(T)^+$ via:

$$\mathrm{ch}(L(\lambda)) = \prod_{j=0}^{s} \mathrm{ch}(L(\lambda_j))^{(j)}$$

where $\mathrm{ch}(L(\lambda_j))^{(j)}$ means the character with all weights multiplied by $p^j$ (the Frobenius twist on the group ring: $e^\mu \mapsto e^{p^j \mu}$).

**Corollary 11.2.9: Dimension Formula**

For $\lambda = \lambda_0 + p\lambda_1 + \cdots + p^s\lambda_s$ with $\lambda_j \in X_1(T)$:

$$\dim L(\lambda) = \prod_{j=0}^{s} \dim L(\lambda_j)$$

The dimension is a *product* of restricted dimensions. In particular, $\dim L(\lambda) \leq \dim L(\mathrm{St})^{s+1} = p^{(s+1)|\Phi^+|}$.

The tensor product structure has a beautiful special case: the *$r$-th Steinberg module*.

**Definition 11.2.10: $r$-th Steinberg Module**

The *$r$-th Steinberg module* is $\mathrm{St}_r = L((p^r-1)\rho)$, the simple module with highest weight $(p^r-1)\rho$.

**Proposition 11.2.11: Properties of $\mathrm{St}_r$**

1. $\mathrm{St}_r \cong L((p-1)\rho) \otimes L((p-1)\rho)^{(1)} \otimes \cdots \otimes L((p-1)\rho)^{(r-1)} = \mathrm{St}_1^{\otimes \text{twisted}}$. That is, $\mathrm{St}_r$ is the tensor product of $r$ Frobenius twists of the ordinary Steinberg module.
2. $\dim \mathrm{St}_r = p^{r|\Phi^+|}$.
3. $\mathrm{St}_r$ is irreducible, projective, and injective.
4. $V((p^r-1)\rho) = L((p^r-1)\rho) = \mathrm{St}_r$ (the Weyl module is irreducible).

***Proof*:**
*(1):* The $p$-adic expansion of $(p^r-1)\rho = (p-1)\rho + p(p-1)\rho + \cdots + p^{r-1}(p-1)\rho$ has all digits equal to $(p-1)\rho \in X_1(T)$ (the Steinberg weight). Steinberg's theorem then gives the tensor product decomposition.

*(2):* $\dim \mathrm{St}_r = (\dim \mathrm{St}_1)^r = (p^{|\Phi^+|})^r = p^{r|\Phi^+|}$ (by corollary 11.2.9).

*(3) and (4):* These follow from the corresponding properties of $\mathrm{St}_1$ (theorem 11.1.4) and the fact that tensor products of projective modules are projective.

The Steinberg modules form a tower: $\mathrm{St}_1 \subset \mathrm{St}_2 \subset \cdots$ (as highest weights $(p-1)\rho \preceq (p^2-1)\rho \preceq \cdots$), with dimensions $p^{|\Phi^+|}, p^{2|\Phi^+|}, \ldots$, growing as powers of $p$. They are the "largest" irreducible modules at each Frobenius level, and they play the role of "free modules" in the modular theory (being projective-injective).

**Example 11.2.12: Dimensions of Simple Modules for $\mathrm{SL}_2$ in Characteristic 3**
For $G = \mathrm{SL}_2$, $p = 3$, the first 15 simple modules have dimensions:

| $d$ | $p$-adic expansion | $L(d) \cong$ | $\dim L(d)$ |
|---|---|---|---|
| 0 | 0 | $L(0)$ | 1 |
| 1 | 1 | $L(1)$ | 2 |
| 2 | 2 | $L(2) = \mathrm{St}_1$ | 3 |
| 3 | $10_3$ | $L(0) \otimes L(1)^{(1)}$ | $1 \cdot 2 = 2$ |
| 4 | $11_3$ | $L(1) \otimes L(1)^{(1)}$ | $2 \cdot 2 = 4$ |
| 5 | $12_3$ | $L(2) \otimes L(1)^{(1)}$ | $3 \cdot 2 = 6$ |
| 6 | $20_3$ | $L(0) \otimes L(2)^{(1)}$ | $1 \cdot 3 = 3$ |
| 7 | $21_3$ | $L(1) \otimes L(2)^{(1)}$ | $2 \cdot 3 = 6$ |
| 8 | $22_3$ | $L(2) \otimes L(2)^{(1)} = \mathrm{St}_2$ | $3 \cdot 3 = 9$ |
| 9 | $100_3$ | $L(0) \otimes L(0)^{(1)} \otimes L(1)^{(2)}$ | $1 \cdot 1 \cdot 2 = 2$ |
| 10 | $101_3$ | $L(1) \otimes L(0)^{(1)} \otimes L(1)^{(2)}$ | $2 \cdot 1 \cdot 2 = 4$ |
| 11 | $102_3$ | $L(2) \otimes L(0)^{(1)} \otimes L(1)^{(2)}$ | $3 \cdot 1 \cdot 2 = 6$ |
| 12 | $110_3$ | $L(0) \otimes L(1)^{(1)} \otimes L(1)^{(2)}$ | $1 \cdot 2 \cdot 2 = 4$ |
| 13 | $111_3$ | $L(1) \otimes L(1)^{(1)} \otimes L(1)^{(2)}$ | $2 \cdot 2 \cdot 2 = 8$ |
| 14 | $112_3$ | $L(2) \otimes L(1)^{(1)} \otimes L(1)^{(2)}$ | $3 \cdot 2 \cdot 2 = 12$ |

Compare with the Weyl module dimensions $\dim V(d) = d + 1$: the Steinberg dimensions are typically much smaller. The Steinberg modules appear at $d = 2, 8, 26, \ldots$ (i.e. $d = 3^r - 1$), with dimensions $3, 9, 27, \ldots = 3^r$.

## Exercise 11.2.1

1. Let $V = L(1)$ be the standard representation of $\mathrm{SL}_2$ (2-dimensional, weights $\{1, -1\}$). Compute the weights of $V^{(1)}$ and $V^{(2)}$ (the first and second Frobenius twists). Verify $V^{(1)} \cong L(p)$ and $V^{(2)} \cong L(p^2)$.

2. Write the $p$-adic expansion of $\lambda = 5\omega_1 + 3\omega_2$ for $\mathrm{SL}_3$ in characteristic $p = 3$. Apply Steinberg's theorem to express $L(\lambda)$ as a tensor product of Frobenius twists of restricted modules. Compute $\dim L(\lambda)$.

3. Verify Steinberg's theorem for $G = \mathrm{SL}_2$, $p = 2$, and $d = 5$: $5 = 1 + 0 \cdot 2 + 1 \cdot 4$ in base 2, so $L(5) \cong L(1) \otimes L(0)^{(1)} \otimes L(1)^{(2)} = L(1) \otimes L(1)^{(2)}$. What is $\dim L(5)$? Compare with $\dim V(5) = 6$.

4. Show that $\mathrm{St}_r \otimes L(\lambda)^{(r)} \cong L((p^r - 1)\rho + p^r\lambda)$ for any dominant $\lambda$. (*Hint*: use Steinberg's theorem on the weight $(p^r - 1)\rho + p^r\lambda$, whose $p$-adic digits are $(p-1)\rho$ for $j < r$ and $\lambda_0$ for $j = r$, etc.)

5. For $G = \mathrm{SL}_2$ in characteristic $p$, show that $\dim L(d) = \prod_j (d_j + 1)$ where $d = \sum d_j p^j$ is the base-$p$ expansion. Deduce that $\dim L(d) \leq p^{\lfloor \log_p d \rfloor + 1}$, and that equality holds if and only if $d = p^r - 1$ (a Steinberg weight).

6. Use Steinberg's theorem to compute $\mathrm{ch}(L(7))$ for $\mathrm{SL}_2$ in characteristic 3: $7 = 1 + 2 \cdot 3$, so $L(7) \cong L(1) \otimes L(2)^{(1)}$. The character is $\mathrm{ch}(L(1)) \cdot \mathrm{ch}(L(2))^{(1)}$ where $(\cdot)^{(1)}$ means replacing $t$ by $t^3$ in the Laurent polynomial. Compute explicitly and list all weights of $L(7)$ with multiplicities.

## 11.3 Lusztig's Conjecture and Beyond

Steinberg's tensor product theorem reduces the character problem to the restricted case: determining $\mathrm{ch}(L(\lambda))$ for $\lambda \in X_1(T)$, the finitely many $p$-restricted dominant weights. This is still a hard problem — the restricted region contains $p^r$ weights, and the composition multiplicities $[V(\lambda) : L(\mu)]$ exhibit subtle dependence on $p$ — but it is at least finite. Lusztig's conjecture, formulated in 1980, proposes an answer in terms of the *Kazhdan–Lusztig polynomials*, combinatorial objects arising from the Hecke algebra of the affine Weyl group. The conjecture has been proved for sufficiently large primes, but it fails for small primes in a way that remains incompletely understood.

### Kazhdan–Lusztig Polynomials

The statement of Lusztig's conjecture requires a brief detour through the combinatorics of Coxeter groups.

The affine Weyl group $W_p$ (introduced in §11.1) is a Coxeter group, generated by the simple reflections $\{s_0, s_1, \dots, s_r\}$ (where $s_1, \dots, s_r$ are the finite Weyl group reflections and $s_0$ is the affine reflection). As for any Coxeter group, Kazhdan and Lusztig (1979) defined a family of polynomials $\{P_{x,w}(q) \in \mathbb{Z}[q] : x, w \in W_p\}$ satisfying:

1. $P_{w,w}(q) = 1$ for all $w$.

2. $P_{x,w}(q) = 0$ unless $x \leq w$ in the Bruhat order.

3. $\deg P_{x,w} \leq \frac{1}{2}(\ell(w) - \ell(x) - 1)$ when $x < w$.

4. The polynomials are determined by a recursion involving the Bruhat order and the length function (we omit the precise recursion, which can be found in [10]).

The values $P_{x,w}(1)$ (evaluating at $q = 1$) are non-negative integers that control the composition multiplicities in Lusztig's conjecture.

## Statement of Lusztig's Conjecture

**Conjecture 11.3.1: Lusztig's Conjecture**

Assume $p \geq 2h - 2$, where $h$ is the *Coxeter number* of the root system (the order of a Coxeter element of $W$; see the table below). Let $\lambda \in X_1(T)$ be a $p$-restricted dominant weight lying in the closure of the fundamental alcove $\overline{C}_0$. Then the character of $L(\lambda)$ is:

$$\mathrm{ch}(L(\lambda)) = \sum_{w \in W_p,\, w\cdot 0 \leq \lambda} (-1)^{\ell(w_\lambda)-\ell(w)} Q_{w,w_\lambda}(1) \cdot \mathrm{ch}(V(w \cdot 0))$$

where $w_\lambda$ is the element of $W_p$ with $w_\lambda \cdot 0 = \lambda$, the sum is over those $w \in W_p$ with $w \cdot 0$ dominant and $\leq \lambda$ in the dominance order, and $Q_{x,w}(q)$ are the *inverse Kazhdan–Lusztig polynomials*.

The formula expresses $\mathrm{ch}(L(\lambda))$ as a $\mathbb{Z}$-linear combination of Weyl module characters $\mathrm{ch}(V(\mu))$ (which are known by the Weyl character formula in all characteristics), with coefficients given by the inverse Kazhdan–Lusztig polynomials evaluated at 1. The formula is the positive-characteristic analogue of the BGG resolution from chapter 8: where the BGG resolution expressed $\mathrm{ch}(L(\lambda))$ as an alternating sum of Verma module characters (with coefficients $\pm 1$), Lusztig's formula replaces the $\pm 1$ by the (generally larger) inverse KL polynomial values.

An equivalent and perhaps more transparent formulation is in terms of the *decomposition numbers* (using the standard KL polynomials rather than their inverses):

**Corollary 11.3.2: Decomposition Numbers from KL Polynomials**

Under the hypotheses of Lusztig's conjecture, the decomposition numbers $[V(\lambda) : L(\mu)]$ (the composition multiplicities of $L(\mu)$ in $V(\lambda)$) are given by:

$$[V(\lambda) : L(\mu)] = P_{w_\mu, w_\lambda}(1)$$

where $w_\mu, w_\lambda \in W_p$ satisfy $w_\mu \cdot 0 = \mu$ and $w_\lambda \cdot 0 = \lambda$.

This is an extraordinary statement: the decomposition matrix of a reductive group in characteristic $p$ (an object of algebra and geometry) is determined by the combinatorics of a Coxeter group (an object of combinatorial group theory). The KL polynomials can be computed algorithmically (by their defining recursion), so the conjecture, when it holds, gives an *effective* answer to the character problem.

## The Coxeter Number

The Coxeter number $h$ appears in the bound $p \geq 2h - 2$. It is a basic invariant of the root system:

| **Type** | $A_n$ | $B_n$ | $C_n$ | $D_n$ | $G_2$ | $F_4$ | $E_6$ | $E_7$ | $E_8$ |
|---|---|---|---|---|---|---|---|---|---|
| $h$ | $n+1$ | $2n$ | $2n$ | $2n-2$ | 6 | 12 | 12 | 18 | 30 |
| $2h-2$ | $2n$ | $4n-2$ | $4n-2$ | $4n-6$ | 10 | 22 | 22 | 34 | 58 |

For $\mathrm{SL}_n$ (type $A_{n-1}$): $h = n$, so the bound is $p \geq 2n - 2$. For $\mathrm{SL}_2$: $p \geq 2$ (all odd primes). For $\mathrm{SL}_3$: $p \geq 4$, i.e. $p \geq 5$. For $E_8$: $p \geq 58$, a substantial restriction.

## Status of the Conjecture

Lusztig's conjecture has a rich and surprising history. We summarize the main developments.

*The conjecture is true for p sufficiently large.* This was proved in stages:

1. Andersen–Jantzen–Soergel [2] proved the conjecture for $p \gg 0$ (larger than an unspecified bound depending on the root system), using the theory of *tilting modules* and a translation to the *quantum group* setting at a root of unity.
2. Fiebig [9, 8] made the bound explicit: the conjecture holds for $p$ larger than an explicitly computable (but astronomically large) bound depending on the root system. For type $A_n$, the bound is roughly of the order of $n^{n^2}$ (far larger than $2h - 2 = 2n$).
3. The "correct" bound remains unknown. It is believed that the conjecture should hold for $p \geq h$ (the Coxeter number), which is much smaller than Fiebig's bound but larger than Lusztig's original $2h - 2$.

*The conjecture fails for small primes.* This was a major surprise:

1. Lusztig [13] originally conjectured the formula for $p \geq h$. The bound $p \geq 2h-2$ appeared in later proofs establishing the conjecture for large primes via the geometry of affine Grassmannians. However, monumental work by Williamson [19] demonstrated that the conjecture actually fails for primes well above these linear bounds.
2. Williamson [19] constructed explicit counterexamples for primes $p$ that grow exponentially with the rank of the group, meaning any linear bound like $p \geq 2h - 2$ or $p \geq h$ is ultimately insufficient for general groups of large rank. For example, the smallest known counterexample for $\mathrm{SL}_n$ occurs for $n$ around 100 and $p$ around 1000.
3. The counterexamples arise from *p-torsion in the intersection cohomology of Schubert varieties*: the KL polynomials compute the correct answer over $\mathbb{Q}$ (for "generic" $p$), but when $p$ divides certain torsion primes in the topology of Schubert varieties, the actual decomposition numbers differ from the KL prediction.

*The current picture.* The field is now focused on finding the "correct" replacement for Lusztig's conjecture at small primes. The leading candidates involve:

1. *p-Kazhdan–Lusztig polynomials*: defined by Williamson and collaborators, these are characteristic-$p$ analogues of the KL polynomials that account for torsion. They are conjectured to give the correct decomposition numbers for *all* $p \geq h$.
2. *Diagrammatic categories*: the theory of *Soergel bimodules* and their diagrammatic calculus (Elias–Williamson) provides a computational framework for the $p$-KL polynomials, making them at least in principle computable.

3. *Geometric approaches*: the geometric Satake equivalence and the theory of *parity sheaves* (Juteau–Mautner–Williamson) provide a geometric framework in which both the classical and $p$-KL theories are unified.

The problem remains wide open for "medium" primes (between $h$ and the Fiebig bound), and the complete determination of the decomposition matrix for any fixed group and prime remains a formidable computational challenge.

## The Jantzen Filtration

Even without a complete answer to the character problem, there is a powerful structural tool that organizes the composition factors of a Weyl module: the *Jantzen filtration*.

**Theorem 11.3.3: The Jantzen Filtration**

Let $\lambda \in X(T)^+$ be a dominant weight. The Weyl module $V(\lambda)$ admits a descending filtration

$$V(\lambda) = V(\lambda)^0 \supsetneq V(\lambda)^1 \supseteq V(\lambda)^2 \supseteq \cdots \supseteq V(\lambda)^N = 0$$

by $G$-submodules, satisfying:

1. $V(\lambda)^0/V(\lambda)^1 \cong L(\lambda)$ (the head of the filtration is the simple module).
2. (The Jantzen sum formula) $\sum_{i\geq 1} \mathrm{ch}(V(\lambda)^i) = \sum_{\alpha\in\Phi^+} \sum_{0<mp\leq\langle\lambda+\rho,\alpha^\vee\rangle} \nu_p(mp) \cdot \mathrm{ch}(V(s_{\alpha,mp}\cdot\lambda))$, where $\nu_p(n)$ is the $p$-adic valuation of $n$ and $s_{\alpha,mp}$ is the affine reflection through the hyperplane $\langle x+\rho,\alpha^\vee\rangle = mp$.

The Jantzen sum formula does not determine the individual layers $V(\lambda)^i/V(\lambda)^{i+1}$, but it determines their *sum* (as virtual characters). This gives strong constraints on the decomposition matrix: the sum formula, combined with the linkage principle and other structural results, often suffices to determine the composition factors in small cases.

**Example 11.3.4: Jantzen Filtration for $\mathrm{SL}_2$**

For $G = \mathrm{SL}_2$ in characteristic $p$ and $\lambda = p$: $V(p) = \mathrm{Sym}^p(k^2)$ has dimension $p+1$. The Jantzen filtration is:

$$V(p) = V(p)^0 \supsetneq V(p)^1 \supsetneq V(p)^2 = 0$$

with $V(p)^0/V(p)^1 = L(p)$ (dimension 2) and $V(p)^1 = L(p-2)$ (dimension $p-1$). The sum formula: $\mathrm{ch}(V(p)^1) = \mathrm{ch}(V(p-2))$ (a single term, since $\langle p+\rho,\alpha^\vee\rangle = p+1$ and the only reflection giving a dominant weight is $s_{\alpha,p}$). Since $V(p-2) = L(p-2)$ (irreducible for $p-2 < p$), this confirms $V(p)^1 \cong L(p-2)$ and $[V(p) : L(p)] = 1$, $[V(p) : L(p-2)] = 1$.

## Further Directions

We close with brief pointers to the deeper aspects of the theory, beyond the scope of this book.

*Tilting modules.* A *tilting module* is a $G$-module that has both a Weyl filtration (a filtration with Weyl module quotients $V(\lambda)$) and a good filtration (a filtration with induced module quotients $H^0(\lambda)$).

The indecomposable tilting modules $\{T(\lambda) : \lambda \in X(T)^+\}$ are parametrized by dominant weights, and their characters are computable from the KL theory. They play the role of "projective covers" in a suitable truncated category and are the key link between the algebraic group theory and the quantum group theory at roots of unity.

*Quantum groups and the Lusztig–Kashiwara theory.* The quantum group $U_q(\mathfrak{g})$ (a $q$-deformation of the universal enveloping algebra) specializes at $q = 1$ to the classical $U(\mathfrak{g})$, and at $q$ a $p$-th root of unity to a structure closely related to $G$ in characteristic $p$. The representation theory at roots of unity was developed by Lusztig and Kashiwara independently, using *crystal bases* and *canonical bases*, and it is through this connection that the KL polynomials enter the characteristic-$p$ theory. The precise link is the *Lusztig character formula* for quantum groups at roots of unity, which is a theorem (proved by Kazhdan–Lusztig and Kashiwara–Tanisaki), and Lusztig's conjecture is the assertion that this formula "descends" to the algebraic group in characteristic $p$.

*The geometric Satake equivalence.* The most conceptual approach to the representation theory of $G$ in characteristic $p$ uses the *geometric Satake equivalence*: an equivalence of tensor categories between $\mathrm{Rep}(G^\vee)$ (representations of the Langlands dual group) and the category of *perverse sheaves* on the affine Grassmannian $\mathrm{Gr}_G = G(\!(t)\!)/G[\![t]\!]$. This equivalence, due to Lusztig, Ginzburg, and Mirković–Vilonen, provides a geometric construction of representations that works in all characteristics and gives a conceptual explanation for the appearance of the KL polynomials.

*References.* For the full development of the modular theory, the standard reference is Jantzen [12]. For the KL theory and its applications, see Humphreys [10] (the classical setting) and Williamson's survey articles for the modern developments. For the geometric perspective, see Mirković–Vilonen and the exposition in Zhu's lectures on the geometric Satake equivalence.

## Exercise 11.3.1

1. Verify the Coxeter numbers $h$ for types $A_3$, $B_3$, $C_3$, and $D_4$ from the table. (*Hint*: $h = 1 + \sum_{i=1}^r n_i$ where $\tilde{\alpha} = \sum n_i \alpha_i$ is the highest root in terms of simple roots. For $A_3$: $\tilde{\alpha} = \alpha_1 + \alpha_2 + \alpha_3$, so $h = 1 + 1 + 1 + 1 = 4$.)
2. For $G = \mathrm{SL}_5$ (type $A_4$), what is the Lusztig bound $p \geq 2h - 2$? List the primes $p$ for which the conjecture is *not* guaranteed by this bound.
3. The KL polynomial $P_{e,s_i}(q) = 1$ for any simple reflection $s_i$ (since $\ell(s_i) - \ell(e) = 1$ and the degree bound forces $P = \text{constant} = 1$). What does this say about the decomposition number $[V(s_i \cdot 0) : L(0)]$ according to Lusztig's conjecture? Interpret this for $G = \mathrm{SL}_2$ and $p \geq 3$: what is $s_0 \cdot 0$ and what is $[V(s_0 \cdot 0) : L(0)]$?
4. Compute the Jantzen filtration of $V(2p)$ for $\mathrm{SL}_2$ in characteristic $p \geq 3$. (*Hint*: $V(2p) = \mathrm{Sym}^{2p}(k^2)$ has dimension $2p + 1$. By Steinberg's theorem, $L(2p) = L(0) \otimes L(2)^{(1)}$, which has dimension 3. The radical $V(2p)^1$ has dimension $2p - 2$, and its composition factors can be determined from the linkage principle and the sum formula.)
5. For $G = \mathrm{SL}_3$ in characteristic $p = 2$, there are 4 restricted dominant weights: $\{0, \omega_1, \omega_2, \omega_1 + \omega_2\}$. Determine which Weyl modules $V(\lambda)$ are irreducible and which are not. (*Hint*: $V(0)$, $V(\omega_1)$, $V(\omega_2)$ are irreducible since their dimensions are $1, 3, 3$, which are small enough to prevent submodules. For $V(\omega_1 + \omega_2)$: $\dim V(\omega_1 + \omega_2) = 8$, and one must check whether the adjoint

representation of $\mathrm{SL}_3$ is reducible in characteristic 2.)

6. (Essay question.) The character problem in characteristic $p$ was "expected" to have a clean answer (Lusztig's conjecture), but Williamson's counterexamples showed that the answer depends on arithmetic properties of $p$ (specifically, torsion in the intersection cohomology of Schubert varieties). In what sense is this a manifestation of the theme "the Lie algebra is an imperfect tool in characteristic $p$" that has run through the book since Chapter 3? How does the failure of Lusztig's conjecture for small $p$ relate to the failure of the exponential map (example 5.3.10), the failure of Cartier's theorem (theorem 3.3.2), and the failure of complete reducibility (theorem 8.3.1)?

# Part III

# Appendix

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# *Bibliography*


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